

Cytokines

THE DEVELOPMENT OF AN EFFECTIVE IMMUNE response involves lymphoid cells, inflammatory cells, and hematopoietic cells. The complex interactions among these cells are mediated by a group of proteins collectively designated **cytokines** to denote their role in cell-to-cell communication. Cytokines are low-molecular-weight regulatory proteins or glycoproteins secreted by white blood cells and various other cells in the body in response to a number of stimuli. These proteins assist in regulating the development of immune effector cells, and some cytokines possess direct effector functions of their own.

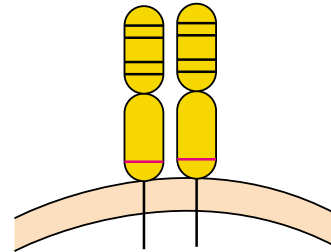
This chapter focuses on the biological activity of cytokines, the structure of cytokines and their receptors, signal transduction by cytokine receptors, the role of cytokine abnormalities in the pathogenesis of certain diseases, and therapeutic uses of cytokines or their receptors. The important role of cytokines in the inflammatory response is described in Chapter 15.

Properties of Cytokines

Cytokines bind to specific receptors on the membrane of target cells, triggering signal-transduction pathways that ultimately alter gene expression in the target cells (Figure 12-1a). The susceptibility of the target cell to a particular cytokine is determined by the presence of specific membrane receptors. In general, the cytokines and their receptors exhibit very high affinity for each other, with dissociation constants ranging from 10^{-10} to 10^{-12} M. Because their affinities are so high, cytokines can mediate biological effects at picomolar concentrations.

A particular cytokine may bind to receptors on the membrane of the same cell that secreted it, exerting **autocrine** action; it may bind to receptors on a target cell in close proximity to the producer cell, exerting **paracrine** action; in a few cases, it may bind to target cells in distant parts of the body, exerting **endocrine** action (Figure 12-1b). Cytokines regulate the intensity and duration of the immune response by stimulating or inhibiting the activation, proliferation, and/or differentiation of various cells and by regulating the secretion of antibodies or other cytokines. As described later, binding of a given cytokine to responsive target cells generally stimulates increased expression of cytokine receptors and secretion of other cytokines, which affect other target cells in turn. Thus, the cytokines secreted by even a small number of lymphocytes activated by antigen can influence

chapter 12



Class I Cytokine Receptors

- Properties of Cytokines
- Cytokine Receptors
- Cytokine Antagonists
- Cytokine Secretion by T_H1 and T_H2 Subsets
- Cytokine-Related Diseases
- Therapeutic Uses of Cytokines and Their Receptors
- Cytokines in Hematopoiesis

the activity of numerous cells involved in the immune response. For example, cytokines produced by activated T_H cells can influence the activity of B cells, T_C cells, natural killer cells, macrophages, granulocytes, and hematopoietic stem cells, thereby activating an entire network of interacting cells.

Cytokines exhibit the attributes of pleiotropy, redundancy, synergy, antagonism, and cascade induction, which permit them to regulate cellular activity in a coordinated, interactive way (Figure 12-2). A given cytokine that has different biological effects on different target cells has a pleiotropic action. Two or more cytokines that mediate similar functions are said to be redundant; redundancy makes it difficult to ascribe a particular activity to a single cytokine. Cytokine synergism occurs when the combined effect of two cytokines on cellular activity is greater than the additive

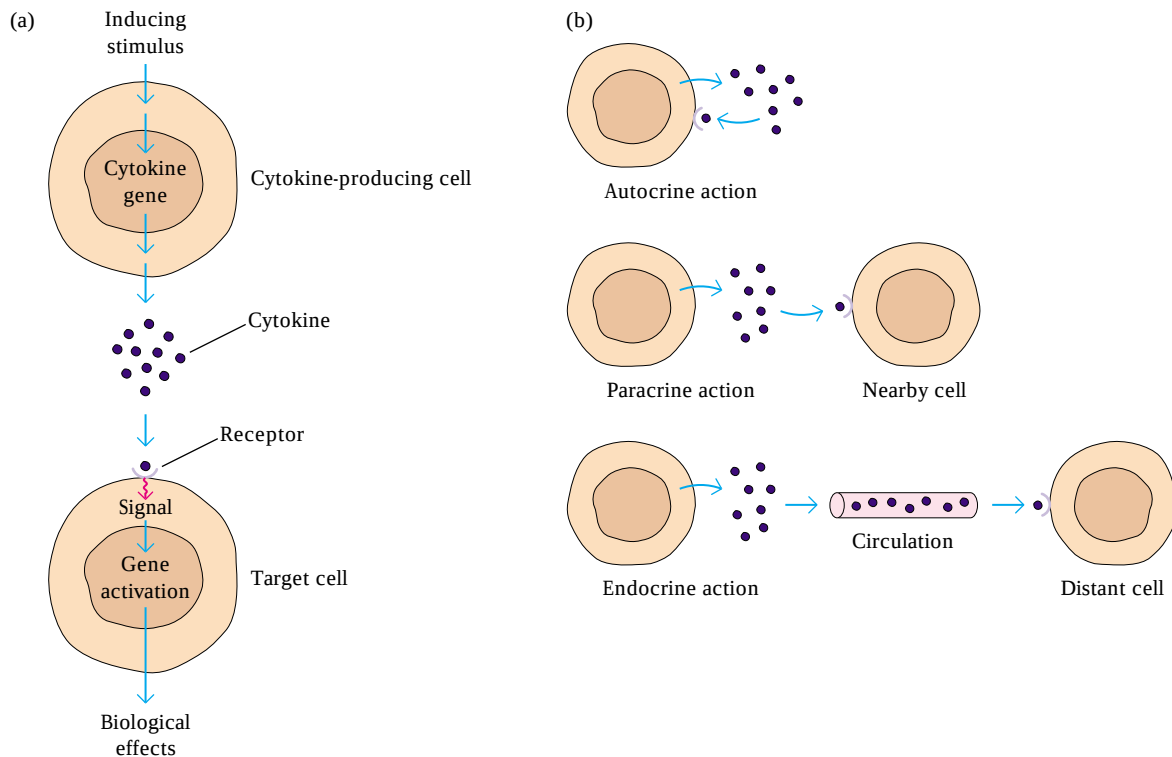


FIGURE 12-1 (a) Overview of the induction and function of cytokines. (b) Most cytokines exhibit autocrine and/or paracrine action; fewer exhibit endocrine action.

effects of the individual cytokines. In some cases, cytokines exhibit antagonism; that is, the effects of one cytokine inhibit or offset the effects of another cytokine. Cascade induction occurs when the action of one cytokine on a target cell induces that cell to produce one or more other cytokines, which in turn may induce other target cells to produce other cytokines.

The term *cytokine* encompasses those cytokines secreted by lymphocytes, substances formerly known as **lymphokines**, and those secreted by monocytes and macrophages, substances formerly known as **monokines**. Although these other two terms continue to be used, they are misleading because secretion of many lymphokines and monokines is not limited to lymphocytes and monocytes as these terms imply, but extends to a broad spectrum of cells and types. For this reason, the more inclusive term *cytokine* is preferred.

Many cytokines are referred to as **interleukins**, a name indicating that they are secreted by some leukocytes and act upon other leukocytes. Interleukins 1–25 have been identified. There is reason to suppose that still other cytokines will be discovered and that the interleukin group will expand further. Some cytokines are known by common names, including the interferons and tumor necrosis factors. Recently gaining prominence is yet another another subgroup

of cytokines, the **chemokines**, a group of low-molecular-weight cytokines that affect chemotaxis and other aspects of leukocyte behavior. These molecules play an important role in the inflammatory response and are described in Chapter 15.

Because cytokines share many properties with hormones and growth factors, the distinction between these three classes of mediators is often blurred. All three are secreted soluble factors that elicit their biological effects at picomolar concentrations by binding to receptors on target cells. Growth factors tend to be produced constitutively, whereas cytokines and hormones are secreted in response to discrete stimuli, and secretion is short-lived, generally ranging from a few hours to a few days. Unlike hormones, which generally act long range in an endocrine fashion, most cytokines act over a short distance in an autocrine or paracrine fashion. In addition, most hormones are produced by specialized glands and tend to have a unique action on one or a few types of target cell. In contrast, cytokines are often produced by, and bind to, a variety of cells.

The activity of cytokines was first recognized in the mid-1960s, when supernatants derived from *in vitro* cultures of lymphocytes were found to contain factors that could regulate proliferation, differentiation, and maturation of allogeneic

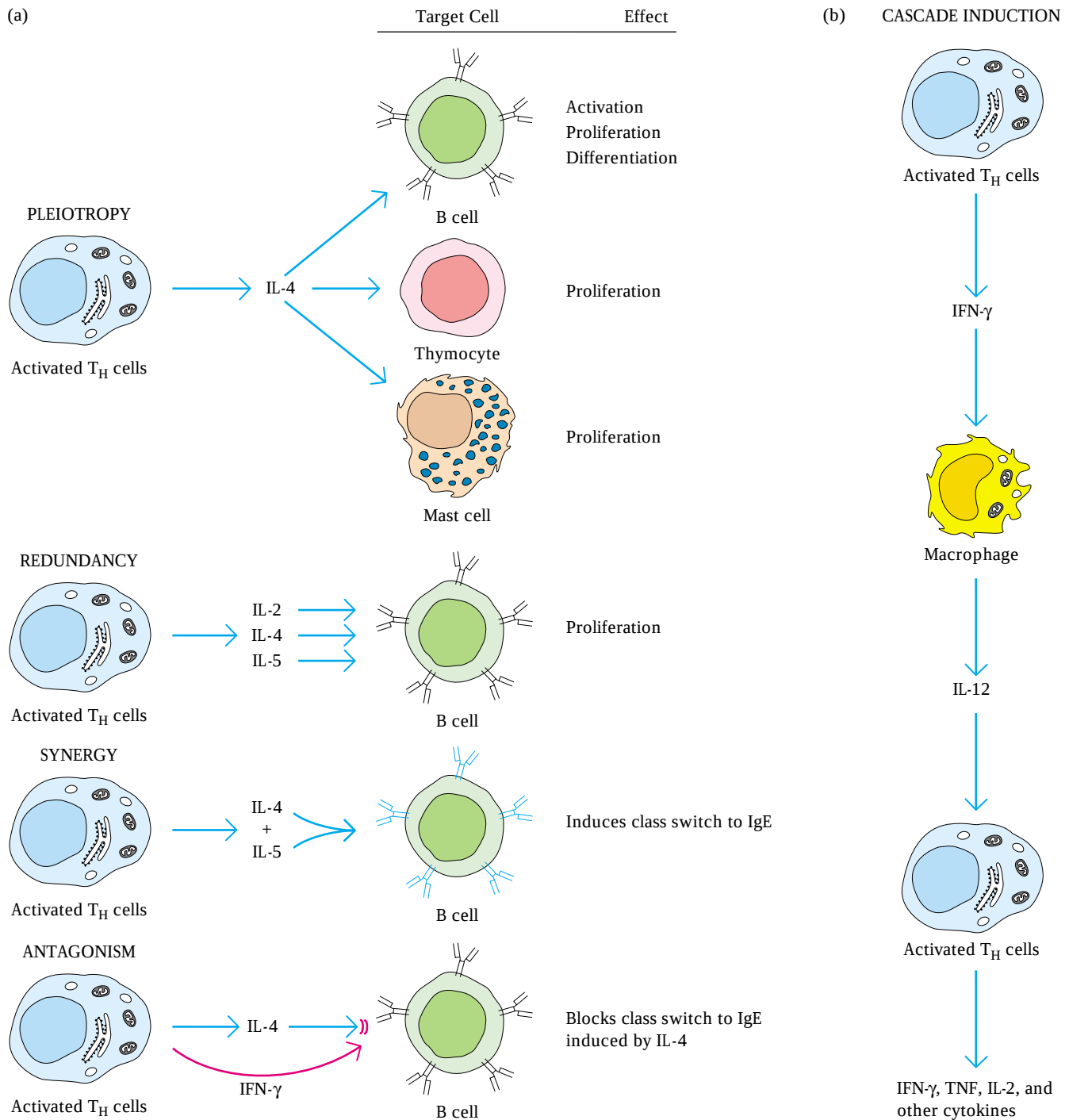


FIGURE 12-2 Cytokine attributes of (a) pleiotropy, redundancy, synergy (synergism), antagonism, and (b) cascade induction.

immune-system cells. Soon after, it was discovered that production of these factors by cultured lymphocytes was induced by activation with antigen or with nonspecific mitogens. Biochemical isolation and purification of cytokines was hampered because of their low concentration in culture super-

natants and the absence of well-defined assay systems for individual cytokines. A great advance was made with the development of gene-cloning techniques during the 1970s and 1980s, which made it possible to produce pure cytokines by expressing the protein from cloned genes. The discovery of

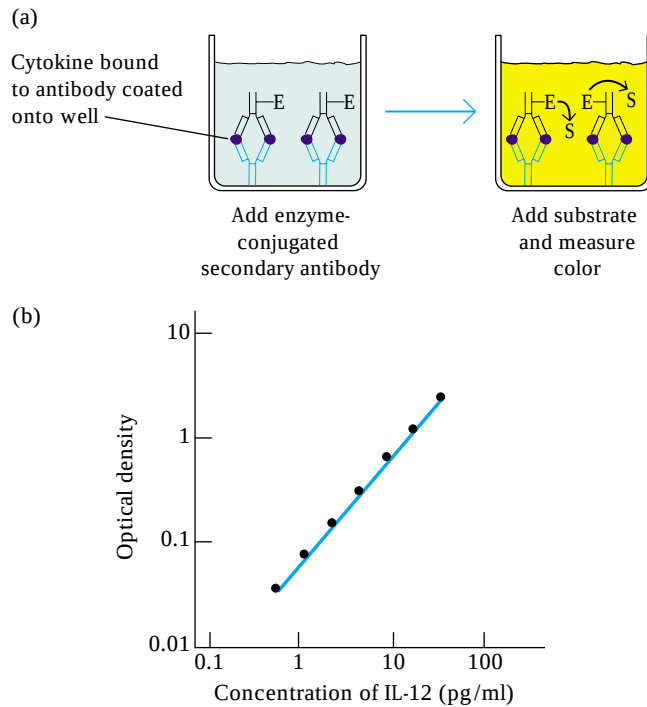


FIGURE 12-3 ELISA assay of a cytokine. (a) The sample containing the cytokine of interest is captured by specific antibody (blue) coated onto wells of a microtiter plate. A second specific antibody (blue), conjugated to an enzyme (E) such as horseradish peroxidase, forms a sandwich with the captured cytokine, immobilizing the enzyme in the microtiter well. A chromogenic substrate (S) is added, and the enzyme generates a color whose intensity is proportional to the amount of cytokine bound to the capture antibody. The optical density of this color produced by the unknown is compared with values on an appropriately determined standard curve. (b) The standard curve shown here is for human interleukin 12 (IL-12). It is clear that this assay is sufficiently sensitive to detect as little as 1 picogram of IL-12. [Part (b) courtesy of R&D Systems.]

cell lines whose growth depended on the presence of a particular cytokine provided researchers with the first simple assay systems. The derivation of monoclonal antibodies specific for each of the more important cytokines has made it possible to develop rapid quantitative immunoassays for each of them (Figure 12-3).

Cytokines Belong to Four Structural Families

Once the genes encoding various cytokines had been cloned, sufficient quantities of purified preparations became available for detailed studies on their structure and function. Cytokines generally have a molecular mass of less than 30 kDa. Structural studies have shown that the cytokines

characterized so far belong to one of four groups: the hematopoietin family, the interferon family, the chemokine family, or the tumor necrosis factor family.

The structures of two members of the hematopoietin family, IL-2 and IL-4, are depicted in Figure 12-4. Although the amino acid sequences of these family members differ considerably, all of them have a high degree of α -helical structure and little or no β -sheet structure. The molecules

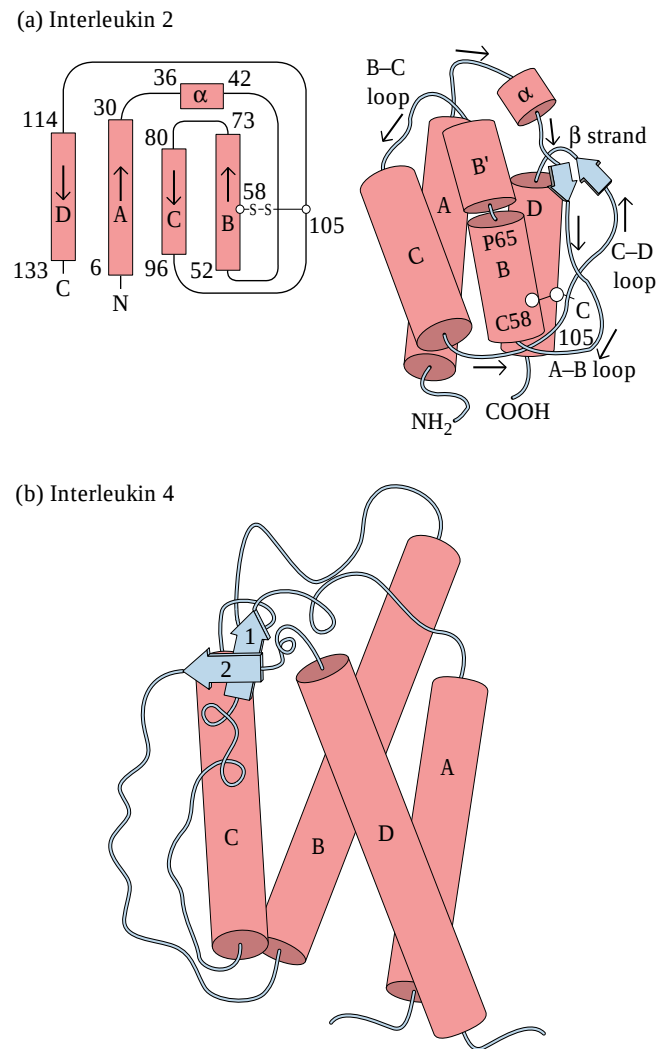


FIGURE 12-4 Several representations of structures in the hematopoietin family. (a) *Left*: Topographical representation of the primary structure of IL-2 showing α -helical regions (α and A-D) and connecting chains of the molecule. *Right*: Proposed three-dimensional model of IL-2. (b) Ribbon model of IL-4 deduced from x-ray crystallographic analysis of the molecule. In (a) and (b) the α helices are shown in red and the β sheets in blue. The structures of other cytokines belonging to the hematopoietin family are thought to be generally similar. [Part (b) from J. L. Boulay and W. E. Paul, 1993, *Curr. Biol.* **3**:573.]



VISUALIZING CONCEPTS

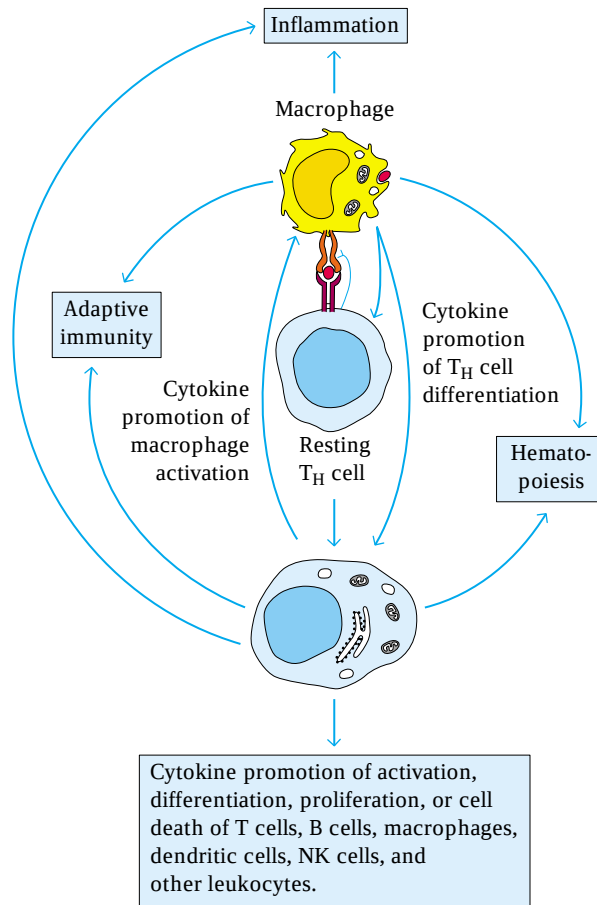


FIGURE 12-5 Interaction of antigen with macrophages and the subsequent activation of resting T_H cells leads to release of nu-

merous cytokines (blue arrows), generating a complex network of interacting cells in the immune response.

share a similar polypeptide fold, with four α -helical regions (A–D) in which the first and second helices and the third and fourth helices run roughly parallel to one another and are connected by loops.

Cytokines Have Numerous Biological Functions

Although a variety of cells can secrete cytokines, the two principal producers are the T_H cell and the macrophage. Cytokines released from these two cell types activate an entire network of interacting cells (Figure 12-5). Among the numerous physiologic responses that require cytokine involvement are development of cellular and humoral immune responses, induction of the inflammatory response, regulation of hematopoiesis, control of cellular proliferation

and differentiation, and the healing of wounds. Although the immune response to a specific antigen may include the production of cytokines, it is important to remember that cytokines act in an antigen-nonspecific manner. That is, they affect whatever cells they encounter that bear appropriate receptors and are in a physiological state that allows them to respond.

Cytokines are involved in a staggeringly broad array of biological activities including innate immunity, adaptive immunity, inflammation, and hematopoiesis. Altogether, the total number of proteins with cytokine activity easily exceeds 100 and research continues to uncover new ones. Table 12-1 summarizes the activities of some cytokines and places them into functional groups. An expanded list of cytokines can be found in the Appendix. It should be kept in mind that most of the listed functions have been identified from analysis of

TABLE 12-1 Functional groups of selected cytokines¹

Cytokine*	Secreted by**	Targets and effects
SOME CYTOKINES OF INNATE IMMUNITY		
Interleukin 1 (IL-1)	Monocytes, macrophages, endothelial cells, epithelial cells	Vasculature (inflammation); hypothalamus (fever); liver (induction of acute phase proteins)
Tumor Necrosis Factor- α (TNF- α)	Macrophages	Vasculature (inflammation); liver (induction of acute phase proteins); loss of muscle, body fat (cachexia); induction of death in many cell types; neutrophil activation
Interleukin 12 (IL-12)	Macrophages, dendritic cells	NK cells; influences adaptive immunity (promotes T _H 1 subset)
Interleukin 6 (IL-6)	Macrophages, endothelial cells	Liver (induces acute phase proteins); influences adaptive immunity (proliferation and antibody secretion of B cell lineage)
Interferon α (IFN- α) (This is a family of molecules)	Macrophages	Induces an antiviral state in most nucleated cells; increases MHC class I expression; activates NK cells
Interferon β (IFN- β)	Fibroblasts	Induces an antiviral state in most nucleated cells; increases MHC class I expression; activates NK cells
SOME CYTOKINES OF ADAPTIVE IMMUNITY		
Interleukin 2 (IL-2)	T cells	T-cell proliferation; can promote AICD. NK cell activation and proliferation; B-cell proliferation
Interleukin 4 (IL-4)	T _H 2 cells; mast cells	Promotes T _H 2 differentiation; isotype switch to IgE
Interleukin 5 (IL-5)	T _H 2 cells	Eosinophil activation and generation
Interleukin 25 (IL-25)	Unknown	Induces secretion of T _H 2 cytokine profile
Transforming growth factor β (TGF- β)	T cells, macrophages, other cell types	Inhibits T-cell proliferation and effector functions; inhibits B-cell proliferation; promotes isotype switch to IgE; inhibits macrophages
Interferon γ (IFN- γ)	T _H 1 cells; CD8 ⁺ cells; NK cells	Activates macrophages; increases expression MHC class I and class II molecules; increases antigen presentation

¹Many cytokines play roles in more than one functional category.

*Only the major cell types providing cytokines for the indicated activity are listed; other cell types may also have the capacity to synthesize the given cytokine.

**Also note that activated cells generally secrete greater amounts of cytokine than unactivated cells.

the effects of recombinant cytokines, often at nonphysiologic concentrations, added individually to in vitro systems. In vivo, however, cytokines rarely, if ever, act alone. Instead, a target cell is exposed to a milieu containing a mixture of cytokines, whose combined synergistic or antagonistic effects can have very different consequences. In addition, cytokines often induce the synthesis of other cytokines, resulting in cascades of activity.

The nonspecificity of cytokines seemingly conflicts with the established specificity of the immune system. What keeps the nonspecific cytokines from activating cells in a nonspecific fashion during the immune response? One way in which specificity is maintained is by careful regulation of the expression of cytokine receptors on cells. Often cytokine receptors are expressed on a cell only after that cell has interacted with antigen. In this way cytokine activation is limited to

antigen-activated lymphocytes. Another means of maintaining specificity may be a requirement for direct interaction between the cytokine-producing cell and the target cell to trigger cytokine secretion, thus ensuring that effective concentrations of the cytokine are released only in the vicinity of the intended target. In the case of the T_H cell, a major producer of cytokines, close cellular interaction occurs when the T-cell receptor recognizes an antigen-MHC complex on an appropriate antigen-presenting cell, such as a macrophage, dendritic cell, or B lymphocyte. Cytokines secreted at the junction of these interacting cells reach high enough local concentrations to affect the target APC but not more distant cells. In addition, the half-life of cytokines in the bloodstream or other extracellular fluids into which they are secreted is usually very short, ensuring that they act for only a limited period of time and thus over a short distance.

Cytokine Receptors

As noted already, to exert their biological effects, cytokines must first bind to specific receptors expressed on the membrane of responsive target cells. Because these receptors are expressed by many types of cells, the cytokines can affect a diverse array of cells. Biochemical characterization of cytokine receptors initially progressed at a very slow pace because their levels on the membrane of responsive cells is quite low. As with the cytokines themselves, cloning of the genes encoding cytokine receptors has led to rapid advances in the identification and characterization of these receptors.

Cytokine Receptors Fall Within Five Families

Receptors for the various cytokines are quite diverse structurally, but almost all belong to one of five families of receptor proteins (Figure 12-6):

- Immunoglobulin superfamily receptors
- Class I cytokine receptor family (also known as the hematopoietin receptor family)
- Class II cytokine receptor family (also known as the interferon receptor family)
- TNF receptor family
- Chemokine receptor family

Many of the cytokine-binding receptors that function in the immune and hematopoietic systems belong to the class I cytokine receptor family. The members of this receptor family have conserved amino acid sequence motifs in the extracellular domain consisting of four positionally conserved cysteine residues (CCCC) and a conserved sequence of tryptophan-serine-(any amino acid)-tryptophan-serine (WSXWS, where X is the nonconserved amino acid). The receptors for all the cytokines classified as hematopoietins belong to the class I cytokine receptor family, which also is called the hematopoietin receptor family. The class II cytokine receptors possess the conserved CCCC motifs, but lack the WSXWS motif present in class I cytokine receptors. Initially only the three interferons, α , β , and γ , were thought to be ligands for these receptors. However, recent work has shown that the IL-10 receptor is also a member of this group.

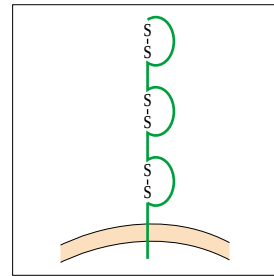
Another feature common to most of the hematopoietin (class I cytokine) and the class II cytokine receptor families is

FIGURE 12-6 Schematic diagrams showing the structural features that define the five types of receptor proteins to which most cytokines bind. The receptors for most of the interleukins belong to the class I cytokine receptor family. C refers to conserved cysteine.

RECEPTOR FAMILY

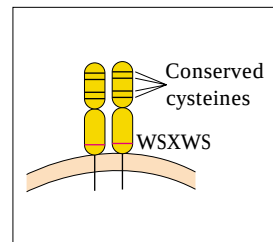
LIGANDS

(a) Immunoglobulin superfamily receptors



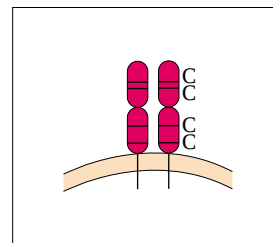
IL-1
M-CSF
C-Kit

(b) Class I cytokine receptors (hematopoietin)



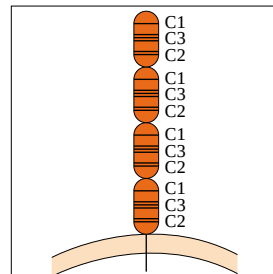
IL-2 IL-13
IL-3 IL-15
IL-4 GM-CSF
IL-5 G-CSF
IL-6 OSM
IL-7 LIF
IL-9 CNTF
IL-11 Growth hormone
IL-12 Prolactin

(c) Class II cytokine receptors (interferon)



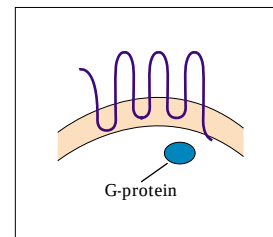
IFN- α
IFN- β
IFN- γ
IL-10

(d) TNF receptors



TNF- α
TNF- β
CD40
Nerve growth factor (NGF)
FAS

(e) Chemokine receptors



IL-8
RANTES
MIP-1
PF4
MCAF
NAP-2

multiple subunits, often including one subunit that binds specific cytokine molecules and another that mediates signal transduction. Note, however, that these two functions are not always confined to one subunit or the other. Engagement of all of the class I and class II cytokine receptors studied to date has been shown to induce tyrosine phosphorylation of the receptor through the activity of protein tyrosine kinases closely associated with the cytosolic domain of the receptors.

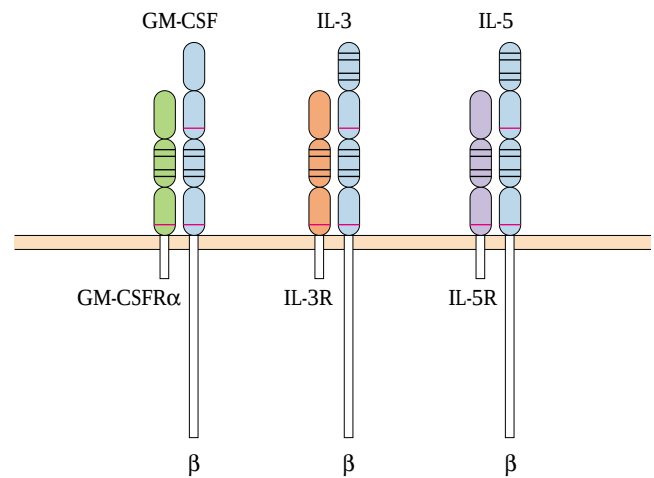
Subfamilies of Class I Cytokine Receptors Have Signaling Subunits in Common

Several subfamilies of class I cytokine receptors have been identified, with all the receptors in a subfamily having an identical signal-transducing subunit. Figure 12-7 schematically illustrates the members of three receptor subfamilies, named after GM-CSF, IL-2, and IL-6.

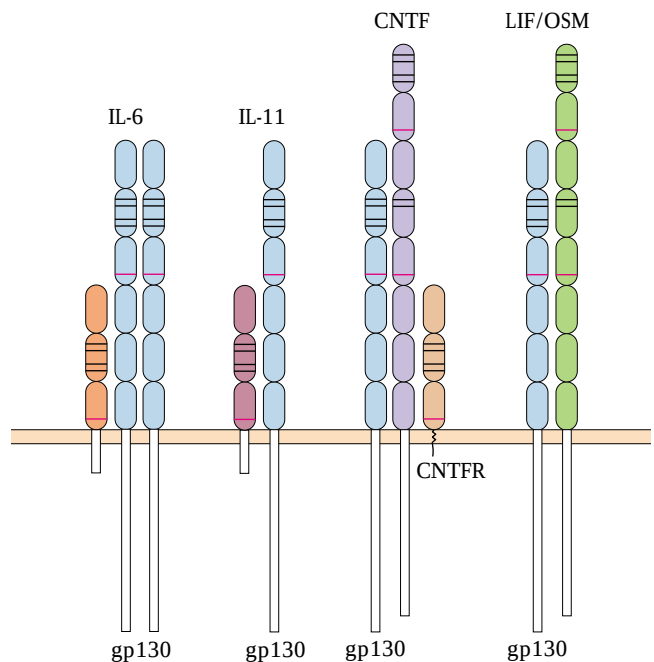
The sharing of signal-transducing subunits among receptors explains the redundancy and antagonism exhibited by some cytokines. Consider the GM-CSF receptor subfamily, which includes the receptors for IL-3, IL-5, and GM-CSF (see Figure 12-7a). Each of these cytokines binds to a unique low-affinity, cytokine-specific receptor consisting of an α subunit only. All three low-affinity subunits can associate noncovalently with a common signal-transducing β subunit. The resulting dimeric receptor not only exhibits increased affinity for the cytokine but also can transduce a signal across the membrane after binding the cytokine (Figure 12-8a). Interestingly, IL-3, IL-5, and GM-CSF exhibit considerable redundancy. IL-3 and GM-CSF both act upon hematopoietic stem cells and progenitor cells, activate monocytes, and induce megakaryocyte differentiation. All three of these cytokines induce eosinophil proliferation and basophil degranulation with release of histamine.

Since the receptors for IL-3, IL-5, and GM-CSF share a common signal-transducing β subunit, each of these cytokines would be expected to transduce a similar activation signal, accounting for the redundancy among their biological effects (Figure 12-8b). In fact, all three cytokines induce the same patterns of protein phosphorylation. Furthermore, IL-3 and GM-CSF exhibit antagonism; IL-3 binding has been shown to be inhibited by GM-CSF, and conversely, binding

(a) GM-CSF receptor subfamily (common β subunit)



(b) IL-6 Receptor subfamily (common gp130 subunit)



(c) IL-2 receptor subfamily (common γ subunit)

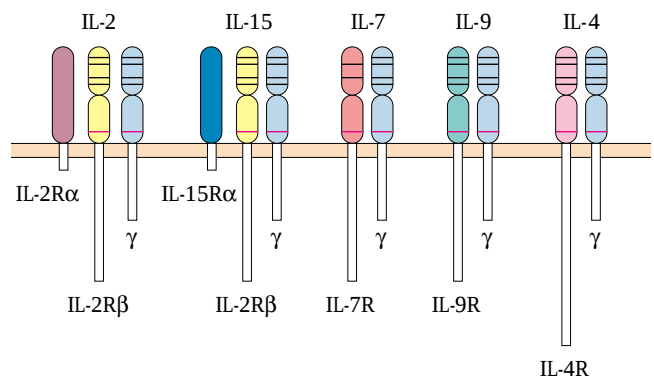


FIGURE 12-7 Schematic diagrams of the three subfamilies of class I cytokine receptors. All members of a subfamily have a common signal-transducing subunit (blue), but a unique cytokine-specific subunit. In addition to the conserved cysteines (double black lines) and WSXWS motifs (red lines) that characterize class I cytokine receptors, immunoglobulin-like domains are present in some of these receptors. CNTF = ciliary neurotrophic factor; LIF/OSM = leukemia-inhibitory factor/oncostatin. [Adapted from K. Sugamura et al., 1996, Annu. Rev. Immunol. 14:179.]

of GM-CSF has been shown to be inhibited by IL-3. Since the signal-transducing β subunit is shared between the receptors for these two cytokines, their antagonism is due to competition for a limited number of β subunits by the cytokine-specific α subunits of the receptors (Figure 12-8c).

A similar situation is found among the IL-6 receptor subfamily, which includes the receptors for IL-6, IL-11, leukemia-inhibitory factor (LIF), oncostatin M (OSM), and ciliary neurotrophic factor (CNTF) (see Figure 12-7b). In this case, a common signal-transducing subunit called gp130 associates with one or two different cytokine-specific subunits. LIF and OSM, which must share certain structural features, both bind to the same α subunit. As expected, the cytokines that bind to receptors in this subfamily display overlapping biological activities: IL-6, OSM, and LIF induce synthesis of acute-phase proteins by liver hepatocytes and differentiation of myeloid leukemia cells into macrophages; IL-6, LIF, and CNTF affect neuronal development, and IL-6, IL-11, and OSM stimulate megakaryocyte maturation and platelet production. The presence of gp130 in all receptors of the IL-6 subfamily explains their common signaling pathways as well as the binding competition for limited gp130 molecules that is observed among these cytokines.

A third signal-transducing subunit defines the IL-2 receptor subfamily, which includes receptors for IL-2, IL-4, IL-7, IL-9, and IL-15 (see Figure 12-7c). The IL-2 and the IL-15 receptors are heterotrimers, consisting of a cytokine-specific α chain and two chains— β and γ —responsible for signal transduction. The IL-2 receptor γ chain functions as the signal-transducing subunit in the other receptors in this subfamily, which are all dimers. Recently, it has been shown that congenital **X-linked severe combined immunodeficiency (XSCID)** results from a defect in the γ -chain gene, which maps to the X chromosome. The immunodeficiencies observed in this disorder are due to the loss of all the cytokine functions mediated by the IL-2 subfamily receptors.

The IL-2R Is One of the Most Thoroughly Studied Cytokine Receptors

Because of the central role of IL-2 and its receptor in the clonal proliferation of T cells, the IL-2 receptor has received intensive study. As noted in the previous section, the complete trimeric receptor comprises three distinct subunits—the α , β , and γ chains. The β and γ chains belong to the class I cytokine receptor family, containing the characteristic CCCC and WSXWS motifs, whereas the α chain has a quite different structure and is not a member of this receptor family (see Figure 12-7c).

The IL-2 receptor occurs in three forms that exhibit different affinities for IL-2: the low-affinity monomeric IL-2R α , the intermediate-affinity dimeric IL-2R $\beta\gamma$, and the high-affinity trimeric IL-2R $\alpha\beta\gamma$ (Figure 12-9). Because the α chain is expressed only by activated T cells, it is often referred to as the TAC (T-cell activation) antigen. A monoclonal anti-

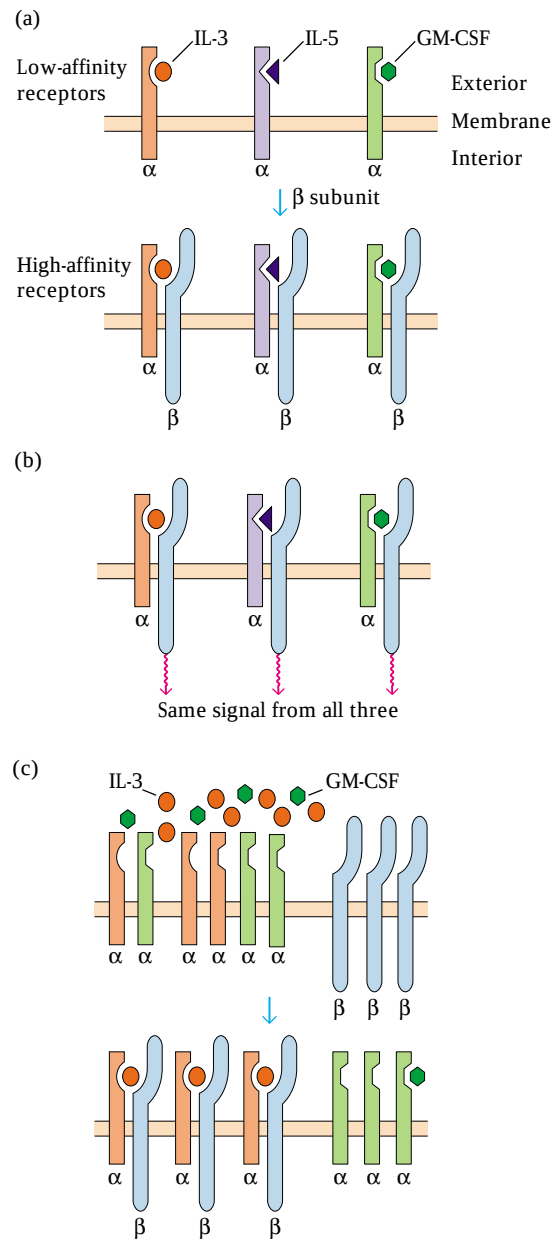


FIGURE 12-8 Interactions between cytokine-specific subunits and a common signal-transducing subunit of cytokine receptors. (a) Schematic diagram of the low-affinity and high-affinity receptors for IL-3, IL-5, and GM-CSF. The cytokine-specific subunits exhibit low-affinity binding and cannot transduce an activation signal. Noncovalent association of each subunit with a common β subunit yields a high-affinity dimeric receptor that can transduce a signal across the membrane. (b) Association of cytokine-specific subunits with a common signaling unit, the β subunit, allows the generation of cytokine-specific signals despite the generation of the same signal by the different cytokine receptors shown. (c) Competition of ligand-binding chains of different receptors for a common subunit can produce antagonistic effects between cytokines. Here binding of IL-3 by α subunits of the IL-3 receptor allows them to out-compete α chains of the GM-CSF receptor for β subunits. [Part (a) adapted from T. Kishimoto et al., 1992, *Science* **258**:593.]

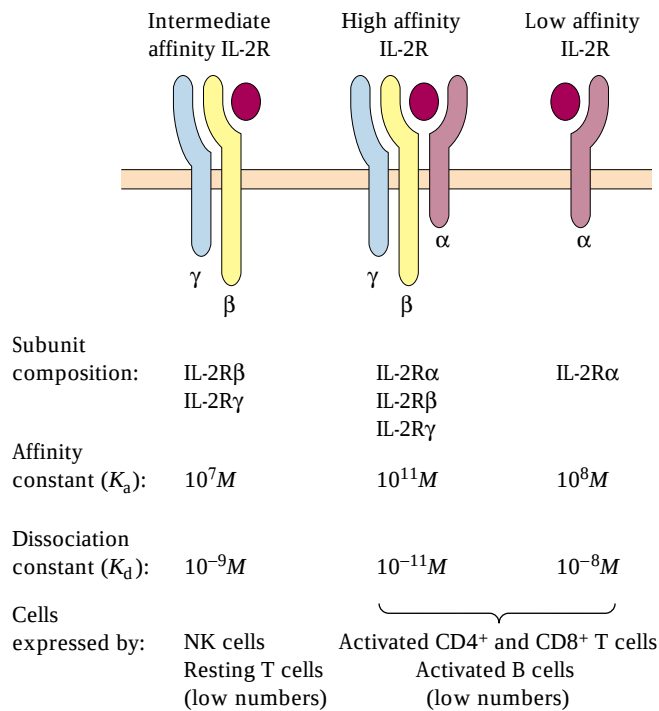


FIGURE 12-9 Comparison of the three forms of the IL-2 receptor. Signal transduction is mediated by the β and γ chains, but all three chains are required for high-affinity binding of IL-2.

body, designated anti-TAC, which binds to the 55-kDa α chain, is often used to identify IL-2R α on cells. Signal transduction by the IL-2 receptor requires both the β and γ chains, but only the trimeric receptor containing the α chain as well binds IL-2 with high affinity. Although the γ chain appears to be constitutively expressed on most lymphoid cells, expression of the α and β chains is more restricted and is markedly enhanced after antigen has activated resting lymphocytes. This phenomenon ensures that only antigen-activated CD4⁺ and CD8⁺ T cells will express the high-affinity IL-2 receptor and proliferate in response to physiologic levels of IL-2. Activated T cells express approximately 5×10^3 high-affinity receptors and ten times as many low-affinity receptors. NK cells express the β and γ subunits constitutively, accounting for their ability to bind IL-2 with an intermediate affinity and to be activated by IL-2.

Engaged Cytokine Receptors Activate Signaling Pathways

While some important cytokine receptors lie outside the class I and class II families, the majority are included within these two families. As mentioned previously, class I and class II cytokine receptors lack signaling motifs (e.g., intrinsic tyrosine kinase domains). Yet, early observations demon-

strated that one of the first events after the interaction of a cytokine with one of these receptors is a series of protein tyrosine phosphorylations. While these results were initially puzzling, they were explained when a unifying model emerged from studies of the molecular events triggered by binding of interferon gamma (IFN- γ) to its receptor, a member of the class II family.

IFN- γ was originally discovered because of its ability to induce cells to block or inhibit the replication of a wide variety of viruses. Antiviral activity is a property it shares with IFN- α and IFN- β . However, unlike these other interferons, IFN- γ plays a central role in many immunoregulatory processes, including the regulation of mononuclear phagocytes, B-cell switching to certain IgG classes, and the support or inhibition of the development of T_H-cell subsets. The discovery of the major signaling pathway invoked by interaction of IFN- γ with its receptor led to the realization that signal transduction through most, if not all, class I and class II cytokine receptors involves the following steps, which are the basis of a unifying signaling model (Figure 12-10).

- *The cytokine receptor is composed of separate subunits, an α chain required for cytokine binding and for signal transduction and a β chain necessary for signaling but with only a minor role in binding.*
- *Different inactive protein tyrosine kinases are associated with different subunits of the receptor. The α chain of the receptor is associated with a novel family of protein tyrosine kinases, the Janus kinase (JAK)* family. The association of the JAK and the receptor subunit occurs spontaneously and does not require the binding of cytokine. However, in the absence of cytokine, JAKs lack protein tyrosine kinase activity.*
- *Cytokine binding induces the association of the two separate cytokine receptor subunits and activation of the receptor-associated JAKs. The ability of IFN- γ , which binds to a class II cytokine receptor, to bring about the association of the ligand-binding chains of its receptor has been directly demonstrated by x-ray crystallographic studies, as shown in Figure 12-11.*
- *Activated JAKs create docking sites for the STAT transcription factors by phosphorylation of specific tyrosine residues on cytokine receptor subunits. Once receptor-associated JAKs are activated, they phosphorylate specific tyrosines in the receptor subunits of the*

*The Roman god Janus had two faces. Kinases of the Janus family have two sites, a binding site at which they link with the cytokine receptor subunit and a catalytic site that, when activated, has protein tyrosine kinase activity. Some biochemists, wearied by the multitude of different protein kinases that have been discovered, claim JAK means Just Another Kinase.

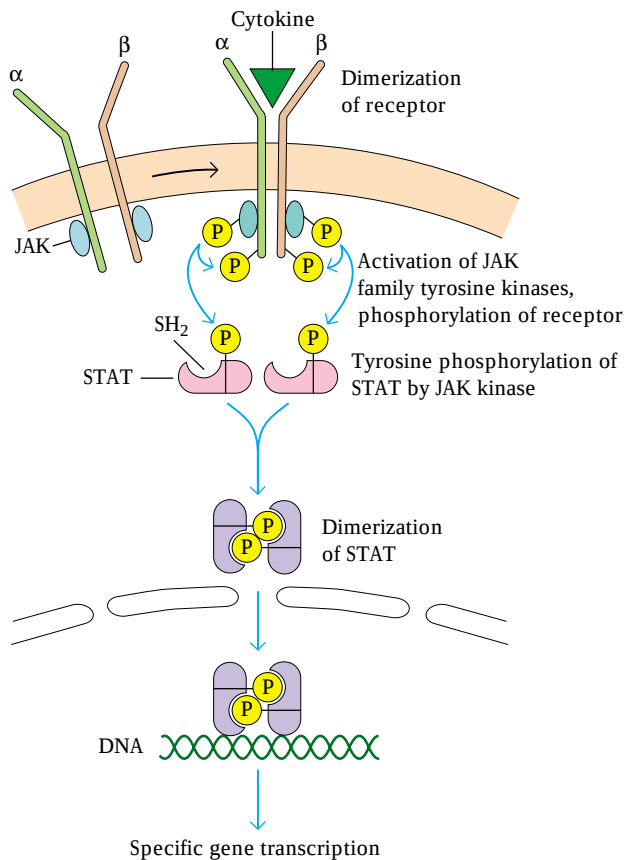


FIGURE 12-10 General model of signal transduction mediated by most class I and class II cytokine receptors. Binding of a cytokine induces dimerization of the receptor subunits, which leads to the activation of receptor-subunit-associated JAK tyrosine kinases by reciprocal phosphorylation. Subsequently, the activated JAKs phosphorylate various tyrosine residues, resulting in the creation of docking sites for STATs on the receptor and the activation of the one or more STAT transcription factors. The phosphorylated STATs dimerize and translocate to the nucleus, where they activate transcription of specific genes.

complex. Members of a family of transcription factors known as **STATs (signal transducers and activators of transcription)** bind to these phosphorylated tyrosine residues. Specific STATs (see Table 12-2) play essential roles in the signaling pathways of a wide variety of cytokines. The binding of STATs to receptor subunits is mediated by the joining of the SH2 domain on the STAT with the docking site created by the JAK-mediated phosphorylation of a particular tyrosine on receptor subunits.

- After undergoing JAK-mediated phosphorylation, STAT transcription factors translocate from receptor docking sites at the membrane to the nucleus, where they initiate

the transcription of specific genes. While docked to receptor subunits, STATs undergo JAK-catalyzed phosphorylation of a key tyrosine. This is followed by the dissociation of the STATs from the receptor subunits and their dimerization. The STAT dimers then translocate into the nucleus and induce the expression of genes containing appropriate regulatory sequences in their promoter regions.

In addition to IFN- γ , a number of other class I and class II ligands have been shown to cause dimerization of their receptors. An important element of cytokine specificity derives from the exquisite specificity of the match between cytokines and their receptors. Another aspect of cytokine specificity is that each particular cytokine (or group of redundant cytokines) induces transcription of a specific subset of genes in a given cell type; the resulting gene products then mediate the various effects typical of that cytokine. The specificity of cytokine effects is then traceable to three factors. First, particular cytokine receptors start particular JAK-STAT pathways. Second, the transcriptional activity of activated STATs

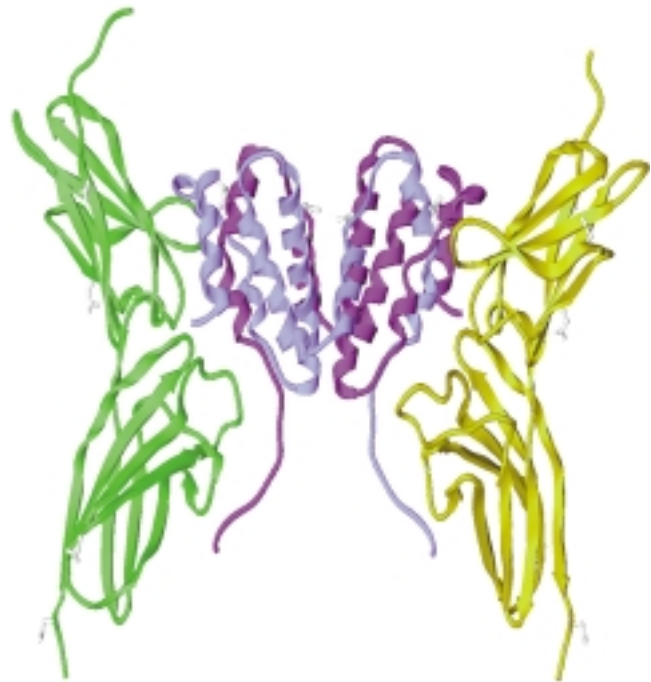


FIGURE 12-11 The complex between IFN- γ and the ligand-binding chains of its receptor. This model is based on the x-ray crystallographic analysis of a crystalline complex of interferon- γ (violet and blue) bound to ligand-binding α chains of the receptor (green and yellow). Note that IFN- γ is shown in its native dimeric form; each member of the dimer engages the α chain of an IFN- γ receptor, thereby bringing about receptor dimerization and signal transduction. [From M. R. Walter et al., 1995, *Nature* **376**:230, courtesy M. Walter, University of Alabama.]

TABLE 12-2

STAT and JAK interaction with selected cytokine receptors during signal transduction

Cytokine receptor	JAK	STAT
IFN- γ	JAK1 and JAK2	Stat1
IFN- α/β	JAK1 and Tyk-2	Stat2
IL-2	JAK1 and JAK3	Stat5
IL-3	JAK2	Stat5
IL-4	JAK1 and JAK3	Stat6
IL-6	JAK1 (and sometimes others)	Stat3
IL-10	JAK1 and Tyk-2*	Stat3
IL-12	JAK2 and Tyk-2*	Stat4

*Despite its name, Tyk-2 is also a Janus kinase.

SOURCE: Adapted from E. A. Bach, M. Aguet, and R. D. Schreiber, 1997, *Annu. Rev. Immun.* **15**:563.

is specific because a particular STAT homodimer or heterodimer will only recognize certain sequence motifs and thus can interact only with the promoters of certain genes. Third, only those target genes whose expression is permitted by a particular cell type can be activated within that variety of cell. That is, in any given cell type only a subset of the potential target genes of a particular STAT may be permitted expression. For example, IL-4 induces one set of genes in T cells, another in B cells, and yet a third in eosinophils.

Cytokine Antagonists

A number of proteins that inhibit the biological activity of cytokines have been reported. These proteins act in one of two ways: either they bind directly to a cytokine receptor but fail to activate the cell, or they bind directly to a cytokine, inhibiting its activity. The best-characterized inhibitor is the IL-1 receptor antagonist (IL-1Ra), which binds to the IL-1 receptor but has no activity. Binding of IL-1Ra to the IL-1 receptor blocks binding of both IL-1 α and IL-1 β , thus accounting for its antagonistic properties. Production of IL-1Ra has been thought by some to play a role in regulating the intensity of the inflammatory response. It has been cloned and is currently being investigated as a potential treatment for chronic inflammatory diseases.

Cytokine inhibitors are found in the bloodstream and extracellular fluid. These soluble antagonists arise from enzymatic cleavage of the extracellular domain of cytokine receptors. Among the soluble cytokine receptors that have been detected are those for IL-2, -4, -6, and -7, IFN- γ and - α , TNF- β , and LIF. Of these, the soluble IL-2 receptor (sIL-2R), which is

released in chronic T-cell activation, is the best characterized. A segment containing the amino-terminal 192 amino acids of the α subunit is released by proteolytic cleavage, forming a 45-kDa soluble IL-2 receptor. The shed receptor can bind IL-2 and prevent its interaction with the membrane-bound IL-2 receptor. The presence of sIL-2R has been used as a clinical marker of chronic T-cell activation and is observed in a number of diseases, including autoimmunity, transplant rejection, and AIDS.

Some viruses also produce cytokine-binding proteins or cytokine mimics. The evolution of such anti-cytokine strategies by microbial pathogens is good biological evidence of the importance of cytokines in organizing and promoting effective anti-microbial immune responses. The poxviruses, for example, have been shown to encode a soluble TNF-binding protein and a soluble IL-1-binding protein. Since both TNF and IL-1 exhibit a broad spectrum of activities in the inflammatory response, these soluble cytokine-binding proteins may prohibit or diminish the inflammatory effects of the cytokines, thereby conferring upon the virus a selective advantage. Epstein-Barr virus produces an IL-10-like molecule (viral IL-10 or vIL-10) that binds to the IL-10 receptor and, like cellular IL-10, suppresses T_H1-type cell-mediated responses (see the next section), which are effective against many intracellular parasites such as viruses. Molecules produced by viruses that mimic cytokines allow the virus to manipulate the immune response in ways that aid the survival of the pathogen. This is an interesting and powerful modification some viruses have undergone in their continuing struggle to overcome the formidable barrier of host immunity. Table 12-3 lists a number of viral products that mimic cytokines or their receptors.

TABLE 12-3

Viral mimics of cytokine and cytokine receptors

Virus	Product
Leporipoxvirus (a myxoma virus)	Soluble IFN- γ receptor
Several poxviruses	Soluble IFN- γ receptor
Vaccinia, smallpox virus	Soluble IL-1 β receptor
Epstein-Barr	IL-10 homolog
Human herpesvirus-8	IL-6 homolog; also homologs of the chemokines MIP-I and MIP-II
Cytomegalovirus	Three different chemokine receptor homologs, one of which binds three different soluble chemokines (RANTES, MCP-1, and MIP-1 α)

Cytokine Secretion by T_H1 and T_H2 Subsets

The immune response to a particular pathogen must induce an appropriate set of effector functions that can eliminate the disease agent or its toxic products from the host. For example, the neutralization of a soluble bacterial toxin requires antibodies, whereas the response to an intracellular virus or to a bacterial cell requires cell-mediated cytotoxicity or delayed-type hypersensitivity. A large body of evidence implicates differences in cytokine-secretion patterns among T_H-cell subsets as determinants of the type of immune response made to a particular antigenic challenge.

CD4⁺ T_H cells exert most of their helper functions through secreted cytokines, which either act on the cells that produce them in an autocrine fashion or modulate the responses of other cells through paracrine pathways. Although CD8⁺ CTLs also secrete cytokines, their array of cytokines generally is more restricted than that of CD4⁺ T_H cells. As briefly discussed in Chapter 10, two CD4⁺ T_H-cell subpopulations designated T_H1 and T_H2, can be distinguished in vitro by the cytokines they secrete. Both subsets secrete IL-3 and GM-CSF but differ in the other cytokines they produce (Table 12-4). T_H1 and T_H2 cells are characterized by the following functional differences:

- The T_H1 subset is responsible for many cell-mediated functions (e.g., delayed-type hypersensitivity and activation of T_C cells) and for the production of opsonization-promoting IgG antibodies (i.e. antibodies that bind to the high-affinity Fc receptors of phagocytes and interact with the complement system). This subset is also associated with the promotion of excessive inflammation and tissue injury.
- The T_H2 subset stimulates eosinophil activation and differentiation, provides help to B cells, and promotes the production of relatively large amounts of IgM, IgE, and noncomplement-activating IgG isotypes. The T_H2 subset also supports allergic reactions.

The differences in the cytokines secreted by T_H1 and T_H2 cells determine the different biological functions of these two subsets. A defining cytokine of the T_H1 subset, IFN- γ , activates macrophages, stimulating these cells to increase microbicidal activity, up-regulate the level of class II MHC, and secrete cytokines such as IL-12, which induces T_H cells to differentiate into the T_H1 subset. IFN- γ secretion by T_H1 cells also induces antibody-class switching to IgG classes (such as IgG2a in the mouse) that support phagocytosis and fixation of complement. TNF- β and IFN- γ are cytokines that mediate inflammation, and it is their secretion that accounts for the association of T_H1 cells with inflammatory phenomena such as delayed hypersensitivity (Chapter 16). T_H1 cells produce IL-2 and IFN- γ cytokines that promote the differentia-

TABLE 12-4

Cytokine secretion and principal functions of mouse T_H1 and T_H2 subsets

Cytokine/function	T _H 1	T _H 2
CYTOKINE SECRETION		
IL-2	+	—
IFN- γ	++	—
TNF- β	++	—
GM-CSF	++	+
IL-3	++	++
IL-4	—	++
IL-5	—	++
IL-10	—	++
IL-13	—	++
FUNCTIONS		
Help for total antibody production	+	++
Help for IgE production	—	++
Help for IgG2a production	++	+
Eosinophil and mast-cell production	—	++
Macrophage activation	++	—
Delayed-type hypersensitivity	++	—
T _C -cell activation	++	—

SOURCE: Adapted from F. Powrie and R. L. Coffman, 1993, *Immunol. Today* 14:270.

tion of fully cytotoxic T_C cells from CD8⁺ precursors. This pattern of cytokine production makes the T_H1 subset particularly suited to respond to viral infections and intracellular pathogens. Finally, IFN- γ inhibits the expansion of the T_H2 population.

The secretion of IL-4 and IL-5 by cells of the T_H2 subset induces production of IgE and supports eosinophil-mediated attack on helminth (roundworm) infections. IL-4 promotes a pattern of class switching that produces IgG that does not activate the complement pathway (IgG1 in mice, for example). IL-4 also increases the extent to which B cells switch from IgM to IgE. This effect on IgE production meshes with eosinophil differentiation and activation by IL-5, because eosinophils are richly endowed with Fc ϵ receptors, which bind IgE. Typically, roundworm infections induce T_H2 responses and evoke anti-roundworm IgE antibody. The antibody bound to the worm binds to the Fc receptors of eosinophils, thus forming an antigen-specific bridge between the worm and the eosinophils. The attack of the eosinophil on the worm is triggered by crosslinking of the Fc ϵ -bound IgE. Despite these beneficial actions of IgE, it is also the Ig

class responsible for allergy. Finally, IL-4 and IL-10 suppress the expansion of T_H1 cell populations.

Because the T_H1 and T_H2 subsets were originally identified in long-term in vitro cultures of cloned T-cell lines, some researchers doubted that they represented true in vivo subpopulations. They suggested instead that these subsets might represent different maturational stages of a single lineage. Also, the initial failure to locate either subset in humans led some to believe that T_H1 , T_H2 , and other subsets of T helper cells did not occur in this species. Further research corrected these views. In many in vivo systems, the full commitment of populations of T cells to either the T_H1 or T_H2 phenotype often signals the endpoint of a chronic infection or allergy. Hence it was difficult to find clear T_H1 or T_H2 subsets in studies employing healthy human subjects, who would not be at this stage of a response. Experiments with transgenic mice demonstrated conclusively that T_H1 and T_H2 cells arise independently. Furthermore, it was possible to demonstrate T_H1 or T_H2 populations in T cells isolated from humans during chronic infectious disease or chronic episodes of allergy. It is also important to emphasize that many helper T cells do not show either a T_H1 or a T_H2 profile; individual cells have shown striking heterogeneity in the T_H -cell population. One of the best described of these is the T_H0 subset, which secretes IL-2, IL-4, IL-5, IFN- γ , and IL-10, as well as IL-3 and GM-CSF.

Numerous reports of studies in both mice and humans now document that the in vivo outcome of the immune response can be critically influenced by the relative levels of T_H1 -like or T_H2 -like activity. Typically, the T_H1 profile of cytokines is higher in response to intracellular pathogens, and the T_H2 profile is higher in allergic diseases and helminthic infections.

The Development of T_H1 and T_H2 Subsets Is Determined by the Cytokine Environment

The cytokine environment in which antigen-primed T_H cells differentiate determines the subset that develops (Figure 12-12). In particular, IL-4 is essential for the development of a T_H2 response, and IFN- γ , IL-12, and IL-18 all are important in the physiology of the development of T_H1 cells. The source of IL-12, one of the key mediators of T_H1 differentiation, is typically macrophages or dendritic cells activated by an encounter with intracellular bacteria, with bacterial products such as LPS, or with a number of other intracellular parasites. T_H1 development is also critically dependent on IFN- γ , which induces a number of changes, including the up-regulation of IL-12 production by macrophages and dendritic cells, and the activation of the IL-12 receptor on activated T cells, which it accomplishes by up-regulating expression of the β chain of the IL-12 receptor. At the beginning of an immune response, IFN- γ is generated by stimulation of T cells and can also come from activated NK cells. Yet another cytokine, IL-18, promotes proliferation and IFN- γ produc-

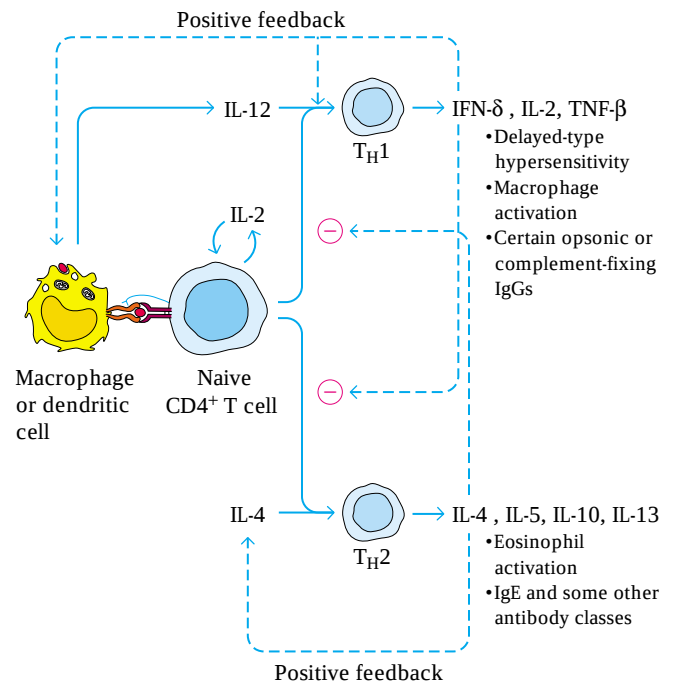


FIGURE 12-12 Cytokine-mediated generation and cross regulation of T_H subsets. Antigen-activated naive $CD4^+$ T cell produces IL-2 and proliferates. If it proliferates in an IL-12 dominated environment, it generates a population of T_H1 cells that secretes a characteristic profile of cytokines including interferon γ . A positive feedback loop is established when IFN- γ secreted by the expanding T_H1 population stimulates dendritic cells or macrophages to produce more IL-12. If the environment is dominated by IL-4, a T_H2 population emerges and secretes a profile of cytokines that promotes eosinophil activation and the synthesis of certain antibody classes. Key cytokines produced by each subset positively regulate the subset that produces it and negatively regulate the other subset. [Adapted from J. Rengarajan, S. Szabo, and L. Glimcher, 2000, *Immunology Today* **21**:479.]

tion by both developing and fully differentiated T_H1 cells and by NK cells. So a regulatory network of cytokines positively controls the generation of T_H1 cells. The critical role played by each of these cytokines and their receptors has been demonstrated in a series of experiments in which either the cytokine or its receptor has been knocked out. Mice in which the genes for any of these critical components have been knocked out fail to generate populations of T_H1 cells.

Just as T_H1 cells require IL-12 and IFN- γ , the generation of T_H2 cells depends critically on IL-4. Exposing naive helper cells to IL-4 at the beginning of an immune response causes them to differentiate into T_H2 cells. In fact, this influence of IL-4 is predominant in directing T_H cells to the T_H2 route. Provided a threshold level of IL-4, T_H2 development is

greatly favored over T_H1 even if IL-12 is present. The critical role of signals from IL-4 in T_H2 development is shown by the observation that knocking out the gene that encodes IL-4 prevents the development of this T-cell subset. Additional evidence supporting the central role of IL-4 comes from an experiment that interrupted the IL-4 signal-transduction pathway. Like so many other cytokines, IL-4 uses a pathway that involves JAK and STAT proteins. The Stat6 transcription factor is the one activated in signaling by IL-4. Consequently, in mice in which the gene for Stat6 has been disrupted (Stat6 knockouts), IL-4 mediated processes are severely inhibited or absent. The observation that Stat6 knockout mice have very few T_H2 cells confirms the importance of IL-4 for the differentiation of this subset.

Cytokine Profiles Are Cross-Regulated

The critical cytokines produced by T_H1 and T_H2 subsets have two characteristic effects on subset development. First, they promote the growth of the subset that produces them; second, they inhibit the development and activity of the opposite subset, an effect known as *cross-regulation*, (see Figure 12-12). For instance, IFN- γ (secreted by the T_H1 subset) preferentially inhibits proliferation of the T_H2 subset, and IL-4 and IL-10 (secreted by the T_H2 subset) down-regulate secretion of IL-12, one of the critical cytokines for T_H1 differentiation, by both macrophages and dendritic cells. Similarly, these cytokines have opposing effects on target cells other than T_H subsets. IFN- γ secreted by the T_H1 subset promotes IgG2a production by B cells but inhibits IgG1 and IgE production. On the other hand, IL-4 secreted by the T_H2 subset promotes production of IgG1 and IgE and suppresses production of IgG2a. The phenomenon of cross-regulation explains the observation that there is often an inverse relationship between antibody production and cell-mediated immunity; that is, when antibody production is high, cell-mediated immunity is low, and vice versa. Furthermore, recent research has shown that IL-4 and IFN- γ make members of the T-cell subset that releases them less responsive to the cytokine that directs differentiation of the other T-cell subset. Thus, IL-4 enhances T_H2 cell development by making T_H cells less susceptible to the cytokine signals that cause these cells to enter a differentiation pathway that would lead to T_H1 development. On the other hand, as explained below, IFN- γ up-regulates the expression of a key regulatory molecule that favors the differentiation and activity of T_H1 cells.

Recent work has given insight into the molecular basis for the cytokine-mediated cross-regulation by which one subset promotes its own expansion and development while inhibiting the development of the opposite subset. Two transcription factors, T-Bet and GATA-3, are key elements in determining subset commitment and cross-regulation. The expression of T-Bet drives cells to differentiate into T_H1 cells and suppresses their differentiation along the T_H2 pathway. Expression of GATA-3 does the opposite, promoting the de-

velopment of naive T cells into T_H2 cells while suppressing their differentiation into T_H1 cells. As shown in Figure 12-13, the expression of T-Bet versus GATA-3 is determined by the cytokines IFN- γ and IL-4. In the presence of IFN- γ , T cells up-regulate the expression of T-Bet and down-regulate GATA-3. This IFN- γ receptor/Stat1-dependent process shifts the cytokine profile to the production of IFN- γ , the signature cytokine of T_H1 cells, and other cytokines typical of the T_H1 set. On the other hand, in a process that involves the IL-4 receptor and Stat6, IL-4 induces the cell to produce IL-4 and other T_H2 cytokines. Further study has revealed that the up-regulation of T-Bet represses the expression of GATA-3. Similarly, expression of GATA-3 down-regulates T-Bet. Consequently, cytokine signals that induce one of these transcription factors set in motion a chain of events that repress the other. At the intracellular level, the differentiation of a T cell along the T_H1 pathway, prevents its development of T_H2 characteristics and vice versa.

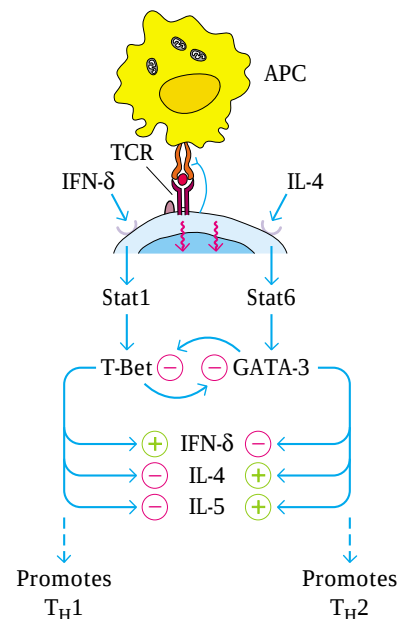


FIGURE 12-13 Cross-regulation at the intracellular level. Signals through the TCR and cytokine receptors determine whether the cell will produce the T_H1 -promoting transcription factor, T-Bet, or the T_H2 -promoting transcription factor, GATA-3. Experimental evidence supports a model in which exposure of cells bearing receptors for IFN- γ to IFN- γ induces the formation of T-Bet, which up-regulates the synthesis of IFN- γ and represses the expression of GATA-3. Exposure of IL-4 R-bearing cells to IL-4 induces the formation of GATA-3, which up-regulates the synthesis of IL-4 and IL-5 but represses the expression of T-Bet. [Adapted from J. Rengarajan, S. Szabo, and L. Glimcher, 2000, *Immunology Today* 21:479.]

The cross-regulation of T_H1 cells by IL-10 secreted from T_H2 cells is not a direct inhibition of the T_H1 cells; instead, IL-10 acts on monocytes and macrophages, interfering with their ability to activate the T_H1 subset. This interference is thought to result from the demonstrated ability of IL-10 to down-regulate the expression of class II MHC molecules on these antigen-presenting cells. IL-10 has other potent immunosuppressant effects on the monocyte-macrophage lineage, such as suppressing the production of nitric oxide and other bactericidal metabolites involved in the destruction of pathogens, and also suppressing the production of various inflammatory mediators (e.g., IL-1, IL-6, IL-8, GM-CSF, G-CSF, and TNF- γ). These suppressive effects on the macrophage serve to further diminish the biologic consequences of T_H1 activation.

The T_H1/T_H2 Balance Determines Disease Outcomes

The progression of some diseases may depend on the balance between the T_H1 and T_H2 subsets. In humans, a well-studied example of this phenomenon is leprosy, which is caused by *Mycobacterium leprae*, an intracellular pathogen that can survive within the phagosomes of macrophages. Leprosy is not a single clinical entity; rather, the disease presents as a spectrum of clinical responses, with two major forms of disease, tuberculoid and lepromatous, at each end of the spectrum. In **tuberculoid leprosy**, a cell-mediated immune response forms granulomas, resulting in the destruction of most of the mycobacteria, so that only a few organisms remain in the tissues. Although skin and peripheral nerves are damaged, tuberculoid leprosy progresses slowly and patients usually survive. In **lepromatous leprosy**, the cell-mediated response is depressed and, instead, humoral antibodies are formed, sometimes resulting in hypergammaglobulinemia. The mycobacteria are widely disseminated in macrophages, often reaching numbers as high as 10^{10} per gram of tissue. Lepromatous leprosy progresses into disseminated infection of the bone and cartilage with extensive nerve damage.

The development of lepromatous or tuberculoid leprosy depends on the balance of T_H1 and T_H2 cells (Figure 12-14). In tuberculoid leprosy, the immune response is characterized by a T_H1 -type response with delayed-type hypersensitivity and a cytokine profile consisting of high levels of IL-2, IFN- γ , and TNF- β . In lepromatous leprosy, there is a T_H2 -type immune response, with high levels of IL-4, IL-5, and IL-10. This cytokine profile explains the diminished cell-mediated immunity and increased production of serum antibody in lepromatous leprosy.

There is also evidence for changes in T_H -subset activity in AIDS. Early in the disease, T_H1 activity is high, but as AIDS progresses, some workers have suggested, there may be a shift from a T_H1 -like to a T_H2 -like response. In addition, some pathogens may influence the activity of the T_H subsets. The Epstein-Barr virus, for instance, produces vIL-10, which has

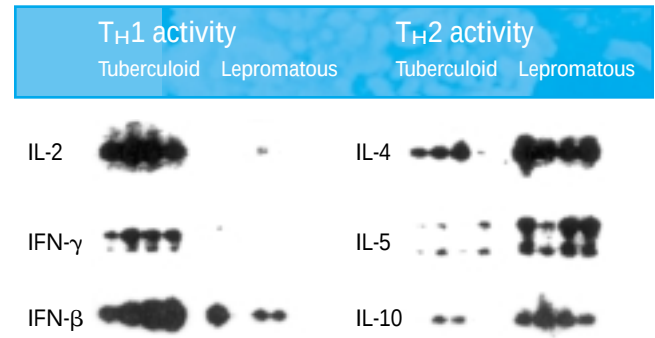


FIGURE 12-14 Correlation between type of leprosy and relative T_H1 or T_H2 activity. Messenger RNA isolated from lesions from tuberculoid and lepromatous leprosy patients was analyzed by Southern blotting using the cytokine probes indicated. Cytokines produced by T_H1 cells predominate in the tuberculoid patients, while cytokines produced by T_H2 cells predominate in the lepromatous patients. [From P. A. Sieling and R. L. Modlin, 1994, *Immunobiology* **191**: 378.]

IL-10-like activity and, like cellular IL-10, tends to suppress T_H1 activity by cross-regulation. Some researchers have speculated that vIL-10 may reduce the cell-mediated response to the Epstein-Barr virus, thus conferring a survival advantage on the virus.

Cytokine-Related Diseases

Defects in the complex regulatory networks governing the expression of cytokines and cytokine receptors have been implicated in a number of diseases. This section describes several diseases resulting from overexpression or underexpression of cytokines or cytokine receptors.

Bacterial Septic Shock Is Common and Potentially Lethal

The role of cytokine overproduction in pathogenesis can be illustrated by bacterial septic shock. This condition may develop a few hours after infection by certain gram-negative bacteria, including *E. coli*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Enterobacter aerogenes*, and *Neisseria meningitidis*. The symptoms of bacterial septic shock, which is often fatal, include a drop in blood pressure, fever, diarrhea, and widespread blood clotting in various organs. This condition afflicts about 500,000 Americans annually and causes more than 70,000 deaths. The annual cost for treating bacterial septic shock is estimated to be in excess of \$5 billion.

Bacterial septic shock apparently develops because bacterial cell-wall **endotoxins** stimulate macrophages to overproduce IL-1 and TNF- α to levels that cause septic shock. In

one study, for example, higher levels of TNF- α were found in patients who died of meningitis than in those who recovered. Furthermore, a condition resembling bacterial septic shock can be produced by injecting mice with recombinant TNF- α in the absence of gram-negative bacterial infection. Several studies offer some hope that neutralization of TNF- α or IL-1 activity with monoclonal antibodies or antagonists may prevent this fatal shock from developing in these bacterial infections. In one study, monoclonal antibody to TNF- α protected animals from endotoxin-induced shock. Another study has shown that injection of a recombinant IL-1 receptor antagonist (IL-1Ra), which prevents binding of IL-1 to the IL-1 receptor, resulted in a three-fold reduction in mortality. It is hoped that these experimental results will lead to clinically useful products for the treatment of bacterial septic shock in humans.

Bacterial Toxic Shock Is Caused by Superantigens

A variety of microorganisms produce toxins that act as **superantigens**. As Chapter 10 described, superantigens bind simultaneously to a class II MHC molecule and to the V β domain of the T-cell receptor, activating all T cells bearing a particular V β domain (Figure 10-16). Because of their unique binding ability, superantigens can activate large numbers of T cells irrespective of their antigenic specificity.

Although less than 0.01% of T cells respond to a given conventional antigen, between 5% and 25% of T cells can respond to a given superantigen. The large proportion of T cells responsive to a particular superantigen results from the limited number of TCR V β genes carried in the germ line. Mice, for example, have about 20 V β genes. Assuming that each V β gene is expressed with equal frequency, then each superantigen would be expected to interact with 1 in 20 T cells, or 5% of the total T-cell population.

A number of bacterial superantigens have been implicated as the causative agent of several diseases such as bacterial toxic shock and food poisoning. Included among these bacterial superantigens are several enterotoxins, exfoliating toxins, and toxic-shock syndrome toxin (TSST1) from *Staphylococcus aureus*; pyrogenic exotoxins from *Streptococcus pyrogenes*; and *Mycoplasma arthritidis* supernatant (MAS). The large number of T cells activated by these superantigens results in excessive production of cytokines. The toxic-shock syndrome toxin, for example, has been shown to induce extremely high levels of TNF and IL-1. As in bacterial septic shock, these elevated concentrations of cytokines can induce systemic reactions that include fever, widespread blood clotting, and shock.

Cytokine Activity Is Implicated in Lymphoid and Myeloid Cancers

Abnormalities in the production of cytokines or their receptors have been associated with some types of cancer. For ex-

ample, abnormally high levels of IL-6 are secreted by cardiac myxoma cells (a benign heart tumor), myeloma and plasmacytoma cells, and cervical and bladder cancer cells. In myeloma cells, IL-6 appears to operate in an autocrine manner to stimulate cell proliferation. When monoclonal antibodies to IL-6 are added to in vitro cultures of myeloma cells, their growth is inhibited. In addition, transgenic mice that express high levels of IL-6 have been found to exhibit a massive, fatal plasma-cell proliferation, called plasmacytosis. Although these plasma cells are not malignant, the high rate of plasma-cell proliferation possibly contributes to the development of cancer.

Chagas' Disease Is Caused by a Parasite

The protozoan *Trypanosoma cruzi* is the causative agent of Chagas' disease, which is characterized by severe immune suppression. The ability of *T. cruzi* to mediate immune suppression can be observed by culturing peripheral-blood T cells in the presence and in the absence of *T. cruzi* and then evaluating their immune reactivity. Antigen, mitogen, or anti-CD3 monoclonal antibody normally can activate peripheral T cells, but in the presence of *T. cruzi*, T cells are not activated by any of these agents. The defect in these lymphocytes has been traced to a dramatic reduction in the expression of the 55-kDa α subunit of the IL-2 receptor. As noted earlier, the high-affinity IL-2 receptor contains α , β , and γ subunits. The α subunit is specific for cytokine binding (see Figure 12-9). Co-culturing of T cells with *T. cruzi* and subsequent staining with fluorescein-labeled anti-TAC, which binds to the α subunit, revealed a 90% decrease in the level of the α subunit.

Although the mechanism by which *T. cruzi* suppresses expression of the α subunit is still unknown, the suppression can be induced across a filter that prevents contact between the lymphocytes and protozoa. This finding suggests that a diffusible factor mediates suppression. Such a factor, once isolated, might have numerous clinical applications for regulating the level of activated T cells in leukemias and autoimmune diseases.

Therapeutic Uses of Cytokines and Their Receptors

The availability of purified cloned cytokines and soluble cytokine receptors offers the prospect of specific clinical therapies to modulate the immune response. A few cytokines—notably, interferons (see Clinical Focus)—and colony-stimulating factors, such as GM-CSF, have proven to be therapeutically useful. However, despite the promise of cytokines as powerful mediators of immune and other biological

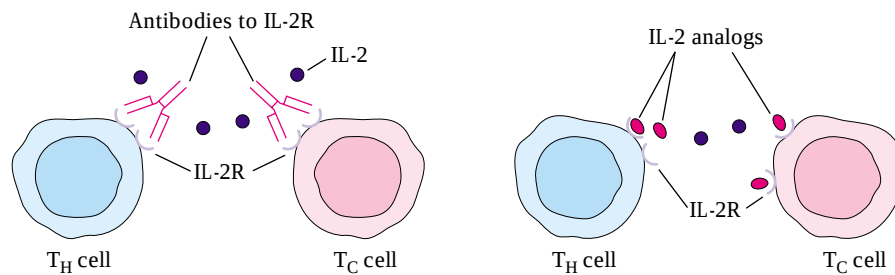
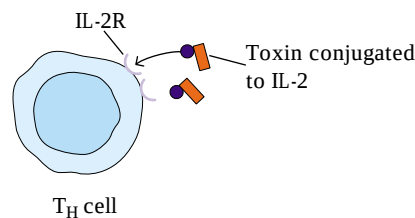
(a) Suppression of T_H -cell proliferation and T_C -cell activation(b) Destruction of activated T_H cells

FIGURE 12-15 Experimental cytokine-related therapeutic agents offer the prospect of selectively modulating the immune response. (a) The anti-IL-2R monoclonal antibody binds to the cytokine receptor.

(b) Conjugation of a toxin with a cytokine results in destruction of cells expressing the cytokine receptor.

responses, not many have made their way into clinical practice. A number of factors are likely to raise difficulties in adapting cytokines for safe and effective routine medical use. One of these is the need to maintain effective dose levels over a clinically significant period of time. During an immune response, interacting cells produce sufficiently high concentrations of cytokines in the vicinity of target cells, but achieving such local concentrations when cytokines must be administered systemically for clinical treatment is difficult. In addition, cytokines often have a very short half-life, so that continuous administration may be required. For example, recombinant human IL-2 has a half-life of only 7–10 min when administered intravenously. Finally, cytokines are extremely potent biological response modifiers and they can cause unpredictable and undesirable side effects. The side effects from administration of recombinant IL-2, for instance, range from mild (e.g., fever, chills, diarrhea, and weight gain) to serious, such as anemia, thrombocytopenia, shock, respiratory distress, and coma. Despite these difficulties, the promise of cytokines for clinical medicine is great and efforts to develop safe and effective cytokine-related strategies continue, particularly in areas such as inflammation, cancer therapy, and modification of the immune response during organ transplantation, infectious disease, and allergy.

Some specific examples of various approaches being explored include cytokine receptor blockade and the use of cytokine analogs and cytokine-toxin conjugates. For instance, proliferation of activated T_H cells and activation of T_C cells can be blocked by anti-TAC, a monoclonal antibody that binds to the α subunit of the high-affinity IL-2 receptor (Figure 12-15a, left panel). Administration of anti-TAC has prolonged the survival of heart transplants in rats. Similar results have been obtained with IL-2 analogs that retain their ability to bind the IL-2 receptor but have lost their biological activity (Figure 12-15a, right panel). Such analogs have been produced by site-directed mutagenesis of cloned IL-2 genes. Finally, cytokines conjugated to various toxins (e.g., the β chain of diphtheria toxin) have been shown to diminish rejection of kidney and heart transplants in animals. Such conjugates containing IL-2 selectively bind to and kill activated T_H cells (Figure 12-15b).

Cytokines in Hematopoiesis

Early work in Australia and Israel demonstrated that soluble factors could support the growth and differentiation of red and white blood cells. The first of these soluble factors to be



CLINICAL FOCUS

Therapy with Interferons

Interferons

are an extraordinary group of proteins whose antiviral activity led to their discovery almost 50 years ago. Subsequent studies showed that interferons have other effects, including the capacity to induce cell differentiation, to inhibit proliferation by some cell types, to inhibit angiogenesis, and to function in various immunoregulatory roles. Their effects on the immune system are important and dramatic. Interferons induce increases in the expression of class I and class II MHC molecules, and augment NK-cell activity. Increased class I expression increases the display of antigen to CD8⁺ cells, a class that includes most of the T_C population. This enhanced display of antigen not only makes the antigen-presenting cells more effective in inducing cytotoxic T-cell populations, it also makes them better targets for attack by T_C cells. In addition to up-regulating class I MHC expression of many cell types, IFN- γ increases the expression of class II MHC molecules on such antigen-presenting cells as macrophages and dendritic cells. This makes them better presenters of antigen to T_H cells. IFN- γ is also a potent inducer of macrophage activation and general promoter of inflammatory responses. Cloning of the genes that encode all three types of interferon, IFN- α , IFN- β , and IFN- γ , has made it possible for the biotechnology industry to produce large amounts of all of these interferons at costs that make their clinical use practical. Some clinical uses of each type of interferon are described here:

- IFN- α (also known by its trade names Roferon and Intron-A) has been used for the treatment of hepatitis C and hepatitis B. It has also found a number of different applications in cancer therapy. A type of B-cell leukemia known as hairy-cell leukemia (because the cells are covered with fine, hairlike cytoplasmic projections) responds well to IFN- α . Chronic myelogenous leukemia, a disease characterized by increased numbers of granulocytes, begins with a slowly developing chronic phase that changes to an accelerated phase and terminates in a blast phase, which is usually resistant to treatment. IFN- α is an effective treatment for this leukemia in the chronic phase (70% response rates have been reported) and some patients (as many as 20% in some studies) undergo complete remission. Kaposi's sarcoma, the cancer most often seen in American AIDS patients, also responds to treatment with IFN- α , and there are reports of a trend toward longer survival and fewer opportunistic infections in patients treated with this agent. IFN- γ has also been used, with varying degrees of success, to treat a variety of malignancies that include non-Hodgkin's lymphoma, cutaneous T-cell lymphoma, and multiple myeloma. Most of the effects mentioned above have been obtained in clinical studies that used
- IFN- β has emerged as the first drug capable of producing clinical improvement in multiple sclerosis (MS). Young adults are the primary target of this autoimmune neurologic disease, in which nerves in the central nervous system (CNS) undergo demyelination. This results in progressive neurologic dysfunction, which leads to significant and, in many cases, severe disability. This disease is often characterized by periods of nonprogression and remission alternating with periods of relapse. Treatment with IFN- β provides longer periods of remission and reduces the severity of relapses. Furthermore, magnetic-resonance-imaging studies of CNS damage in treated and untreated patients revealed that MS-induced damage was less in a group of IFN- β -treated patients than in untreated ones.
- IFN- γ has found application in the clinic as an agent for the treatment of chronic granulomatous disease (CGD). This disease is hereditary and quite rare. Its central feature is a serious impairment of the ability of phagocytic cells to kill ingested microbes. Patients with CGD are beset with recurring infections by a number of bacteria (*Staphylococcus aureus*, *Klebsiella*, *Pseudomonas*, and others) and fungi such as *Aspergillus* and *Candida*. Before interferon therapy, standard treatment for the disease was attempts to avoid infection, aggressive administration of

IFN- α alone. Future clinical trials in which IFN- α is used in combinations with other agents may improve the effectiveness of this interferon in cancer therapy.

characterized, erythropoietin, was isolated from the urine of anemic patients and shown to support the development of red blood cells. Subsequently, many cytokines have been shown to play essential roles in hematopoiesis (see

Table 12-5). During hematopoiesis, cytokines act as developmental signals that direct commitment of progenitor cells into and through particular lineages. As shown in Figure 12-16, a myeloid progenitor in the presence erythropoietin

antibiotics, and surgical drainage of abscesses. A failure to generate microbicidal oxidants (H_2O_2 , superoxide, and others) is the basis of CGD, and the administration of $IFN-\gamma$ significantly reverses this defect. Therapy of CGD patients with $IFN-\gamma$ significantly reduces the incidence of infections. Also, the infections that are contracted are less severe and the average number of days spent by patients in the hospital goes down.

- $IFN-\gamma$ has also been shown to be effective in the treatment of

osteopetrosis, (*not* osteoporosis) a life-threatening congenital disorder characterized by overgrowth of bone which results in blindness and deafness. Another problem presented by this disease is that the buildup of bone reduces the amount of space available for bone marrow and the decrease in hematopoiesis results in fewer red blood cells and anemia. The decreased generation of white blood cells causes an increased susceptibility to infection.

The use of interferons in clinical practice is likely to expand as more is learned

about their effects in combination with other therapeutic agents. Although, in common with other cytokines, interferons are powerful modifiers of biological responses, fortunately, the side effects accompanying their use are much milder. Typical side effects include flu-like symptoms, such as headache, fever, chills, and fatigue. These symptoms can largely be managed with acetaminophen (Tylenol) and diminish in intensity during continued treatment. Although interferon toxicity is usually not severe, serious manifestations such as anemia and depressed platelet and white-blood-cell counts have been seen.

Cytokine-Based Therapies In Clinical Use

Agent	Nature of agent	Clinical application
Enbrel	Chimeric TNF-receptor/IgG constant region	Rheumatoid arthritis
Remicade	Monoclonal antibody against TNF- α receptor	Rheumatoid arthritis
Interferon α -2a	Antiviral cytokine	Hepatitis B Hairy cell leukemia Kaposi's sarcoma
Interferon α -2b	Antiviral cytokine	Hepatitis C Melanoma
Interferon β	Antiviral cytokine	Multiple sclerosis
Actimmune	Interferon γ	Chronic granulomatous disease (CGD) Osteopetrosis
Neupogen	G-CSF (hematopoietic cytokine)	Stimulates production of neutrophils Reduction of infection in cancer patients treated with chemotherapy
Leukine	GM-CSF (hematopoietic cytokine)	Stimulates production of myeloid cells after bone-marrow transplantation
Neumega	Interleukin 11 (IL-11), a hematopoietic cytokine	Stimulates production of platelets
Epogen	Erythropoietin (hematopoietic cytokine)	Stimulates red-blood-cell production

would proceed down a pathway that leads to the production of erythrocytes; suitable concentrations of a group of cytokines including IL-3, GM-CSF, IL-1, and IL-6 will cause it to enter differentiation pathways that lead to the generation of

monocytes, neutrophils, and other leukocytes of the myeloid group. The participation of leukocytes in immune responses often results in their death and removal. However, both adaptive and innate immune responses generate cytokines that

TABLE 12-5 Haematopoietic cytokines

Haematopoietic growth factor	Sites of production	Main functions
Erythropoietin	Kidney, liver	Erythrocyte production
G-CSF	Endothelial cells, fibroblasts, macrophages	Neutrophil production
Thrombopoietin	Liver, kidney	Platelet production
M-CSF	Fibroblasts, endothelial cells, macrophages	Macrophage and osteoclast production
SCF/ <i>c-kit</i> ligand	Bone marrow stromal cells, constitutively	Stem cell, progenitor cells survival/division; mast cell differentiation
Flt-3 ligand	Fibroblasts, endothelial cells	Early progenitor cell expansion; pre-B cells
GM-CSF	T cells (T_H1 and T_H2), macrophages, mast cells	Macrophage, granulocyte production; dendritic cell maturation and activation
IL-3	T cells (T_H1 and T_H2), macrophages	Stem cells and myeloid progenitor cell growth; mast cells
IL-5	Activated helper T cells – T_H2 response only	Eosinophil production murine B-cell growth
IL-6	Activated T cells monocytes, fibroblasts, endothelial cells	Progenitor cell stimulation; platelet production; immunoglobulin production in B cells
IL-11	As above	As LIF
IL-7		T-cell survival

G-CSF, granulocyte colony-stimulating factor; GM-CSF, granulocyte-macrophage colony-stimulating factor; IL, interleukin; M-CSF, macrophage colony-stimulating factor; SCF, stem cell factor. Adapted from D. Thomas and A. Lopez, 2001.

Encyclopedia of Life Sciences: Haematopoietic growth factors, Nature Publishing Group.

stimulate and support the production of leukocytes. The steps at which a number of cytokines participate in hematopoiesis is shown in Figure 12-16.

SUMMARY

- Cytokines are low-molecular-weight proteins that are produced and secreted by a variety of cell types. They play major roles in the induction and regulation of the cellular interactions involving cells of the immune, inflammatory and hematopoietic systems.
- The biological activities of cytokines exhibit pleiotropy, redundancy, synergy, antagonism, and, in some instances, cascade induction.
- There are over 200 different cytokines, most of which fall into one of the following families: hematopoietins, interferons, chemokines, and tumor necrosis factors.
- Cytokines act by binding to cytokine receptors, most of which can be classified as immunoglobulin superfamily receptors, class I cytokine receptors, class II cytokine receptors, members of the TNF receptor family, and chemokine receptors.
- A cytokine can only act on a cell that expresses a receptor for it. The activity of particular cytokines is directed to specific cells by regulation of the cell's profile of cytokine receptors.

- Cytokine-induced multimerization of class I and class II cytokine receptors activates a JAK/STAT signal-transduction pathway.
- Antigen stimulation of T_H cells in the presence of certain cytokines can lead to the generation of subpopulations of helper T cells known as T_H1 and T_H2 . Each subset displays characteristic and different profiles of cytokine secretion.
- The cytokine profile of T_H1 cells supports immune responses that involve the marshalling of phagocytes, CTLs, and NK cells to eliminate intracellular pathogens. T_H2 cells produce cytokines that support production of particular immunoglobulin isotypes and IgE-mediated responses.
- Therapies based on cytokines and cytokine receptors have entered clinical practice.

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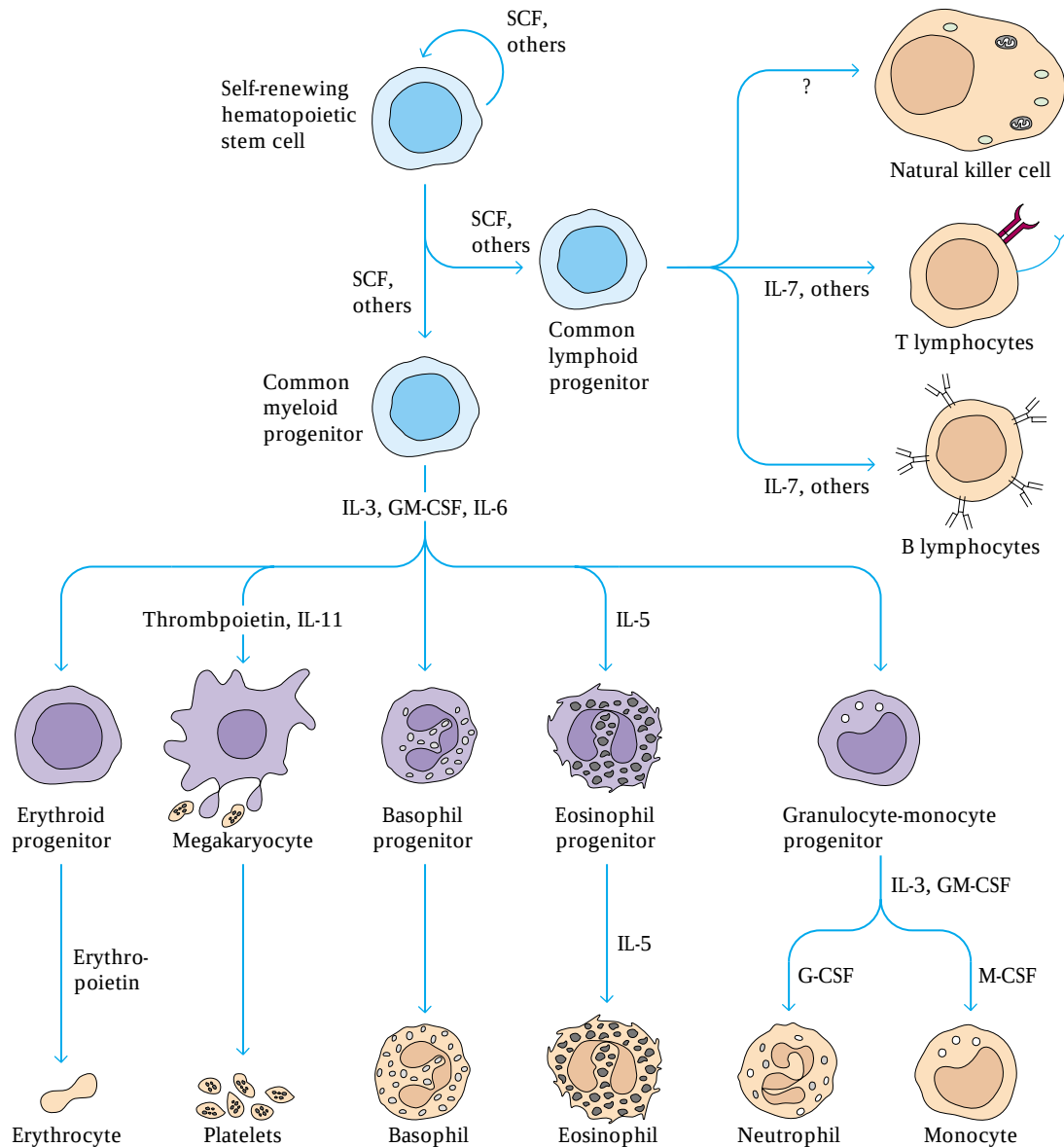


FIGURE 12-16 Hematopoietic cytokines and hematopoiesis. A variety of cytokines are involved in supporting the growth and directing the differentiation of hematopoietic cells. Note that additional factors may

be required for some of the developmental pathways shown in the diagram. CFU = colony-forming unit, a cell capable of generating a colony of cells from which the fully differentiated cell type emerges.

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USEFUL WEB SITES

<http://www.rndsystems.com/>

The cytokine minireviews found at R&D Systems Web site provide extensive, detailed, well-referenced, and often strikingly illustrated reviews of many cytokines and their receptors.

<http://www.ncbi.nlm.nih.gov/80/LocusLink/index.html>

LocusLink provides access to sequence and descriptive information about genetic loci of cytokines and other proteins. It also references papers discussing the basic biology (function and structure) of the gene or protein of interest.

Study Questions

CLINICAL FOCUS QUESTION Cytokines are proving to be powerful drugs, but their use is accompanied by side effects that can be harmful to patients. What are some of the side effects produced by Actimmune, Roferon, and interferon beta? (Hint: Manufacturer's Web sites often provide detailed information on the side effects of drugs they produce.)

1. Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.
 - a. The high-affinity IL-2 receptor consists of two transmembrane proteins.
 - b. The anti-TAC monoclonal antibody recognizes the IL-1 receptor on T cells.
 - c. All cytokine-binding receptors contain two or three subunits.
 - d. Expression of the β subunit of the IL-2 receptor is indicative of T-cell activation.
 - e. Some cytokine receptors possess domains with tyrosine kinase activity that function in signal transduction.
 - f. All members of each subfamily of the class I cytokine (hematopoietin) receptors share a common signal-transducing subunit.
2. When IL-2 is secreted by one T cell in a peripheral lymphoid organ, do all the T cells in the vicinity proliferate in response to the IL-2 or only some of them? Explain.
3. Briefly describe the similarities and differences among cytokines, growth factors, and hormones.
4. Indicate which subunit(s) of the IL-2 receptor are expressed by the following types of cells:
 - a. _____ Resting T cells
 - b. _____ Activated T cells
 - c. _____ Activated T cells + cyclosporin A
 - d. _____ Resting T_C cells
 - e. _____ CTLs
 - f. _____ NK cells
5. Superantigens have been implicated in several diseases and have been useful as research tools.
 - a. What properties of superantigens distinguish them from conventional antigens?
 - b. By what mechanism are bacterial superantigens thought to cause symptoms associated with food poisoning and toxic-shock syndrome?
 - c. Does the activity of superantigens exhibit MHC restriction?
6. IL-3, IL-5, and GM-CSF exhibit considerable redundancy in their effects. What structural feature of the receptors for these cytokines might explain this redundancy?
7. Considerable evidence indicates the existence of two T_H-cell subsets, differing in the pattern of cytokines they secrete.
 - a. What type of immune response is mediated by the T_H1 subset? What type of antigen challenge is likely to induce a T_H1-mediated response?
 - b. What type of immune response is mediated by the T_H2 subset? What type of antigen challenge is likely to induce a T_H2-mediated response?

chapter 13

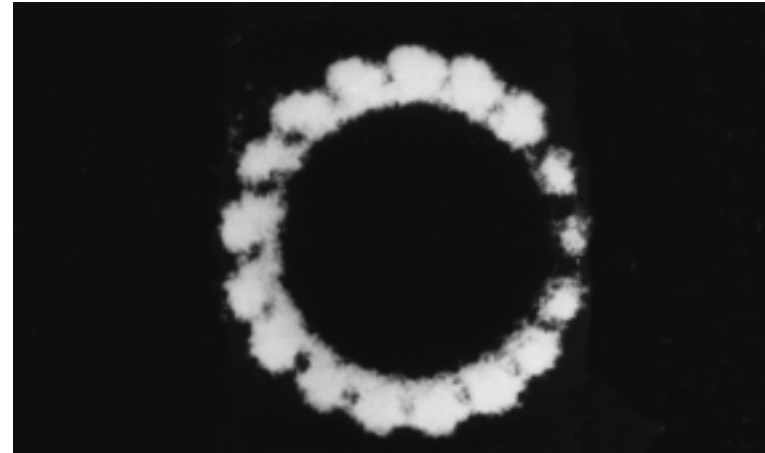
The Complement System

THE COMPLEMENT SYSTEM IS THE MAJOR EFFECTOR of the humoral branch of the immune system. Research on complement began in the 1890s, when Jules Bordet at the Institut Pasteur in Paris showed that sheep antiserum to the bacterium *Vibrio cholerae* caused lysis of the bacteria and that heating the antiserum destroyed its bacteriolytic activity. Surprisingly, the ability to lyse the bacteria was restored to the heated serum by adding fresh serum that contained no antibodies directed against the bacterium and was unable to kill the bacterium by itself. Bordet correctly reasoned that bacteriolytic activity requires two different substances: first, the specific antibacterial antibodies, which survive the heating process, and a second, heat-sensitive component responsible for the lytic activity. Bordet devised a simple test for the lytic activity, the easily detected lysis of antibody-coated red blood cells, called **hemolysis**. Paul Ehrlich in Berlin independently carried out similar experiments and coined the term *complement*, defining it as “the activity of blood serum that completes the action of antibody.” In ensuing years, researchers discovered that the action of complement was the result of interactions of a large and complex group of proteins.

This chapter describes the complement components and their activation pathways, the regulation of the complement system, the effector functions of various complement components, and the consequences of deficiencies in them. A Clinical Focus section describes consequences of a defect in proteins that regulate complement activity.

The Functions of Complement

Research on complement now includes more than 30 soluble and cell-bound proteins. The biological activities of this system affect both innate and acquired immunity and reach far beyond the original observations of antibody-mediated lysis of bacteria and red blood cells. Structural comparisons of the proteins involved in complement pathways place the origin of this system in primitive organisms possessing the most rudimentary innate immune systems. By contrast, the realization that interaction of cellular



Poly-C9 Complex

- The Functions of Complement
- The Complement Components
- Complement Activation
- Regulation of the Complement System
- Biological Consequences of Complement Activation
- Complement Deficiencies

receptors with complement proteins controls B-cell activities gives this system a role in the highly developed acquired immune system. Thus we have a system that straddles innate and acquired immunity, contributing to each in a variety of ways.

After initial activation, the various complement components interact, in a highly regulated cascade, to carry out a number of basic functions (Figure 13-1) including:

- Lysis of cells, bacteria, and viruses
- Opsonization, which promotes phagocytosis of particulate antigens
- Binding to specific complement receptors on cells of the immune system, triggering specific cell functions, inflammation, and secretion of immunoregulatory molecules
- Immune clearance, which removes immune complexes from the circulation and deposits them in the spleen and liver

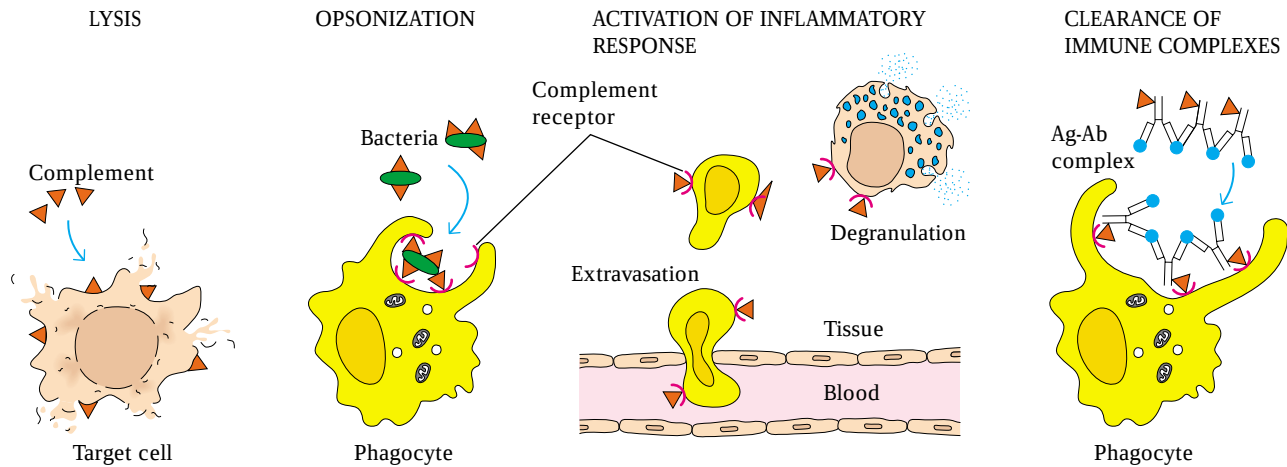


FIGURE 13-1 The multiple activities of the complement system. Serum complement proteins and membrane-bound complement receptors partake in a number of immune activities: lysis of foreign cells by antibody-dependent or antibody-independent pathways; opsonization or uptake of particulate antigens, including bacteria, by

phagocytes; activation of inflammatory responses; and clearance of circulating immune complexes by cells in the liver and spleen. Soluble complement proteins are schematically indicated by a triangle and receptors by a semi-circle; no attempt is made to differentiate among individual components of the complement system here.

The Complement Components

The proteins and glycoproteins that compose the complement system are synthesized mainly by liver hepatocytes, although significant amounts are also produced by blood monocytes, tissue macrophages, and epithelial cells of the gastrointestinal and genitourinary tracts. These components constitute 5% (by weight) of the serum globulin fraction. Most circulate in the serum in functionally inactive forms as proenzymes, or *zymogens*, which are inactive until proteolytic cleavage, which removes an inhibitory fragment and exposes the active site. The complement-reaction sequence starts with an enzyme cascade.

Complement components are designated by numerals (C1–C9), by letter symbols (e.g., factor D), or by trivial names (e.g., homologous restriction factor). Peptide fragments formed by activation of a component are denoted by small letters. In most cases, the smaller fragment resulting from cleavage of a component is designated “a” and the larger fragment designated “b” (e.g., C3a, C3b; note that C2 is an exception: C2a is the larger cleavage fragment). The larger fragments bind to the target near the site of activation, and the smaller fragments diffuse from the site and can initiate localized inflammatory responses by binding to specific receptors. The complement fragments interact with one another to form functional complexes. Those complexes that have enzymatic activity are designated by a bar over the number or symbol (e.g., C4b2a, C3bBb).

Complement Activation

Figure 13-2 on page 301 outlines the pathways of complement activation. The early steps, culminating in formation of

C5b, can occur by the **classical pathway**, the **alternative pathway**, or the **lectin pathway**. The final steps that lead to a membrane attack are the same in all pathways.

The Classical Pathway Begins with Antigen-Antibody Binding

Complement activation by the classical pathway commonly begins with the formation of soluble antigen-antibody complexes (immune complexes) or with the binding of antibody to antigen on a suitable target, such as a bacterial cell. IgM and certain subclasses of IgG (human IgG1, IgG2, and IgG3) can activate the classical complement pathway. The initial stage of activation involves C1, C2, C3, and C4, which are present in plasma in functionally inactive forms. Because the components were named in order of their discovery and before their functional roles had been determined, the numbers in their names do not always reflect the order in which they react.

The formation of an antigen-antibody complex induces conformational changes in the Fc portion of the IgM molecule that expose a binding site for the C1 component of the complement system. C1 in serum is a macromolecular complex consisting of C1q and two molecules each of C1r and C1s, held together in a complex (C1q₂s₂) stabilized by Ca²⁺ ions. The C1q molecule is composed of 18 polypeptide chains that associate to form six collagen-like triple helical arms, the tips of which bind to exposed C1q-binding sites in the C_H2 domain of the antibody molecule (Figure 13-3, on page 302). Each C1r and C1s monomer contains a catalytic domain and an interaction domain; the latter facilitates interaction with C1q or with each other.

Each C1 molecule must bind by its C1q globular heads to at least two Fc sites for a stable C1-antibody interaction to

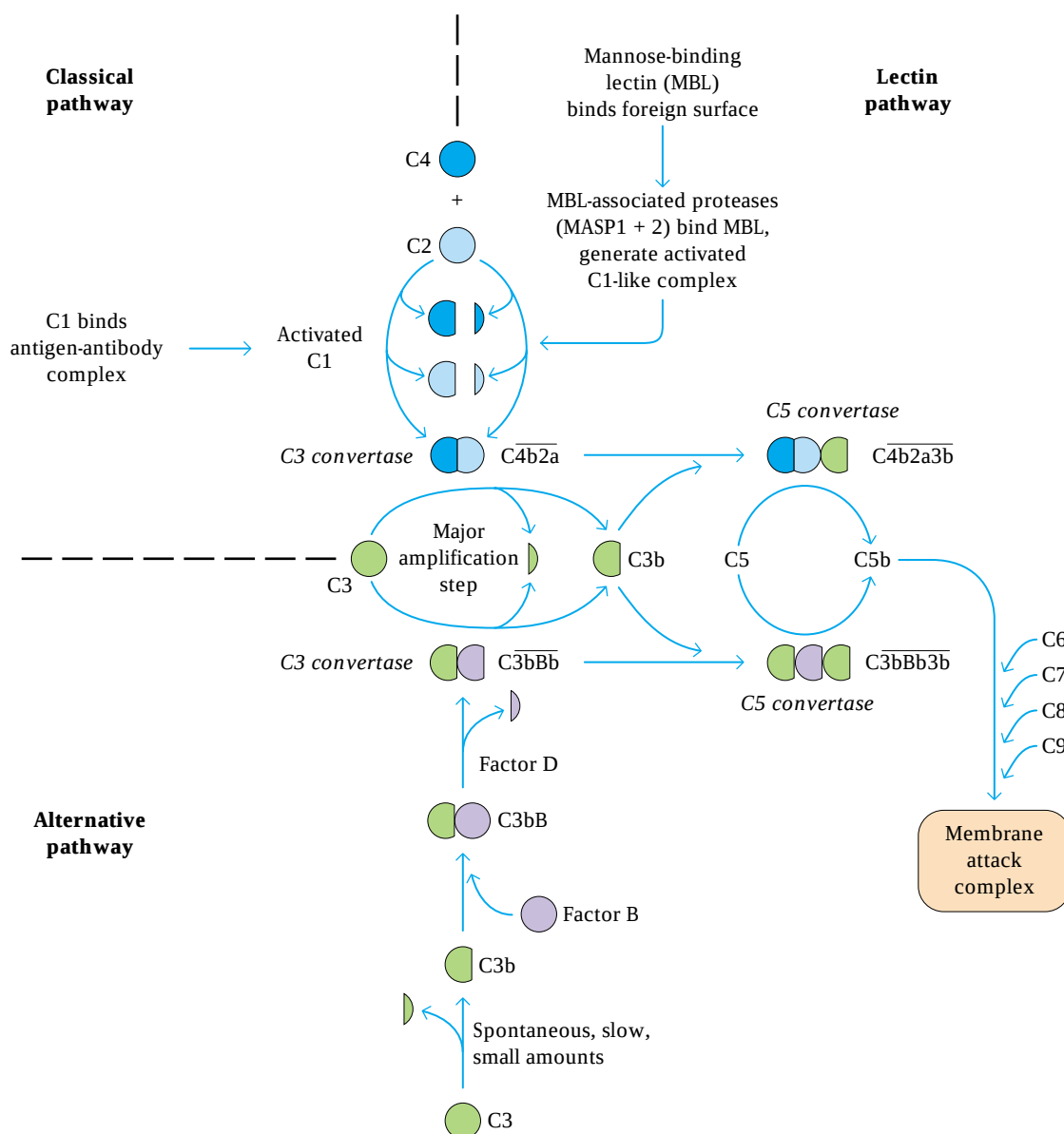


FIGURE 13-2 Overview of the complement activation pathways. The classical pathway is initiated when C1 binds to antigen-antibody complexes. The alternative pathway is initiated by binding of spontaneously generated C3b to activating surfaces such as microbial cell walls. The lectin pathway is initiated by binding of the serum protein MBL to the surface of a pathogen. All three pathways generate C3 and C5 convertases and bound C5b, which is converted into a mem-

brane-attack complex (MAC) by a common sequence of terminal reactions. Hydrolysis of C3 is the major amplification step in all pathways, generating large amounts of C3b, which forms part of C5 convertase. C3b also can diffuse away from the activating surface and bind to immune complexes or foreign cell surfaces, where it functions as an opsonin.

occur. When pentameric IgM is bound to antigen on a target surface it assumes the so-called “staple” configuration, in which at least three binding sites for C1q are exposed. Circulating IgM, however, exists as a planar configuration in which the C1q-binding sites are not exposed (Figure 13-4, on page 302) and therefore cannot activate the complement cascade. An IgG molecule, on the other hand, contains only a single C1q-binding site in the C_{H2} domain of the Fc, so that firm C1q binding is achieved only when two IgG molecules are within 30–40 nm of each other on a target surface or in a complex,

providing two attachment sites for C1q. This difference accounts for the observation that a single molecule of IgM bound to a red blood cell can activate the classical complement pathway and lyse the red blood cell while some 1000 molecules of IgG are required to assure that two IgG molecules are close enough to each other on the cell surface to initiate C1q binding.

The intermediates in the classical activation pathway are depicted schematically in Figure 13-5 (page 303). Binding of C1q to Fc binding sites induces a conformational change in

(text continues on page 304)

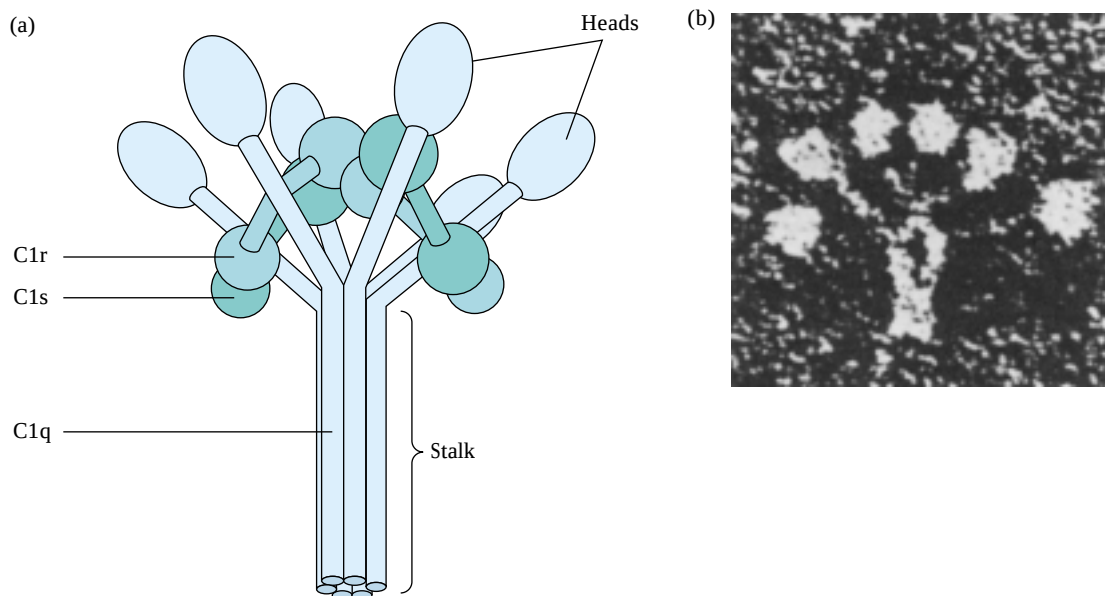


FIGURE 13-3 Structure of the C1 macromolecular complex. (a) Diagram of C1q₂S₂ complex. A C1q molecule consists of 18 polypeptide chains arranged into six triplets, each of which contains one A, one B, and one C chain. Each C1r and C1s monomer contains a cat-

alytic domain with enzymatic activity and an interaction domain that facilitates binding with C1q or with each other. (b) Electron micrograph of C1q molecule showing stalk and six globular heads. [Part (b) from H. R. Knobel *et al.*, 1975, *Eur. J. Immunol.* **5**:78.]

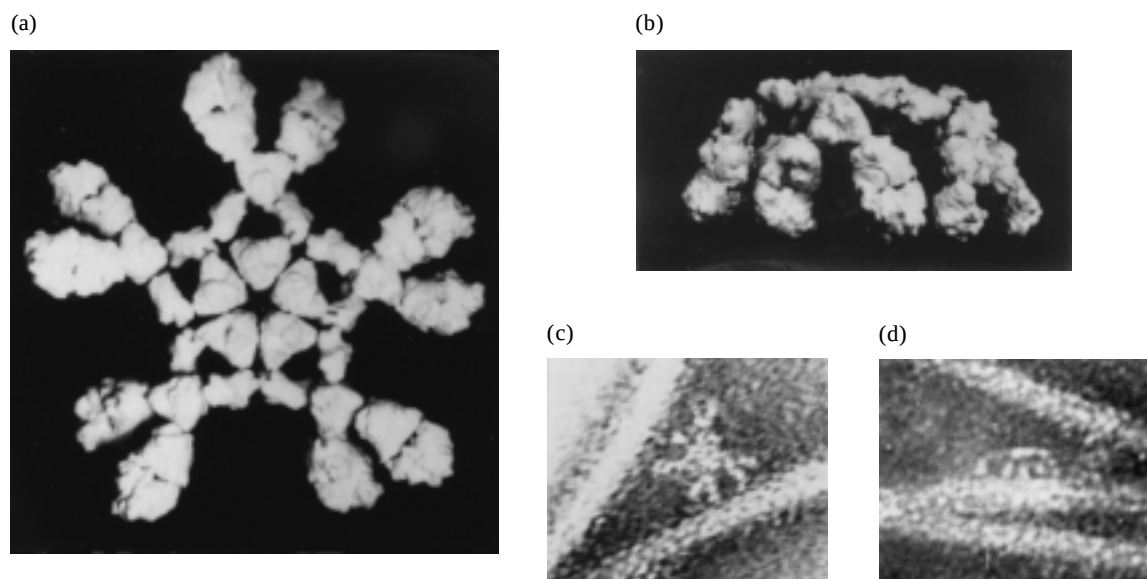


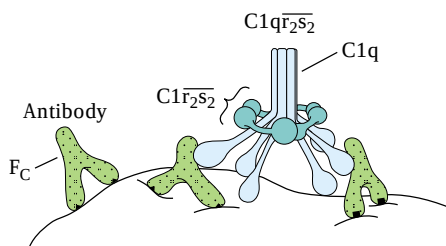
FIGURE 13-4 Models of pentameric IgM in planar form (a) and "staple" form (b). Several C1q-binding sites in the Fc region are accessible in the staple form, whereas none are exposed in the planar form. Electron micrographs of IgM antflagellum antibody

bound to flagella, showing the planar form (c) and staple form (d). [From A. Feinstein *et al.*, 1981, *Monogr. Allergy*, **17**:28, and 1981, *Ann. N.Y. Acad. Sci.* **190**:1104.]

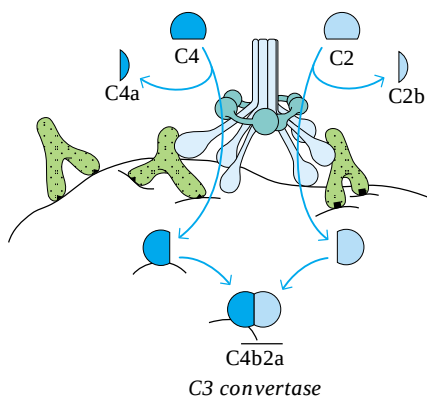


VISUALIZING CONCEPTS

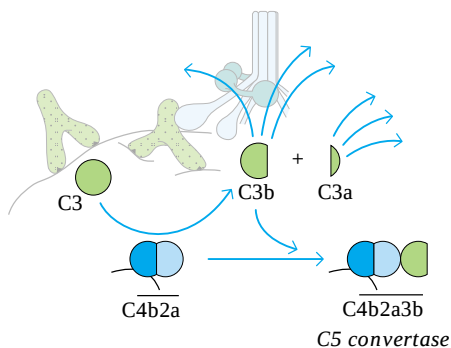
- ① C1q binds antigen-bound antibody. C1r activates autocatalytically and activates the second C1r; both activate C1s



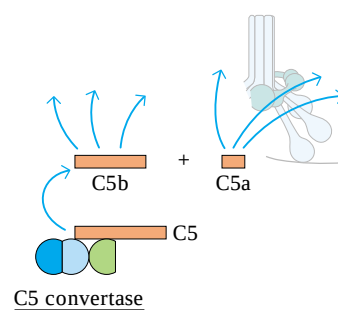
- ② C1s cleaves C4 and C2. Cleaving C4 exposes the binding site for C2. C4 binds the surface near C1 and C2 binds C4, forming C3 convertase



- ③ C3 convertase hydrolyzes many C3 molecules. Some combine with C3 convertase to form C5 convertase



- ④ The C3b component of C5 convertase binds C5, permitting C4b2a to cleave C5



- ⑤ C5b binds C6, initiating the formation of the membrane-attack complex

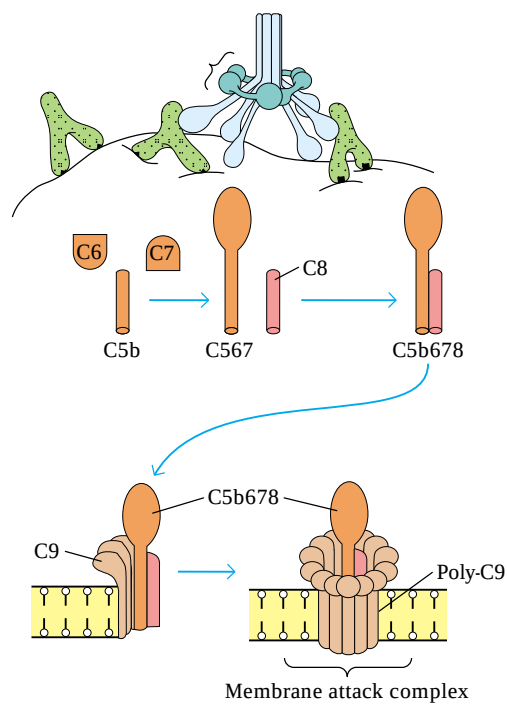


FIGURE 13-5 Schematic diagram of intermediates in the classical pathway of complement activation. The completed membrane-

attack complex (MAC, bottom right) forms a large pore in the membrane.

C1r that converts C1r to an active serine protease enzyme, C1r, which then cleaves C1s to a similar active enzyme, C1s. C1s has two substrates, C4 and C2. The C4 component is a glycoprotein containing three polypeptide chains α , β , and γ . C4 is activated when C1s hydrolyzes a small fragment (C4a) from the amino terminus of the α chain, exposing a binding site on the larger fragment (C4b). The C4b fragment attaches to the target surface in the vicinity of C1, and the C2 proenzyme then attaches to the exposed binding site on C4b, where the C2 is then cleaved by the neighboring C1s; the smaller fragment (C2b) diffuses away. The resulting C4b2a complex is called C3 convertase, referring to its role in converting the C3 into an active form. The smaller fragment from C4 cleavage, C4a, is an anaphylatoxin, or mediator of inflammation, which does not participate directly in the complement cascade; the anaphylatoxins, which include the smaller fragments of C4, C3, and C5 are described below.

The native C3 component consists of two polypeptide chains, α and β . Hydrolysis of a short fragment (C3a) from the amino terminus of the α chain by the C3 convertase generates C3b (Figure 13-6). A single C3 convertase molecule can generate over 200 molecules of C3b, resulting in tremendous amplification at this step of the sequence. Some of the C3b binds to C4b2a to form a trimolecular complex C4b2a3b, called C5 convertase. The C3b component of this complex binds C5 and alters its conformation, so that the C4b2a component can cleave C5 into C5a, which diffuses away, and C5b, which attaches to C6 and initiates formation of the membrane-attack complex in a sequence described later. Some of the C3b generated by C3 convertase activity does not associate with C4b2a; instead it diffuses away and then coats immune complexes and particulate antigens, functioning as an opsonin as described in the Clinical Focus. C3b may also bind directly to cell membranes.

The Alternative Pathway Is Antibody-Independent

The alternative pathway generates bound C5b, the same product that the classical pathway generates, but it does so without the need for antigen-antibody complexes for initiation. Because no antibody is required, the alternative pathway is a component of the innate immune system. This major pathway of complement activation involves four serum proteins: C3, factor B, factor D, and properdin. The alternative pathway is initiated in most cases by cell-surface constituents that are foreign to the host (Table 13-1). For example, both gram-negative and gram-positive bacteria have cell-wall constituents that can activate the alternative pathway. The intermediates in the alternative pathway for generating C5b are shown schematically in Figure 13-7 (page 306).

In the classical pathway, C3 is rapidly cleaved to C3a and C3b by the enzymatic activity of the C3 convertase. In the alternative pathway, serum C3, which contains an unstable thioester bond, is subject to slow spontaneous hydrolysis to yield C3a and C3b. The C3b component can bind to foreign surface antigens (such as those on bacterial cells or viral particles) or even to the host's own cells (see Figure 13-6c). The membranes of most mammalian cells have high levels of sialic acid, which contributes to the rapid inactivation of bound C3b molecules on host cells; consequently this binding rarely leads to further reactions on the host cell membrane. Because many foreign antigenic surfaces (e.g., bacterial cell walls, yeast cell walls, and certain viral envelopes) have only low levels of sialic acid, C3b bound to these surfaces remains active for a longer time. The C3b present on the surface of the foreign cells can bind another serum protein called factor B to form a complex stabilized by Mg^{2+} . Binding to C3b exposes a site on factor B that serves as the sub-

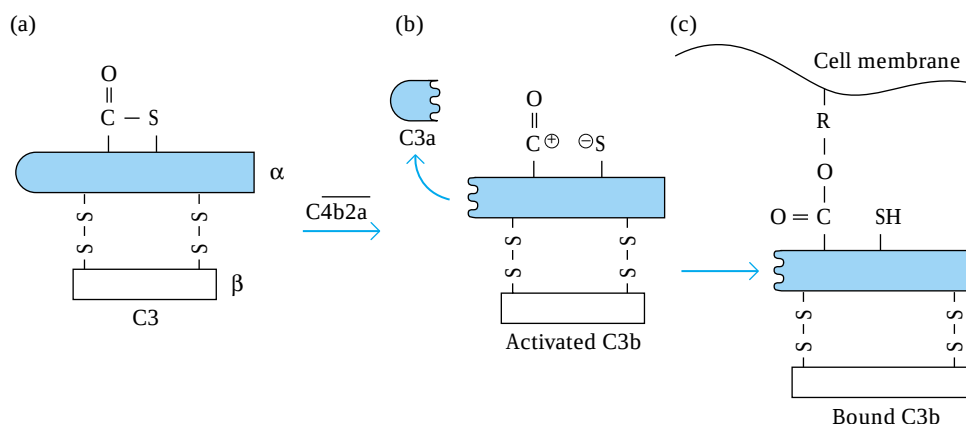


FIGURE 13-6 Hydrolysis of C3 by C3 convertase C4b2a (a) Native C3. (b) Activated C3 showing site of cleavage by C4b2a resulting in production of the C3a and C3b fragments. (c) A labile internal thioester bond in C3 is activated as C3b is formed, allowing the C3b

fragment to bind to free hydroxyl or amino groups (R) on a cell membrane. Bound C3b exhibits various biological activities, including binding of C5 and binding to C3b receptors on phagocytic cells.

TABLE 13-1 Initiators of the alternative pathway of complement activation

PATHOGENS AND PARTICLES OF MICROBIAL ORIGIN
Many strains of gram-negative bacteria
Lipopolysaccharides from gram-negative bacteria
Many strains of gram-positive bacteria
Teichoic acid from gram-positive cell walls
Fungal and yeast cell walls (zymosan)
Some viruses and virus-infected cells
Some tumor cells (Raji)
Parasites (trypanosomes)
NONPATHOGENS
Human IgG, IgA, and IgE in complexes
Rabbit and guinea pig IgG in complexes
Cobra venom factor
Heterologous erythrocytes (rabbit, mouse, chicken)
Anionic polymers (dextran sulfate)
Pure carbohydrates (agarose, inulin)

SOURCE: Adapted from M. K. Pangburn, 1986, in *Immunobiology of the Complement System*, Academic Press.

strate for an enzymatically active serum protein called factor D. Factor D cleaves the C3b-bound factor B, releasing a small fragment (Ba) that diffuses away and generating C3bBb. The C3bBb complex has C3 convertase activity and thus is analogous to the C4b2a complex in the classical pathway. The C3 convertase activity of C3bBb has a half-life of only 5 minutes unless the serum protein properdin binds to it, stabilizing it and extending the half-life of this convertase activity to 30 minutes.

The C3bBb generated in the alternative pathway can activate unhydrolyzed C3 to generate more C3b autocatalytically. As a result, the initial steps are repeated and amplified, so that more than 2×10^6 molecules of C3b can be deposited on an antigenic surface in less than 5 minutes. The C3 convertase activity of C3bBb generates the C3bBb3b complex, which exhibits C5 convertase activity, analogous to the C4b2a3b complex in the classical pathway. The nonenzymatic C3b component binds C5, and the Bb component subsequently hydrolyzes the bound C5 to generate C5a and C5b (see Figure 13-7); the latter binds to the antigenic surface.

The Lectin Pathway Originates With Host Proteins Binding Microbial Surfaces

Lectins are proteins that recognize and bind to specific carbohydrate targets. (Because the lectin that activates comple-

ment binds to mannose residues, some authors designate this the MBlectin pathway or mannan-binding lectin pathway.) The lectin pathway, like the alternative pathway, does not depend on antibody for its activation. However, the mechanism is more like that of the classical pathway, because after initiation, it proceeds, through the action of C4 and C2, to produce a C5 convertase (see Figure 13-2).

The lectin pathway is activated by the binding of man-nose-binding lectin (MBL) to mannose residues on glyco-proteins or carbohydrates on the surface of microorganisms including certain *Salmonella*, *Listeria*, and *Neisseria* strains, as well as *Cryptococcus neoformans* and *Candida albicans*. MBL is an **acute phase protein** produced in inflammatory responses. Its function in the complement pathway is similar to that of C1q, which it resembles in structure. After MBL binds to the surface of a cell or pathogen, MBL-associated serine proteases, MASP-1 and MASP-2, bind to MBL. The active complex formed by this association causes cleavage and activation of C4 and C2. The MASP-1 and -2 proteins have structural similarity to C1r and C1s and mimic their activities. This means of activating the C2–C4 components to form a C5 convertase without need for specific antibody binding represents an important innate defense mechanism comparable to the alternative pathway, but utilizing the elements of the classical pathway except for the C1 proteins.

The Three Complement Pathways Converge at the Membrane-Attack Complex

The terminal sequence of complement activation involves C5b, C6, C7, C8, and C9, which interact sequentially to form a macromolecular structure called the **membrane-attack complex (MAC)**. This complex forms a large channel through the membrane of the target cell, enabling ions and small molecules to diffuse freely across the membrane.

The end result of activating the classical, alternative, or lectin pathways is production of an active C5 convertase. This enzyme cleaves C5, which contains two protein chains, α and β . After binding of C5 to the nonenzymatic C3b component of the convertase, the amino terminus of the α chain is cleaved. This generates the small C5a fragment, which diffuses away, and the large C5b fragment, which binds to the surface of the target cell and provides a binding site for the subsequent components of the membrane-attack complex (see Figure 13-5, step 5). The C5b component is extremely labile and becomes inactive within 2 minutes unless C6 binds to it and stabilizes its activity.

Up to this point, all the complement reactions take place on the hydrophilic surface of membranes or on immune complexes in the fluid phase. As C5b6 binds to C7, the resulting complex undergoes a hydrophilic-amphiphilic structural transition that exposes hydrophobic regions, which serve as binding sites for membrane phospholipids. If the reaction occurs on a target-cell membrane, the hydrophobic binding sites enable the C5b67 complex to insert into the phospholipid bilayer. If, however, the reaction occurs on an immune



VISUALIZING CONCEPTS

- 1 C3 hydrolyzes spontaneously, C3b fragment attaches to foreign surface
- 2 Factor B binds C3a, exposes site acted on by Factor D. Cleavage generates C3bBb, which has C3 convertase activity
- 3 Binding of properdin stabilizes convertase
- 4 Convertase generates C3b; some binds to C3 convertase activating C5' convertase. C5b binds to antigenic surface

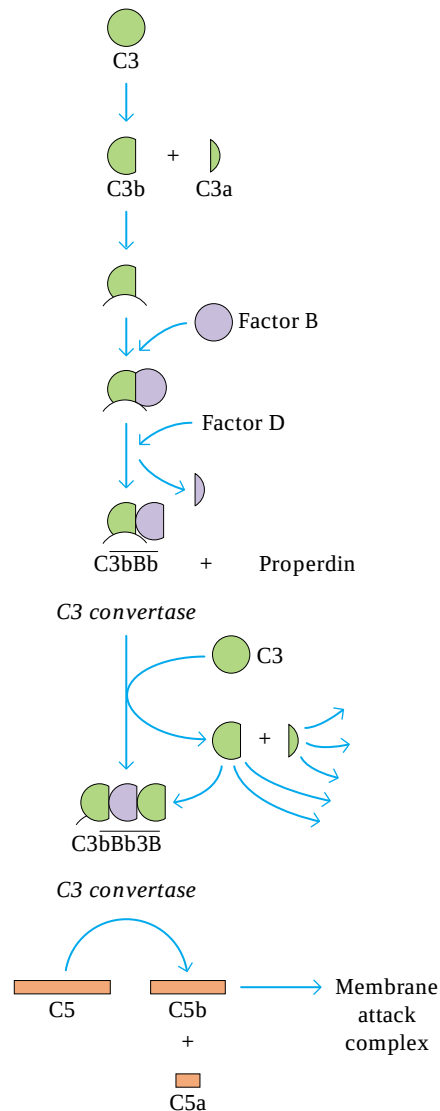


FIGURE 13-7 Schematic diagram of intermediates in the formation of bound C5b by the alternative pathway of complement activation. The C3bBb complex is stabilized by binding of prop-

erdin. Conversion of bound C5b to the membrane-attack complex occurs by the same sequence of reactions as in the classical pathway (see Figure 13-5).

complex or other noncellular activating surface, then the hydrophobic binding sites cannot anchor the complex and it is released. Released C5b67 complexes can insert into the membrane of nearby cells and mediate “innocent-bystander” lysis. Regulator proteins normally prevent this from occurring, but in certain diseases cell and tissue damage may result from innocent-bystander lysis. A hemolytic disorder resulting from deficiency in a regulatory protein is explained in the Clinical Focus section and an autoimmune process in which immune

complexes mediate tissue damage will be considered in Chapter 20.

Binding of C8 to membrane-bound C5b67 induces a conformational change in C8, so that it too undergoes a hydrophilic-amphiphilic structural transition, exposing a hydrophobic region, which interacts with the plasma membrane. The C5b678 complex creates a small pore, 10 Å in diameter; formation of this pore can lead to lysis of red blood cells but not of nucleated cells. The final step in formation of

the MAC is the binding and polymerization of C9, a perforin-like molecule, to the C5b678 complex. As many as 10–17 molecules of C9 can be bound and polymerized by a single C5b678 complex. During polymerization, the C9 molecules undergo a hydrophilic-amphiphilic transition, so that they too can insert into the membrane. The completed MAC, which has a tubular form and functional pore size of 70–100 Å, consists of a C5b678 complex surrounded by a poly-C9 complex (Figure 13-8). Since ions and small molecules can diffuse freely through the central channel of the MAC, the cell cannot maintain its osmotic stability and is killed by an influx of water and loss of electrolytes.

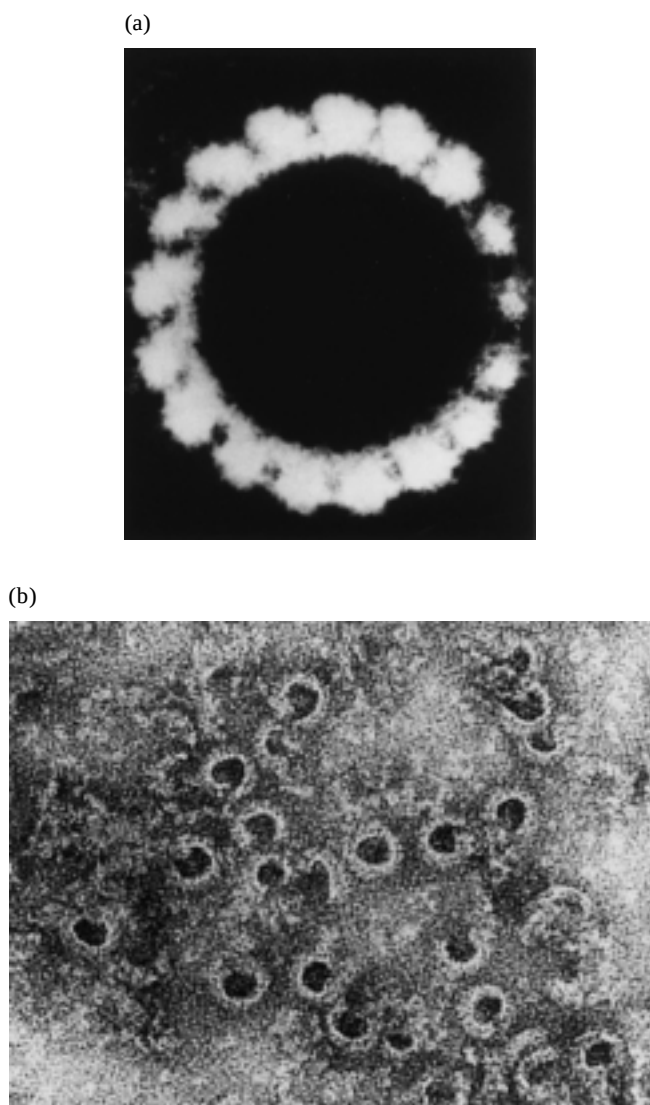


FIGURE 13-8 (a) Photomicrograph of poly-C9 complex formed by in vitro polymerization of C9. (b) Photomicrograph of complement-induced lesions on the membrane of a red blood cell. These lesions result from formation of membrane-attack complexes. [Part (a) from E. R. Podack, 1986, in *Immunobiology of the Complement System*, Academic Press; part (b) from J. Humphrey and R. Dourmashkin, 1969, *Adv. Immunol.* **11**:75.]

Regulation of the Complement System

Because many elements of the complement system are capable of attacking host cells as well as foreign cells and microorganisms, elaborate regulatory mechanisms have evolved to restrict complement activity to designated targets. A general mechanism of regulation in all complement pathways is the inclusion of highly labile components that undergo spontaneous inactivation if they are not stabilized by reaction with other components. In addition, a series of regulatory proteins can inactivate various complement components (Table 13-2). For example, the glycoprotein C1 inhibitor (C1Inh) can form a complex with C1r₂s₂, causing it to dissociate from C1q and preventing further activation of C4 or C2 (Figure 13-9a(1)).

The reaction catalyzed by the C3 convertase enzymes of the classical, lectin, and alternative pathways is the major amplification step in complement activation, generating hundreds of molecules of C3b. The C3b generated by these enzymes has the potential to bind to nearby cells, mediating damage to the healthy cells by causing their opsonization by phagocytic cells bearing C3b receptors or by induction of the membrane-attack complex. Damage to normal host cells is prevented because C3b undergoes spontaneous hydrolysis by the time it has diffused 40 nm away from the C4b2a or C3bBb convertase enzymes, so that it can no longer bind to its target site. The potential destruction of healthy host cells by C3b is further limited by a family of related proteins that regulate C3 convertase activity in the classical and alternative pathways. These regulatory proteins all contain repeating amino acid sequences (or motifs) of about 60 residues, termed *short consensus repeats* (SCRs). All these proteins are encoded at a single location on chromosome 1 in humans, known as the *regulators of complement activation* (RCA) gene cluster.

In the classical and lectin pathways, three structurally distinct RCA proteins act similarly to prevent assembly of C3 convertase (Figure 13-9a(2)). These regulatory proteins include soluble C4b-binding protein (C4bBP) and two membrane-bound proteins, complement receptor type 1 (CR1) and membrane cofactor protein (MCP). Each of these regulatory proteins binds to C4b and prevents its association with C2a. Once C4bBP, CR1, or MCP is bound to C4b, another regulatory protein, factor I, cleaves the C4b into bound C4d and soluble C4c (Figure 13-9a(3)). A similar regulatory sequence operates to prevent assembly of the C3 convertase C3bBb in the alternative pathway. In this case CR1, MCP, or a regulatory component called factor H binds to C3b and prevents its association with factor B (Figure 13-9a(4)). Once CR1, MCP, or factor H is bound to C3b, factor I cleaves the C3b into a bound iC3b fragment and a soluble C3f fragment. Further cleavage of iC3b by factor I releases C3c and leaves C3dg bound to the membrane (Figure 13-9a(5)). The molecular events involved in regulation of cell-bound C4b and C3b are depicted in Figure 13-10 (page 310).

TABLE 13-2 Proteins that regulate the complement system

Protein	Type of protein	Pathway affected	Immunologic function
C1 inhibitor (C1Inh)	Soluble	Classical	Serine protease inhibitor: causes C1r ₂ S ₂ to dissociate from C1q
C4b-binding protein (C4bBP)*	Soluble	Classical and lectin	Blocks formation of C3 convertase by binding C4b; cofactor for cleavage of C4b by factor I
Factor H*	Soluble	Alternative	Blocks formation of C3 convertase by binding C3b; cofactor for cleavage of C3b by factor I
Complement-receptor type 1 (CR1)* Membrane-cofactor protein (MCP)*	Membrane bound	Classical, alternative, and lectin	Block formation of C3 convertase by binding C4b or C3b; cofactor for factor I-catalyzed cleavage of C4b or C3b C3bBb
Decay-accelerating factor (DAE or CD55)*			
Factor-I	Soluble	Classical, alternative, and lectin	Serine protease: cleaves C4b or C3b using C4bBP, CR1, factor H, DAE, or MCP as cofactor
S protein	Soluble	Terminal	Binds soluble C5b67 and prevents its insertion into cell membrane
Homologous restriction factor (HRF) Membrane inhibitor of reactive lysis (MIRL or CD59)*	Membrane bound	Terminal	Bind to C5b678 on autologous cells, blocking binding of C9
Anaphylatoxin inactivator			
	Soluble	Effector	Inactivates anaphylatoxin activity of C3a, C4a, and C5a by carboxypeptidase N removal of C-terminal Arg

*An RCA (regulator of complement activation) protein. In humans, all RCA proteins are encoded on chromosome 1 and contain short consensus repeats.

Several RCA proteins also act on the assembled C3 convertase, causing it to dissociate; these include the previously mentioned C4bBP, CR1, and factor H. In addition, decay-accelerating factor (DAF or CD55), which is a glycoprotein anchored covalently to a glycopospholipid membrane protein, has the ability to dissociate C3 convertase. The consequences of DAF deficiency are described in the Clinical Focus section. Each of these RCA proteins accelerates decay (dissociation) of C3 convertase by releasing the component with enzymatic activity (C2a or Bb) from the cell-bound component (C4b or C3b). Once dissociation of the C3 convertase occurs, factor I cleaves the remaining membrane-bound C4b or C3b component, irreversibly inactivating the convertase (Figure 13-9b).

Regulatory proteins also operate at the level of the membrane-attack complex. The potential release of the C5b67 complex poses a threat of innocent-bystander lysis to healthy cells. A number of serum proteins counter this threat by binding to released C5b67 and preventing its insertion into the membrane of nearby cells. A serum protein called S pro-

tein can bind to C5b67, inducing a hydrophilic transition and thereby preventing insertion of C5b67 into the membrane of nearby cells (Figure 13-9c(1)).

Complement-mediated lysis of cells is more effective if the complement is from a species different from that of the cells being lysed. This phenomenon depends on two membrane proteins that block MAC formation. These two proteins, present on the membrane of many cell types, are *homologous restriction factor* (HRF) and *membrane inhibitor of reactive lysis* (MIRL or CD59). Both HRF and MIRL protect cells from nonspecific complement-mediated lysis by binding to C8, preventing assembly of poly-C9 and its insertion into the plasma membrane (Figure 13-9c(2)). However, this inhibition occurs only if the complement components are from the same species as the target cells. For this reason, MIRL and HRF are said to display homologous restriction, for which the latter was named. As discussed in Chapter 21, homologous restriction poses a barrier to the use of organs from other species for clinical transplantation.

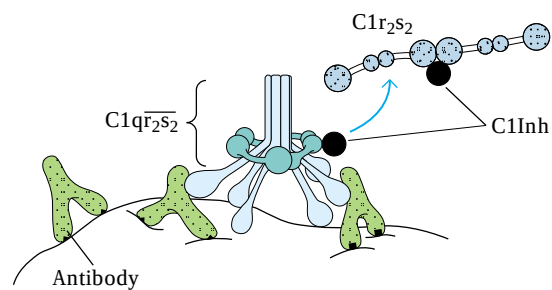


VISUALIZING CONCEPTS

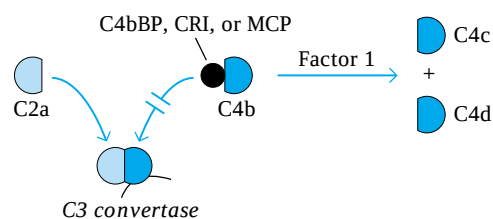
Regulation of the Complement System

(a) Before assembly of convertase activity

- 1 C1 inhibitor (C1Iab) binds C1r₂s₂, causing dissociation from C1q

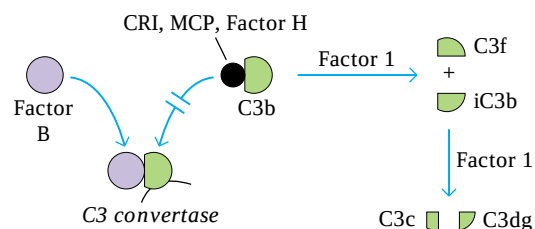


- 2 Association of C4b and C2a is blocked by binding C4b-binding protein (C4bBP), complement receptor type I, or membrane cofactor protein (MCP)



- 3 Inhibitor-bound C4b is cleaved by Factor 1

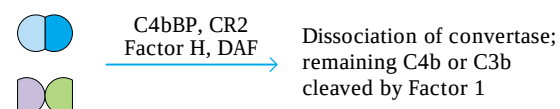
- 4 In alternative pathway, CR1, MCP, or Factor H prevent binding of C3b and Factor B



- 5 Inhibitor-bound C3b is cleaved by Factor 1

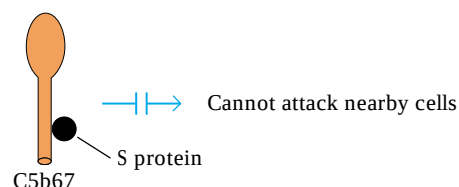
(b) After assembly of convertase

C3 convertases are dissociated by C4bBP, CR1, Factor H, and decay-accelerating factor (DAF)

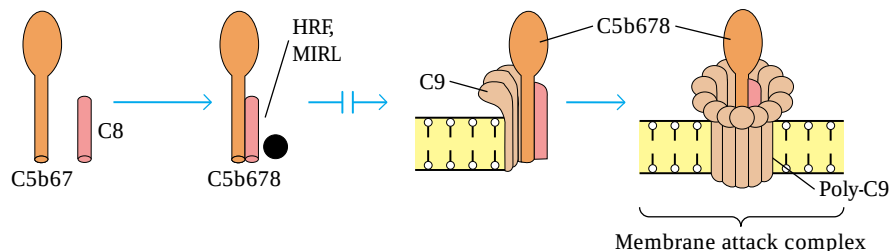


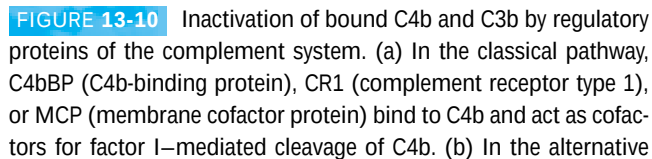
(c) Regulation at assembly of membrane-attack complex (MAC)

- 1 S protein prevents insertion of C5b67 MAC component into the membrane



- 2 Homologous restriction factor (HRF) and membrane inhibitor of reactive lysis (MIRL or CD59) bind C8₁, preventing assembly of poly-C9 and blocking formation of MAC

**FIGURE 13-9** Regulation of the complement system by regulatory proteins (black).



Biological Consequences of Complement Activation

Many of the biological activities of the complement system depend on the binding of complement fragments to complement receptors, which are expressed by various cells. In addition, some complement receptors play an important role in regulating complement activity by binding biologically active complement components and degrading them into inactive products. The complement receptors and their primary ligands, which include various complement components and their proteolytic breakdown products, are listed in Table 13-4 (page 312).

The membrane-attack complex formed by complement activation can lyse gram-negative bacteria, parasites, viruses, erythrocytes, and nucleated cells. Because the alternative and lectin pathways of activation generally occur without an initial antigen-antibody interaction, these pathways serve as important innate immune defenses against infectious microorganisms. The requirement for an initial antigen-antibody reaction in the classical pathway supplements these nonspecific innate defenses with a more specific defense mechanism. In some instances, the requirement for antibody in the activating event may be supplied by so-called natural antibodies, which are raised against common components of ubiquitous microbes.

The importance of cell-mediated immunity in host defense against viral infections has been emphasized in previous chapters. Nevertheless, antibody and complement do play a role in host defense against viruses and are often crucial in containing viral spread during acute infection and in protecting against reinfection. Most—perhaps



CLINICAL FOCUS

Paroxymal Nocturnal Hemoglobinuria: a Defect in Regulation of Complement Lysis

Common conditions associated with deficiency in the complement components include increased susceptibility to bacterial infections and systemic lupus erythematosus which is related to the inability to clear immune complexes. Deficiency in the proteins that regulate complement activity can cause equally serious disorders. An example is paroxymal nocturnal hemoglobinuria, or PNH, which manifests as increased fragility of erythrocytes, leading to chronic hemolytic anemia, pancytopenia (loss of blood cells of all types) and venous thrombosis (formation of blood clots). The name PNH derives from the presence of hemoglobin in the urine, most commonly observed in the first urine passed after a night's sleep. The cause of PNH is a general defect in synthesis of cell-surface proteins, which affects the expression of two regulators of complement, DAF (decay accelerating factor or CD55) and MIRL (membrane inhibitor of reactive lysis or CD59).

DAF and MIRL are cell-surface proteins that function as inhibitors of com-

plement-mediated cell lysis, but act at different stages of the process. DAF inhibits cell lysis by causing dissociation and inactivation of the C3 convertases of the classical, lectin, and alternative pathways (see Figure 13-9b). MIRL acts later in the pathway by binding to the C5b678 complex, which inhibits C9 binding and prevents formation of the pores that destroy the cell under attack. Both proteins are expressed on erythrocytes as well as a number of other hematopoietic cell types. Deficiency in these proteins leads to highly increased sensitivity of host cells to the lytic effects of the host's complement activity. PNH, the clinical consequence of deficiency in DAF and MIRL, is a chronic disease with a mean survival time between 10 and 15 years. The most common causes of mortality in PNH are venous thrombosis affecting hepatic veins and progressive bone-marrow failure.

An obvious question about this rare but serious disease concerns the fact that two different proteins are involved in its pathogenesis. The simultaneous occurrence of a genetic defect in each of

them would be rarer than the 1 in 100,000 incidence of PNH. The answer is that neither protein itself is defective in PNH; the defect lies in a posttranslational modification of the peptide anchor that binds them to the cell membrane. While most proteins that are expressed on the surface of cells have hydrophobic sequences that traverse the lipid bilayer in the cell membrane, some proteins are bound by glycolipid anchors (glycosyl phosphatidylinositol, or GPI) attached to amino acid residues in the protein. Without the ability to form GPI anchors, proteins that attach in this manner will be absent from the cell surface, including both DAF and MIRL.

The defect identified in PNH lies early in the enzymatic path to formation of a GPI anchor and resides in the *pig-a* gene (phosphatidylinositol glycan complementation class A gene). Transfection of cells from PNH patients with an intact *pig-a* gene restored the cells' resistance to host complement lysis. Examination of *pig-a* sequences in PNH patients reveals a number of different defects in this X-linked gene, indicating somatic rather than genetic origin of the defect. This description of PNH underscores the fact that the complement system is a powerful defender of the host but also a dangerous one. Complex systems of regulation are necessary to protect host cells from the activated complement complexes generated to lyse intruders.

all—enveloped viruses are susceptible to complement-mediated lysis. The viral envelope is largely derived from the plasma membrane of infected host cells and is therefore susceptible to pore formation by the membrane-attack complex. Among the pathogenic viruses susceptible to lysis by complement-mediated lysis are herpesviruses, orthomyxoviruses, paramyxoviruses, and retroviruses.

The complement system is generally quite effective in lysing gram-negative bacteria (Figure 13-11). However, some gram-negative bacteria and most gram-positive bacteria have mechanisms for evading complement-mediated

damage (Table 13-5). For example, a few gram-negative bacteria can develop resistance to complement-mediated lysis that correlates with the virulence of the organism. In *Escherichia coli* and *Salmonella*, resistance to complement is associated with the smooth bacterial phenotype, which is characterized by the presence of long polysaccharide side chains in the cell-wall lipopolysaccharide (LPS) component. It has been proposed that the increased LPS in the wall of resistant strains may prevent insertion of the MAC into the bacterial membrane, so that the complex is released from the bacterial cell rather than forming a pore. Strains of *Neisseria*

TABLE 13-3 Summary of biological effects mediated by complement products

Effect	Complement product mediating*
Cell lysis	C5b–9, the membrane-attack complex (MAC)
Inflammatory response	
Degranulation of mast cells and basophils [†]	C3a, C4a, and C5a (anaphylatoxins)
Degranulation of eosinophils	C3a, C5a
Extravasation and chemotaxis of leukocytes at inflammatory site	C3a, C5a , C5b67
Aggregation of platelets	C3a, C5a
Inhibition of monocyte/macrophage migration and induction of their spreading	Bb
Release of neutrophils from bone marrow	C3c
Release of hydrolytic enzymes from neutrophils	C5a
Increased expression of complement receptors type 1 and 3 (CR1 and CR3) on neutrophils	C5a
Opsonization of particulate antigens, increasing their phagocytosis	C3b , C4b, iC3b
Viral neutralization	C3b, C5b–9 (MAC)
Solubilization and clearance of immune complexes	C3b

*Boldfaced component is most important in mediating indicated effect.

[†]Degranulation leads to release of histamine and other mediators that induce contraction of smooth muscle and increased permeability of vessels.

gonorrheae resistant to complement-mediated killing have been associated with disseminated gonococcal infections in humans. Some evidence suggests that the membrane proteins of resistant *Neisseria* strains undergo noncovalent interactions with the MAC that prevent its insertion into the outer membrane of the bacterial cells. These examples of resistant gram-negative bacteria are the exception; most gram-negative bacteria are susceptible to complement-mediated lysis.

Gram-positive bacteria are generally resistant to complement-mediated lysis because the thick peptidoglycan layer in their cell wall prevents insertion of the MAC into the inner membrane. Although complement activation can occur on the cell membrane of encapsulated bacteria such as *Streptococcus pneumoniae*, the capsule prevents interaction between C3b deposited on the membrane and the CR1 on phagocytic cells. Some bacteria possess an elastase that inactivates C3a and C5a, preventing these split products from inducing an

TABLE 13-4 Complement-binding receptors

Receptor	Major ligands	Activity	Cellular distribution
CR1 (CD35)	C3b, C4b	Blocks formation of C3 convertase; binds immune complexes to cells	Erythrocytes, neutrophils, monocytes, macrophages, eosinophils, follicular dendritic cells, B cells, some T cells
CR2 (CD21)	C3d, C3dg,* iC3b	Part of B-cell coreceptor; binds Epstein-Barr virus	B cells, follicular dendritic cells, some T cells
CR3 (CD11b/18)	iC3b	Bind cell-adhesion molecules on neutrophils, facilitating their extravasation; bind immune complexes, enhancing their phagocytosis	Monocytes, macrophages, neutrophils, natural killer cells, some T cells
CR4 (CD11c/18)			
C3a/C4a receptor	C3a, C4a	Induces degranulation of mast cells and basophils	Mast cells, basophils, granulocytes
C5a receptor	C5a	Induces degranulation of mast cells and basophils	Mast cells, basophils, granulocytes, monocytes, macrophages, platelets, endothelial cells

*Cleavage of C3dg by serum proteases generates C3d and C3g.

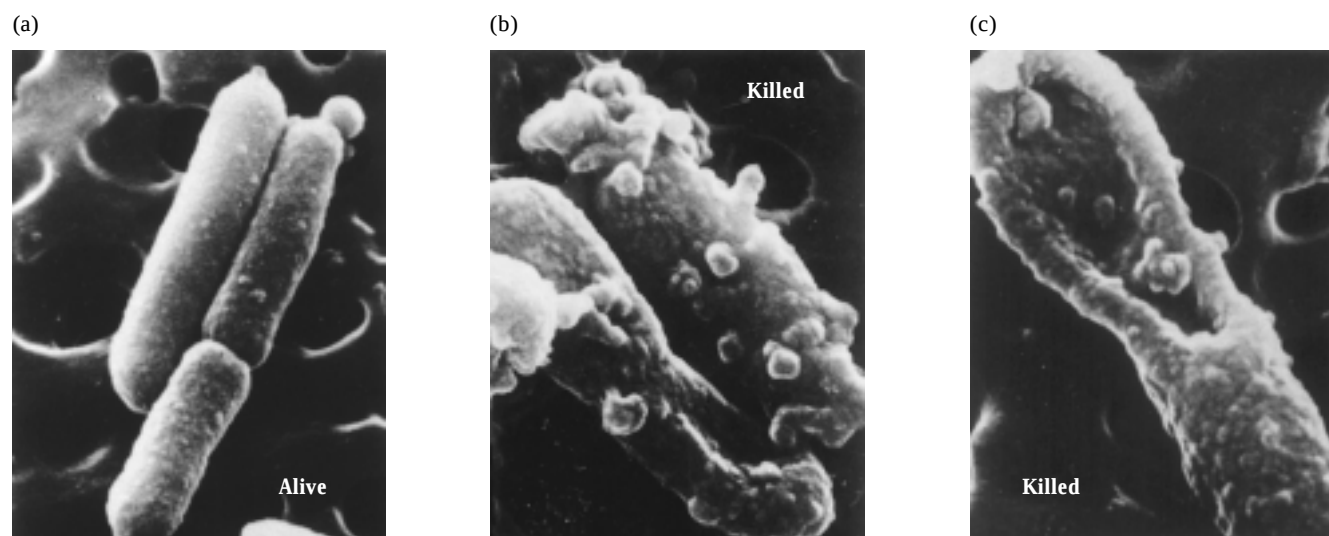


FIGURE 13-11 Scanning electron micrographs of *E. coli* showing (a) intact cells and (b, c) cells killed by complement-mediated lysis.

Note membrane blebbing on lysed cells. [From R. D. Schreiber et al., 1979, *J. Exp. Med.* **149**:870.]

inflammatory response. In addition to these mechanisms of evasion, various bacteria, viruses, fungi, and protozoans contain proteins that can interrupt the complement cascade on their surfaces, thus mimicking the effects of the normal complement regulatory proteins C4bBP, CR1, and DAF.

Lysis of nucleated cells requires formation of multiple membrane attack complexes, whereas a single MAC can lyse a red blood cell. Many nucleated cells, including the majority of cancer cells, can endocytose the MAC. If the complex is removed soon enough, the cell can repair any membrane

TABLE 13-5 Microbial evasion of complement-mediated damage

Microbial component	Mechanism of evasion	Examples
GRAM-NEGATIVE BACTERIA		
Long polysaccharide chains in cell-wall LPS	Side chains prevent insertion of MAC into bacterial membrane	Resistant strains of <i>E. coli</i> and <i>Salmonella</i>
Outer membrane protein	MAC interacts with membrane protein and fails to insert into bacterial membrane	Resistant strains of <i>Neisseria gonorrhoeae</i>
Elastase	Anaphylatoxins C3a and C5a are inactivated by microbial elastase	<i>Pseudomonas aeruginosa</i>
GRAM-POSITIVE BACTERIA		
Peptidoglycan layer of cell wall	Insertion of MAC into bacterial membrane is prevented by thick layer of peptidoglycan	<i>Streptococcus</i>
Bacterial capsule	Capsule provides physical barrier between C3b deposited on bacterial membrane and CR1 on phagocytic cells	<i>Streptococcus pneumoniae</i>
OTHER MICROBES		
Proteins that mimic complement regulatory proteins	Protein present in various bacteria, viruses, fungi, and protozoans inhibit the complement cascade	Vaccinia virus, herpes simplex, Epstein-Barr virus, <i>Trypanosoma cruzi</i> , <i>Candida albicans</i>

KEY: CR1 = type 1 complement receptor; LPS = lipopolysaccharide; MAC = membrane-attack complex (C5b–9).

damage and restore its osmotic stability. An unfortunate consequence of this effect is that complement-mediated lysis by antibodies specific for tumor-cell antigens, which offers a potential weapon against cancer, may be rendered ineffective by endocytosis of the MAC (see Chapter 22).

Cleavage Products of Complement Components Mediate Inflammation

The complement cascade is often viewed in terms of the final outcome of cell lysis, but various peptides generated during formation of the MAC play a decisive role in the development of an effective inflammatory response (see Table 13-3). The smaller fragments resulting from complement cleavage, C3a, C4a, and C5a, called **anaphylatoxins**, bind to receptors on mast cells and blood basophils and induce degranulation, with release of histamine and other pharmacologically active mediators. The anaphylatoxins also induce smooth-muscle contraction and increased vascular permeability. Activation of the complement system thus results in influxes of fluid that carries antibody and phagocytic cells to the site of antigen entry. The activities of these highly reactive anaphylatoxins are regulated by a serum protease called carboxypeptidase N, which cleaves an Arg residue from the C terminus of the molecules, yielding so-called *des-Arg* forms. The *des-Arg* forms of C3a and C4a are completely inactive while that of C5a retains about 10% of its chemotactic activity and 1% of its ability to cause smooth muscle contraction.

C3a, C5a, and C5b67 can each induce monocytes and neutrophils to adhere to vascular endothelial cells, extravasate through the endothelial lining of the capillary, and migrate toward the site of complement activation in the tissues. C5a is most potent in mediating these processes, effective in picomolar quantities. The role of complement in leukocyte chemotaxis is discussed more fully in Chapter 15.

C3b and C4b Binding Facilitates Opsonization

C3b is the major **opsonin** of the complement system, although C4b and iC3b also have opsonizing activity. The amplification that occurs with C3 activation results in a coating of C3b on immune complexes and particulate antigens. Phagocytic cells, as well as some other cells, express complement receptors (CR1, CR3, and CR4) that bind C3b, C4b, or iC3b (see Table 13-4). Antigen coated with C3b binds to cells bearing CR1. If the cell is a phagocyte (e.g., a neutrophil, monocyte, or macrophage), phagocytosis will be enhanced (Figure 13-12). Activation of phagocytic cells by various agents, including C5a anaphylatoxin, has been shown to increase the number of CR1s from 5000 on resting phagocytes to 50,000 on activated cells, greatly facilitating their phagocytosis of C3b-coated antigen. Recent studies indicate that complement fragment C3b acts as an adjuvant when coupled with protein antigens. C3b targets the antigen directly to the phagocyte, enhancing the initiation of antigen processing and accelerating specific antibody production.

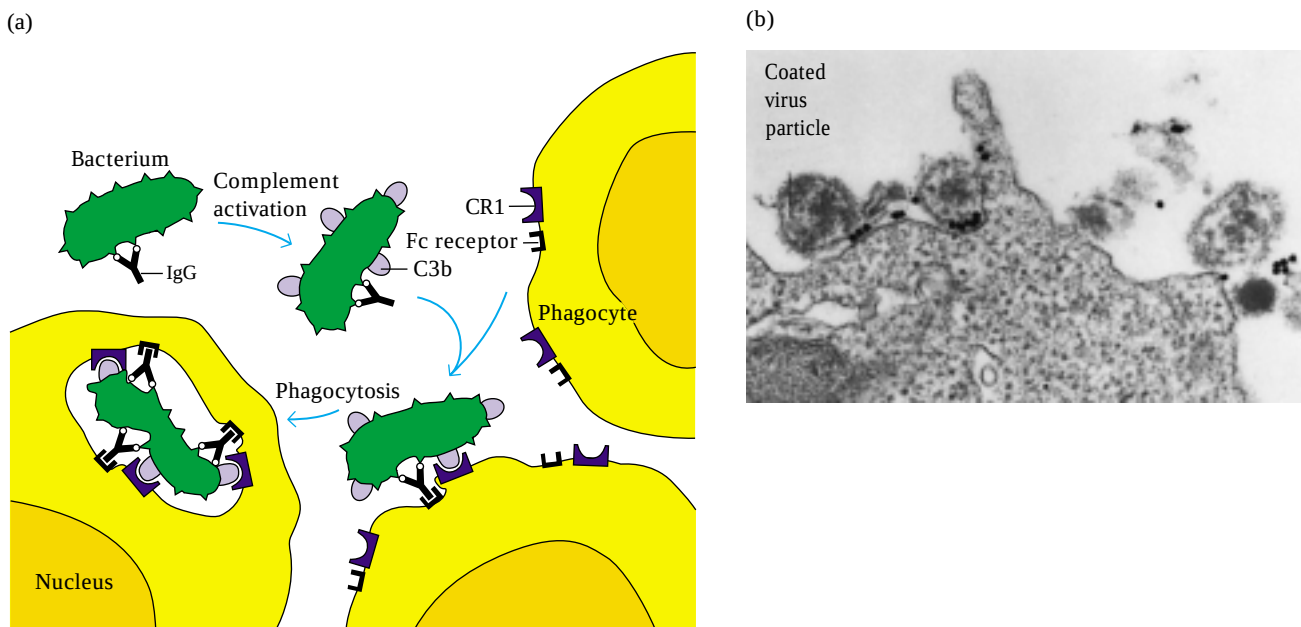


FIGURE 13-12 (a) Schematic representation of the roles of C3b and antibody in opsonization. (b) Electron micrograph of Epstein-Barr virus coated with antibody and C3b and bound to the Fc and

C3b receptor (CR1) on a B lymphocyte. [Part (b) from N. R. Cooper and G. R. Nemerow, 1986, in *Immunobiology of the Complement System*, Academic Press.]

The Complement System Also Neutralizes Viral Infectivity

For most viruses, the binding of serum antibody to the repeating subunits of the viral structural proteins creates particulate immune complexes ideally suited for complement activation by the classical pathway. Some viruses (e.g., retroviruses, Epstein-Barr virus, Newcastle disease virus, and rubella virus) can activate the alternative, lectin, or even the classical pathway in the absence of antibody.

The complement system mediates viral neutralization by a number of mechanisms. Some degree of neutralization is achieved through the formation of larger viral aggregates, simply because these aggregates reduce the net number of infectious viral particles. Although antibody plays a role in the formation of viral aggregates, *in vitro* studies show that the C3b component facilitates aggregate formation in the presence of as little as two molecules of antibody per virion. For example, polyoma virus coated with antibody is neutralized when serum containing activated C3 is added.

The binding of antibody and/or complement to the surface of a viral particle creates a thick protein coating that can be visualized by electron microscopy (Figure 13-13). This coating neutralizes viral infectivity by blocking attachment to susceptible host cells. The deposits of antibody and complement on viral particles also facilitate binding of the viral particle to cells possessing Fc or type 1 complement receptors (CR1). In the case of phagocytic cells, such binding can be followed by phagocytosis and intracellular destruction of the ingested viral particle. Finally, complement is effective in lysing most, if not all, enveloped viruses, resulting in fragmentation of the envelope and disintegration of the nucleocapsid.

The Complement System Clears Immune Complexes from Circulation

The importance of the complement system in clearing immune complexes is seen in patients with the autoimmune disease systemic lupus erythematosus (SLE). These individuals produce large quantities of immune complexes and suffer tissue damage as a result of complement-mediated lysis and the induction of type II or type III hypersensitivity (see Chapter 16). Although complement plays a significant role in the development of tissue damage in SLE, the paradoxical finding is that deficiencies in C1, C2, C4, and CR1 predispose an individual to SLE; indeed, 90% of individuals who completely lack C4 develop SLE. The complement deficiencies are thought to interfere with effective solubilization and clearance of immune complexes; as a result, these complexes persist, leading to tissue damage by the very system whose deficiency was to blame.

The coating of soluble immune complexes with C3b is thought to facilitate their binding to CR1 on erythrocytes. Although red blood cells express lower levels of CR1 ($\sim 5 \times 10^2$ per cell) than granulocytes do ($\sim 5 \times 10^4$ per cell), there are about 10^3 red blood cells for every white blood cell; therefore, erythrocytes account for about 90% of the CR1 in the blood. For this reason, erythrocytes play an important role in binding C3b-coated immune complexes and carrying these complexes to the liver and spleen. In these organs, immune complexes are stripped from the red blood cells and are phagocytosed, thereby preventing their deposition in tissues (Figure 13-14). In SLE patients, deficiencies in C1, C2, and C4 each contribute to reduced levels of C3b on immune complexes and hence inhibit their clearance. The lower levels of CR1 expressed on the erythrocytes of SLE patients also may interfere with the proper binding and clearance of immune complexes.

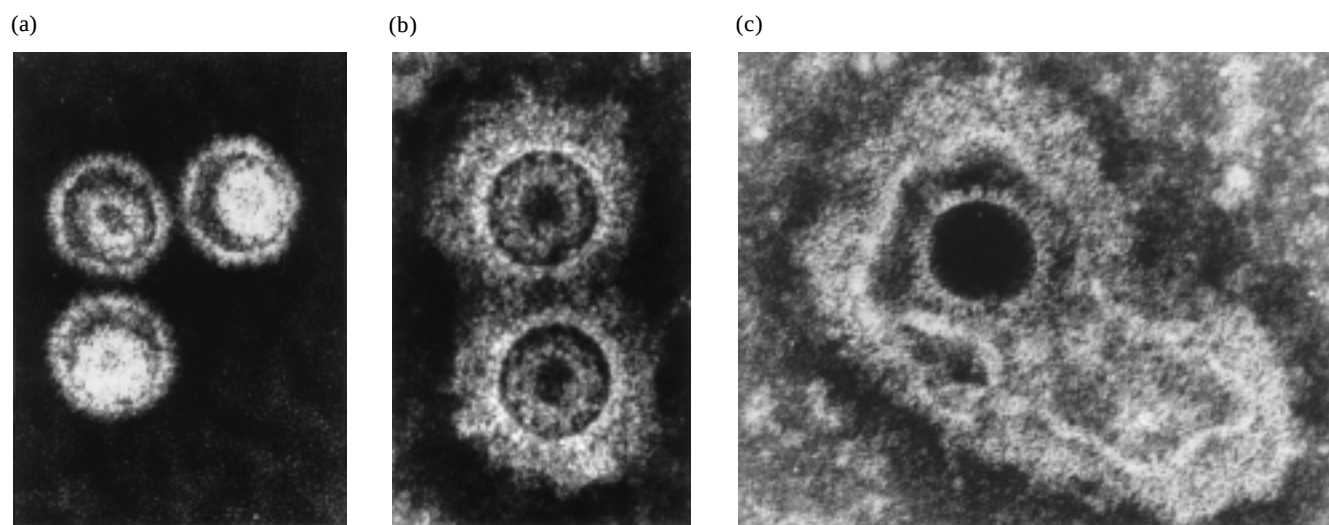


FIGURE 13-13 Electron micrographs of negatively stained preparations of Epstein-Barr virus. (a) Control without antibody. (b) Antibody-coated particles. (c) Particles coated with antibody and complement.

[From N. R. Cooper and G. R. Nemerow, 1986, in *Immunobiology of the Complement System*, Academic Press.]

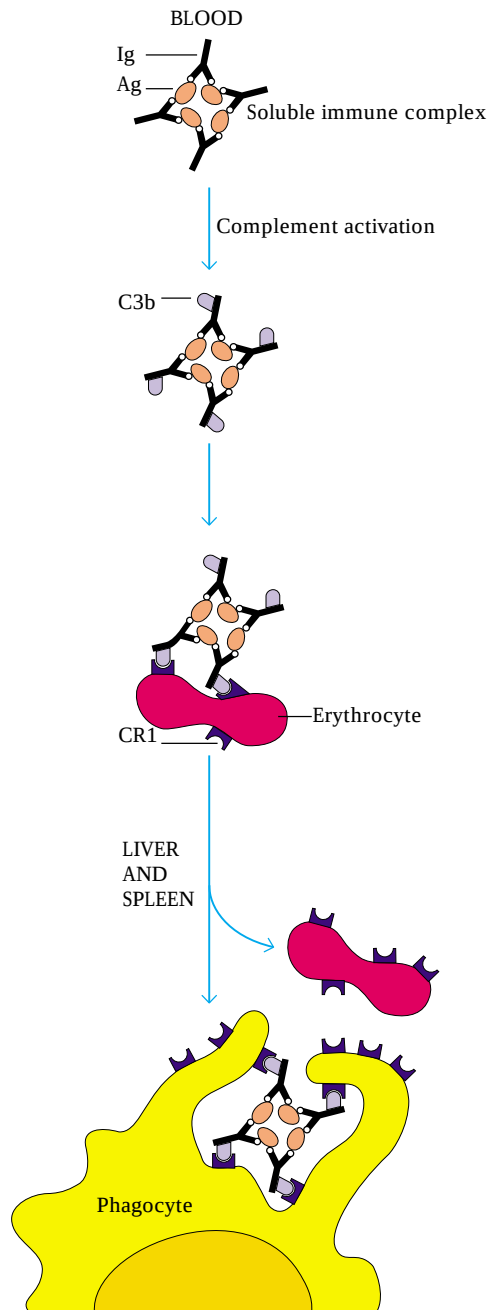


FIGURE 13-14 Clearance of circulating immune complexes by reaction with receptors for complement products on erythrocytes and removal of these complexes by receptors on macrophages in the liver and spleen. Because erythrocytes have fewer receptors than macrophages, the latter can strip the complexes from the erythrocytes as they pass through the liver or spleen. Deficiency in this process can lead to renal damage due to accumulation of immune complexes.

complex diseases, individuals with such complement deficiencies may suffer from recurrent infections by pyogenic (pus-forming) bacteria such as streptococci and staphylococci. These organisms are gram-positive and therefore resistant to the lytic effects of the MAC. Nevertheless, the early complement components ordinarily prevent recurrent infection by mediating a localized inflammatory response and opsonizing the bacteria. Deficiencies in factor D and properdin—early components of the alternative pathway—appear to be associated with *Neisseria* infections but not with immune-complex disease.

Patients with C3 deficiencies have the most severe clinical manifestations, reflecting the central role of C3 in activation of C5 and formation of the MAC. The first patient identified with a C3 deficiency was a child suffering from frequent severe bacterial infections and initially diagnosed as having agammaglobulinemia. After tests revealed normal immunoglobulin levels, a deficiency in C3 was discovered. This case highlights the critical function of the complement system in converting a humoral antibody response into an effective defense mechanism. The majority of patients with C3 deficiency have recurrent bacterial infections and may have immune-complex diseases.

Individuals with homozygous deficiencies in the components involved in the MAC develop recurrent meningococcal and gonococcal infections caused by *Neisseria* species. In normal individuals, these gram-negative bacteria are generally susceptible to complement-mediated lysis or are cleared by the opsonizing activity of C3b. MAC-deficient individuals rarely have immune-complex disease, which suggests that they produce enough C3b to clear immune complexes. Interestingly, a deficiency in C9 results in no clinical symptoms, suggesting that the entire MAC is not always necessary for complement-mediated lysis.

Congenital deficiencies of complement regulatory proteins have also been reported. The C1 inhibitor (C1Inh) regulates activation of the classical pathway by preventing excessive C4 and C2 activation by C1. Deficiency of C1Inh is an autosomal dominant condition with a frequency of 1 in 1000. The deficiency gives rise to a condition called hereditary angioedema, which manifests clinically as localized edema of the tissue, often following trauma, but sometimes with no known cause. The edema can be in subcutaneous tissues or within the bowel, where it causes abdominal pain, or in the upper respiratory tract, where it causes obstruction of the airway.

Studies in humans and experimental animals with homozygous deficiencies in complement components have

Complement Deficiencies

Genetic deficiencies have been described for each of the complement components. Homozygous deficiencies in any of the early components of the classical pathway (C1q, C1r, C1s, C4, and C2) exhibit similar symptoms, notably a marked increase in immune-complex diseases such as systemic lupus erythematosus, glomerulonephritis, and vasculitis. These deficiencies highlight the importance of the early complement reactions in generating C3b, and the critical role of C3b in solubilization and clearance of immune complexes. In addition to immune-

been the major source of information concerning the role of individual complement components in immunity. These findings have been greatly extended by studies using knock-out mice genetically engineered to lack expression of specific complement components. Investigations of in vivo complement activity in these animals has allowed dissection of the complex system of complement proteins and the assignment of precise biologic roles to each.

SUMMARY

- The complement system comprises a group of serum proteins, many of which exist in inactive forms.
- Complement activation occurs by the classical, alternative, or lectin pathways, each of which is initiated differently.
- The three pathways converge in a common sequence of events that leads to generation of a molecular complex that causes cell lysis.
- The classical pathway is initiated by antibody binding to a cell target; reactions of IgM and certain IgG subclasses activate this pathway.
- Activation of the alternative and lectin pathways is antibody-independent. These pathways are initiated by reaction of complement proteins with surface molecules of microorganisms.
- In addition to its key role in cell lysis, the complement system mediates opsonization of bacteria, activation of inflammation, and clearance of immune complexes.
- Interactions of complement proteins and protein fragments with receptors on cells of the immune system control both innate and acquired immune responses.
- Because of its ability to damage the host organism, the complement system requires complex passive and active regulatory mechanisms.
- Clinical consequences of inherited complement deficiencies range from increases in susceptibility to infection to tissue damage caused by immune complexes.

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USEFUL WEB SITES

<http://www.complement-genetics.uni-mainz.de/>

The Complement Genetics Homepage from the University of Mainz gives chromosomal locations and information on genetic deficiencies of complement proteins.

<http://www.cehs.siu.edu/fix/medmicro/cfix.htm>

A clever graphic representation of the basic assay for complement activity using red blood cell lysis, from D. Fix at University of Southern Illinois, Carbondale.

<http://www.gla.ac.uk/Acad/Immunology/compsyst.htm>

Notes from D. F. Lappin at University of Glasgow, UK, on the complement system. The site includes a listing of all complement proteins and their molecular properties.

Study Questions

CLINICAL FOCUS QUESTION Explain why complement disorders involving regulatory components such as PNH may be more serious than deficiencies in the active complement components.

1. Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.
 - a. A single molecule of bound IgM can activate the C1q component of the classical complement pathway.



- b. C3a and C3b are fragments of C3.
 - c. The C4 and C2 complement components are present in the serum in a functionally inactive proenzyme form.
 - d. Nucleated cells tend to be more resistant to complement-mediated lysis than red blood cells.
 - e. Enveloped viruses cannot be lysed by complement because their outer envelope is resistant to pore formation by the membrane-attack complex.
 - f. C4-deficient individuals have difficulty eliminating immune complexes.
2. Explain why serum IgM cannot activate complement by itself.
 3. Would you expect a C1 or C3 complement deficiency to be more serious clinically? Why?
 4. Some microorganisms produce enzymes that can degrade the Fc portion of antibody molecules. Why would such enzymes be advantageous for the survival of microorganisms that possess them?
 5. Complement activation can occur via the classical, alternative, or lectin pathway.
 - a. How do the three pathways differ in the substances that can initiate activation?
 - b. Which portion of the overall activation sequence differs in the three pathways? Which portion is similar?
 - c. How do the biological consequences of complement activation via these pathways differ?
 6. Enucleated cells, such as red blood cells, are more susceptible to complement-mediated lysis than nucleated cells.
 - a. Explain why the red blood cells of an individual are not normally destroyed as the result of innocent-bystander lysis by complement.
 - b. Under what conditions might complement cause lysis of an individual's own red blood cells?
 7. Briefly explain the mechanism of action of the following complement regulatory proteins. Indicate which pathway(s) each protein regulates.
 - a. C1 inhibitor (C1Inh)
 - b. C4b-binding protein (C4bBP)
 - c. Homologous restriction factor (HRF)
 - d. Decay-accelerating factor (DAF)
 - e. Factor H
 - f. Membrane cofactor protein (MCP)
 8. For each complement component(s) or reaction (a–l), select the most appropriate description listed below (1–13). Each description may be used once, more than once, or not at all.

Complement Component(s)/Reactions

- a. _____ C3b
- b. _____ C1, C4, C2, and C3
- c. _____ C9
- d. _____ C3, factor B, and factor D
- e. _____ C1q
- f. _____ C4b2a3b
- g. _____ C5b, C6, C7, C8, and C9
- h. _____ $C3 \rightarrow C3a + C3b$

- i. _____ C3a, C5a, and C5b67
- j. _____ C3a, C4a, and C5a
- k. _____ C4b2a
- l. _____ $C3b + B \rightarrow C3bBb + Ba$

Descriptions

- (1) Reaction that produces major amplification during activation
 - (2) Are early components of alternative pathway
 - (3) Compose the membrane-attack complex
 - (4) Mediates opsonization
 - (5) Are early components of classical pathway
 - (6) Has perforin-like activity
 - (7) Binds to Fc region of antibodies
 - (8) Have chemotactic activity
 - (9) Has C3 convertase activity
 - (10) Induce degranulation of mast cells (are anaphylatoxins)
 - (11) Has C5 convertase activity
 - (12) Reaction catalyzed by factor D
 - (13) Reaction catalyzed by C1qr₂S₂
9. You have prepared knockout mice with mutations in the genes that encode various complement components. Each knockout strain cannot express one of the complement components listed across the top of the table below. Predict the effect of each mutation on the steps in complement activation and on the complement effector functions indicated in the table below using the following symbols: NE = no effect; D = process/function decreased but not abolished; A = process/function abolished.

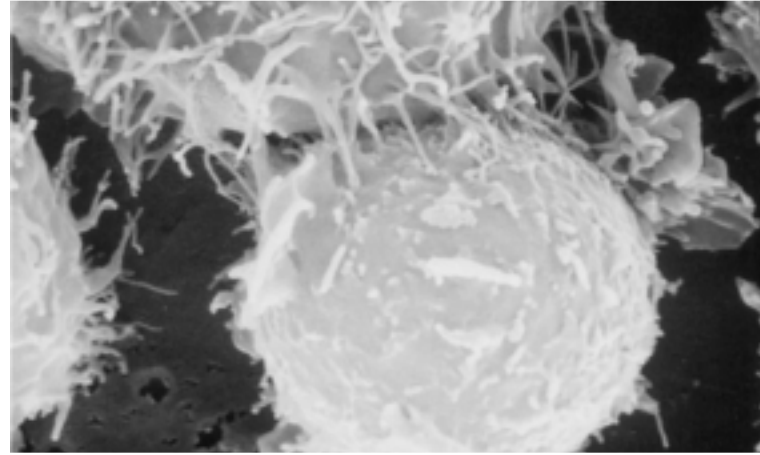
	Component knocked out						
	C1q	C4	C3	C5	C6	C9	Factor B
COMPLEMENT ACTIVATION							
Formation of C3 convertase in classical pathway							
Formation of C3 convertase in alternative pathway							
Formation of C5 convertase in classical pathway							
Formation of C5 convertase in alternative pathway							
EFFECTOR FUNCTIONS							
C3b-mediated opsonization							
Neutrophil chemotaxis							
Cell lysis							

Cell-Mediated Effector Responses

THE CELL-MEDIATED AND HUMORAL BRANCHES OF the immune system assume different roles in protecting the host. The effectors of the humoral branch are secreted antibodies, highly specific molecules that can bind and neutralize antigens on the surface of cells and in the extracellular spaces. The primary domain of antibody protection lies outside the cell. If antibodies were the only agents of immunity, pathogens that managed to evade them and colonize the intracellular environment would escape the immune system. This is not the case. The principal role of cell-mediated immunity is to detect and eliminate cells that harbor intracellular pathogens. Cell-mediated immunity also can recognize and eliminate cells, such as tumor cells, that have undergone genetic modifications so that they express antigens not typical of normal cells.

Both antigen-specific and -nonspecific cells can contribute to the cell-mediated immune response. Specific cells include $CD8^+$ cytotoxic T lymphocytes (T_C cells or CTLs) and cytokine-secreting $CD4^+$ T_H cells that mediate delayed-type hypersensitivity (DTH). The discussion of DTH reactions and the role of $CD4^+$ T cells in their orchestration appears in Chapter 16. Nonspecific cells include NK cells and non-lymphoid cell types such as macrophages, neutrophils, and eosinophils. The activity of both specific and nonspecific components usually depends on effective local concentrations of various cytokines. T cells, NK cells, and macrophages are the most important sources of the cytokines that organize and support cell-mediated immunity. Finally, although humoral and cell-mediated immunity have many distinctive features, they are not completely independent. Cells such as macrophages, NK cells, neutrophils, and eosinophils can use antibodies as receptors to recognize and target cells for killing. Also, chemotactic peptides generated by the activation of complement in response to antigen-antibody complexes can contribute to assembling the cell types required for a cell-mediated response.

In the preceding chapters, various aspects of the humoral and cell-mediated effector responses have been described. This chapter addresses cytotoxic effector mechanisms mediated by T_C cells, NK cells, antibody-dependent cell-mediated cytotoxicity (ADCC), and the experimental assay of cytotoxicity.



Big CTL Attacks Little Tumor Cell

- Effector Responses
- General Properties of Effector T Cells
- Cytotoxic T Cells
- Natural Killer Cells
- Antibody-Dependent Cell-Mediated Cytotoxicity
- Experimental Assessment of Cell-Mediated Cytotoxicity

Effector Responses

The importance of cell-mediated immunity becomes evident when the system is defective. Children with DiGeorge syndrome, who are born without a thymus and therefore lack the T-cell component of the cell-mediated immune system, generally are able to cope with infections of extracellular bacteria, but they cannot effectively eliminate intracellular pathogens. Their lack of functional cell-mediated immunity results in repeated infections with viruses, intracellular bacteria, and fungi. The severity of the cell-mediated immunodeficiency in these children is such that even the attenuated virus present in a vaccine, capable of only limited growth in normal individuals, can produce life-threatening infections.

Cell-mediated immune responses can be divided into two major categories according to the different effector populations that are mobilized. One group comprises effector cells

that have direct cytotoxic activity. These effectors eliminate foreign cells and altered self-cells by mounting a cytotoxic reaction that lyses their target. The various cytotoxic effector cells can be grouped into two general categories: one comprises antigen-specific cytotoxic T lymphocytes (CTLs) and nonspecific cells, such as natural killer (NK) cells and macrophages. The target cells to which these effectors are directed include allogeneic cells, malignant cells, virus-infected cells, and chemically conjugated cells. The other group is a subpopulation of effector $CD4^+$ T cells that mediates delayed-type hypersensitivity reactions (see Chapter 16). The next section reviews the general properties of effector T cells and how they differ from naive T cells.

General Properties of Effector T Cells

The three types of effector T cells— $CD4^+$, T_H1 and T_H2 cells, and $CD8^+$ CTLs—exhibit several properties that set them apart from naive helper and cytotoxic T cells (Table 14-1). In particular, effector cells are characterized by their less stringent activation requirements, increased expression of cell-adhesion molecules, and production of both membrane-bound and soluble effector molecules.

The Activation Requirements of T Cells Differ

T cells at different stages of differentiation may respond with different efficiencies to signals mediated by the T-cell receptor and may consequently require different levels of a second set of co-stimulatory signals. As described in Chapter 10, activation of naive T cells and their subsequent proliferation and differentiation into effector T cells require both a primary signal, delivered when the TCR complex and CD4 or CD8 coreceptor interact with a foreign peptide–MHC molecule complex, and a co-stimulatory signal, delivered by interaction between particular membrane molecules on the T cell and the antigen-presenting cell. In contrast, antigen-experienced effector cells and memory cells (as opposed to

naive T cells) are able to respond to TCR-mediated signals with little, if any co-stimulation.

The reason for the different activation requirements of naive and activated T cells is an area of continuing research, but some clues have been found. One is that many populations of naive and effector T cells express different isoforms of CD45, designated CD45RA and CD45RO, which are produced by alternative splicing of the RNA transcript of the CD45 gene. This membrane molecule mediates TCR signal transduction by catalyzing dephosphorylation of a tyrosine residue on the protein tyrosine kinases Lck and Fyn, activating these kinases and triggering the subsequent steps in T-cell activation (see figures 10-10 and 10-11). The CD45RO isoform, which is expressed on effector T cells, associates with the TCR complex and its coreceptors, CD4 and CD8, much better than does the CD45RA isoform, which is expressed by naive T cells. Memory T cells have both isoforms, but the CD45RO is predominant. As a result, effector and memory T cells are more sensitive to TCR-mediated activation by a peptide–MHC complex. They also have less stringent requirements for co-stimulatory signals and therefore are able to respond to peptide–MHC complexes displayed on target cells or antigen-presenting cells that lack the co-stimulatory B7 molecules.

Cell-Adhesion Molecules Facilitate TCR-Mediated Interactions

CD2 and the integrin LFA-1 are cell-adhesion molecules on the surfaces of T cells that bind, respectively, to LFA-3 and ICAMs (intracellular cell-adhesion molecules) on antigen-presenting cells and various target cells (see Figure 9-13). The level of LFA-1 and CD2 is twofold to fourfold higher on effector T cells than on naive T cells, enabling the effector T cells to bind more effectively to antigen-presenting cells and to various target cells that express low levels of ICAMs or LFA-3.

As Chapter 9 showed, the initial interaction of an effector T cell with an antigen-presenting cell or target cell is weak, allowing the TCR to scan the membrane for specific peptides

TABLE 14-1

Comparison of naive and effector T cells

Property	Naive T cells	Effector T cells
Co-stimulatory signal (CD28-B7 interaction)	Required for activation	Not required for activation
CD45 isoform	CD45RA	CD45RO
Cell-adhesion molecules (CD2 and LFA-1)	Low	High
Trafficking patterns	HEVs* in secondary lymphoid tissue	Tertiary lymphoid tissues; inflammatory sites

*HEV = high endothelial venules, sites in blood vessel used by lymphocytes for extravasation.

TABLE 14-2 Effector molecules produced by effector T cells

Cell type	Soluble effectors	Membrane-bound effectors
CTL	Cytotoxins (perforins and granzymes), IFN- γ , TNF- β	Fas ligand (FASL)
T _H 1	IL-2, IL-3, TNF- β , IFN- γ , GM-CSF (high)	Tumor necrosis factor β (TNF- β)
T _H 2	IL-3, IL-4, IL-5, IL-6, IL-10, IL-13, GM-CSF (low)	CD40 ligand

presented by self-MHC molecules. If no peptide-MHC complex is recognized by the effector cell, it will disengage from the APC or target cell. Recognition of a peptide-MHC complex by the TCR, however, produces a signal that increases the affinity of LFA-1 for ICAMs on the APC or target-cell membrane, prolonging the interaction between the cells. For example, T_H1 effector cells remain bound to macrophages that display peptide–class II MHC complexes; T_H2 effector cells remain bound to B cells that display peptide–class II MHC complexes; and CTL effector cells bind tightly to virus-infected target cells that display peptide–class I MHC complexes.

Effector T Cells Express a Variety of Effector Molecules

Unlike naive T cells, effector T cells express certain effector molecules, which may be membrane bound or soluble (Table 14-2). The membrane-bound molecules belong to the tumor necrosis factor (TNF) family of membrane proteins and include the Fas ligand (FASL) on CD8⁺ CTLs, TNF- β on T_H1 cells, and the CD40 ligand on T_H2 cells. Each of the effector T-cell populations also secretes distinct panels of soluble effector molecules. CTLs secrete cytotoxins (perforins and granzymes) as well as two cytokines, IFN- γ and TNF- β . As described in Chapter 12, the T_H1 and T_H2 subsets secrete largely nonoverlapping sets of cytokines.

Each of these membrane-bound and secreted molecules plays an important role in various T-cell effector functions. The Fas ligand, perforins, and granzymes, for example, mediate target-cell destruction by the CTL; membrane-bound TNF- β and soluble IFN- γ and GM-CSF promote macrophage activation by the T_H1 cell; and the membrane-bound CD40 ligand and soluble IL-4, IL-5, and IL-6 all play a role in B-cell activation by the T_H2 cell.

Cytotoxic T Cells

Cytotoxic T lymphocytes, or CTLs, are generated by immune activation of T cytotoxic (T_C) cells. These effector cells have lytic capability and are critical in the recognition and elimination of altered self-cells (e.g., virus-infected cells and

tumor cells) and in graft-rejection reactions. In general, CTLs are CD8⁺ and are therefore class I MHC restricted, although in rare instances CD4⁺ class II–restricted T cells have been shown to function as CTLs. Since virtually all nucleated cells in the body express class I MHC molecules, CTLs can recognize and eliminate almost any altered body cell.

The CTL-mediated immune response can be divided into two phases, reflecting different aspects of the response. The first phase activates and differentiates naive T_C cells into functional effector CTLs. In the second phase, effector CTLs recognize antigen–class I MHC complexes on specific target cells, which leads them to destroy the target cells.

Effector CTLs Are Generated from CTL Precursors

Naive T_C cells are incapable of killing target cells and are therefore referred to as CTL precursors (CTL-Ps) to denote their functionally immature state. Only after a CTL-P has been activated will the cell differentiate into a functional CTL with cytotoxic activity. Generation of CTLs from CTL-Ps appears to require at least three sequential signals (Figure 14-1):

- An antigen-specific signal 1 transmitted by the TCR complex upon recognition of a peptide–class I MHC molecule complex
- A co-stimulatory signal transmitted by the CD28-B7 interaction of the CTL-P and the antigen-presenting cell
- A signal induced by the interaction of IL-2 with the high-affinity IL-2 receptor, resulting in proliferation and differentiation of the antigen-activated CTL-P into effector CTLs

Unactivated CTL-Ps do not express IL-2 or IL-2 receptors, do not proliferate, and do not display cytotoxic activity. Antigen activation induces a CTL-P to begin expressing the IL-2 receptor and to a lesser extent IL-2, the principal cytokine required for proliferation and differentiation of activated CTL-Ps into effector CTLs. In some cases, the amount of IL-2 secreted by an antigen-activated CTL-P may be sufficient to induce its own proliferation and differentiation; this

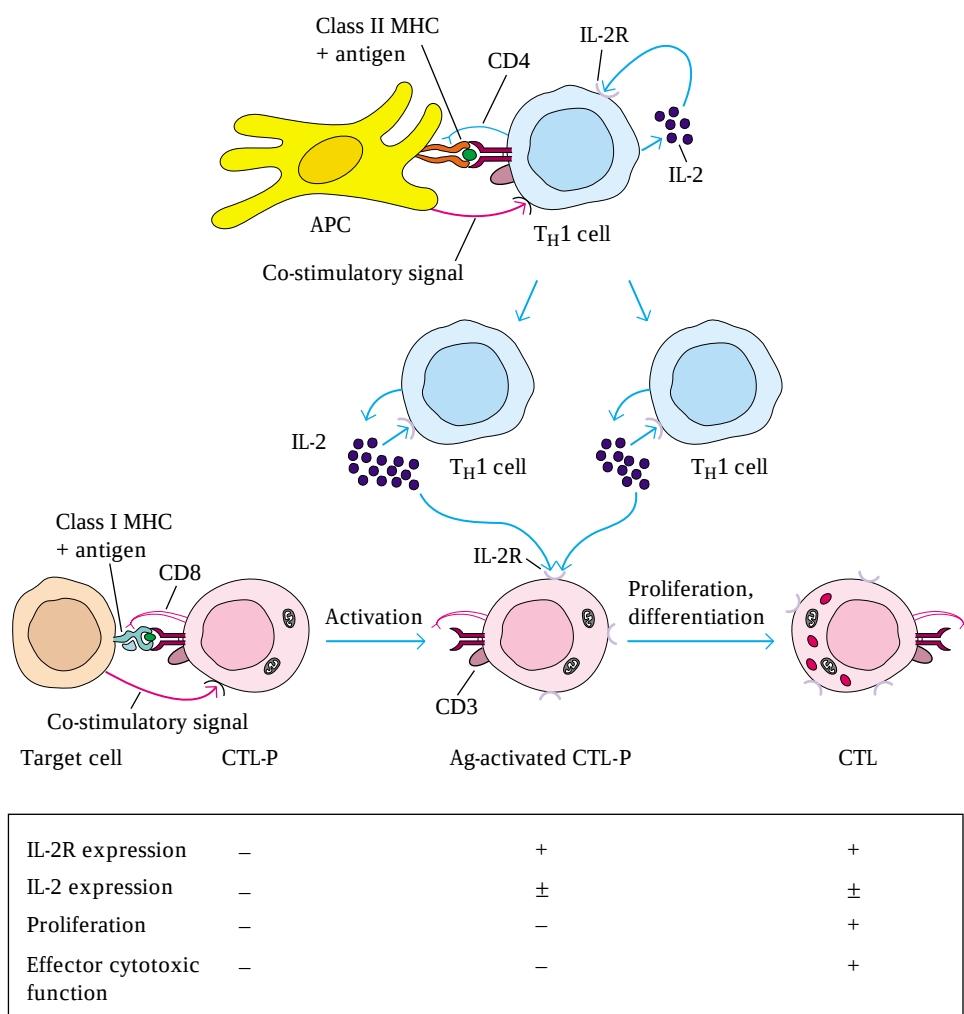


FIGURE 14-1 Generation of effector CTLs. Upon interaction with antigen–class I MHC complexes on appropriate target cells, CTL-Ps begin to express IL-2 receptors (IL-2R) and lesser amounts of IL-2. Proliferation and differentiation of antigen-activated CTL-Ps generally require

additional IL-2 secreted by T_H1 cells resulting from antigen activation and proliferation of $CD4^+$ T cells. In the subsequent effector phase, CTLs destroy specific target cells.

is particularly true of memory CTL-Ps, which have lower activation requirements than naive cells do (Figure 14-2a). In general, though, most activated CTL-Ps require additional IL-2 produced by proliferating T_H1 cells to proliferate and differentiate into effector CTLs. The fact that the IL-2 receptor is not expressed until after a CTL-P has been activated by antigen plus a class I MHC molecule favors the clonal expansion and acquisition of cytotoxicity by only the antigen-specific CTL-Ps. The proliferation and differentiation of both antigen-activated T_H1 cells and CTL-Ps depend on IL-2. In IL-2 knockout mice, the absence of IL-2 has been shown to abolish CTL-mediated cytotoxicity. After clearance of antigen, the level of IL-2 declines, which induces T_H1 cells and CTLs to undergo programmed cell death by apoptosis. In this way,

the immune response is rapidly terminated, lessening the likelihood of nonspecific tissue damage from the inflammatory response. The role of T_H1 cells in the generation of CTLs from naive CTL-Ps is not completely understood, and it is unlikely that a T_H1 cell and CTL-P interact directly. However, IL-2 and co-stimulation are important in the transformation of naive CTL-Ps into effector cells, and T_H1 cells can be mediators in the provision of these essential requirements. As shown in Figure 14-2b, the interaction of helper cells with antigen-presenting cells can result in production of IL-2 by the T_H1 cell. The paracrine action of this cytokine on nearby naive CTL-Ps whose TCRs are engaged can cause them to proliferate and differentiate into active CTLs. Additionally, T_H1 can induce the up-regulation of co-stimulatory molecules on the

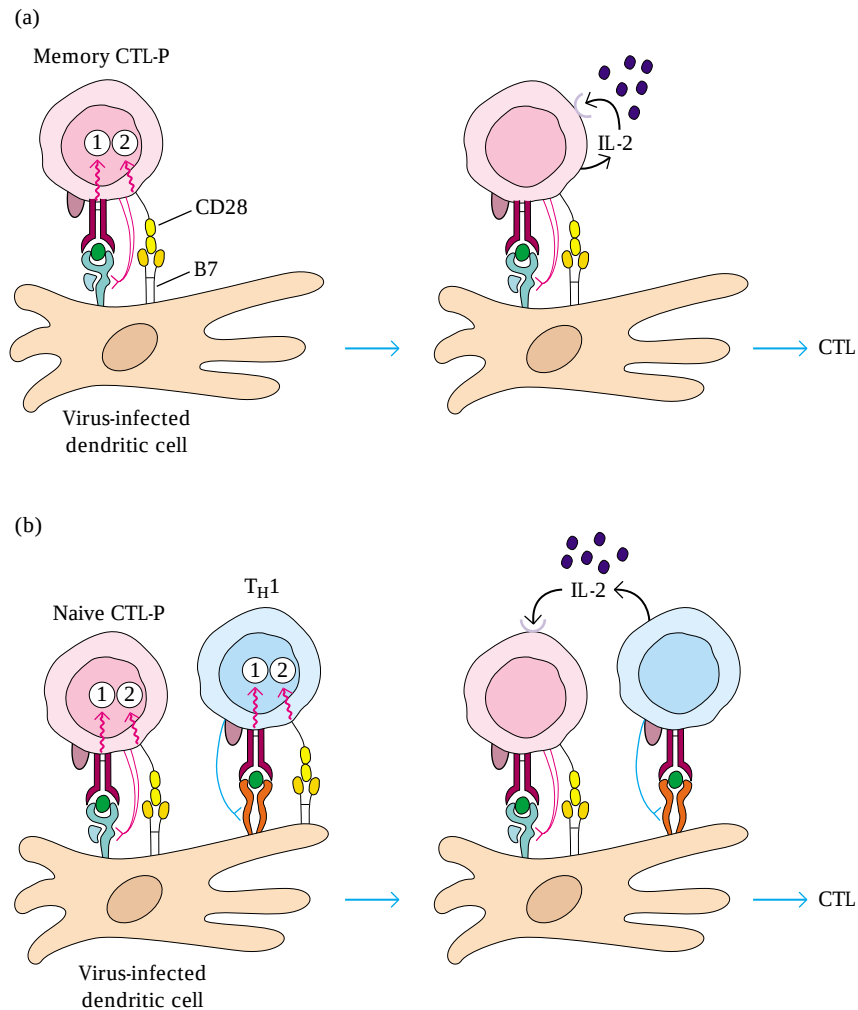


FIGURE 14-2 Proliferation of memory CTL-Ps may not require help from T_H cells. (a) Antigen-activated memory CTL-Ps appear to secrete sufficient IL-2 to stimulate their own proliferation and differentiation into effector CTLs. They also may not require the CD28-B7 co-stimulatory signal for

activation. (b) A T_H cell may provide the IL-2 necessary for proliferation of an antigen-activated naive CTL-P when it binds to the same APC as the CTL-P. Also, T_H cells may alter the behavior of APCs in a number of ways, such as increasing the display of co-stimulatory molecules by the APC.

surface of antigen-presenting cells. In this manner, T_H1 cells help CTL-P division and differentiation by causing the generation of adequate levels of co-stimulation.

CD8⁺ CTLs Can Be Tracked with MHC Tetramer Technology

MHC tetramers are laboratory-generated complexes of four MHC class I molecules bound to a specific peptide and linked to a fluorescent molecule. A given MHC-tetramer-peptide complex binds only CD8⁺ T cells that have TCRs specific for the particular peptide-MHC complex that makes up the tetramer. Thus, when a particular tetramer is added to a cell population containing T cells (spleen cells or lymph-node cells, for example), cells that bear TCRs specific for

the tetramer become fluorescently labeled (Figure 14-3). Using flow cytometry, it is then possible to determine the proportion of cells in a population that have TCRs specific for a particular antigen by counting the number of fluorescently labeled cells in a cell population. This very sensitive approach can detect antigen-specific T cells even when their frequency in the CD8⁺ population is as low as 0.1%. Furthermore, one can directly measure the increase in antigen-specific CD8⁺ T cells in response to exposure to pathogens such as viruses or cancer-associated antigens. In a related application, researchers infected mice with vesicular stomatitis virus (VSV) and systematically examined the distribution of CD8⁺ cells specific for a VSV-derived peptide-MHC complex throughout the entire body. This study demonstrated that during acute infection with VSV, the distribution of

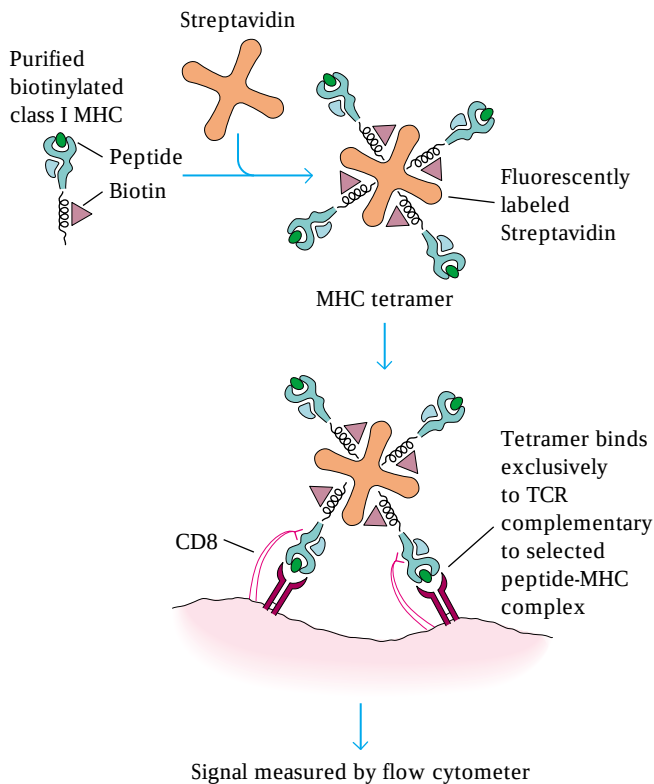


FIGURE 14-3 MHC tetramers. A homogeneous population of peptide-bound class I MHC molecules (HLA-A1 bound to an HIV-derived peptide, for example) is conjugated to biotin and mixed with fluorescently labeled Streptavidin. Four biotinylated MHC-peptide complexes bind to the high affinity binding sites of Streptavidin to form a tetramer. Addition of the tetramer to a population of T cells results in exclusive binding of the fluorescent tetramer to those CD8⁺ T cells with TCRs complementary to the peptide-MHC complexes of the tetramer. This results in the labeling of the subpopulation of T cells that are specific for the target antigen, making them readily detectable by flow cytometry. [Adapted in part from P. Klenerman, V. Cerundolo, and P. R. Dunbar, 2002, *Nature Reviews/Immunology* 2:264.]

VSV-specific CD8⁺ cells is far from uniform (Figure 14-4); large populations of antigen-specific cells are not limited to the lymphoid system, but can be found in the liver and kidney, too.

CTLs Kill Cells in Two Ways

The effector phase of a CTL-mediated response involves a carefully orchestrated sequence of events that begin with the embrace of the target cell by the attacking cell (Figure 14-5). Long-term cultures of CTL clones have been used to identify many of the membrane molecules and membrane events involved in this process. As described below, studies with

mouse strains carrying mutations that affect the ability of CTLs to induce death have led to the identification of the necessary molecules.

The primary events in CTL-mediated death are conjugate formation, membrane attack, CTL dissociation, and target-cell destruction (Figure 14-6). When antigen-specific CTLs are incubated with appropriate target cells, the two cell types interact and undergo conjugate formation. Formation of a CTL–target cell conjugate is followed within several minutes by a Ca²⁺-dependent, energy-requiring step in which the CTL programs the target cell for death. The CTL then dissociates from the target cell and goes on to bind to another target cell. Within a variable period of time (up to a few hours) after CTL dissociation, the target cell dies by apoptosis. Each of the steps in this process has been studied in detail with cloned CTLs.

The TCR-CD3 membrane complex on a CTL recognizes antigen in association with class I MHC molecules on a target cell. After this antigen-specific recognition, the integrin

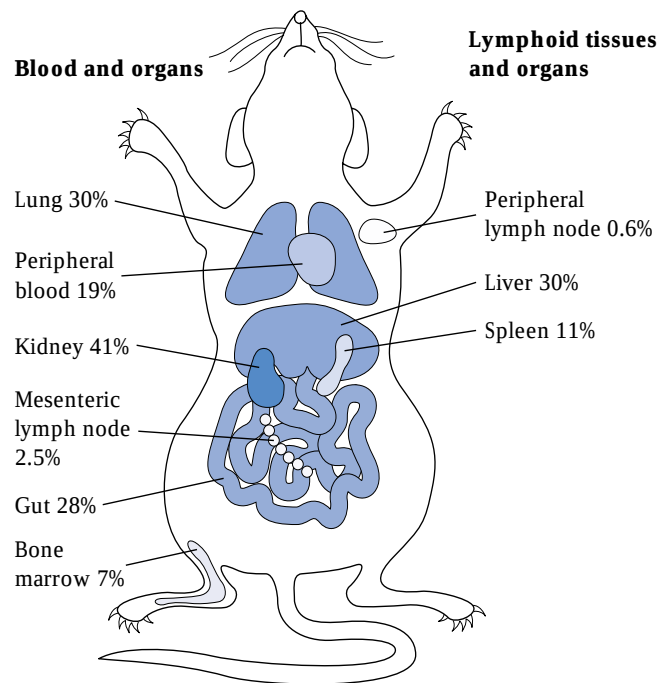


FIGURE 14-4 Localizing antigen specific CD8⁺ T-cell populations in vivo. Mice were infected with vesicular stomatitis virus (VSV) and during the course of the acute stage of the infection, cell populations were isolated from the tissues indicated in the figure and incubated with tetramers containing VSV-peptide/MHC complexes. Flow cytometric analysis allowed determination of the percentages of CD8⁺ T cells that were VSV-specific in each of the populations examined. [Adapted from P. Klenerman, V. Cerundolo, and P. R. Dunbar, 2002, *Nature Reviews/Immunology* 2:269.]

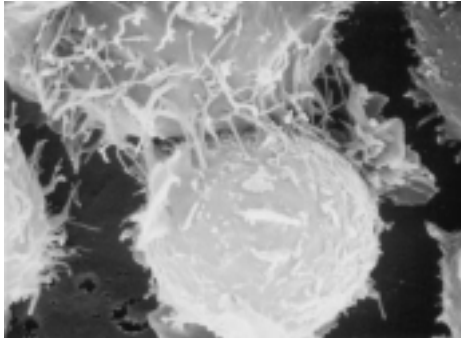


FIGURE 14-5 Scanning electron micrograph of tumor-cell attack by a CTL. The CTL makes contact with a smaller tumor cell. [From J. D. E. Young and Z. A. Cohn, 1988, *Sci. Am.* **258**(1):38.]

receptor LFA-1 on the CTL membrane binds to ICAMs on the target-cell membrane, resulting in the formation of a conjugate. Antigen-mediated CTL activation converts LFA-1 from a low-avidity state to a high-avidity state (Figure 14-7). Because of this phenomenon, CTLs adhere to and form conjugates only with appropriate target cells that display antigenic

peptides associated with class I MHC molecules. LFA-1 persists in the high-avidity state for only 5–10 min after antigen-mediated activation, and then it returns to the low-avidity state. This downshift in LFA-1 avidity may facilitate dissociation of the CTL from the target cell.

Electron microscopy of cultured CTL clones reveals the presence of intracellular electron-dense storage granules. These granules have been isolated by fractionation and shown to mediate target-cell damage by themselves. Analysis of their contents revealed 65-kDa monomers of a pore-forming protein called **perforin** and several serine proteases called **granzymes** (or **fragmentins**). CTLs lack cytoplasmic granules and perforin; upon activation, cytoplasmic granules appear, bearing newly expressed perforin monomers.

Immediately after formation of a CTL–target cell conjugate, the Golgi stacks and storage granules reorient within the cytoplasm of the CTL to concentrate near the junction with the target cell (Figure 14-8). Evidence suggests that perforin monomers and the granzyme proteases are then released from the granules by exocytosis into the junctional space between the two cells. As the perforin monomers contact the target-cell membrane, they undergo a conformational change, exposing an amphipathic domain that inserts into the target-cell membrane; the monomers then polymerize (in the presence of

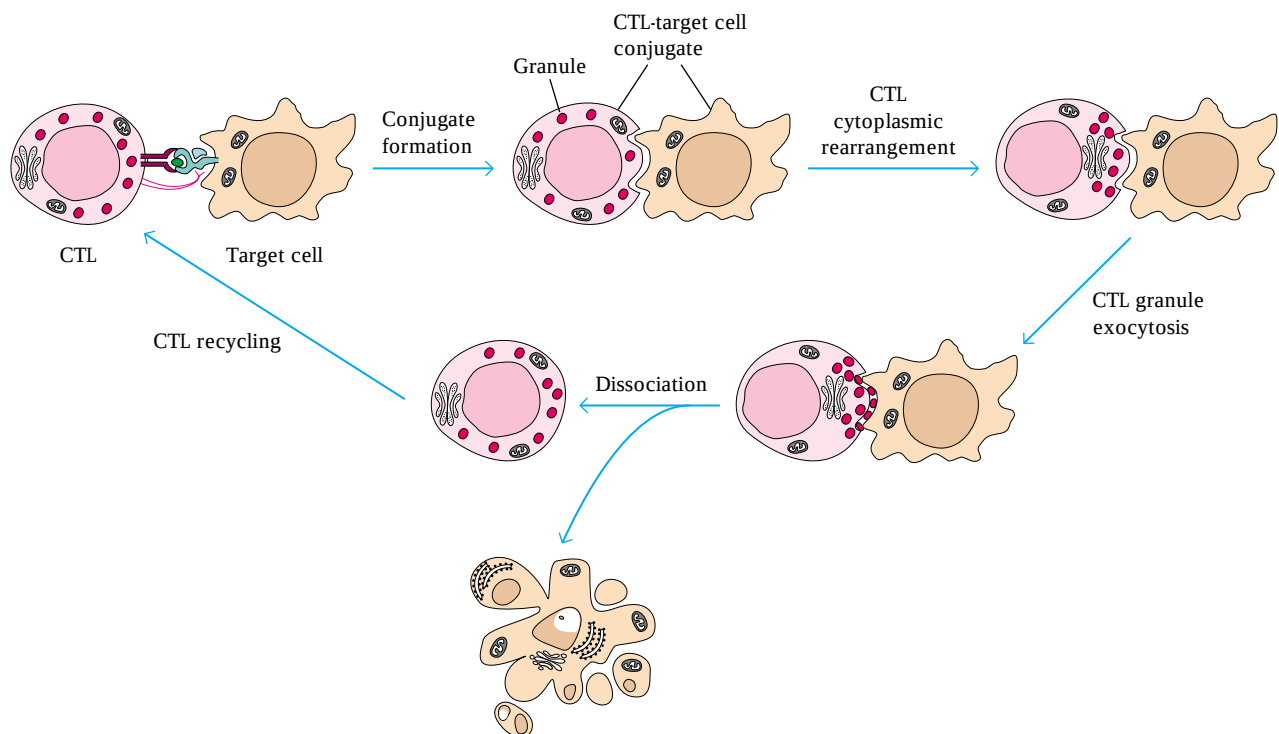


FIGURE 14-6 Stages in CTL-mediated killing of target cells. T-cell receptors on a CTL interact with processed antigen-class I MHC complexes on an appropriate target cell, leading to formation of a CTL/target-cell conjugate. The Golgi stacks and granules in the CTL

reorient towards the point of contact with the target cell, and the granule's contents are released by exocytosis. After dissociation of the conjugate, the CTL is recycled and the target cell dies by apoptosis. [Adapted from P. A. Henkart, 1985, *Annu. Rev. Immunol.* **3**:31.]

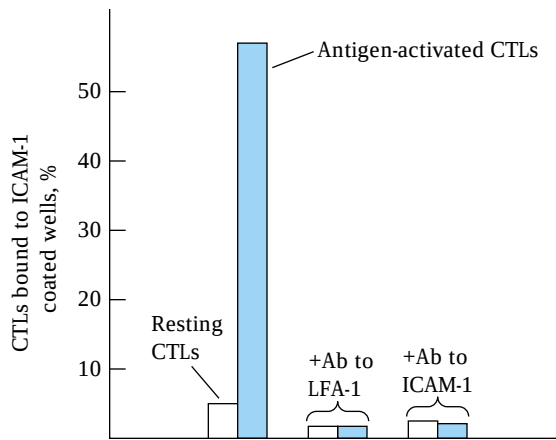


FIGURE 14-7 Effect of antigen activation on the ability of CTLs to bind to the intercellular cell-adhesion molecule ICAM-1. Resting mouse CTLs were first incubated with anti-CD3 antibodies. Crosslinkage of CD3 molecules on the CTL membrane by anti-CD3 has the same activating effect as interaction with antigen–class I MHC complexes on a target cell. Adhesion was assayed by binding radiolabeled CTLs to microwells coated with ICAM-1. Antigen activation increased CTL binding to ICAM-1 more than 10-fold. The presence of excess monoclonal antibody to LFA-1 or ICAM-1 in the microwell abolished binding, demonstrating that both molecules are necessary for adhesion. [Based on M. L. Dustin and T. A. Springer, 1989, *Nature* **341**:619.]

Ca^{2+}) to form cylindrical pores with an internal diameter of 5–20 nm (Figure 14-9a). A large number of perforin pores are visible on the target-cell membrane in the region of conjugate formation (Figure 14-9b). Interestingly, perforin exhibits some sequence homology with the terminal C9 component of the complement system, and the membrane pores formed by perforin are similar to those observed in complement-mediated lysis. The importance of perforin to CTL-mediated killing is demonstrated by perforin-deficient knockout mice, which are unable to eliminate lymphocytic choriomeningitis virus (LCMV) even though they mount a significant CD8^+ immune response to the virus.

Pore formation in the cell membrane of the target is one way that perforin mediates granzyme entry; another is the perforin-assisted pathway. Many target cells have a molecule known as the mannose 6-phosphate receptor on their surface that also binds to granzyme B. Granzyme B/mannose 6-phosphate receptor complexes are internalized and appear inside vesicles. In this case, perforin is necessary for releasing granzyme B from the vesicle into the cytoplasm of the target cell.

Once it enters the cytoplasm of the target cell, granzyme B initiates a cascade of reactions that result in the fragmenta-

tion of the target-cell DNA into oligomers of 200 bp; this type of DNA fragmentation is typical of apoptosis. Since granzymes are proteases, they cannot directly mediate DNA fragmentation. Rather, they activate an apoptotic pathway within the target cell. This apoptotic process does not require mRNA or protein synthesis in either the CTL or the target cell. Within 5 min of CTL contact, target cells begin to exhibit DNA fragmentation. Interestingly, viral DNA within infected target cells has also been shown to be fragmented during this process. This observation shows that CTL-mediated killing not only kills virus-infected cells but can also destroy the viral DNA in those cells. It has been suggested that the rapid onset of DNA fragmentation after CTL contact may prevent continued viral replication and assembly in the period before the target cell is destroyed.

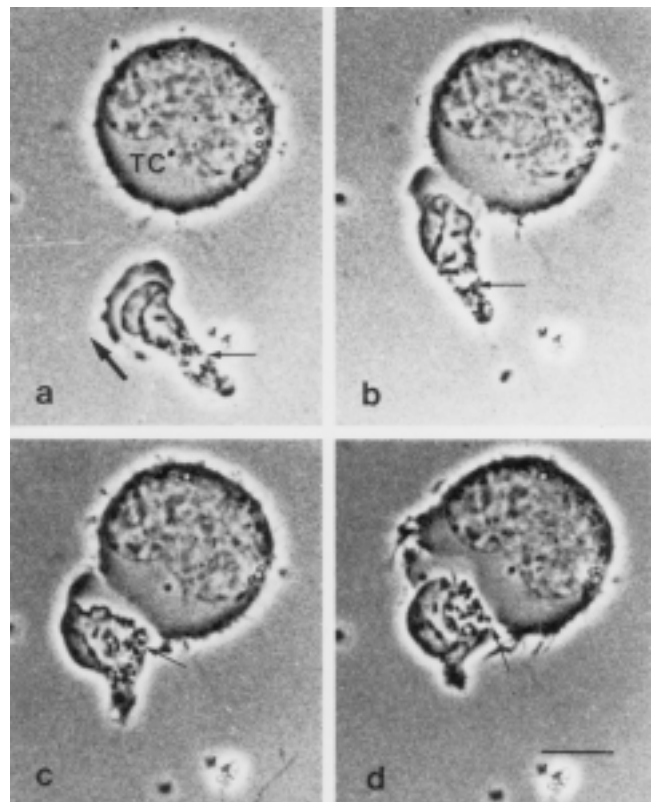


FIGURE 14-8 Formation of a conjugate between a CTL and target cell and reorientation of CTL cytoplasmic granules as recorded by time-lapse cinematography. (a) A motile mouse CTL (thin arrow) approaches an appropriate target cell (TC). The thick arrow indicates direction of movement. (b) Initial contact of the CTL and target cell has occurred. (c) Within 2 min of initial contact, the membrane-contact region has broadened and the rearrangement of dark cytoplasmic granules within the CTL (thin arrow) is under way. (d) Further movement of dark granules toward the target cell is evident 10 min after initial contact. [From J. R. Yanelli et al., 1986, *J. Immunol.* **136**:377.]

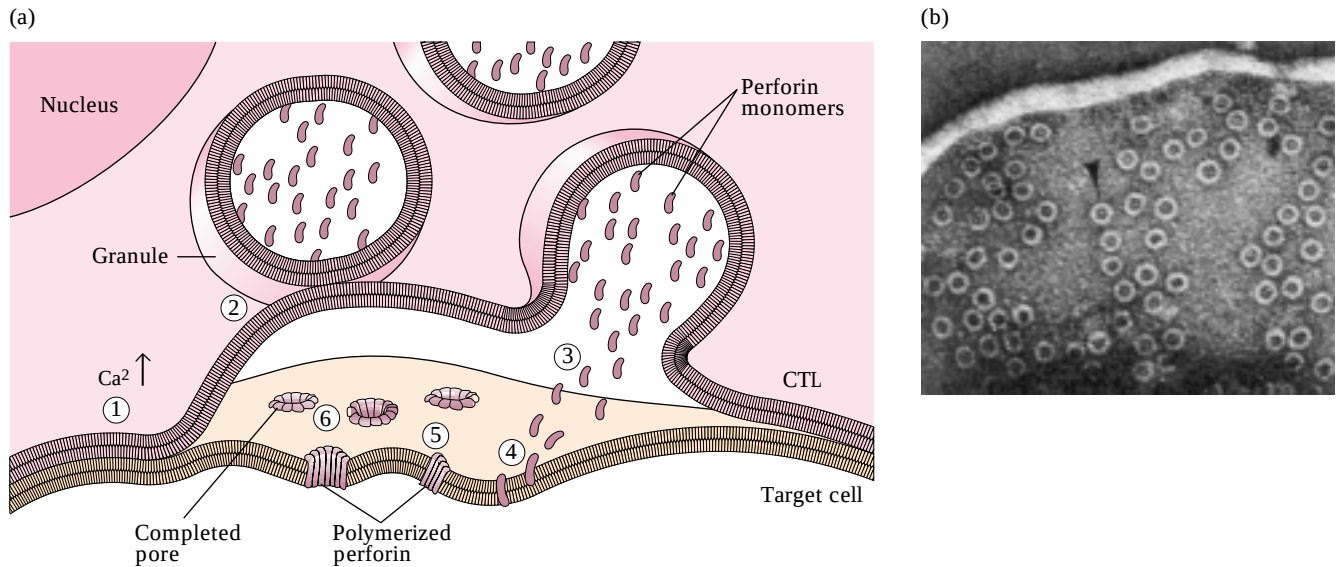


FIGURE 14-9 CTL-mediated pore formation in target-cell membrane. (a) In this model, a rise in intracellular Ca^{2+} triggered by CTL-target cell interaction (1) induces exocytosis, in which the granules fuse with the CTL cell membrane (2) and release monomeric perforin into the small space between the two cells (3). The released perforin monomers undergo a Ca^{2+} -induced conformational change that al-

lows them to insert into the target-cell membrane (4). In the presence of Ca^{2+} , the monomers polymerize within the membrane (5), forming cylindrical pores (6). (b) Electron micrograph of perforin pores on the surface of a rabbit erythrocyte target cell. [Part (a) adapted from J. D. E. Young and Z. A. Cohn, 1988, *Sci. Am.* **258**(1):38; part (b) from E. R. Podack and G. Dennert, 1983, *Nature* **301**:442.]

Some potent CTL lines have been shown to lack perforin and granzymes. In these cases, cytotoxicity is mediated by Fas. As described in Chapter 10, this transmembrane protein, which is a member of the TNF-receptor family, can deliver a death signal when crosslinked by its natural ligand, a member of the tumor necrosis family called Fas ligand (see Figure 10-19). Fas ligand (FasL) is found on the membrane of CTLs, and the interaction of FasL with Fas on a target cell triggers apoptosis.

Key insight into the role of perforin and the Fas-FasL system in CTL-mediated cytolysis came from experiments with mutant mice. These experiments used two types of mutant mice, the perforin knockout mice mentioned above and a strain of mice known as *lpr* (Figure 14-10). Mice that are homozygous for the *lpr* mutation express little or no Fas and, consequently, cells from these mice cannot be killed by interaction with Fas ligand. If lymphocytes from normal H-2^b mice are incubated with killed cells from H-2^k mice, anti-H-2^k CTLs are generated. These H-2^b CTLs will kill target cells from normal H-2^k mice or from H-2^k animals that are homozygous for the *lpr* mutation. Incubation of H-2^b cells of perforin knockout mice with killed cells from H-2^k mice resulted in CTLs that killed wild-type target cells but failed to induce lysis in target cells from H-2^k mice homozygous for the *lpr* mutation.

The results of these experiments taken together with other studies allowed the investigators to make the following inter-

pretation. CTLs raised from normal mice can kill target cells by a perforin-mediated mechanism, by a mechanism involving engagement of target-cell Fas with Fas ligand displayed on the CTL membrane, or, in some cases perhaps, by a combination of both mechanisms. Such CTLs can kill target cells that lack membrane Fas by using the perforin mechanism alone. On the other hand, CTLs from perforin-knockout mice can kill only by the Fas-FasL mechanism. Consequently, CTLs from perforin-knockout mice can kill Fas-bearing normal target cells but not *lpr* cells, which lack Fas. These workers also concluded that all of the CTL-mediated killing they observed could be traced to the action of perforin-dependent killing, Fas-mediated killing, or a combination of the two. No other mechanism was detected.

This experiment taken together with a number of other studies shows that two mechanisms are responsible for initiating all CTL-mediated apoptotic death of target cells:

- Directional delivery of cytotoxic proteins (perforin and granzymes) that are released from CTLs and enter target cells
- Interaction of the membrane-bound Fas ligand on CTLs with the Fas receptor on the surface of target cells

Either of these initiating events results in the activation of a signaling pathway that culminates in the death of the target cell

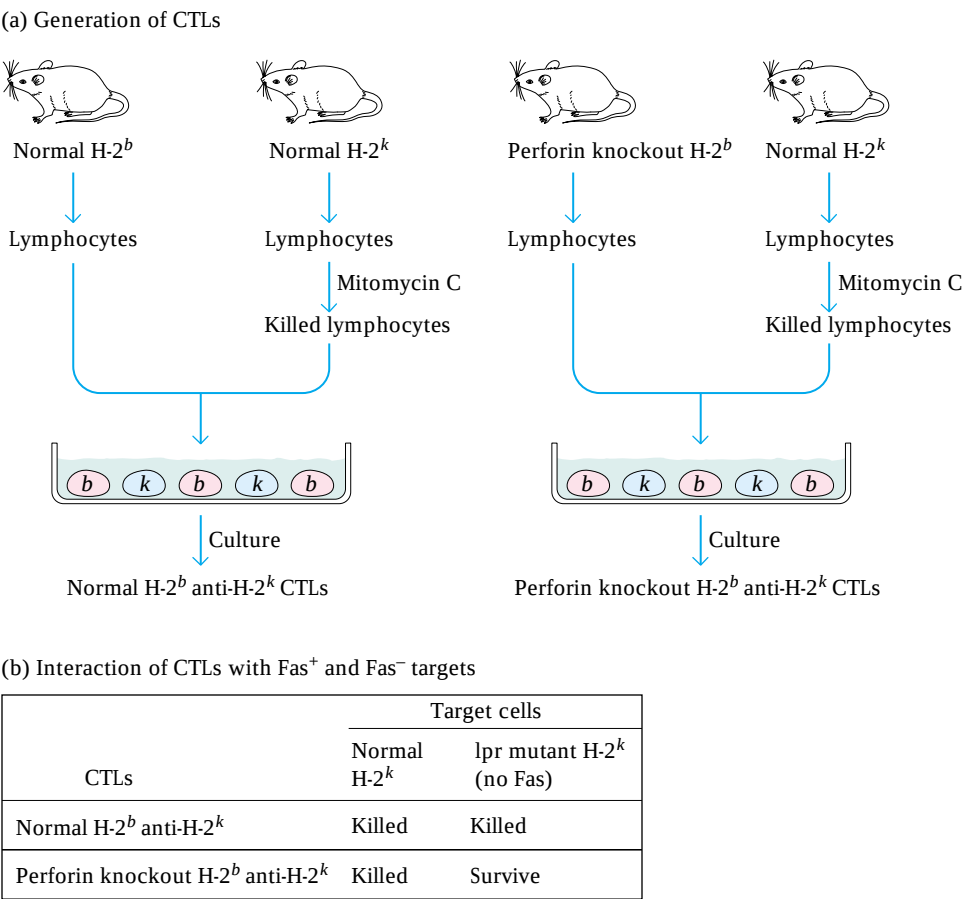


FIGURE 14-10 Experimental demonstration that CTLs use Fas and perforin pathways. (a) Generation of CTLs. Lymphocytes were harvested from mice of H-2^b and H-2^k MHC haplotypes. H-2^k haplotype cells were killed by treatment with mitomycin C and co-cultured with H-2^b haplotype cells to stimulate the generation of H-2^k CTLs. If the H-2^b lymphocytes were derived from normal mice, they gave rise to CTLs that had both perforin and Fas ligand. If the CTLs were raised

by stimulation of lymphocytes from perforin knockout (KO) mice, they expressed Fas ligand but not perforin. (b) Interaction of CTLs with Fas⁺ and Fas⁻ targets. Normal H-2^b anti-H-2^k CTLs that express both Fas ligand and perforin kill normal H-2^k target cells and H-2^k lpr mutant cells, which do not express Fas. In contrast, H-2^b anti-H-2^k CTLs from perforin KO mice kill Fas⁺ normal cells by engagement of Fas with Fas ligand but are unable to kill the lpr cells, which lack Fas.

by apoptosis (Figure 14-11). A feature of cell death by apoptosis is the involvement of the caspase family of cysteine proteases, which cleave after an aspartic acid residue. The name **caspase** incorporates all of these elements (cysteine, *aspartate* protease). Normally, caspases are present in the cell as inactive proenzymes—procaspases—that require proteolytic cleavage for conversion to the active forms. More than a dozen different caspases have been found, each with its own specificity. Cleavage of a procaspase produces an active initiator caspase, which cleaves other procaspases, thereby activating their proteolytic activity. The end result is the systematic and orderly disassembly of the cell that is the hallmark of apoptosis.

CTLs use granzymes and Fas ligand to initiate caspase cascades in their targets. The granzymes introduced into the target cell from the CTL mediate proteolytic events that activate an initiator caspase. Similarly, the engagement of Fas on a tar-

get cell by Fas ligand on the CTL causes the activation of an initiator caspase in the target cell. Fas is associated with a protein known as FADD (*Fas-associated protein with death domain*), which in turn associates with a procaspase form of caspase 8. Upon Fas crosslinking, procaspase 8 is converted to caspase 8 and initiates an apoptotic caspase cascade. The end result of both the perforin/granzyme and Fas-mediated pathways is the activation of dormant death pathways that are present in the target cell. As one immunologist has so aptly put it, CTLs don't so much kill target cells as persuade them to commit suicide.

Natural Killer Cells

Natural killer cells were discovered quite by accident when immunologists were measuring in vitro activity of tumor-

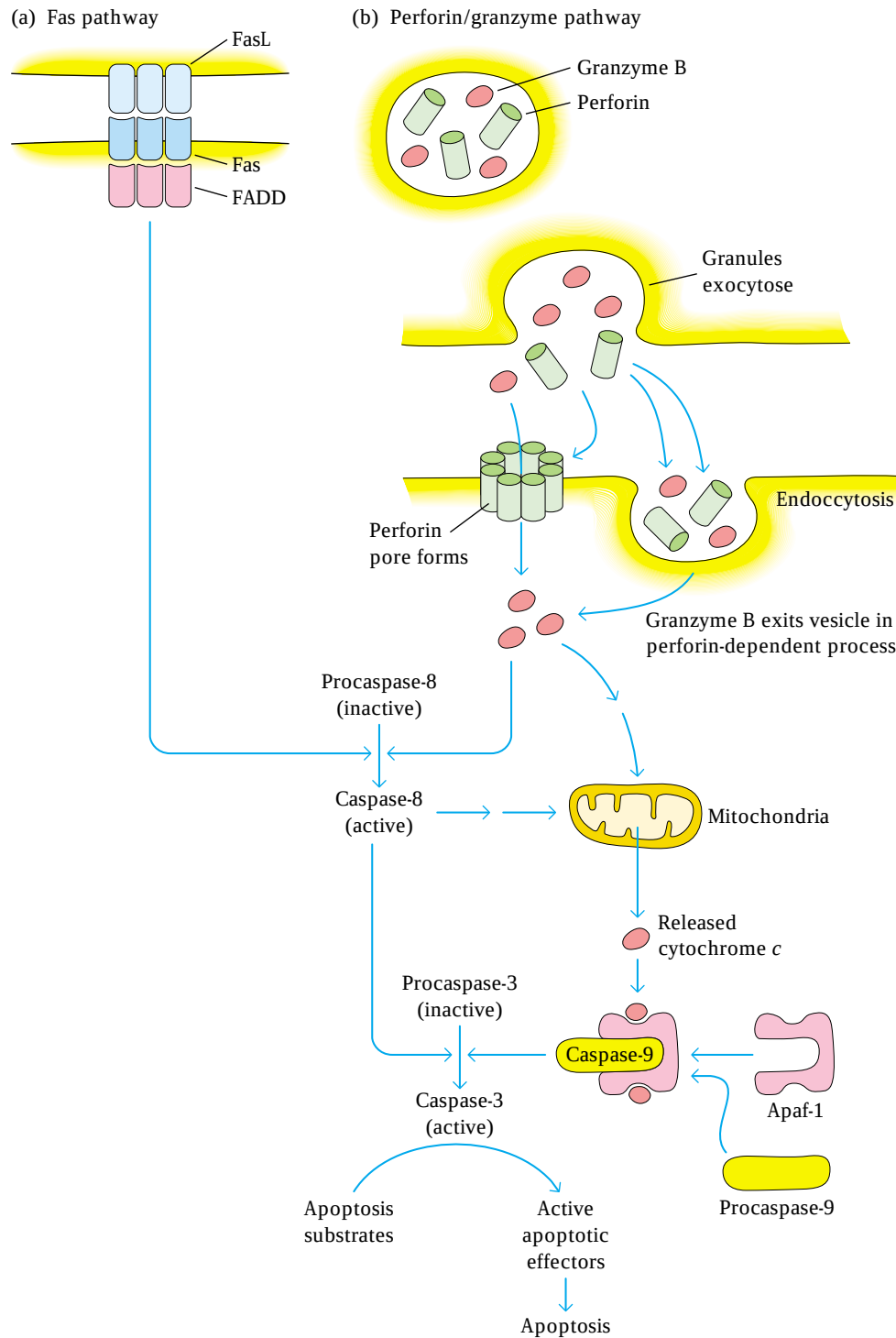


FIGURE 14-11 Two pathways of target-cell apoptosis stimulated by CTLs. (a) The Fas pathway. Ligation of trimeric Fas units by CTL-borne Fas ligand leads to the association of the death domains of Fas with FADD, which in turn results in a series of reactions leading to apoptosis of the target cell. (b) The perforin/granzyme pathway. Granule exocytosis releases granzymes and perforin from the CTL into the space between the CTL and the target cell. Granzyme B enters the target cell in two ways: via perforin-generated pores, or by

binding to mannose 6-phosphate receptors that are subsequently endocytosed. Granzyme B is then released into the cytoplasm in a perforin-dependent process. Cleavage of procaspase 8 by granzyme B activates a caspase cascade that results in the apoptotic death of the cell, and interaction of granzyme B with other targets can invoke mitochondrially mediated death pathways. [Adapted from M. Barry and C. Bleackley, 2002, *Nature Reviews Immunology* 2:401.]

specific cells taken from mice with tumors. Normal unimmunized mice and mice with unrelated tumors served as negative controls. Much to the consternation of the investigators, the controls showed significant lysis of the tumor cells, too. Characterization of this nonspecific tumor-cell killing revealed that a population of large granular lymphocytes was responsible. The cells, which were named natural killer (NK) cells for their nonspecific cytotoxicity, make up 5%–10% of the recirculating lymphocyte population. These cells are involved in immune defenses against viruses and tumors. Because NK cells produce a number of immunologically important cytokines, they play important roles in immune regulation and influence both innate and adaptive immunity. In particular, IFN- γ production by NK cells can affect the participation of macrophages in innate immunity by activation of the phagocytic and microbicidal activities. IFN- γ derived from NK cells can influence the T_H1 versus T_H2 commitment of helper T cell populations by its inhibitory effects on T_H2 expansion, and stimulate T_H1 development via induction of IL-12 by macrophages and dendritic cells. The Chediak-Higashi syndrome described in the Clinical Focus illustrates the disastrous consequences of a lack of NK cells.

NK cells are involved in the early response to infection with certain viruses and intracellular bacteria. NK activity is stimulated by IFN- α , IFN- β , and IL-12. In the course of a viral infection, the level of these cytokines rapidly rises, followed closely by a wave of NK cells that peaks in about 3 days (Figure 14-12). NK cells are the first line of defense against virus infection, controlling viral replication during the time required for activation, proliferation, and differentiation of CTL-P cells into functional CTLs at about day 7. The importance of NK cells in defense against viral infections is illustrated by the case of a young woman who completely lacked these cells. Even though this patient had normal T- and B-cell counts, she suffered severe varicella virus infections and a life-threatening cytomegalovirus infection.

NK Cells and T Cells Share Some Features

NK cells are lymphoid cells derived from bone marrow that share a common early progenitor with T cells, but their detailed lineage remains to be worked out. They express some membrane markers that are found on monocytes and granulocytes, as well as some that are typical of T cells. Different NK cells express different sets of membrane molecules. It is not known whether this heterogeneity reflects subpopulations of NK cells or different stages in their activation or maturation. Among the membrane molecules expressed by NK cells are CD2, the 75-kDa β subunit of the IL-2 receptor, and, on almost all NK cells, CD16 (or Fc γ RIII), a receptor for the Fc region of IgG. Cell depletion with monoclonal anti-CD16 antibody removes almost all NK-cell activity from peripheral blood.

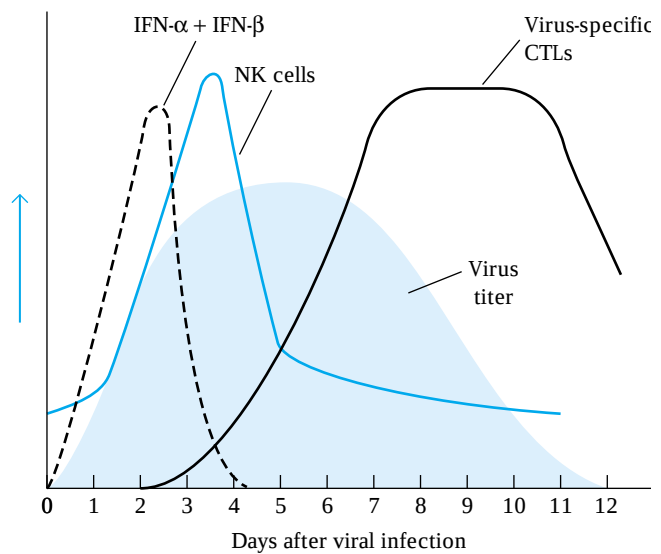


FIGURE 14-12 Time course of viral infection. IFN- α and IFN- β (dashed curve) are released from virus-infected cells soon after infection. These cytokines stimulate the NK cells, quickly leading to a rise in the NK-cell population (blue curve) from the basal level. NK cells help contain the infection during the period required for generation of CTLs (black curve). Once the CTL population reaches a peak, the virus titer (blue area) rapidly decreases.

Despite some similarities of NK cells to T lymphocytes, they do not develop exclusively in the thymus. Nude mice, which lack a thymus and have few or no T cells, have functional NK-cell populations. Unlike T cells and B cells, NK cells do not undergo rearrangement of receptor genes. This is demonstrated by the observation that NK cells develop in mice in which the recombinase genes *RAG-1* or *RAG-2* have been knocked out. Furthermore, while no T or B cells are found in SCID mice, functional populations of NK cells can be readily demonstrated. The power of NK cells and other protective mechanisms of innate immunity to protect animals totally lacking in adaptive immunity is nicely illustrated by the family of *RAG-1* knockout mice shown in Figure 14-13.

Killing by NK Cells Is Similar to CTL-Mediated Killing

Natural killer cells appear to kill tumor cells and virus-infected cells by processes similar to those employed by CTLs. NK cells bear FasL on their surface and readily induce death in Fas-bearing target cells. The cytoplasm of NK cells contains numerous granules containing perforin and granzymes. Unlike CTLs, which need to be activated before granules appear, NK cells are constitutively cytotoxic, always having large granules in their cytoplasm. After an NK cell adheres to a target

cell, degranulation occurs with release of perforin and granzymes at the junction of the interacting cells. The roles of perforin and granzymes in NK-mediated killing of target cells by apoptosis are believed to be similar to their roles in the CTL-mediated process.

Despite these similarities, NK cells differ from CTLs in several significant ways. First, NK cells do not express antigen-specific T-cell receptors or CD3. In addition, recognition of target cells by NK cells is not MHC restricted; that is, in many cases the same levels of NK-cell activity are observed with syngeneic and allogeneic tumor cells. Moreover, although prior priming enhances CTL activity, NK-cell activity does not increase after a second injection with the same tumor cells. In other words, the NK-cell response generates no immunologic memory.

NK Cells Have Both Activation and Inhibition Receptors

Given that NK cells do not express antigen-specific receptors, the mechanism by which NK cells recognize altered self-cells and distinguish them from normal body cells baffled immunologists for years. The solution to the problem emerged with the realization that NK cells employ two different categories of receptors, one that delivers inhibition signals to NK cells, and another that delivers activation signals. Initially, it was thought that there were two receptors, one that activated and another that inhibited NK cells—the so-called two-receptor model. It is now clear that there are many different

cell-surface receptors for activation signals and a number of different kinds for inhibitory ones. Consequently, it is more appropriate to think in terms of an *opposing-signals model* rather than a *two-receptor model*. It is the balance between activating signals and inhibitory signals that is believed to enable NK cells to distinguish healthy cells from infected or cancerous ones. It is important to be aware that additional NK-activating signals can be delivered by soluble factors. These include cytokines such as α and β interferons, TNF- α , IL-12, and IL-15.

The exact nature of the membrane-bound receptors on NK cells that produce activation is not completely clear. Antibody crosslinking of many molecules found on the surface of NK cells can activate these cells artificially, but the natural ligands for many of these putative activation receptors (ARs) are not known. Some of the candidate ARs are members of a class of carbohydrate-binding proteins known as **C-type lectins**, so named because they have calcium-dependent carbohydrate-recognition domains. NKR-P1 is an example of a C-type lectin found on NK cells that has activation properties. In addition to lectins, other molecules on NK cells might be involved in activation, including CD2 (receptor for the adhesion molecule LFA-3), and the Fc γ III receptor, CD16. Although CD16 is responsible for antibody-mediated recognition and killing of target cells by NK cells, it is probably not involved in non-antibody-dependent killing. In addition to the molecules already mentioned, three additional proteins, NKp30, NKp44, and NKp46, appear to play significant roles in the activation of human NK cells.

Clues to the sources of inhibitory signals came from studies of the killing of tumor cells and virus-infected cells by NK cells. It was noticed that the preferential killing of mouse tumor cells compared with normal cells correlated with a lack of expression of MHC molecules by the tumor cells. Experiments with human cells showed that NK cells lysed a B-cell line that was MHC deficient because it had been transformed by Epstein-Barr virus. However, when this cell line was transformed with human HLA genes so that it expressed high levels of MHC molecules, NK cells failed to lyse it. These observations led to the idea that NK cells target for killing cells that have aberrant MHC expression. Since many virus-infected and tumor cells have reduced MHC expression, this model made good physiological sense. Vindication of this proposal has come from the discovery of receptors on NK cells that produce inhibitory signals when they recognize MHC molecules on potential target cells. These inhibitory receptors on the NK cell then prevent NK-cell killing, proliferation, and cytokine release.

Two major groups of inhibitory receptors have been found on NK cells. One of these is a family of *C-type-lectin–inhibitory receptors* (CLIR), and the other is a group of Ig-superfamily–inhibitory receptors (ISIR) known as the *killer-cell–inhibitory receptors* (KIR). Even though these groups are chemically quite different, they are together referred to as the *inhibitory-receptor superfamily* (IRS). In humans, the



FIGURE 14-13 Family of RAG-1 KO mice. These mice have no adaptive immunity because they lack T and B cells. However, NK cells and other mechanisms of innate immunity provide sufficient protection against infection that, if maintained in clean conditions, these mice can reproduce and raise healthy offspring. However, they are more susceptible to infection than normal mice and have reduced lifespans. [From the laboratory of R. A. Goldsby.]



CLINICAL FOCUS

Chediak-Higashi Syndrome

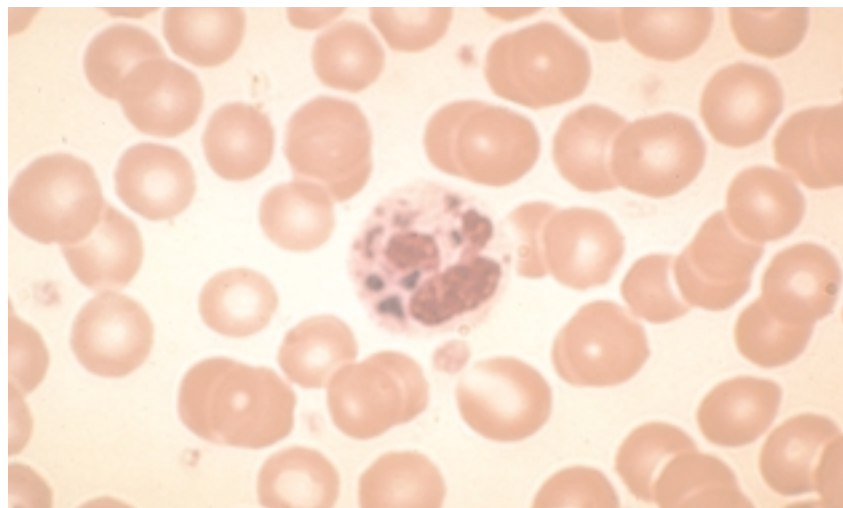
The Chediak-Higashi syndrome (CHS) is a rare inherited disorder that inflicts a diversity of maladies on those afflicted by it. Identifying features of the disease include progressive neurological dysfunction, an increased tendency to develop leukemia and lymphoma, and depigmentation of hair, skin and eyes. Almost 90% of those afflicted have severe immunological deficiency, displaying defective natural-killer-cell function and deficits in neutrophil activity. These abnormalities in the leukocyte population are reflected in a greatly heightened susceptibility to infection, traceable in part to neutrophils that are deficient in chemotactic and bactericidal activities, and to dysfunctional populations of natural killer cells. The result is a greatly shortened life span; many Chediak-Higashi patients succumb to the disease in childhood. Microscopic examination of leukocytes from CHS patients reveals giant lysosomes that are characteristic of this disease.

Only those homozygous for a mutant form of a gene known as *CHS-1/LYST* (lysosomal trafficking regulator) develop Chediak-Higashi syndrome. A corresponding mutation has been found in beige mice and the mouse analogue of human *CHS-1/LYST*. The mouse and human homologues both encode a very large polypeptide of 2,186 amino acids. Beige mice display a pattern of symptoms very much like those seen in humans, and their granulocytes, like those of afflicted

humans, display the huge cytoplasmic granules that are a morphological hallmark of the disease. Studies of the disease in beige mice complement those in humans, and have led to the conclusion that severe defects in the formation, fusion, or trafficking of intracellular vesicles probably underlie its devastating pathology.

Bone marrow transplantation (BMT) is the only effective therapy for the defective natural killer activity, aberrant macrophage activation, and susceptibility to bacterial infections that plague those afflicted with Chediak-Higashi syndrome. However, this is a risky and complex therapy. A look at the experience of 10 CHS children who underwent BMT for their disease is informative. BMT is best done

with marrow from a donor whose HLA type is identical to that of the recipient. Unfortunately, it may be difficult or impossible to obtain HLA-matched bone marrow, and 3 of the patients had to settle for HLA-nonidentical marrow. After a median interval of 6.5 years post-transplantation, 6 of the 7 patients who had received marrow from HLA-identical donors were alive, but only 1 of the 3 recipients of HLA-nonidentical marrow survived. The clinical picture in the survivors was markedly improved. They were no longer hypersusceptible to bacterial infection, displayed significant NK-cell activity, and did not suffer from uncontrolled and pathological macrophage activation. However, the albinism and lack of eye pigmentation were not improved by BMT. HLA-identical BMT is thus accepted as a curative treatment for Chediak-Higashi syndrome, but reliance on HLA-nonidentical transplantation is experimental and carries very high risk.



A neutrophil with the giant lysosomes characteristic of Chediak-Higashi syndrome. (Courtesy of American Society of Hematology Slide Bank, 3rd edition.)

C-type-lectin-inhibitory receptor is CD94/NKG2, a disulfide-bonded heterodimer made up of two glycoproteins, one of which is CD94 and the other a member of the NKG2 family. The CD94/NKG2 receptors recognize HLA-E on potential target cells. Because HLA-E is not transported to the surface of a cell unless it has bound a peptide derived from HLA-A,

HLA-B, or HLA-C, the amount of HLA-E on the surface serves as indicator of the overall level of class I MHC biosynthesis in the cells. These inhibitory CD94/NKG2 receptors are thus not specific for a particular HLA allele and will send inhibitory signals to the NK cell, with the net result that killing of potential target cells is inhibited if they are express-

ing adequate levels of class I. In contrast, KIR receptors, of which more than 50 family members have been found, are specific for one or a limited number of polymorphic products of particular HLA loci. Unlike B and T cells, NK cells are not limited to expressing a single KIR, but may express several, each specific for a different MHC molecule or for a set of closely related MHC molecules. For example, individual clones of human NK cells expressing a CD94/NKG2 receptor and as many as six different KIR receptors have been found. Because signals from inhibitory receptors have veto power over signals from activating receptors, a negative signal from any inhibitory receptor, whether of the CD94/NKG2 or KIR type, can block the lysis of target cells by NK cells. Thus, cells expressing normal levels of unaltered MHC class I molecules tend to escape all forms of NK-cell-mediated killing.

In the opposing-signals model of NK-cell regulation that is emerging from studies of NK cells (Figure 14-14), activating receptors engage ligands on the target cell. These ligands may be abnormal patterns of glycosylation on the surface of tumor or virus-infected cells. Recognition of these determinants by ARs on NK cells would signal NK cells to kill the target cells. Ligand engagement by NKR-P1-type lectin receptors, or a number of other ARs, such as CD16, or in some cases CD2, generates signals that direct the NK cell to kill the target cell. Any of these killing signals can be overridden by a signal from inhibitory receptors. As we have already seen, members of the inhibitory superfamily of receptors (ISRs) provide a signal that decisively overrides activation signals when these inhibitory receptors detect normal levels of MHC class I expression on potential target cells. This prevents the death of the target cell. It also prevents NK-cell proliferation and the induction of secretion of cytokines such as IFN- γ and TNF- α . The overall consequence of the opposing-signals model is to spare cells that express critical indicators of normal self, the MHC class I molecules, and to kill cells that lack indicators of self (absence of normal levels of class I MHC).

Antibody-Dependent Cell-Mediated Cytotoxicity

A number of cells that have cytotoxic potential express membrane receptors for the Fc region of the antibody molecule. When antibody is specifically bound to a target cell, these receptor-bearing cells can bind to the antibody Fc region, and thus to the target cells, and subsequently cause lysis of the target cell. Although these cytotoxic cells are nonspecific for antigen, the specificity of the antibody directs them to specific target cells. This type of cytotoxicity is referred to as **antibody-dependent cell-mediated cytotoxicity (ADCC)**.

Among the cells that can mediate ADCC are NK cells, macrophages, monocytes, neutrophils, and eosinophils. Antibody-dependent cell-mediated killing of cells infected with the measles virus can be observed *in vitro* by adding anti-measles

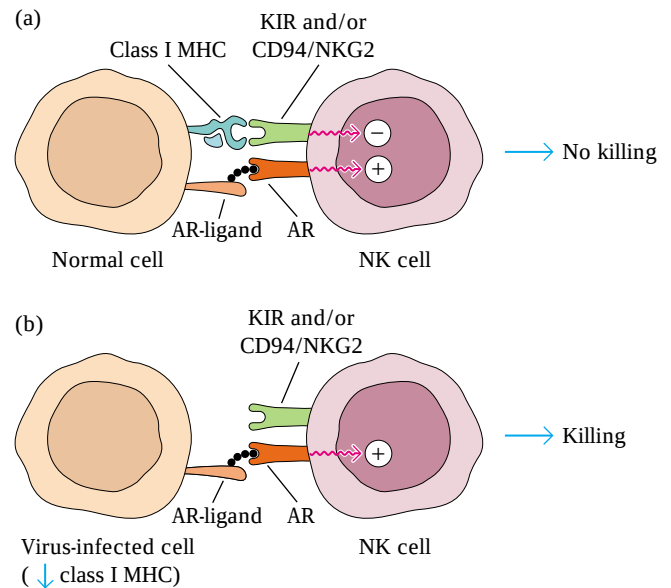


FIGURE 14-14 Opposing-signals model of how cytotoxic activity of NK cells is restricted to altered self-cells. An activation receptor (AR) on NK cells interacts with its ligand on normal and altered self-cells, inducing an activation signal that results in killing. However, engagement of inhibitory NK cell receptors such as KIR and CD94/NKG2 by class I MHC molecules delivers an inhibitory signal that counteracts the activation signal. Expression of class I molecules on normal cells thus prevents their destruction by NK cells. Because class I expression is often decreased on altered self-cells, the killing signal predominates, leading to their destruction.

antibody together with macrophages to a culture of measles-infected cells. Similarly, cell-mediated killing of helminths, such as schistosomes or blood flukes, can be observed *in vitro* by incubating larvae (schistosomules) with antibody to the schistosomules together with eosinophils.

Target-cell killing by ADCC appears to involve a number of different cytotoxic mechanisms, but not complement-mediated lysis (Figure 14-15). When macrophages, neutrophils, or eosinophils bind to a target cell by way of the Fc receptor, they become more active metabolically; as a result, the lytic enzymes in their cytoplasmic lysosomes or granules increase. Release of these lytic enzymes at the site of the Fc-mediated contact may result in damage to the target cell. In addition, activated monocytes, macrophages, and NK cells have been shown to secrete tumor necrosis factor (TNF), which may have a cytotoxic effect on the bound target cell. Since both NK cells and eosinophils contain perforin in cytoplasmic granules, their target-cell killing also may involve perforin-mediated membrane damage similar to the mechanism described for CTL-mediated cytotoxicity.

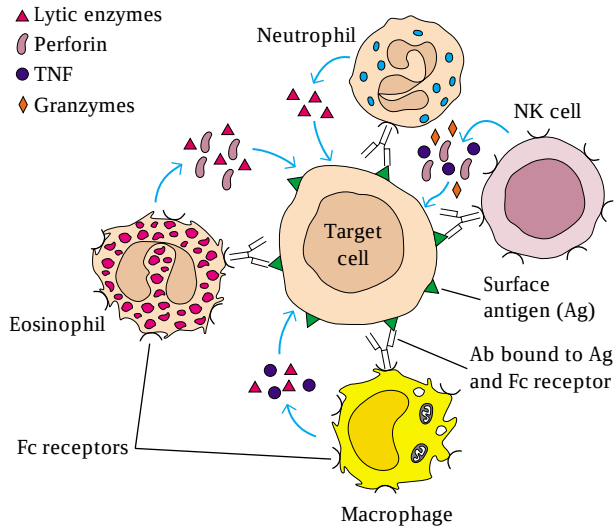


FIGURE 14-15 Antibody-dependent cell-mediated cytotoxicity (ADCC). Nonspecific cytotoxic cells are directed to specific target cells by binding to the Fc region of antibody bound to surface antigens on the target cells. Various substances (e.g., lytic enzymes, TNF, perforin, granzymes) secreted by the nonspecific cytotoxic cells then mediate target-cell destruction.

Experimental Assessment of Cell-Mediated Cytotoxicity

Three experimental systems have been particularly useful for measuring the activation and effector phases of cell-mediated cytotoxic responses. The *mixed-lymphocyte reaction* (MLR) is an in vitro system for assaying T_H -cell proliferation in a cell-mediated response; *cell-mediated lympholysis* (CML) is an in vitro assay of effector cytotoxic function; and the *graft-versus-host reaction* (GVH) in experimental animals provides an in vivo system for studying cell-mediated cytotoxicity.

Co-Culturing T Cells with Foreign Cells Stimulates MLR

During the 1960s, early in the history of modern cellular immunology, it was observed that when rat lymphocytes were cultured on a monolayer of mouse fibroblast cells, the rat lymphocytes proliferated and destroyed the mouse fibroblasts. In 1970 it was discovered that functional CTLs could also be generated by co-culturing allogeneic spleen cells in a system termed the mixed-lymphocyte reaction (MLR). The T lymphocytes in an MLR undergo extensive blast transformation and cell proliferation. The degree of proliferation can be assessed by adding [^3H] thymidine to the culture medium and monitoring uptake of label into DNA in the course of repeated cell divisions.

Both populations of allogeneic T lymphocytes proliferate in an MLR unless one population is rendered unresponsive by treatment with mitomycin C or lethal x-irradiation (Figure 14-16). In the latter system, called a one-way MLR, the unresponsive population provides stimulator cells that express alloantigens foreign to the responder T cells. Within 24–48 h, the responder T cells begin dividing in response to the alloantigens of the stimulator cells, and by 72–96 h an expanding population of functional CTLs is generated. With this experimental system, functional CTLs can be generated entirely in vitro, after which their activity can be assessed with various effector assays.

The significant role of T_H cells in the one-way MLR can be demonstrated by use of antibodies to the T_H -cell membrane marker CD4. In a one-way MLR, responder T_H cells recognize allogeneic class II MHC molecules on the stimulator cells and proliferate in response to these differences. Removal of the $CD4^+$ T_H cells from the responder population with anti-CD4 plus complement abolishes the MLR and prevents generation of CTLs. In addition to T_H cells, accessory cells such as macrophages also are necessary for the MLR to proceed. When adherent cells (mostly macrophages) are removed from the stimulator population, the proliferative response in the MLR is abolished and functional CTLs are no longer generated. It is now known that the function of these macrophages is to activate the class II MHC-restricted T_H cells, whose proliferation is measured in the MLR. In the absence of T_H -cell activation, there is no proliferation.

CTL Activity Can Be Demonstrated by CML

Development of the cell-mediated lympholysis (CML) assay was a major experimental advance that contributed to understanding of the mechanism of target-cell killing by CTLs. In this assay, suitable target cells are labeled intracellularly with chromium-51 (^{51}Cr) by incubating the target cells with $\text{Na}_2^{51}\text{CrO}_4$. After the ^{51}Cr diffuses into a cell, it binds to cytoplasmic proteins, reducing passive diffusion of the label out of the cell. When specifically activated CTLs are incubated for 1–4 h with such labeled target cells, the cells lyse and the ^{51}Cr is released. The amount of ^{51}Cr released correlates directly with the number of target cells lysed by the CTLs. By means of this assay, the specificity of CTLs for allogeneic cells, tumor cells, virus-infected cells, and chemically modified cells has been demonstrated (Figure 14-17).

The T cells responsible for CML were identified by selectively depleting different T-cell subpopulations by means of antibody-plus-complement lysis. In general, the activity of CTLs exhibits class I MHC restriction. That is, they can kill only target cells that present antigen associated with syngeneic class I MHC molecules. Occasionally, however, class II-restricted $CD4^+$ T cells have been shown to function as CTLs.

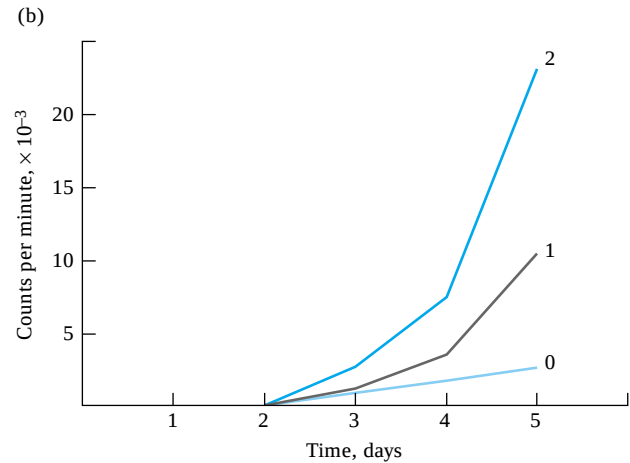
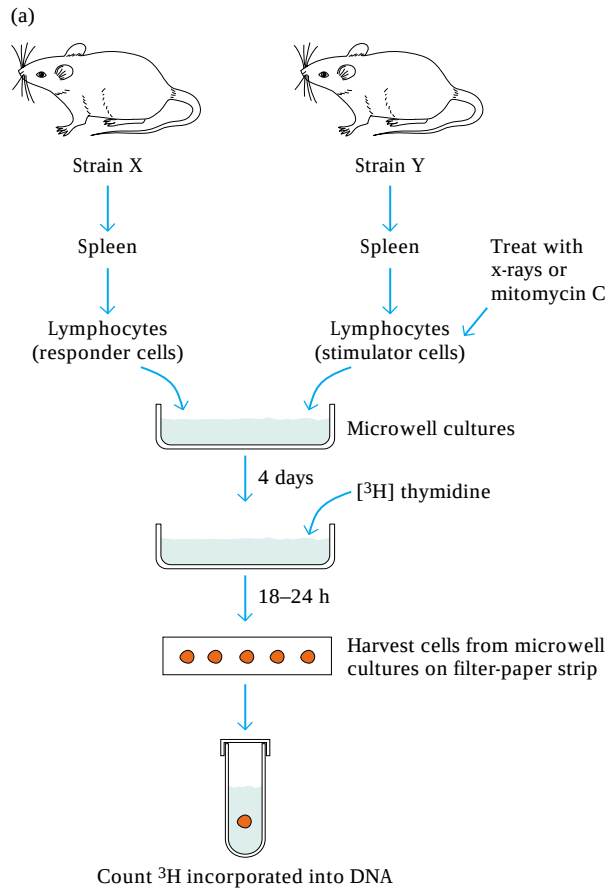


FIGURE 14-16 One-way mixed-lymphocyte reaction (MLR). (a) This assay measures the proliferation of lymphocytes from one strain (responder cells) in response to allogeneic cells that have been x-irradiated or treated with mitomycin C to prevent proliferation (stimulator cells). The amount of [³H] thymidine incorporated into the DNA is directly proportional to the extent of responder-cell proliferation. (b) The amount of [³H]-thymidine uptake in a one-way MLR depends on the degree of differences in class II MHC molecules between the stimulator and responder cells. Curve 0 = no class II MHC differences; curve 1 = one class II MHC difference; curve 2 = two class II MHC differences. These results demonstrate that the greater the class II MHC differences, the greater the proliferation of responder cells.

The GVH Reaction Is an Indication of Cell-Mediated Cytotoxicity

The graft-versus-host (GVH) reaction develops when immunocompetent lymphocytes are injected into an allogeneic recipient whose immune system is compromised. Because the donor and recipient are not genetically identical, the grafted lymphocytes begin to attack the host, and the host's compromised state prevents an immune response against the graft. In humans, GVH reactions often develop after transplantation of bone marrow into patients who have had radiation exposure or who have leukemia, immunodeficiency diseases, or autoimmune anemias. The clinical manifestations of the GVH reaction include diarrhea, skin lesions, jaundice, spleen enlargement, and death. Epithelial cells of the skin and gastrointestinal tract often become necrotic, causing the skin and intestinal lining to be sloughed.

Experimentally, GVH reactions develop when immunocompetent lymphocytes are transferred into an allogeneic neonatal or x-irradiated animal. The recipients, especially neonatal ones, often exhibit weight loss. The grafted lymphocytes generally are carried to a number of organs, including

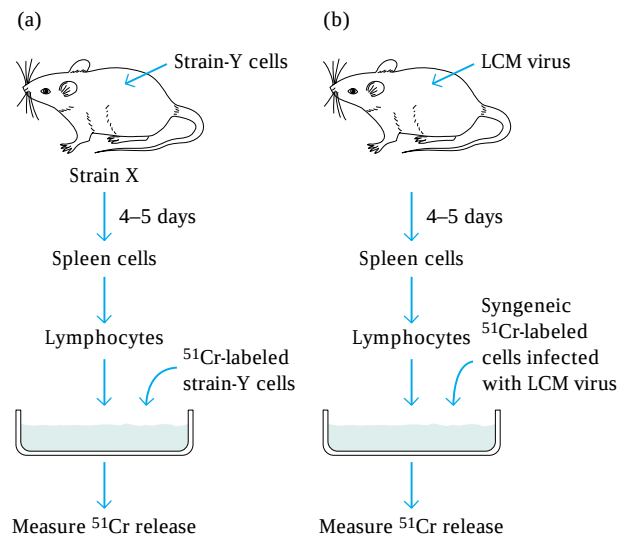


FIGURE 14-17 In vitro cell-mediated lympholysis (CML) assay. This assay can measure the activity of cytotoxic T lymphocytes (CTLs) against allogeneic cells (a) or virus-infected cells (b). In both cases the release of ⁵¹Cr into the supernatant indicates the presence of CTLs that can lyse the target cells.

the spleen, where they begin to proliferate in response to the allogeneic MHC antigens of the host. This proliferation induces an influx of host cells and results in visible spleen enlargement, or splenomegaly. The intensity of a GVH reaction can be assessed by calculating the *spleen index* as follows:

Spleen index =

$$\frac{\text{weight of experimental spleen/total body weight}}{\text{weight of control spleen/total body weight}}$$

A spleen index of 1.3 or greater is considered to be indicative of a positive GVH reaction. Spleen enlargement results from proliferation of both CD4⁺ and CD8⁺ T-cell populations. NK cells also have been shown to play a role in the GVH reaction, and these cells may contribute to some of the skin lesions and intestinal-wall damage observed.

SUMMARY

- The cell-mediated branch of the immune system involves two types of antigen-specific effector cells: cytotoxic T lymphocytes (CTLs) and CD4⁺ T cells that mediate DTH reactions (discussed in Chapter 17). Compared with naive T_H and T_C cells, the effector cells are more easily activated, express higher levels of cell-adhesion molecules, exhibit different trafficking patterns, and produce both soluble and membrane effector molecules.
- The first phase of the CTL-mediated immune response involves the activation and differentiation of T_C cells, called CTL precursors (CTL-Ps).
- Antigen-specific CD8⁺ populations can be identified and tracked by labeling with MHC tetramers.
- The second phase of the CTL-mediated response involves several steps: TCR-MHC mediated recognition of target cells, formation of CTL/target-cell conjugates, reorientation of CTL cytoplasmic granules toward the target cell, granule release, formation of pores in the target-cell membrane, dissociation of CTL from the target, and the death of the target cell.
- CTLs induce cell death via two mechanisms: the perforin-granzyme pathway and the Fas/FasL pathway.
- Various nonspecific (non-MHC dependent) cytotoxic cells (NK cells, neutrophils, eosinophils, macrophages) can also kill target cells. Many of these cells bind to the Fc region of antibody on target cells and subsequently release lytic enzymes, perforin, or TNF, which damage the target-cell membrane, a process, called antibody-dependent cell-mediated cytotoxicity (ADCC).
- NK cells mediate lysis of tumor cells and virus-infected cells by perforin-induced pore formation, a mechanism similar to one of those employed by CTLs.

- The expression of relatively high levels of class I MHC molecules on normal cells protects them against NK cell-mediated killing. NK cell killing is regulated by the balance between positive signals generated by the engagement of activating receptors (NKR-P1 and others) and negative signals from inhibitory receptors (CD94/NKG2 and the KIR family).

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USEFUL WEB SITES

<http://www.cellsalive.com/ctl.htm>

This Cells Alive subsite has a time-lapse video of cytotoxic T lymphocytes (CTLs) recognizing, attacking, and killing a much larger influenza-infected target.

<http://www.antibodyassay.com/nkcell.htm>

The AAL Reference Laboratories web site provides the rationale for clinical tests of NK cell function and tells how such tests are performed.

Study Questions

CLINICAL FOCUS QUESTION Would you expect the Chediak-Higashi syndrome to have its immediate effects on innate or adaptive immunity? Considering what was learned in Chapter 12, which T cell-mediated adaptive responses would you predict would be more significantly affected by CHS? Justify your answer.

1. Indicate whether each of the following statements is true or false. If you believe a statement is false, explain why.
 - a. Cytokines can regulate which branch of the immune system is activated.



- b. Both CTLs and NK cells release perforin after interacting with target cells.
- c. Antigen activation of naive CTL-Ps requires a co-stimulatory signal delivered by interaction of CD28 and B7.
- d. CTLs use a single mechanism to kill target cells.
- e. The secretion of certain critical cytokines is the basis of the role played by T cells in DTH reactions.

2. You have a monoclonal antibody specific for LFA-1. You perform CML assays of a CTL clone, using target cells for which the clone is specific, in the presence and absence of this antibody. Predict the relative amounts of ^{51}Cr released in the two assays. Explain your answer.

3. You decide to co-culture lymphocytes from the strains listed in the table below in order to observe the mixed-lymphocyte reaction (MLR). In each case, indicate which lymphocyte population(s) you would expect to proliferate.

Population 1	Population 2	Proliferation
C57BL/6 (H-2 ^b)	CBA (H-2 ^k)	
C57BL/6 (H-2 ^b)	CBA (H-2 ^k) mitomycin C-treated	
C57BL/6 (H-2 ^b)	(CBA × C57BL/6) F ₁ (H-2 ^{k/b})	
C57BL/6 (H-2 ^b)	C57L (H-2 ^b)	

4. In the mixed-lymphocyte reaction (MLR), the uptake of [^3H]thymidine often is used to assess cell proliferation.

- a. Which cell type proliferates in the MLR?
- b. How could you prove the identity of the proliferating cell?
- c. Explain why production of IL-2 also can be used to assess cell proliferation in the MLR.

5. Indicate whether each of the properties listed below is exhibited by T_H cells, CTLs, both T_H cells and CTLs, or neither cell type.

- a. _____ Can make IFN- γ
- b. _____ Can make IL-2
- c. _____ Is class I MHC restricted
- d. _____ Expresses CD8
- e. _____ Is required for B-cell activation
- f. _____ Is cytotoxic for target cells
- g. _____ Is the main proliferating cell in an MLR
- h. _____ Is the effector cell in a CML assay
- i. _____ Is class II MHC restricted
- j. _____ Expresses CD4
- k. _____ Expresses CD3
- l. _____ Adheres to target cells by LFA-1
- m. _____ Can express the IL-2 receptor
- n. _____ Expresses the $\alpha\beta$ T-cell receptor
- o. _____ Is the principal target of HIV
- p. _____ Responds to soluble antigens alone

- q. _____ Produces perforin
- r. _____ Expresses the CD40 ligand on its surface

6. Mice from several different inbred strains were infected with LCM virus, and several days later their spleen cells were isolated. The ability of the primed spleen cells to lyse LCM-infected, ^{51}Cr -labeled target cells from various strains was determined. In the accompanying table, indicate with a (+) or (−) whether the spleen cells listed in the left column would cause ^{51}Cr release from the target cells listed in the headings across the top of the table.

Source of primed spleen cells	^{51}Cr release from LCM-infected target cells			
	B10.D2 (H-2 ^d)	B10 (H-2 ^b)	B10.BR (H-2 ^k)	(BALB/c × B10) F ₁ (H-2 ^{b/d})
B10.D2 (H-2 ^d)				
B10 (H-2 ^b)				
BALB/c (H-2 ^d)				
BALB/c × B10 (H-2 ^{b/d})				

7. A mouse is infected with influenza virus. How could you assess whether the mouse has T_H and T_C cells specific for influenza?

8. Explain why NK cells from a given host will kill many types of virus-infected cells but do not kill normal cells from that host.

9. Consider the following genetically altered mice and predict the outcome of the indicated procedures. H-2^d mice in which both perforin and Fas ligand have been knocked out are immunized with LCM virus. One week after immunization, T cells from these mice are harvested and tested for cytotoxicity on the following:

- a. Target cells from normal LCM-infected H-2^b mice
- b. Target cells from normal H-2^d mice
- c. Target cells from H-2^d mice in which both perforin and Fas have been knocked out
- d. Target cells from LCM-infected normal H-2^d mice
- e. Target cells from H-2^d mice in which both perforin and FasL have been knocked out

10. You wish to determine the levels of class I-restricted T cells in an HIV-infected individual that are specific for a peptide that is generated from gp120, a component of the virus. Assume that you know the HLA type of the subject. What method would you use and how would you perform the analysis? Please be as specific as you can.

Leukocyte Migration and Inflammation

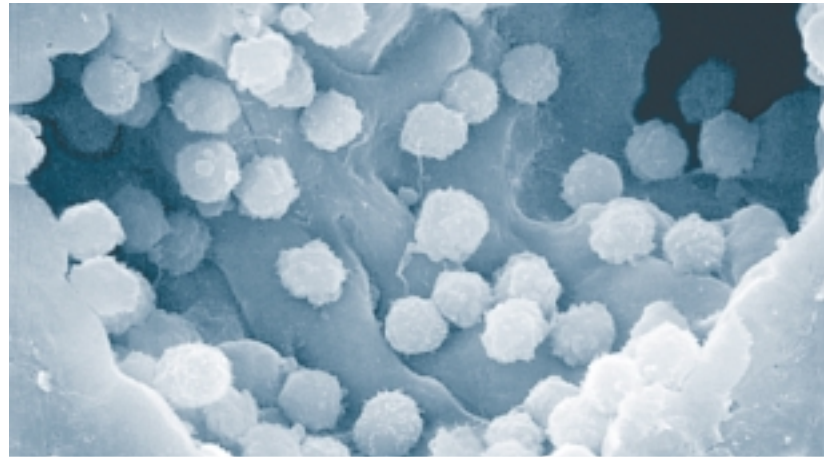
MANY TYPES OF LEUKOCYTES MOVE FROM ONE part of the body to another. This is especially true of lymphocytes, which circulate continually in the blood and lymph and, in common with other types of leukocytes, migrate into the tissues at sites of infection or tissue injury. This recirculation not only increases the chance that lymphocytes specific for a particular antigen will encounter that antigen but also is critical to development of an **inflammatory response**. Inflammation is a complex response to local injury or other trauma; it is characterized by redness, heat, swelling, and pain. Inflammation involves various immune-system cells and numerous mediators. Assembling and regulating inflammatory responses would be impossible without the controlled migration of leukocyte populations. This chapter covers the molecules and processes that play a role in leukocyte migration, various molecules that mediate inflammation, and the characteristic physiologic changes that accompany inflammatory responses.

Lymphocyte Recirculation

Lymphocytes are capable of a remarkable level of recirculation, continually moving through the blood and lymph to the various lymphoid organs (Figure 15-1). After a brief transit time of approximately 30 min in the bloodstream, nearly 45% of all lymphocytes are carried from the blood directly to the spleen, where they reside for approximately 5 h. Almost equal numbers (42%) of lymphocytes exit from the blood into various peripheral lymph nodes, where they reside for about 12 h. A smaller number of lymphocytes (10%) migrate to tertiary extralymphoid tissues by crossing between endothelial cells that line the capillaries. These tissues normally have few, if any, lymphoid cells but can import them during an inflammatory response. The most immunologically active tertiary extralymphoid tissues are those that interface with the external environment, such as the skin and various mucosal epithelia of the gastrointestinal, pulmonary, and genitourinary tracts.

The process of continual lymphocyte recirculation allows maximal numbers of antigenically committed lymphocytes to encounter antigen. An individual lymphocyte may make a complete circuit from the blood to the tissues and lymph

chapter 15



Lymphocytes Attached to the Surface of a High-Endothelial Venule

- Lymphocyte Recirculation
- Cell-Adhesion Molecules
- Neutrophil Extravasation
- Lymphocyte Extravasation
- Chemokines—Key Mediators of Inflammation
- Other Mediators of Inflammation
- The Inflammatory Process
- Anti-Inflammatory Agents

and back again as often as 1–2 times per day. Since only about one in 10^5 lymphocytes recognizes a particular antigen, it would appear that a large number of T or B cells must contact antigen on a given antigen-presenting cell within a short time in order to generate a specific immune response. The odds of the small percentage of lymphocytes committed to a given antigen actually making contact with that antigen when it is present are elevated by the extensive recirculation of lymphocytes. The likelihood of such contacts is profoundly increased also by factors that regulate, organize, and direct the circulation of lymphocytes and antigen-presenting cells.

Cell-Adhesion Molecules

The vascular endothelium serves as an important “gate-keeper,” regulating the movement of blood-borne molecules

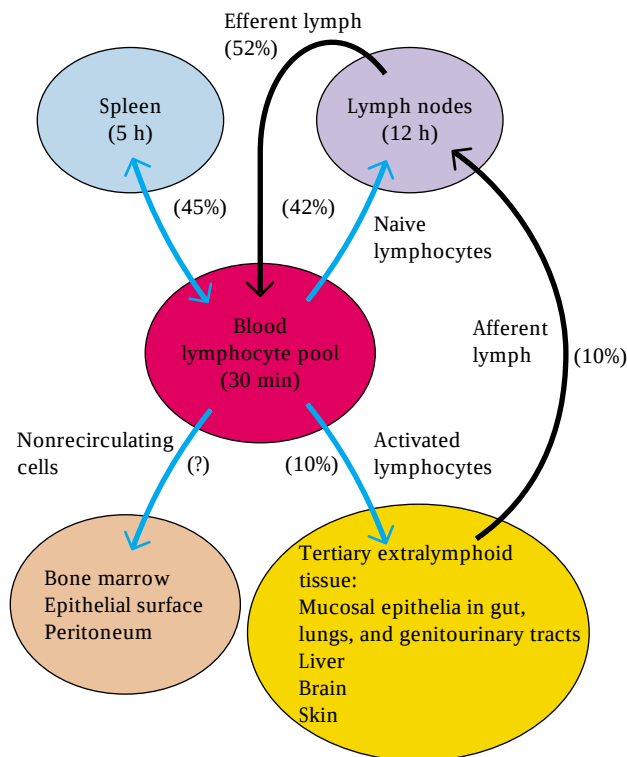


FIGURE 15-1 Lymphocyte recirculation routes. The percentage of the lymphocyte pool that circulates to various sites and the average transit times in the major sites are indicated. Lymphocytes migrate from the blood into lymph nodes through specialized areas in post-capillary venules called high-endothelial venules (HEVs). Although most lymphocytes circulate, some sites appear to contain lymphocytes that do not. [Adapted from A. Ager, 1994, *Trends Cell Biol.* 4:326.]

and leukocytes into the tissues. In order for circulating leukocytes to enter inflamed tissue or peripheral lymphoid organs, the cells must adhere to and pass between the endothelial cells lining the walls of blood vessels, a process called **extravasation**. Endothelial cells express leukocyte-specific **cell-adhesion molecules (CAMs)**. Some of these membrane proteins are expressed constitutively; others are expressed only in response to local concentrations of cytokines produced during an inflammatory response. Recirculating lymphocytes, monocytes, and granulocytes bear receptors that bind to CAMs on the vascular endothelium, enabling these cells to extravasate into the tissues.

In addition to their role in leukocyte adhesion to vascular endothelial cells, CAMs on leukocytes also serve to increase the strength of the functional interactions between cells of the immune system. Various adhesion molecules have been shown to contribute to the interactions between T_H cells and APCs, T_H and B cells, and CTLs and target cells.

A number of endothelial and leukocyte CAMs have been cloned and characterized, providing new details about the extravasation process. Most of these CAMs belong to four families of proteins: the selectin family, the mucin-like family, the integrin family, and the immunoglobulin (Ig) superfamily (Figure 15-2).

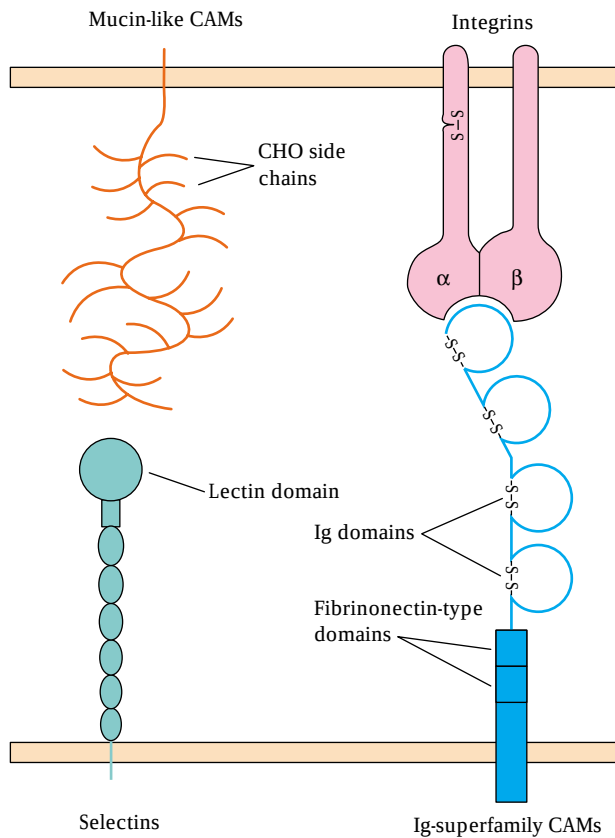
SELECTINS The **selectin** family of membrane glycoproteins has a distal lectin-like domain that enables these molecules to bind to specific carbohydrate groups. Selectins interact primarily with sialylated carbohydrate moieties, which are often linked to mucin-like molecules. The selectin family includes three molecules, designated L, E, and P. Most circulating leukocytes express L-selectin, whereas E-selectin and P-selectin are expressed on vascular endothelial cells. Selectin molecules are responsible for the initial stickiness of leukocytes to vascular endothelium.

MUCINS The **mucins** are a group of serine- and threonine-rich proteins that are heavily glycosylated. Their extended structure allows them to present sialylated carbohydrate ligands to selectins. For example, L-selectin on leukocytes recognizes sialylated carbohydrates on two mucin-like molecules (CD34 and GlyCAM-1) expressed on certain endothelial cells of lymph nodes. Another mucin-like molecule (PSGL-1) found on neutrophils interacts with E- and P-selectin expressed on inflamed endothelium.

INTEGRINS The **integrins** are heterodimeric proteins (consisting of an α and a β chain) that are expressed by leukocytes and facilitate both adherence to the vascular endothelium and other cell-to-cell interactions. The integrins are grouped into categories according to which β subunit they contain. Different integrins are expressed by different populations of leukocytes, allowing these cells to bind to different CAMs that belong to the immunoglobulin superfamily expressed along the vascular endothelium. As described later, some integrins must be activated before they can bind with high affinity to their ligands. The importance of integrin molecules in leukocyte extravasation is demonstrated by **leukocyte-adhesion deficiency (LAD)**, an autosomal recessive disease described later in this chapter (see the Clinical Focus). It is characterized by recurrent bacterial infections and impaired healing of wounds.

ICAMS Several adhesion molecules contain a variable number of immunoglobulin-like domains and thus are classified in the **immunoglobulin superfamily**. Included in this group are ICAM-1, ICAM-2, ICAM-3, and VCAM, which are expressed on vascular endothelial cells and bind to various integrin molecules. An important cell-adhesion molecule called MAdCAM-1 has both Ig-like domains and mucin-like domains. This molecule is expressed on mucosal endothelium and directs lymphocyte entry into mucosa. It binds to integrins by its immunoglobulin-like domain and to selectins by its mucin-like domain.

(a) General structure of CAM families



(b) Selected CAMs belonging to each family

Mucin-like CAMs:	Selectins:
GlyCAM-1	L-selectin
CD34	P-selectin
PSGL-1	E-selectin
MAdCAM-1	
Ig-superfamily CAMs:	Integrins:
ICAM-1, -2, -3	$\alpha 4 \beta 1$ (VLA-4, LPAM-2)
VCAM-1	$\alpha 4 \beta 7$ (LPAM-1)
LFA-2 (CD2)	$\alpha 6 \beta 1$ (VLA-6)
LFA-3 (CD58)	$\alpha L \beta 2$ (LFA-1)
MAdCAM-1	$\alpha M \beta 2$ (Mac-1)
	$\alpha X \beta 2$ (CR4, p150/95)

FIGURE 15-2 Schematic diagrams depicting the general structures of the four families of cell-adhesion molecules (a) and a list of representative molecules in each family (b). The lectin domain in selectins interacts primarily with carbohydrate (CHO) moieties on mucin-like molecules. Both component chains in integrin molecules contribute to the binding site, which interacts with an Ig domain in CAMs belonging to the Ig superfamily. MAdCAM-1 contains both mucin-like and Ig-like domains and can bind to both selectins and integrins.

Neutrophil Extravasation

As an inflammatory response develops, various cytokines and other inflammatory mediators act upon the local blood vessels, inducing increased expression of endothelial CAMs. The vascular endothelium is then said to be **activated**, or **inflamed**. Neutrophils are generally the first cell type to bind to inflamed endothelium and extravasate into the tissues. To accomplish this, neutrophils must recognize the inflamed endothelium and adhere strongly enough so that they are not swept away by the flowing blood. The bound neutrophils must then penetrate the endothelial layer and migrate into the underlying tissue. Monocytes and eosinophils extravasate by a similar process, but the steps have been best established for the neutrophil, so we focus on neutrophils here.

The process of neutrophil extravasation can be divided into four sequential steps: (1) rolling, (2) activation by chemoattractant stimulus, (3) arrest and adhesion, and (4) transendothelial migration (Figure 15-3a). In the first step, neutrophils attach loosely to the endothelium by a low-affinity selectin-carbohydrate interaction. During an inflammatory response, cytokines and other mediators act upon the local endothelium, inducing expression of adhesion molecules of the selectin family. These E- and P-selectin molecules bind to mucin-

like cell-adhesion molecules on the neutrophil membrane or with a sialylated lactosaminoglycan called sialyl Lewis^x (Figure 15-3b). This interaction tethers the neutrophil briefly to the endothelial cell, but the shear force of the circulating blood soon detaches the neutrophil. Selectin molecules on another endothelial cell again tether the neutrophil; this process is repeated so that the neutrophil tumbles end-over-end along the endothelium, a type of binding called *rolling*.

As the neutrophil rolls, it is activated by various **chemoattractants**; these are either permanent features of the endothelial cell surface or secreted locally by cells involved in the inflammatory response. Among the chemoattractants are members of a large family of chemoattractive cytokines called **chemokines**. Two chemokines involved in the activation process are interleukin 8 (IL-8) and macrophage inflammatory protein (MIP-1 β). However, not all chemoattractants belong to the chemokine group. Other chemoattractants are platelet-activating factor (PAF), the complement split products C5a, C3a, and C5b67 and various *N*-formyl peptides produced by the breakdown of bacterial proteins during an infection. Binding of these chemoattractants to receptors on the neutrophil membrane triggers an activating signal mediated by G proteins associated with the receptor. This signal induces a conformational change in the integrin molecules in the neu-

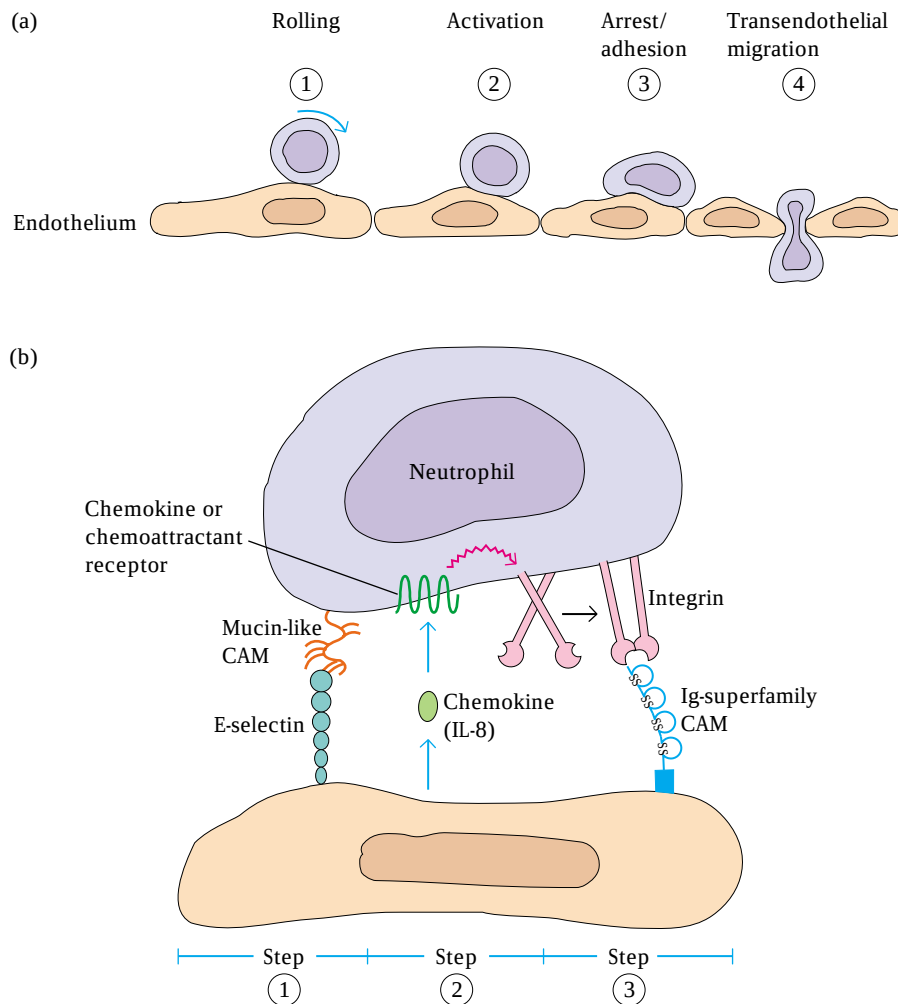


FIGURE 15-3 (a) The four sequential but overlapping steps in neutrophil extravasation. (b) Cell-adhesion molecules and chemokines involved in the first three steps of neutrophil extravasation. Initial rolling is mediated by binding of E-selectin molecules on the vascular endothelium to sialylated carbohydrate moieties on mucin-like CAMs. A chemokine such as IL-8 then binds to a G-protein–linked receptor on the neutrophil, triggering an activating signal. This signal induces a conformational change in the integrin molecules, enabling them to adhere firmly to Ig-superfamily molecules on the endothelium.

trophil membrane, increasing their affinity for the Ig-superfamily adhesion molecules on the endothelium. Subsequent interaction between integrins and Ig-superfamily CAMs stabilizes adhesion of the neutrophil to the endothelial cell, enabling the cell to adhere firmly to the endothelial cell.

Subsequently, the neutrophil migrates through the vessel wall into the tissues. The steps in transendothelial migration and how it is directed are still largely unknown; they may be mediated by further activation by chemoattractants and subsequent integrin–Ig-superfamily interactions or by a separate migration stimulus.

Lymphocyte Extravasation

Various subsets of lymphocytes exhibit directed extravasation at inflammatory sites and secondary lymphoid organs. The recirculation of lymphocytes thus is carefully controlled to ensure that appropriate populations of B and T cells are recruited into different tissues. As with neutrophils, extrava-

sation of lymphocytes involves interactions among a number of cell-adhesion molecules (Table 15-1). The overall process is similar to what happens during neutrophil extravasation and comprises the same four stages of contact and rolling, activation, arrest and adhesion, and, finally, transendothelial migration.

High-Endothelial Venules Are Sites of Lymphocyte Extravasation

Some regions of vascular endothelium in postcapillary venules of various lymphoid organs are composed of specialized cells with a plump, cuboidal (“high”) shape; such regions are called **high-endothelial venules**, or **HEVs** (Figure 15-4a, b). Their cells contrast sharply in appearance with the flattened endothelial cells that line the rest of the capillary. Each of the secondary lymphoid organs, with the exception of the spleen, contains HEVs. When frozen sections of lymph nodes, Peyer’s patches, or tonsils are incubated with lymphocytes and washed to remove unbound cells, over 85% of the



TABLE 15-1 Some interactions between cell-adhesion molecules implicated in leukocyte extravasation*

Receptor on cells	Expression	Ligands on endothelium	Step involving interaction [†]	Main function
CLA or ESL-1	Effector T cells	E-selectin	Tethering/rolling	Homing to skin and migration into inflamed tissue
L-selectin	All leukocytes	GlyCAM-1, CD34, MAdCAM-1	Tethering/rolling	Lymphocyte recirculation via HEVs to peripheral lymph nodes and migration into inflamed tertiary sites
LFA-1 (α L β 2)	Leukocyte subsets	ICAM-1, 2, 3	Adhesion/arrest	General role in lymphocyte extravasation via HEVs and leukocyte migration into inflamed tissue
LPAM-1 (α 4 β 7)	Effector T cells, monocytes	MAdCAM-1, VCAM-1	Rolling/adhesion	Homing of T cells to gut via mucosal HEV; migration into inflamed tissue
Mac-1 (α M β 2)	Monocytes	VCAM-1	—	Monocyte migration into inflamed tissue
PSGL-1	Neutrophils	E- and P-selectin	Tethering/rolling	Neutrophil migration into inflamed tissue
VLA-4 (α 4 β 1)	Neutrophils, T cells, monocytes	VCAM-1, MAdCAM-1, fibronectin	Rolling/adhesion	General role in leukocyte migration into inflamed tissue
VLA-6 (α 6 β 1)	T cells	Laminin	—	Homing of progenitor T cells to thymus; possible role in T-cell homing to nonmucosal sites

*Most endothelial and leukocyte CAMs belong to four groups of proteins as shown in Figure 15-2. In general, molecules in the integrin family bind to Ig-superfamily CAMs, and molecules in the selectin family bind to mucin-like CAMs. Members of the selectin and mucin-like families can be expressed on both leukocytes and endothelial cells, whereas integrins are expressed only on leukocytes, and Ig-superfamily CAMs are expressed only on endothelium.

[†]See Figures 15-3a and 15-7 for an illustration of steps in the extravasation process.

bound cells are found adhering to HEVs, even though HEVs account for only 1%–2% of the total area of the frozen section (Figure 15-4c).

It has been estimated that as many as 1.4×10^4 lymphocytes extravasate every second through HEVs into a single lymph node. The development and maintenance of HEVs in lymphoid organs is influenced by cytokines produced in response to antigen capture. For example, HEVs fail to develop in animals raised in a germ-free environment. The role of antigenic activation of lymphocytes in the maintenance of HEVs has been demonstrated by surgically blocking the afferent lymphatic vasculature to a node, so that antigen entry to the node is blocked. Within a short period of time, the HEVs show impaired function and eventually revert to a more flattened morphology.

High-endothelial venules express a variety of cell-adhesion molecules. Like other vascular endothelial cells, HEVs express CAMs of the selectin family (E- and P-selectin), the mucin-

like family (GlyCAM-1 and CD34), and the immunoglobulin superfamily (ICAM-1, ICAM-2, ICAM-3, VCAM-1, and MAdCAM-1). Some of these adhesion molecules are distributed in a tissue-specific manner. These tissue-specific adhesion molecules have been called **vascular addressins (VAs)** because they serve to direct the extravasation of different populations of recirculating lymphocytes to particular lymphoid organs.

Lymphocyte Homing Is Directed by Receptor Profiles and Signals

The general process of lymphocyte extravasation is similar to neutrophil extravasation. An important feature distinguishing the two processes is that different subsets of lymphocytes migrate differentially into different tissues. This process is called **trafficking**, or **homing**. The different trafficking patterns of lymphocyte subsets are mediated by unique combi-

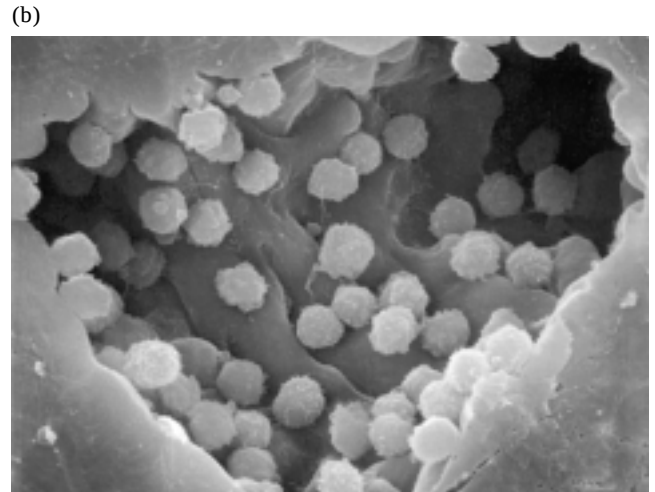
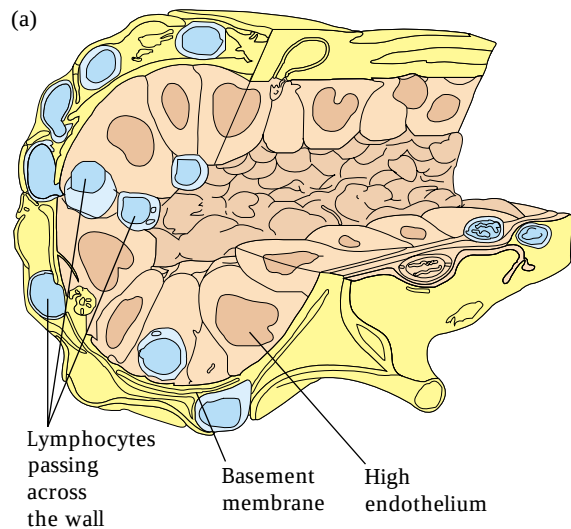


FIGURE 15-4 (a) Schematic cross-sectional diagram of a lymph-node postcapillary venule with high endothelium. Lymphocytes are shown in various stages of attachment to the HEV and in migration across the wall into the cortex of the node. (b) Scanning electron micrograph showing numerous lymphocytes bound to the surface of a high-endothelial venule. (c) Micrograph of frozen sections of lymphoid tissue. Some 85% of the lymphocytes (darkly stained) are bound to HEVs (in cross section), which comprise only 1%–2% of the total area of the tissue section. [Part (a) adapted from A. O. Anderson and N. D. Anderson, 1981, in *Cellular Functions in Immunity and Inflammation*, J. J. Oppenheim et al. (eds.), Elsevier, North-Holland; part (b) from S. D. Rosen and L. M. Stoolman, 1987, *Vertebrate Lectins*, Van Nostrand Reinhold; part (c) from S. D. Rosen, 1989, *Curr. Opin. Cell Biol.* **1**:913.]

nations of adhesion molecules and chemokines; receptors that direct the circulation of various populations of lymphocytes to particular lymphoid and inflammatory tissues are called **homing receptors**. Researchers have identified a number of lymphocyte and endothelial cell-adhesion molecules that participate in the interaction of lymphocytes with HEVs and with endothelium at tertiary sites or sites of inflammation (see Table 15-1). As is described later, in the section on chemokines, these molecules play a major role in determining the heterogeneity of lymphocyte circulation patterns.

Naive Lymphocytes Recirculate to Secondary Lymphoid Tissue

A naive lymphocyte is not able to mount an immune response until it has been activated to become an effector cell. Activation of a naive cell occurs in specialized microenvironments within secondary lymphoid tissue (e.g., peripheral

lymph nodes, Peyer's patches, tonsils, and spleen). Within these microenvironments, dendritic cells capture antigen and present it to the naive lymphocyte, resulting in its activation. Naive cells do not exhibit a preference for a particular type of secondary lymphoid tissue but instead circulate indiscriminately to secondary lymphoid tissue throughout the body by recognizing adhesion molecules on HEVs.

The initial attachment of naive lymphocytes to HEVs is generally mediated by the binding of the homing receptor L-selectin to adhesion molecules such as GlyCAM-1 and CD34 on HEVs (Figure 15-5a). The trafficking pattern of naive cells is designed to keep these cells constantly recirculating through secondary lymphoid tissue, whose primary function is to trap blood-borne or tissue-borne antigen.

Once naive lymphocytes encounter antigen trapped in a secondary lymphoid tissue, they become activated and enlarge into lymphoblasts. Activation takes about 48 h, and during this time the blast cells are retained in the paracortical

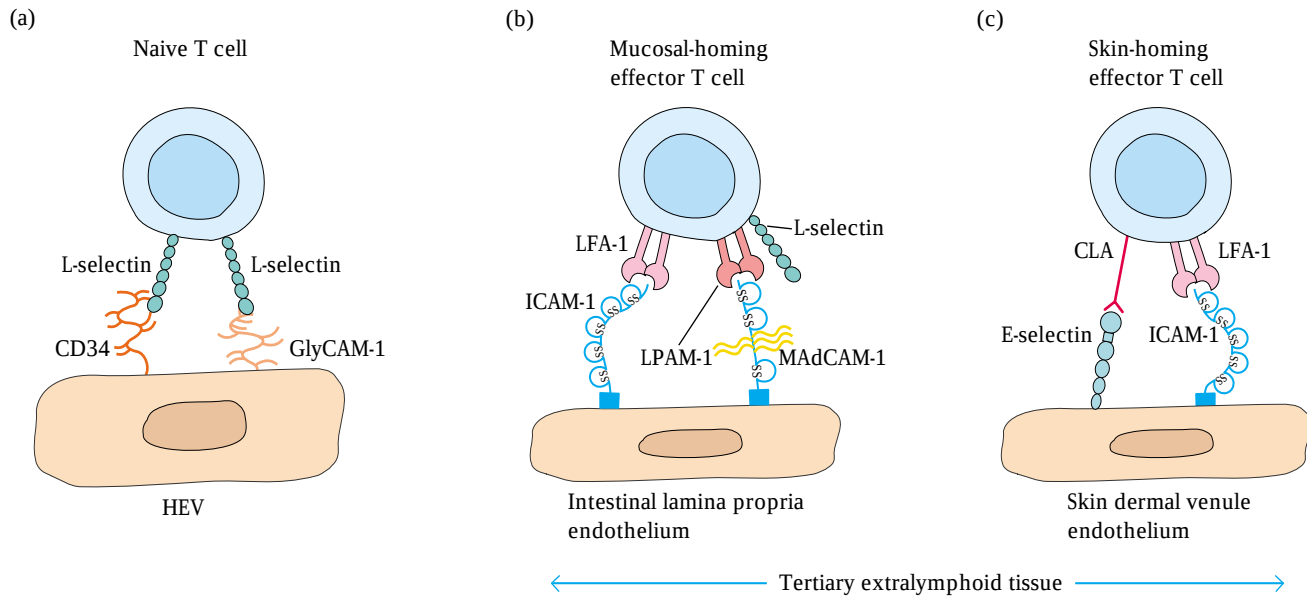


FIGURE 15-5 Examples of homing receptors and vascular addressins involved in selective trafficking of naive and effector T cells. (a) Naive T cells tend to home to secondary lymphoid tissues through their HEV regions. The initial interaction involves the homing receptor L-selectin and mucin-like cell-adhesion molecules such as CD34 or GlyCAM-1 ex-

pressed on HEV cells. (b, c) Various subsets of effector T cells express high levels of particular homing receptors that allow them to home to endothelium in particular tertiary extralymphoid tissues. The initial interactions in homing of effector T cells to mucosal and skin sites are illustrated.

region of the secondary lymphoid tissue. During this phase, called the shut-down phase, the antigen-specific lymphocytes cannot be detected in the circulation (Figure 15-6). Rapid proliferation and differentiation of naive cells occurs during the shut-down phase. The effector and memory cells that are generated by this process then leave the lymphoid tissue and begin to recirculate.

Effector and Memory Lymphocytes Adopt Different Trafficking Patterns

The trafficking patterns of effector and memory lymphocytes differ from those of naive lymphocytes. Effector cells tend to home to regions of infection by recognizing inflamed vascular endothelium and chemoattractant molecules that are generated during the inflammatory response. Memory lymphocytes, on the other hand, home selectively to the type of tissue in which they first encountered antigen. Presumably this ensures that a particular memory cell will return to the tissue where it is most likely to reencounter a subsequent threat by the antigen it recognizes.

Effector and memory cells express increased levels of certain cell-adhesion molecules, such as LFA-1, that interact with ligands present on tertiary extralymphoid tissue (such as skin and mucosal epithelia) and at sites of inflammation, allowing effector and memory cells to enter these sites. Naive cells lack corresponding cell-adhesion molecules and do not

home to these sites. Inflamed endothelium expresses a number of adhesion molecules, including E- and P-selectin and the Ig-superfamily molecules VCAM-1 and ICAM-1, that bind to the receptors expressed at high levels on memory and effector cells.

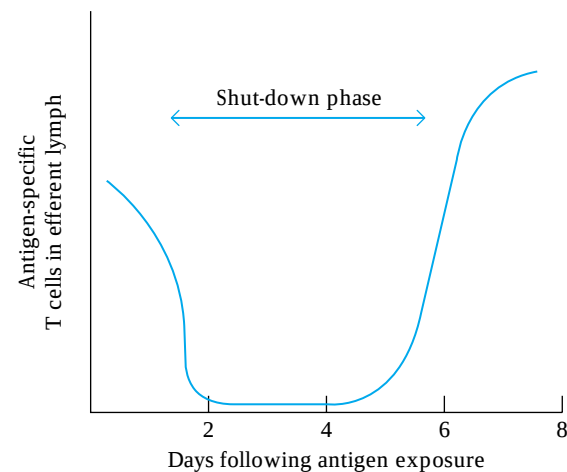


FIGURE 15-6 T-cell activation in the paracortical region of a lymph node results in the brief loss of lymphocyte recirculation. During this shut-down phase, antigen-specific T cells cannot be detected leaving the node in the efferent lymph.

Unlike naive lymphocytes, subsets of the memory and effector populations exhibit tissue-selective homing behavior. Such tissue specificity is imparted not by a single adhesion receptor but by different combinations of adhesion molecules. For example, a mucosal homing subset of memory/effector cells has high levels of the integrins LPAM-1 ($\beta\alpha4\beta7$) and LFA-1 ($\alpha\text{Lb}2$), which bind to MAdCAM and various ICAMs on intestinal lamina propria venules (see Figure 15-5b). However, these cells avoid direction to secondary lymphoid tissues because they have low levels of the L-selectin that would facilitate their entry into secondary lymphoid tissue. A second subset of memory/effector cells displays preferential homing to the skin. This subset also expresses low levels of L-selectin but displays high levels of cutaneous lymphocyte antigen (CLA) and LFA-1, which bind to E-selectin and ICAMs on dermal venules of the skin (see Figure 15-5c). Although effector and memory cells that express reduced levels of L-selectin do not tend to home through HEVs into peripheral lymph nodes, they can enter peripheral lymph nodes through the afferent lymphatic vessels.

Adhesion-Molecule Interactions Play Critical Roles in Extravasation

The extravasation of lymphocytes into secondary lymphoid tissue or regions of inflammation is a multistep process involving a cascade of adhesion-molecule interactions similar to those involved in neutrophil emigration from the blood-

stream. Figure 15-7 depicts the typical interactions in extravasation of naive T cells across HEVs into lymph nodes. The first step is usually a selectin-carbohydrate interaction similar to that seen with neutrophil adhesion. Naive lymphocytes initially bind to HEVs by L-selectin, which serves as a homing receptor that directs the lymphocytes to particular tissues expressing a corresponding mucin-like vascular addressin such as CD34 or GlyCAM-1. Lymphocyte rolling is less pronounced than that of neutrophils. Although the initial selectin-carbohydrate interaction is quite weak, the slow rate of blood flow in postcapillary venules, particularly in regions of HEVs, reduces the likelihood that the shear force of the flowing blood will dislodge the tethered lymphocyte.

In the second step, an integrin-activating stimulus is mediated by chemokines that are either localized on the endothelial surface or secreted locally. The thick glycocalyx covering of the HEVs may function to retain these soluble chemoattractant factors on the HEVs. If, as some have proposed, HEVs secrete lymphocyte-specific chemoattractants, it would explain why neutrophils do not extravasate into lymph nodes at the HEVs even though they express L-selectin. Chemokine binding to G-protein-coupled receptors on the lymphocyte leads to activation of integrin molecules on the membrane, as occurs in neutrophil extravasation. Once activated, the integrin molecules interact with Ig-superfamily adhesion molecules (e.g., ICAM-1), so the lymphocyte adheres firmly to the endothelium. The molecular mechanisms involved in the final step, transendothelial migration, are poorly understood.

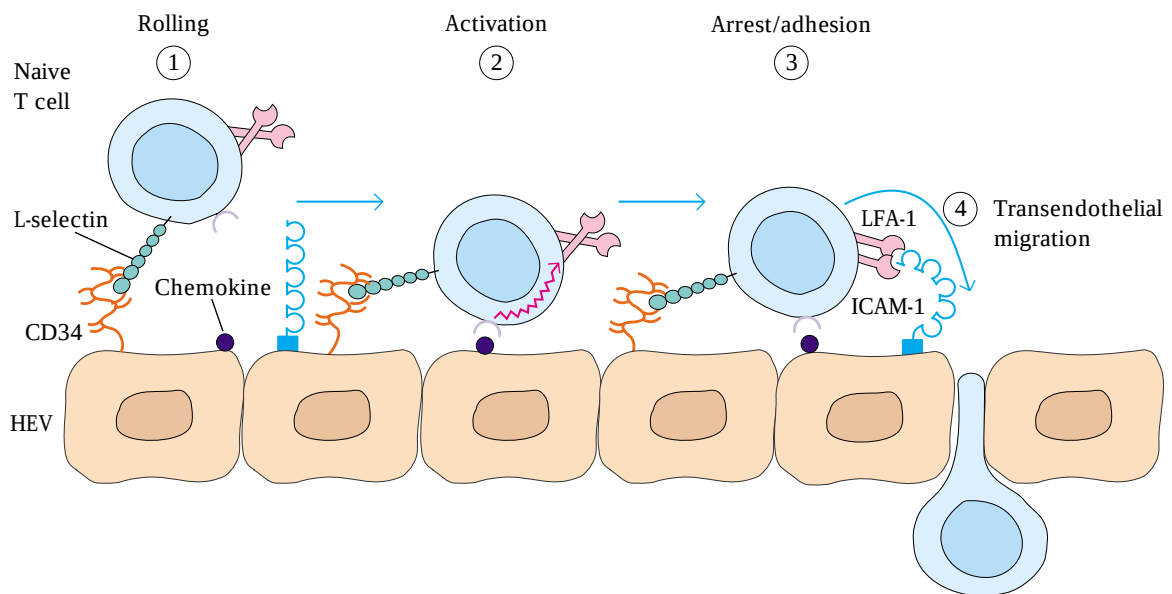


FIGURE 15-7 Steps in extravasation of a naive T cell through a high-endothelial venule into a lymph node. Extravasation of lymphocytes includes the same basic steps as neutrophil extravasation but some of the

cell-adhesion molecules differ. Activation of the integrin LFA-1, induced by chemokine binding to the lymphocyte, leads to firm adhesion followed by migration between the endothelial cells into the tissue.

Chemokines—Key Mediators of Inflammation

Chemokines are a superfamily of small polypeptides, most of which contain 90–130 amino acid residues. They selectively, and often specifically, control the adhesion, chemotaxis, and activation of many types of leukocyte populations and subpopulations. Consequently, they are major regulators of leukocyte traffic. Some chemokines are primarily involved in inflammatory processes, others are constitutively expressed and play important homeostatic or developmental roles. “Housekeeping” chemokines are produced in lymphoid organs and tissues or in non-lymphoid sites such as skin, where they direct normal trafficking of lymphocytes, such as determining the correct positioning of leukocytes newly generated by hematopoiesis and arriving from bone marrow. The thymus constitutively expresses chemokines, and normal B cell lymphopoiesis is also dependent on appropriate chemokine expression. Chemokine-mediated effects are not limited to the immune system. Mice that lack either the chemokine CXCL12 (also called SDF-1) or its receptor (see Table 15-2) show major defects in the development of the brain and the heart. Members of the chemokine family have also been shown to play regulatory roles in the development of blood vessels (angiogenesis), and wound healing.

The inflammatory chemokines are typically induced in response to infection. Contact with pathogens or the action of proinflammatory cytokines, such as TNF- α , up-regulate the expression of inflammatory cytokines at sites of developing inflammation. Chemokines cause leukocytes to move into various tissue sites by inducing the adherence of these cells to the vascular endothelium. After migrating into tissues, leukocytes are attracted toward high localized concentrations of chemokines resulting in the targeted recruitment of phagocytes and effector lymphocyte populations to inflammatory sites. The assembly of leukocytes at sites of infection, orchestrated by chemokines, is an essential part of mounting an appropriately focused response to infection.

More than 50 chemokines and at least 15 chemokine receptors have been described (Table 15-2). The chemokines possess four conserved cysteine residues and based on the position of two of the four invariant cysteine residues, almost all fall into one or the other of two distinctive subgroups:

- **C-C subgroup** chemokines, in which the conserved cysteines are contiguous;
- **C-X-C subgroup** chemokines, in which the conserved cysteines are separated by some other amino acid (X).

Chemokine action is mediated by receptors whose polypeptide chain traverses the membrane seven times. There are two subgroups of receptors, CC receptors (CCRs), which recognize CC chemokines, and CXC receptors (CXCRs), which recognize CXC chemokines. As with cytokines, the interac-

tion between chemokines and their receptors is of high affinity ($K_a > 10^9$) and high specificity. However, as Table 15-2 shows, most receptors bind more than one chemokine. For example, CXCR2 recognizes at least six different chemokines, and many chemokines can bind to more than one receptor.

When a receptor binds an appropriate chemokine, it activates heterotrimeric large G proteins, initiating a signal-transduction process that generate such potent second messengers as cAMP, IP₃, Ca²⁺, and activated small G pro-

TABLE 15-2 Human chemokines and their receptors*	
Chemokine receptors	Chemokines bound by receptor
CXC SUBGROUP	
CXCR1	IL-8, GCP-2
CXCR2	IL-8, Gro- α , Gro- β , Gro- γ , NAP-2, ENA-78
CXCR3	IP-10, Mig, I-TAC
CXCR4	SDF-1, PBSF
CXCR5	BCA-1
CC SUBGROUP	
CCR1	MIP-1, RANTES, MCP-2, MIP-5
CCR2	MCP-1, MCP-2, MCP-3
CCR3	Eotaxin, RANTES, MCP-2, MCP-3, MCP-4, Eotaxin-2, MIP-5
CCR4	TARC, RANTES
CCR5	MIP-1 α RANTES, MIP-1 β
CCR6	Exodus-1
CCR7	ELC
CCR8	1-309
CCR10	MCP-1, MCP-2, MCP-3, RANTES
BOTH CC AND CXC SUBGROUPS	
DARC (the Duffy antigen of RBCs)	Binds to a number of CC and CXC chemokines

*This table lists most known chemokine receptors but not all chemokines. The full names for a number of the chemokines abbreviated in the table are as follows: ELC (Ebl1 ligand chemokine); ENA-78 (epithelial-cell-derived neutrophil-activating protein); GCP-2 (granulocyte chemotactic protein 2); Gro- α , β , γ (growth-related oncogene α , β , γ); MCP-1, 2, 3, or 4 (monocyte chemoattractant protein 1, 2, 3, or 4); Mig (monokine induced by interferon γ); MIP-1 α , 1 β , or 5 (macrophage inflammatory protein 1 α , 1 β , or 5); NAP-2 (neutrophil-activating protein 2); RANTES (regulated upon activation, normal T-cell expressed and secreted); TARC (thymus- and activation-regulated chemokine.)

SOURCE: Adapted from Nelson and Krensky, 1998, *Curr. Opin. Immunol.* 10:265, and Baggiolini, 1998, *Nature* 392:565.

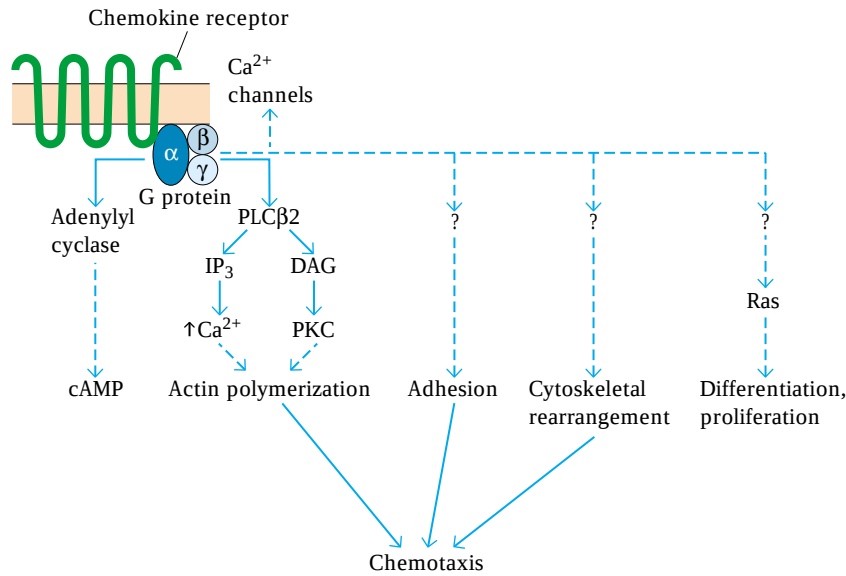


FIGURE 15-8 Chemokines signal through receptors coupled with heterotrimeric large G proteins. Binding of a chemokine to its receptor activates many signal-transduction pathways, resulting in a variety of modifications in the physiology of the target cell. If the signal-transduction pathway is not known or incompletely worked out, dashed lines and question marks are used here to represent probable pathways. [Adapted from Premack et al., 1996, *Nature Medicine* 2:1174.]

teins (Figure 15-8). Dramatic changes are effected by the chemokine-initiated activation of these signal transduction pathways. Within seconds, the addition of an appropriate chemokine to leukocytes causes abrupt and extensive changes in shape, the promotion of greater adhesiveness to endothelial walls by activation of leukocyte integrins, and the generation of microbicidal oxygen radicals in phagocytes. These signal-transduction pathways promote other changes such as the release of granular contents, proteases in neutrophils and macrophages, histamine from basophils, and cytotoxic proteins from eosinophils.

Chemokine-Receptor Profiles Mediate Leukocyte Activity

Among major populations of human leukocytes, neutrophils express CXCR1, -2, and -4; eosinophils have CCR1 and CCR3 (Figure 15-9). While resting naive T cells display few types of chemokine receptors, some activated T cells have CCR1, -2, -3, and -5, CXCR3 and -4, and possibly others. Clearly, a cell can respond to a chemokine only if it possesses a receptor that recognizes it. Consequently, differences in the expression of chemokine receptors by leukocytes coupled with the

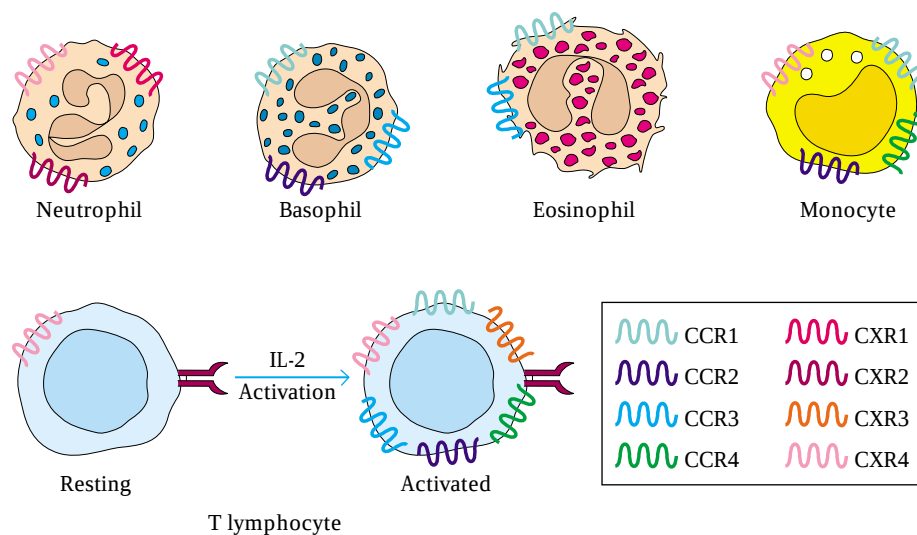


FIGURE 15-9 Patterns of expression of some principal chemokine receptors on different classes of human leukocytes. So far the great-

est variety of chemokine receptors has been observed on activated T lymphocytes. [Adapted from M. Baggiolini, 1998, *Nature* 392:565.]

production of distinctive profiles of chemokines by destination tissues and sites provide rich opportunities for the differential regulation of activities of different leukocyte populations. Indeed, differences in patterns of chemokine-receptor expression occur within leukocyte populations as well as between different ones. Recall that T_H1 and T_H2 subsets of T_H cells can be distinguished by their different patterns of cytokine production. These subsets also display different profiles of chemokine receptors. T_H2 cells express CCR3 and -4, and a number of other receptors not expressed by T_H1 cells. On the other hand, T_H1 cells express CCR1, -3, and -5, but most T_H2 cells do not.

Other Mediators of Inflammation

In addition to chemokines, a variety of other mediators released by cells of the innate and acquired immune systems trigger or enhance specific aspects of the inflammatory response. They are released by tissue mast cells, blood platelets, and a variety of leukocytes, including neutrophils, monocytes/macrophages, eosinophils, basophils, and lymphocytes. In addition to these sources, plasma contains four interconnected mediator-producing systems: the kinin system, the clotting system, the fibrinolytic system, and the complement system. The first three systems share a common intermediate, Hageman factor, as illustrated in Figure 15-10. When tissue damage occurs, these four systems are activated to form a web of interacting systems that generate a number of mediators of inflammation.

The Kinin System Is Activated by Tissue Injury

The kinin system is an enzymatic cascade that begins when a plasma clotting factor, called Hageman factor, is activated following tissue injury. The activated Hageman factor then activates prekallikrein to form kallikrein, which cleaves kininogen to produce **bradykinin** (see Figure 15-10). This inflammatory mediator is a potent basic peptide that increases vascular permeability, causes vasodilation, induces pain, and induces contraction of smooth muscle. Kallikrein also acts directly on the complement system by cleaving C5 into C5a and C5b. The C5a complement component is an anaphylatoxin that induces mast-cell degranulation, resulting in the release of a number of inflammatory mediators from the mast cell.

The Clotting System Yields Fibrin-Generated Mediators of Inflammation

Another enzymatic cascade that is triggered by damage to blood vessels yields large quantities of thrombin. Thrombin acts on soluble fibrinogen in tissue fluid or plasma to produce insoluble strands of **fibrin** and **fibrinopeptides**. The

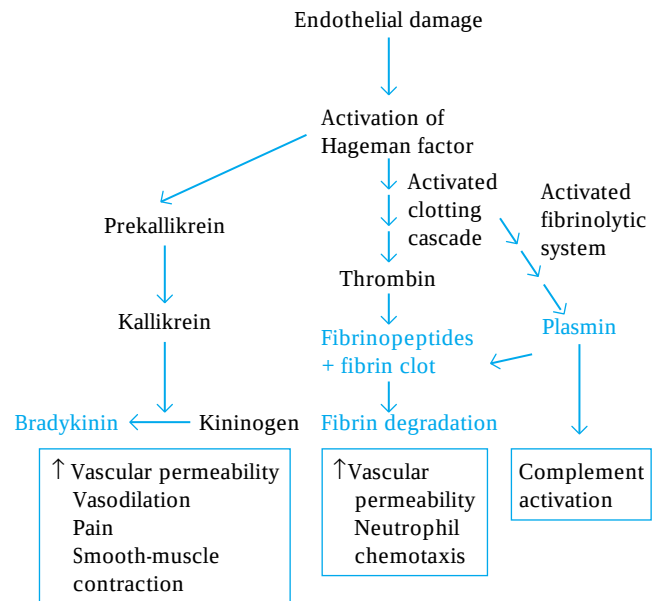


FIGURE 15-10 Tissue damage induces formation of plasma enzyme mediators by the kinin system, the clotting system, and the fibrinolytic system. These mediators cause vascular changes, among the earliest signs of inflammation, and various other effects. Plasmin not only degrades fibrin clots but also activates the classical complement pathway.

insoluble fibrin strands crisscross one another to form a **clot**, which serves as a barrier to the spread of infection. The clotting system is triggered very rapidly after tissue injury to prevent bleeding and limit the spread of invading pathogens into the bloodstream. The fibrinopeptides act as inflammatory mediators, inducing increased vascular permeability and neutrophil chemotaxis.

The Fibrinolytic System Yields Plasmin-Generated Mediators of Inflammation

Removal of the fibrin clot from the injured tissue is achieved by the fibrinolytic system. The end product of this pathway is the enzyme **plasmin**, which is formed by the conversion of plasminogen. Plasmin, a potent proteolytic enzyme, breaks down fibrin clots into degradation products that are chemotactic for neutrophils. Plasmin also contributes to the inflammatory response by activating the classical complement pathway.

The Complement System Produces Anaphylatoxins

Activation of the complement system by both classical and alternative pathways results in the formation of a number of

complement split products that serve as important mediators of inflammation (see Chapter 13). Binding of the **anaphylatoxins** (C3a, C4a, and C5a) to receptors on the membrane of tissue mast cells induces degranulation with release of histamine and other pharmacologically active mediators. These mediators induce smooth-muscle contraction and increase vascular permeability. C3a, C5a, and C5b67 act together to induce monocytes and neutrophils to adhere to vascular endothelial cells, extravasate through the endothelial lining of the capillary, and migrate toward the site of complement activation in the tissues. Activation of the complement system thus results in influxes of fluid that carry antibody and phagocytic cells to the site of antigen entry.

Some Lipids Act as Inflammatory Mediators

Following membrane perturbations, phospholipids in the membrane of several cell types (e.g., macrophages, monocytes, neutrophils, and mast cells) are degraded into arachidonic acid and lyso-platelet-activating factor (Figure 15-11). The latter is subsequently converted into platelet-activating factor (PAF), which causes platelet activation and has many inflammatory effects, including eosinophil chemotaxis and the activation and degranulation of neutrophils and eosinophils.

Metabolism of arachidonic acid by the cyclooxygenase pathway produces **prostaglandins** and **thromboxanes**. Different prostaglandins are produced by different cells: monocytes and macrophages produce large quantities of PGE₂ and PGF₂; neutrophils produce moderate amounts of PGE₂; mast

cells produce PGD₂. Prostaglandins have diverse physiological effects, including increased vascular permeability, increased vascular dilation, and induction of neutrophil chemotaxis. The thromboxanes cause platelet aggregation and constriction of blood vessels.

Arachidonic acid is also metabolized by the lipoxygenase pathway to yield the four **leukotrienes**: LTB₄, LTC₄, LTD₄, and LTE₄. Three of these (LTC₄, LTD₄, and LTE₄) together make up what was formerly called **slow-reacting substance of anaphylaxis (SRS-A)**; these mediators induce smooth-muscle contraction. LTB₄ is a potent chemoattractant of neutrophils. The leukotrienes are produced by a variety of cells, including monocytes, macrophages, and mast cells.

Some Cytokines Are Important Inflammatory Mediators

A number of cytokines play a significant role in the development of an acute or chronic inflammatory response. IL-1, IL-6, TNF- α , IL-12, and many chemokines exhibit redundant and pleiotropic effects that together contribute to the inflammatory response. Some of the effects mediated by IL-1, IL-6, and TNF- α are listed in Table 15-3. In addition, IFN- γ contributes to the inflammatory response, acting later in the acute response and contributing in a major way to chronic inflammation by attracting and activating macrophages. IL-12 induces the differentiation of the proinflammatory T_H1 subset. The role of several of these inflammatory cytokines in the development of acute and chronic inflammation will be described more fully in the next section.

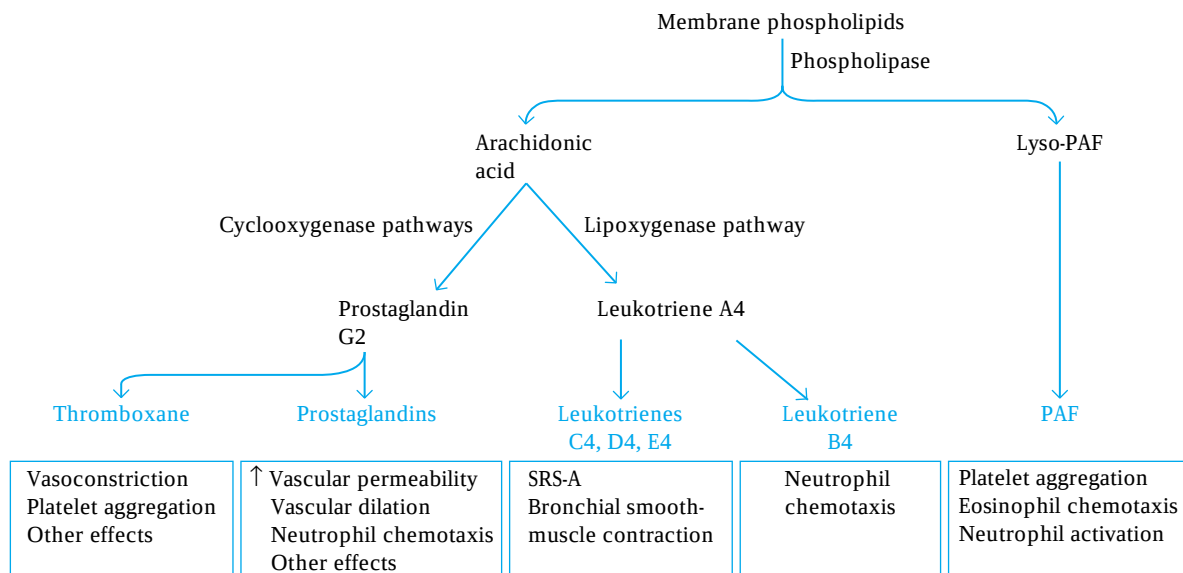


FIGURE 15-11 The breakdown of membrane phospholipids generates important mediators of inflammation, including thromboxane, prostaglandins, leukotrienes, and platelet-activating factor (PAF).

TABLE 15-3 Redundant and pleiotropic effects of IL-1, TNF- α , and IL-6

Effect	IL-1	TNF- α	IL-6
Endogenous pyrogen fever	+	+	+
Synthesis of acute-phase proteins by liver	+	+	+
Increased vascular permeability	+	+	+
Increased adhesion molecules on vascular endothelium	+	+	–
Fibroblast proliferation	+	+	–
Platelet production	+	–	+
Chemokine induction (e.g., IL-8)	+	+	–
Induction of IL-6	+	+	–
T-cell activation	+	+	+
B-cell activation	+	+	+
Increased immunoglobulin synthesis	–	–	+

The Inflammatory Process

Inflammation is a physiologic response to a variety of stimuli such as infections and tissue injury. In general, an acute inflammatory response has a rapid onset and lasts a short while. Acute inflammation is generally accompanied by a systemic reaction known as the acute-phase response, which is characterized by a rapid alteration in the levels of several plasma proteins. In some diseases persistent immune activation can result in chronic inflammation, which often has pathologic consequences.

Neutrophils Play an Early and Important Role in Inflammation

In the early stages of an inflammatory response, the predominant cell type infiltrating the tissue is the neutrophil. Neutrophil infiltration into the tissue peaks within the first 6 h of an inflammatory response, with production of neutrophils in the bone marrow increasing to meet this need. A normal adult produces more than 10^{10} neutrophils per day, but during a period of acute inflammation, neutrophil production may increase as much as tenfold.

The neutrophils leave the bone marrow and circulate within the blood. In response to mediators of acute inflammation, vascular endothelial cells increase their expression of E- and P-selectin. Thrombin and histamine induce increased expression of P-selectin; cytokines such as IL-1 or TNF- α induce increased expression of E-selectin. The circulating neutrophils express mucins such as PSGL-1 or the tetrasaccharides sialyl Lewis^a and sialyl Lewis^x, which bind to E- and P-selectin.

As described earlier, this binding mediates the attachment or tethering of neutrophils to the vascular endothelium, allowing the cells to roll in the direction of the blood flow. During this time, chemokines such as IL-8 or other chemoattractants act upon the neutrophils, triggering a G-protein-mediated activating signal that leads to a conformational change in the integrin adhesion molecules, resulting in neutrophil adhesion and subsequent transendothelial migration (see Figure 15-3).

Once in tissues, the activated neutrophils also express increased levels of receptors for chemoattractants and consequently exhibit **chemotaxis**, migrating up a gradient of the chemoattractant. Among the inflammatory mediators that are chemotactic for neutrophils are several chemokines, complement split products (C3a, C5a, and C5b67), fibrinopeptides, prostaglandins, and leukotrienes. In addition, molecules released by microorganisms, such as formyl methionyl peptides, are also chemotactic for neutrophils. Activated neutrophils express increased levels of Fc receptors for antibody and receptors for complement, enabling these cells to bind more effectively to antibody- or complement-coated pathogens, thus increasing phagocytosis.

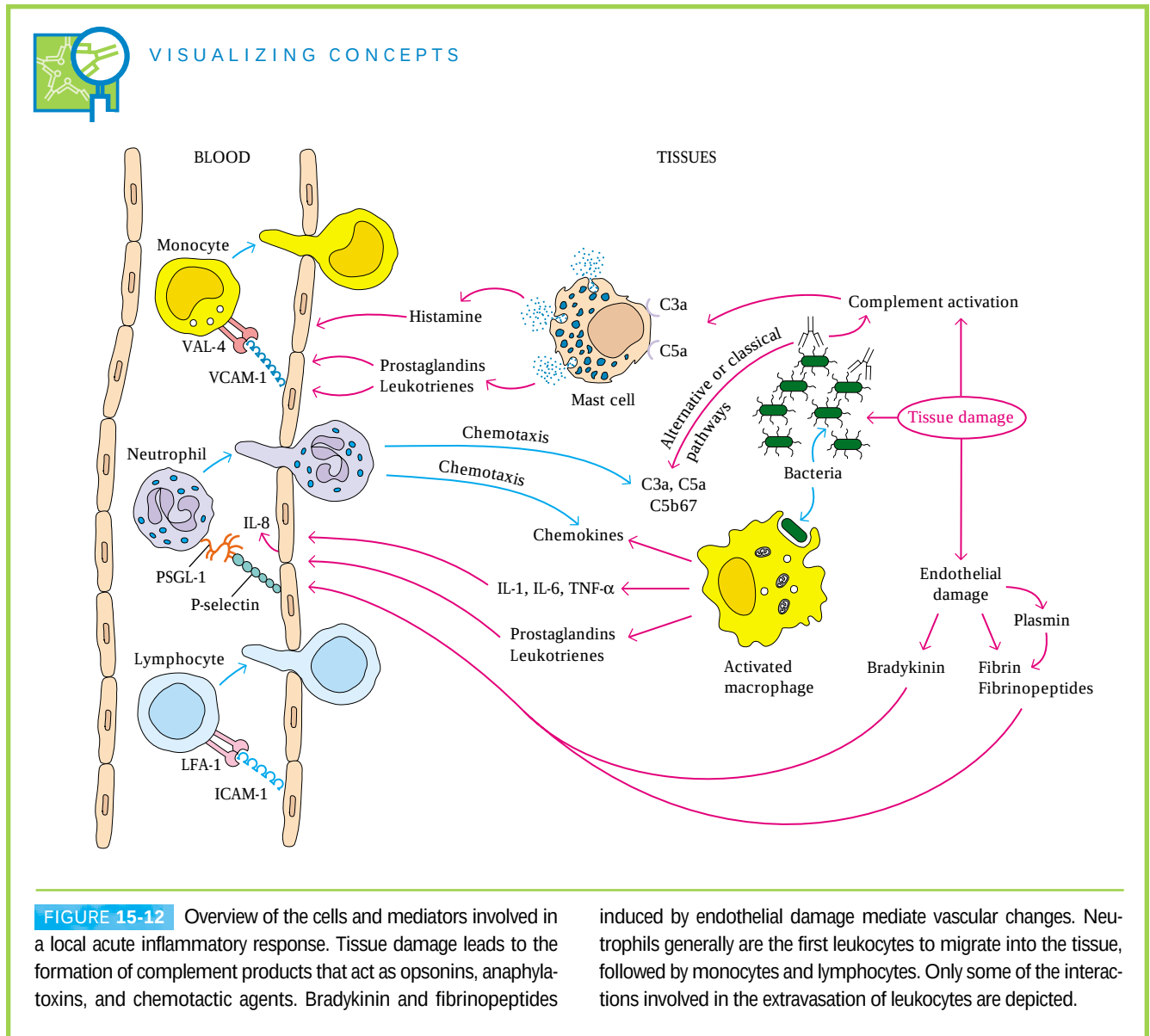
The activating signal also stimulates metabolic pathways to a respiratory burst, which produces **reactive oxygen intermediates** and **reactive nitrogen intermediates** (see Chapter 2). Release of some of these reactive intermediates and the release of mediators from neutrophil primary and secondary granules (proteases, phospholipases, elastases, and collagenases) play an important role in killing various pathogens. These substances also contribute to the tissue damage that can result from an inflammatory response. The accumulation of dead cells and microorganisms, together with accumulated fluid and various proteins, makes up what is known as pus.

Inflammatory Responses May Be Localized or Systemic

Infection or tissue injury induces a complex cascade of non-specific events, known as the inflammatory response, that provides early protection by restricting the tissue damage to the site of infection or tissue injury. The acute inflammatory response involves both localized and systemic responses.

LOCALIZED INFLAMMATORY RESPONSE

The hallmarks of a localized acute inflammatory response, first described almost 2000 years ago, are swelling (*tumor*), redness (*rubor*), heat (*calor*), pain (*dolor*), and loss of function. Within minutes after tissue injury, there is an increase in vascular diameter (vasodilation), resulting in an increase in the volume of blood in the area and a reduction in the flow of blood. The increased blood volume heats the tissue and causes it to redden. Vascular permeability also increases, leading to leakage of fluid from the blood vessels, particularly at postcapillary venules. This results in an accumulation of fluid (**edema**) in the tissue and, in some instances, extravasation of leukocytes, contribut-



ing to the swelling and redness in the area. When fluid exudes from the bloodstream, the kinin, clotting, and fibrinolytic systems are activated (see Figure 15-10). Many of the vascular changes that occur early in a local response are due to the direct effects of plasma enzyme mediators such as bradykinin and fibrinopeptides, which induce vasodilation and increased vascular permeability. Some of the vascular changes are due to the indirect effects of the complement anaphylatoxins (C3a, C4a, and C5a), which induce local mast-cell degranulation with release of histamine. Histamine is a potent mediator of inflammation, causing vasodilation and smooth-muscle contraction. The prostaglandins can also contribute to the vasodilation and increased vascular permeability associated with the acute inflammatory response.

Within a few hours of the onset of these vascular changes, neutrophils adhere to the endothelial cells, and migrate out of the blood into the tissue spaces (Figure 15-12). These neutrophils phagocytose invading pathogens and release mediators that contribute to the inflammatory response. Among the mediators are the macrophage inflammatory proteins (MIP-1 α and MIP-1 β), chemokines that attract macrophages to the site of inflammation. Macrophages arrive about 5–6 hours after an inflammatory response begins. These macrophages are activated cells that exhibit increased phagocytosis and increased release of mediators and cytokines that contribute to the inflammatory response.

Activated tissue macrophages secrete three cytokines (IL-1, IL-6, and TNF- α) that induce many of the localized and

systemic changes observed in the acute inflammatory response (see Table 15-3). All three cytokines act locally, inducing coagulation and an increase in vascular permeability. Both TNF- α and IL-1 induce increased expression of adhesion molecules on vascular endothelial cells. For instance, TNF- α stimulates expression of E-selectin, an endothelial adhesion molecule that selectively binds adhesion molecules on neutrophils. IL-1 induces increased expression of ICAM-1 and VCAM-1, which bind to integrins on lymphocytes and monocytes. Circulating neutrophils, monocytes, and lymphocytes recognize these adhesion molecules on the walls of blood vessels, adhere, and then move through the vessel wall into the tissue spaces. IL-1 and TNF- α also act on macrophages and endothelial cells to induce production of the chemokines that contribute to the influx of neutrophils by increasing their adhesion to vascular endothelial cells and by acting as potent chemotactic factors. In addition, IFN- γ and TNF- α activate macrophages and neutrophils, promoting increased phagocytic activity and increased release of lytic enzymes into the tissue spaces.

A local acute inflammatory response can occur without the overt involvement of the immune system. Often, however, cytokines released at the site of inflammation facilitate both the adherence of immune-system cells to vascular endothelial cells and their migration through the vessel wall into the tissue spaces. The result is an influx of lymphocytes, neutrophils, monocytes, eosinophils, basophils, and mast cells to the site of tissue damage, where these cells participate in clearance of the antigen and healing of the tissue.

The duration and intensity of the local acute inflammatory response must be carefully regulated to control tissue damage and facilitate the tissue-repair mechanisms that are necessary for healing. TGF- β has been shown to play an important role in limiting the inflammatory response. It also promotes accumulation and proliferation of fibroblasts and the deposition of an extracellular matrix that is required for proper tissue repair.

Clearly, the processes of leukocyte adhesion are of great importance in the inflammatory response. A failure of proper leukocyte adhesion can result in disease, as exemplified by leukocyte-adhesion deficiency (see Clinical Focus on page 358).

SYSTEMIC ACUTE-PHASE RESPONSE

The local inflammatory response is accompanied by a systemic response known as the **acute-phase response** (Figure 15-13). This response is marked by the induction of fever, increased synthesis of hormones such as ACTH and hydrocortisone, increased production of white blood cells (leukocytosis), and production of a large number of **acute-phase proteins** in the liver. The increase in body temperature inhibits the growth of a number of pathogens and appears to enhance the immune response to the pathogen.

C-reactive protein is a prototype acute-phase protein whose serum level increases 1000-fold during an acute-phase

response. It is composed of five identical polypeptides held together by noncovalent interactions. C-reactive protein binds to a wide variety of microorganisms and activates complement, resulting in deposition of the opsonin C3b on the surface of microorganisms. Phagocytic cells, which express C3b receptors, can then readily phagocytose the C3b-coated microorganisms.

Many systemic acute-phase effects are due to the combined action of IL-1, TNF- α and IL-6 (see Figure 15-13). Each of these cytokines acts on the hypothalamus to induce a fever response. Within 12–24 h of the onset of an acute-phase inflammatory response, increased levels of IL-1, TNF- α and IL-6 (as well as leukemia inhibitory factor (LIF) and oncostatin M (OSM)) induce production of acute-phase proteins by hepatocytes. TNF- α also acts on vascular endothelial cells and macrophages to induce secretion of colony-stimulating factors (M-CSF, G-CSF, and GM-CSF). These CSFs stimulate hematopoiesis, resulting in transient increases in the number of white blood cells needed to fight the infection.

The redundancy in the ability of at least five cytokines (TNF- α , IL-1, IL-6, LIF, and OSM) to induce production of acute-phase proteins by the liver results from the induction of a common transcription factor, NF-IL6, after each of these cytokines interacts with its receptor. Amino-acid sequencing of cloned NF-IL6 revealed that it has a high degree of sequence identity with C/EBP, a liver-specific transcription factor (Figure 15-14a). Both NF-IL6 and C/EBP contain a leucine-zipper domain and a basic DNA-binding domain, and both proteins bind to the same nucleotide sequence in the promoter or enhancer of the genes encoding various liver proteins. C/EBP, which stimulates production of albumin and transthyretin, is expressed constitutively by hepatocytes. As an inflammatory response develops and the cytokines interact with their respective receptors on liver hepatocytes, expression of NF-IL6 increases and that of C/EBP decreases (Figure 15-14b). The inverse relationship between these two transcription factors accounts for the observation that serum levels of proteins such as albumin and transthyretin decline while those of acute-phase proteins increase during an inflammatory response.

Chronic Inflammation Develops When Antigen Persists

Some microorganisms are able to evade clearance by the immune system, for example by possessing cell-wall components that enable them to resist phagocytosis. Such organisms often induce a chronic inflammatory response, resulting in significant tissue damage. Chronic inflammation also occurs in a number of autoimmune diseases in which self-antigens continually activate T cells. Finally, chronic inflammation also contributes to the tissue damage and wasting associated with many types of cancer.

The accumulation and activation of macrophages is the hallmark of chronic inflammation. Cytokines released by the



VISUALIZING CONCEPTS

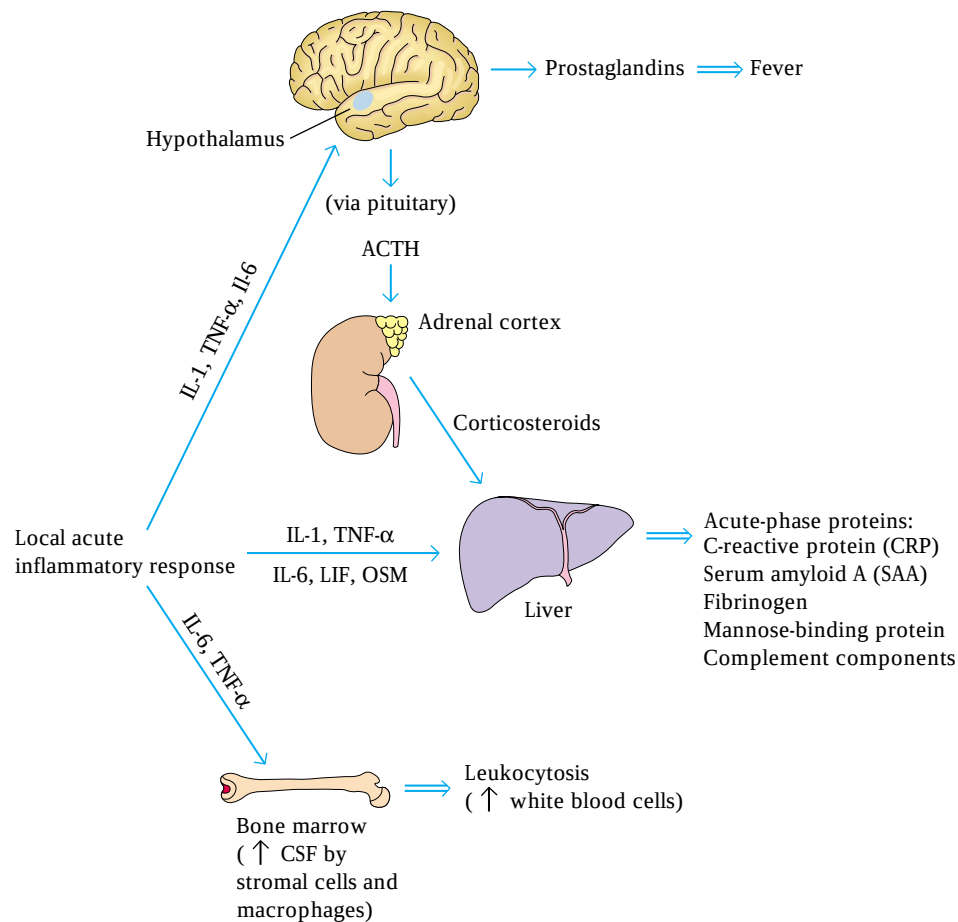


FIGURE 15-13 Overview of the organs and mediators involved in a systemic acute-phase response. IL-1, IL-6, and TNF- α , which are produced by activated macrophages at the site of inflamma-

tion, are particularly important in mediating acute-phase effects. LIF = leukemia inhibitory factor; OSM = oncostatin M.

chronically activated macrophages also stimulate fibroblast proliferation and collagen production. A type of scar tissue develops at sites of chronic inflammation by a process called **fibrosis**, a wound-healing reaction that can interfere with normal tissue function. Chronic inflammation may also lead to formation of a **granuloma**, a tumor-like mass consisting of a central area of activated macrophages surrounded by activated lymphocytes. The center of the granuloma often contains multinucleated giant cells formed by the fusion of activated macrophages. These giant cells typically are surrounded by large modified macrophages that resemble epithelial cells and therefore are called epithelioid cells.

Roles of IFN- γ and TNF- α in Chronic Inflammation

Two cytokines in particular, IFN- γ and TNF- α , play a central role in the development of chronic inflammation. T_H1 cells, NK cells, and T_C cells release IFN- γ , while activated macrophages secrete TNF- α .

Members of the interferon family of glycoproteins (IFN- α and IFN- β) are released from virus-infected cells and confer antiviral protection on neighboring cells. Exactly which interferon is produced depends on the type of cell infected. IFN- α is produced by leukocytes, IFN- β , often called fibroblast

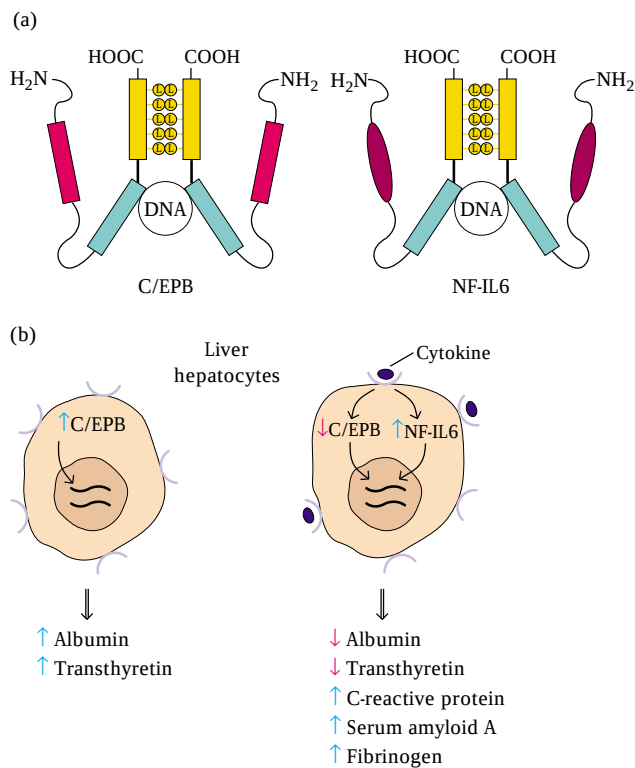


FIGURE 15-14 Comparison of the structure and function of C/EBP and NF-IL6. (a) Both transcription factors are dimeric proteins containing a leucine-zipper domain (light orange) and a basic DNA-binding domain (blue). (b) C/EBP is expressed constitutively in liver hepatocytes and promotes transcription of albumin and transthyretin genes. During an inflammatory response, binding of IL-1, IL-6, TNF- α , LIF, or OSM to receptors on liver hepatocytes induces production of NF-IL6, which promotes transcription of the genes encoding various acute-phase proteins. Concurrently, C/EBP levels decrease and the levels of albumin and transthyretin consequently decrease.

interferon, is made largely by fibroblasts. IFN- γ is produced exclusively by T cells and NK cells. However, IFN- γ has a number of pleiotropic activities that distinguish it from IFN- α and IFN- β and contribute to the inflammatory response (Figure 15-15). One of the most striking effects of IFN- γ is its ability to activate macrophages. Activated macrophages exhibit increased expression of class II MHC molecules, increased cytokine production, and increased microbicidal activity compared with nonactivated macrophages; thus, they are more effective in antigen presentation and killing of intracellular microbial pathogens. In a chronic inflammatory response, however, the large numbers of activated macrophages release various hydrolytic enzymes and reactive oxygen and nitrogen

intermediates, which are responsible for much of the damage to surrounding tissue.

One of the principal cytokines secreted by activated macrophages is TNF- α . The activity of this cytokine was first observed around the turn of the century by the surgeon William Coley. He noted that when cancer patients developed certain bacterial infections, the tumors would become necrotic. In the hope of providing a cure for cancer, Coley began to inject cancer patients with supernatants derived from various bacterial cultures. These culture supernatants, called “Coley’s toxins,” did induce hemorrhagic necrosis in the tumors but had numerous undesirable side effects, making them unsuitable for cancer therapy. Decades later, the active component of Coley’s toxin was shown to be a lipopolysaccharide (endotoxin) component of the bacterial cell wall. This endotoxin does not itself induce tumor necrosis but instead induces macrophages to produce TNF- α . This cytokine has a direct cytotoxic effect on tumor cells but not on normal cells (Figure 15-16a). Potential immunotherapeutic approaches using TNF- α for the treatment of cancer are examined in Chapter 22.

Several lines of evidence indicate that TNF- α also contributes to much of the tissue wasting that characterizes chronic inflammation. For example, mice carrying a TNF- α transgene become severely wasted (Figure 15-16b). In studies by A. Cerami and coworkers, rabbits were found to lose nearly half of their body mass within 2 months of being infected with trypanosomes. These workers subsequently discovered that a macrophage-derived factor was responsible for the profound wasting; they called the factor cachetin. Cloning of the genes for TNF- α and cachetin revealed that they were the same protein.

Activation of macrophages by IFN- α promotes increased transcription of the TNF- α gene and increases the stability of TNF- α mRNA. Both effects result in increased TNF- α production. TNF- α acts synergistically with IFN- γ to initiate a chronic inflammatory response. Both cytokines together induce much greater increases in ICAM-1, E-selectin, and class I MHC molecules than either cytokine alone. The increase in intercellular adhesion molecules facilitates the recruitment of large numbers of cells in a chronic inflammatory response.

CHRONIC INFLAMMATORY DISEASES

Recent studies suggest that regions of plump endothelial cells resembling HEVs appear along the vasculature in tertiary extralymphoid sites of chronic infection. These HEV-like regions, which appear to be sites of lymphocyte extravasation into the inflamed tissue, express several mucins (e.g., GlyCAM-1, MADCAM-1, and CD34) that are often displayed on normal HEVs. Several cytokines, notably IFN- γ and TNF- α , that are associated with chronic inflammation may play a role in the induction of HEV-like regions along the vasculature.

These HEV-like regions have been observed in a number of chronic inflammatory diseases in humans, including

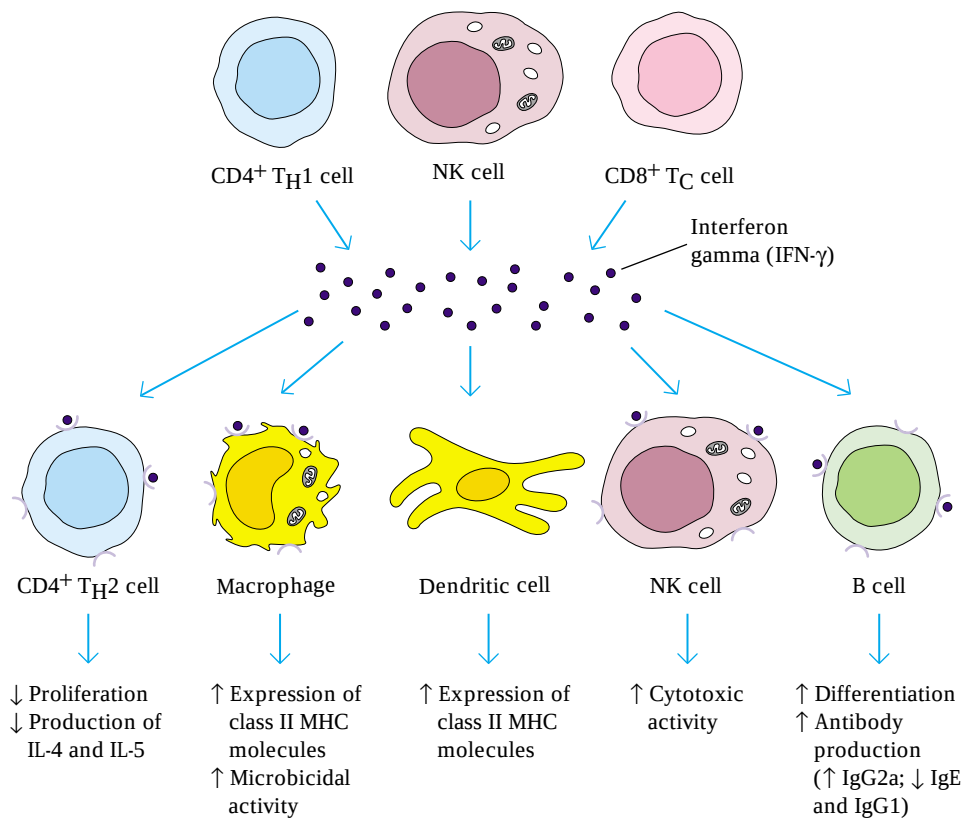


FIGURE 15-15 Summary of pleiotropic activity of interferon gamma (IFN-γ). The activation of macrophages induced by IFN-γ plays a critical role in chronic inflammation. This cytokine is secreted by T_H1 cells,

NK cells, and T_C cells and acts on numerous cell types. [Adapted from *Research News*, 1993, *Science* **259**:1693.]

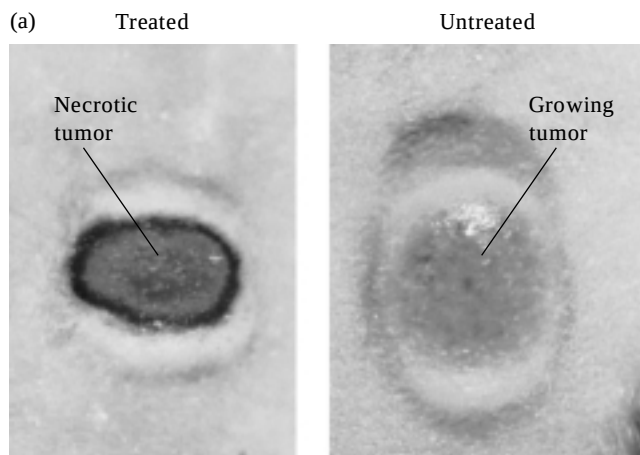


FIGURE 15-16 Biological activities of TNF-α. (a) A cancerous tumor in a mouse injected with endotoxin (*left*) shows hemorrhagic necrosis, unlike a tumor in an untreated mouse (*right*). Endotoxin induces the production of TNF-α, which then acts to destroy the tumor.



(b) Transgenic mouse (*top*) bearing a TNF-α transgene becomes anorectic and severely wasted. Normal mouse is shown on the bottom. [Part (a) from L. J. Old, 1988, *Sci. Am.* **258**:59; part (b) from B. Beutler, 1993, *Hosp. Prac.* (April 15):45.]



CLINICAL FOCUS

Leukocyte-Adhesion Deficiency (LAD) in Humans and Cattle

The immune system uses inflammation to assemble the components of an effective response and focus these resources at the site of infection. Inflammation is complex, involving vasodilation, increased vascular permeability, exudation of plasma proteins, and a gathering of inflammatory cells. Chemoattractants are key elements in calling leukocytes to sites of inflammation. These include chemokines such as IL-8, monocyte chemoattractant protein 1 (MCP-1), macrophage inflammatory protein 1 (MIP-1) and peptide fragments, such as C5a, generated during complement fixation. Chemoattractants signal passing leukocytes to adhere tightly to the vascular surface, and, using adhesive interactions for traction, these cells push their way between endothelial cells and gain entry into the surrounding tissue.

Once in, they are guided by gradients of chemoattractants to the sites of the inflammatory responses and become participants in the process. The key players in the adhesive interactions that are central to adhesion and extravasation are heterodimeric integrin molecules on the surface of the migrating leukocytes. There are a number of integrins, among which are LFA-1 (composed of CD11a and CD18); Mac-1, also called CR3 (components: CD11b and CD18); and p150/95 or CR4 (components: CD11c and CD18). When leukocytes encounter the appropriate chemokine or other chemoattractant, their complement of membrane integrin molecules undergoes a conformational change that transforms them from a slightly adhesive to a highly adhesive state.

In 1979, a paper entitled “Delayed separation of the umbilical cord, widespread

infections, and defective neutrophil mobility” appeared in *Lancet*, a British medical journal. This was the first in a series of reports that have appeared over the years describing patients afflicted with a rare autosomal recessive disease in which the first indication is quite often omphalitis, a swelling and reddening around the stalk of the umbilical cord. Although no more susceptible to virus infections than normal controls, those afflicted with this disorder suffer recurrent and often chronic bacterial infections, and sites where one would expect to find pus are instead pus-free. This observation is particularly striking because the patients are not deficient in granulocytes; in fact, they typically have greatly elevated numbers of granulocytes in the circulation. Detailed immunological work-ups of these patients showed that Ig levels were in the normal range and that they had nearly normal B-, T-, and NK-cell function. However, examination of leukocyte migration in response to tissue damage revealed the root cause of the disease in these patients.

One method of evaluating leukocyte migration involves gently scraping the

rheumatoid arthritis, Crohn’s disease, ulcerative colitis, Graves’ disease, Hashimoto’s thyroiditis, and diabetes mellitus (Table 15-4). Development of this HEV-like vasculature is likely to facilitate a large-scale influx of leukocytes, contribut-

ing to chronic inflammation. These observations suggest that an effective approach for treating chronic inflammatory diseases may be to try to control the development of these HEV-like regions.

TABLE 15-4 Chronic inflammatory diseases associated with HEV-like vasculature

Disease	Affected organ	Plump endothelium	Mucin-like CAMs on endothelium*
Crohn’s disease	Gut	+	+
Diabetes mellitus	Pancreas	+	+
Graves’ disease	Thyroid	+	+
Hashimoto’s thyroiditis	Thyroid	+	+
Rheumatoid arthritis	Synovium	+	+
Ulcerative colitis	Gut	+	+

*Includes GlyCAM-1, MAdCAM-1, and CD34.
SOURCE: Adapted from J. P. Girard and T. A. Springer, 1995, *Immunol. Today* 16:449.

skin from a small area of the arm; the cell populations that move into the abraded area are then sampled by capturing some of those cells on a glass coverslip placed onto the wounded skin. A series of glass coverslips is sequentially placed, incubated, and removed over a period of several hours. Typically, each coverslip is left in place for two hours, and the procedure is repeated four times over an eight-hour period. Examination of the coverslips under a microscope reveals whether leukocytes have adhered to the coverslips. In normal individuals, the response of the immune system to tissue injury is to deliver leukocytes to the damaged area, and one finds these cells on the coverslips. However, in the patients described here, the coverslips were largely negative for leukocytes. Examination of white blood cells in these patients revealed an absence of CD18, an essential component of a number of integrins. A key element in the migration of leukocytes is integrin-mediated cell adhesion, and these patients suffer from an inability of their leukocytes to undergo adhesion-dependent migration into sites of inflammation. Hence, this syndrome

has been named leukocyte-adhesion deficiency (LAD).

Bacterial infections in these patients can be treated with antibiotics, but they recur. Furthermore, there are antibiotic-resistant strains of many pathogenic bacteria, and LAD patients must live under this microbial Sword of Damocles, never knowing when the life-saving thread of antibiotics will fail. If a suitable bone-marrow donor can be found (almost always a close relative), however, there is a curative strategy. The LAD patient's hematopoietic system is destroyed, perhaps by treatment with cytotoxic chemicals, and then bone-marrow transplantation is performed. If successful, this procedure provides the patient with leukocytes that have normal levels of functional integrin and display the full range of migratory capacities.

This disease is not limited to humans. A strikingly similar version known as bovine leukocyte adhesion disease (BLAD) occurs in cattle. The cause of BLAD in these animals is identical to the cause of LAD in human patients—the lack of a functional integrin subunit. What is different in some dairy herds is the incidence of the disease; though rare in

humans, it can occur at economically important frequencies in cattle. This is a consequence of the high degree of inbreeding that exists in populations of dairy cattle. Typically, dairy herds are sired by the artificial insemination of semen from very few bulls. As a consequence of this practice, by the 1980s, almost 1 in 20 dairy bulls could be traced back to a single Holstein bull who happened to be heterozygous for BLAD. Such a high frequency of this recessive trait in the sire population dramatically raised the frequency of this disease in dairy herds. During the early 1990s, in some countries, the incidence of the BLAD gene was as high as 10% in a number of dairy herds. The gene for bovine CD18 has been cloned, which has allowed the design of a PCR-based assay for the aberrant forms of this gene. It is now possible to routinely screen sires and recipients for the BLAD allele. As a result, bulls that are carriers of the BLAD gene have been identified and eliminated from the breeding pool. This has led to a dramatic reduction in the frequency of new BLAD cases as well as in the overall frequency of the BLAD allele in dairy-herd populations.

Anti-Inflammatory Agents

Although development of an effective inflammatory response can play an important role in the body's defense, the response can sometimes be detrimental. Allergies, autoimmune diseases, microbial infections, transplants, and burns may initiate a chronic inflammatory response. Various therapeutic approaches are available for reducing long-term inflammatory responses and thus the complications associated with them.

Antibody Therapies Reduce Leukocyte Extravasation

Because leukocyte extravasation is an integral part of the inflammatory response, one approach for reducing inflammation is to impede this process. Theoretically, one way to reduce leukocyte extravasation is to block the activity of various adhesion molecules with antibodies. In animal models,

for example, antibodies to the integrin LFA-1 have been used to reduce neutrophil buildup in inflammatory tissue. Antibodies to ICAM-1 have also been used, with some success, in preventing the tissue necrosis associated with burns and in reducing the likelihood of kidney-graft rejection in animal models. The results with antibodies specific for these adhesins have been so encouraging that a combination of antibodies (anti-ICAM-1 and anti-LFA-1) was used in clinical trials on human kidney-transplant patients. A combination of two anti-adhesins had to be used because failure to block both LFA-1 and ICAM-1 results in rejection.

Corticosteroids Are Powerful Anti-Inflammatory Drugs

The corticosteroids, which are cholesterol derivatives, include prednisone, prednisolone, and methylprednisolone. These potent anti-inflammatory agents exert various effects that result in a reduction in the numbers and activity of immune-system cells. They are regularly used in anti-inflammatory therapy.

Corticosteroid treatment causes a decrease in the number of circulating lymphocytes as the result either of steroid-induced lysis of lymphocytes (lympholysis) or of alterations in lymphocyte-circulation patterns. Some species (e.g., hamster, mouse, rat, and rabbit) are particularly sensitive to corticosteroid-induced lympholysis. In these animals, corticosteroid treatment at dosages as low as 10^{-7} M causes such widespread lympholysis that the weight of the thymus is reduced by 90%; the spleen and lymph nodes also shrink visibly. Immature thymocytes in these species appear to be particularly sensitive to corticosteroid-mediated killing. In rodents, corticosteroids induce programmed cell death of immature thymocytes, whereas mature thymocytes are resistant to this activity. Within 2 h following *in vitro* incubation with corticosteroids, immature thymocytes begin to show the characteristic morphology of apoptosis, and 90% of the chromatin is degraded into the characteristic nucleosome ladder by 24 h after treatment. The steps involved in the induction of apoptosis by corticosteroids remain to be determined. In humans, guinea pigs, and monkeys, corticosteroids do not induce apoptosis but instead affect lymphocyte-circulation patterns, causing a decrease in thymic weight and a marked decrease in the number of circulating lymphocytes.

Like other steroid hormones, the corticosteroids are lipophilic and thus can cross the plasma membrane and bind to receptors in the cytosol. The resulting receptor-hormone complexes are transported to the nucleus, where they bind to specific regulatory DNA sequences, regulating transcription up or down. The corticosteroids have been shown to induce increased transcription of the NF- κ B inhibitor (I- κ B). Binding of this inhibitor to NF- κ B in the cytosol prevents the translocation of NF- κ B into the nucleus and consequently prevents NF- κ B activation of a number of genes, including genes involved in T-cell activation and cytokine production.

Corticosteroids also reduce both the phagocytic and killing ability of macrophages and neutrophils, and this effect may contribute to their anti-inflammatory action. In addition, they reduce chemotaxis, so that fewer inflammatory cells are attracted to the site of T_H -cell activation. In the presence of corticosteroids, expression of class II MHC molecules and IL-1 production by macrophages is dramatically reduced; such reductions would be expected to lead to corresponding reductions in T_H -cell activation. Finally, corticosteroids also stabilize the lysosomal membranes of participating leukocytes, so that decreased levels of lysosomal enzymes are released at the site of inflammation.

NSAIDs Combat Pain and Inflammation

Since the time of Hippocrates, extracts of willow bark have been used for relief of pain. The active ingredient, salicylate, which is found in aspirin, is just one of many nonsteroidal anti-inflammatory drugs (NSAIDs). NSAIDs are the most frequently used medication for treating pain and inflammation. Clinically, NSAIDs have been shown to be effective for treat-

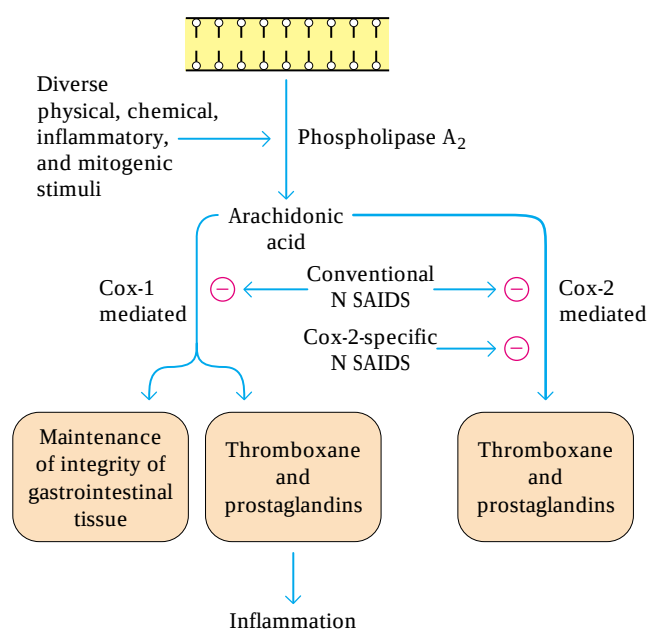


FIGURE 15-17 Inhibition of cyclooxygenase 1 and 2 by NSAIDs. A variety of agents trigger the release of arachidonic acid from the cell membrane by the action of phospholipase A₂. The subsequent action of Cox-1 and Cox-2 initiates the conversion of arachidonic acid to a variety of lipid mediators of inflammation and many other processes. Many NSAIDs inhibit both enzymatic pathways, but those with greater specificity for the Cox-2 arm produce antiinflammatory effects with fewer side effects. [Adapted from G. A. FitzGerald and C. N. Patrono, 2001, *New England Journal of Medicine* **345**:433.]

ment of many acute and chronic inflammatory reactions. The major mechanism by which these drugs exert anti-inflammatory effects is by inhibiting the cyclooxygenase pathway that produces prostaglandins and thromboxanes from arachidonic acid. The reduction in prostaglandin production limits the increase in vascular permeability and neutrophil chemotaxis in the inflammatory response. As shown in Figure 15-17, the cyclooxygenase pathway is mediated by two enzymes, cyclooxygenase 1 and cyclooxygenase 2 (Cox-1 & Cox-2).

Although NSAIDs such as aspirin, Tylenol, ibuprofen, Naproxen, and others are routinely prescribed for the treatment of ailments as diverse as arthritis, sprains, tissue injury, and back pain, the duration of their use is limited by gastrointestinal side effects that include unease and abdominal pain and in more serious cases bleeding or perforation of the stomach or upper GI tract. Investigation of the mechanism of NSAIDs has provided a basis for the beneficial and deleterious effects of many NSAIDs. Studies have shown that, although most NSAIDs inhibit both Cox-1 and Cox-2, it is the inhibition of Cox-2 that is responsible for the anti-inflammatory effects of NSAIDs. On the other hand, inhibi-

tion of Cox-1 by these agents causes damage to the GI tract but does not have significant anti-inflammatory benefits. This realization led to the design and development of a new generation of NSAIDs that specifically inhibit Cox-2 but have little effect on Cox-1 activity. The action of these highly targeted drugs is shown in Figure 15-17.

SUMMARY

- Lymphocytes undergo constant recirculation between the blood, lymph, lymphoid organs, and tertiary extralymphoid tissues, increasing the chances that the small number of lymphocytes specific for a given antigen (about 1 in 10^5 cells) will actually encounter that antigen.
- Migration of leukocytes into inflamed tissue or into lymphoid organs requires interaction between cell-adhesion molecules (CAMs) on the vascular endothelium and those on the circulating cells.
- Most CAMs fall into one of four protein families: the selectins, the mucin-like family, integrins, or the Ig superfamily. Selectins and mucin-like CAMs interact with each other, and members of each family are expressed on both leukocytes and endothelial cells. Integrins, expressed on leukocytes, interact with Ig-superfamily CAMs, expressed on endothelial cells.
- Extravasation of both neutrophils and lymphocytes involves four steps: rolling, activation, arrest and adhesion, and transendothelial migration. Neutrophils are generally the first cell type to move from the bloodstream into inflammatory sites.
- Unlike neutrophils, various lymphocyte populations exhibit differential extravasation into various tissues. Homing receptors on lymphocytes interact with tissue-specific adhesion molecules, called vascular addressins, on high-endothelial venules (HEVs) in lymphoid organs and on the endothelium in tertiary extralymphoid tissues.
- Naive lymphocytes home to secondary lymphoid organs, extravasating across HEVs, whereas effector lymphocytes selectively home to inflamed vascular endothelium.
- Inflammation is a physiologic response to a variety of stimuli such as tissue injury and infection. An acute inflammatory response involves both localized and systemic effects. The localized response begins when tissue and endothelial damage induces formation of plasma enzyme mediators that lead to vasodilation and increased vascular permeability.
- Several types of mediators play a role in the inflammatory response. Chemokines act as chemoattractants and activating molecules during leukocyte extravasation. Plasma enzyme mediators include bradykinin and fibrinopeptides, which increase vascular permeability; plasmin is a proteolytic enzyme that degrades fibrin clots into chemo-

tactic products and activates complement; and various complement products act as anaphylatoxins, opsonins, and chemotactic molecules for neutrophils and monocytes. Lipid inflammatory mediators include thromboxanes, prostaglandins, leukotrienes, and platelet-activating factor. Three cytokines, IL-1, IL-6, and TNF- α , mediate many of the local and systemic features of the acute inflammatory response

- Activation of tissue macrophages and degranulation of mast cells lead to release of numerous inflammatory mediators, some of which induce the acute-phase response, which includes fever, leukocytosis, and production of corticosteroids and acute-phase proteins.
- A chronic inflammatory response may accompany allergies, autoimmune diseases, microbial infections, transplants, and burns. Drug-based therapies employing corticosteroids and a variety of nonsteroidal anti-inflammatory drugs (NSAIDs) are the most commonly used medications for pain and inflammation.

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USEFUL WEB SITES

<http://www.mdsystems.com>

The Cytokine Mini-Reviews Section (Site Map/Reviews & Technical Notes) of the R&D Systems site contains extensive,



detailed, and well-illustrated reviews of many chemokines and chemokine receptors.

<http://www.ncbi.nlm.nih.gov/Omim/>

Online Mendelian Inheritance in Man is a catalog of human genes and genetic disorders. It contains pictures and references on many diseases, including LAD.

Study Questions

CLINICAL FOCUS QUESTION Why does a defect in CD18 result in an increased vulnerability to bacterial infection? Please address, as precisely as you can, the cell biology of cell migration.

1. Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.

- Chemokines are chemoattractants for lymphocytes but not other leukocytes.
- Integrins are expressed on both leukocytes and endothelial cells.
- Leukocyte extravasation involves multiple interactions between cell-adhesion molecules.
- Most secondary lymphoid organs contain high-endothelial venules (HEVs).
- Mucin-like CAMs interact with selectins.
- An acute inflammatory response involves only localized effects in the region of tissue injury or infection.
- MAdCAM-1 is an endothelial adhesion molecule that binds to L-selectin and to several integrins.
- Granuloma formation is a common symptom of local inflammation.

2. Various inflammatory mediators induce expression of ICAMs on a wide variety of tissues. What effect might this induction have on the localization of immune cells?

3. Extravasation of neutrophils and of lymphocytes occurs by generally similar mechanisms, although some differences distinguish the two processes.

- List in order the four basic steps in leukocyte extravasation.
- At which sites are neutrophils most likely to extravasate? Why?
- Different lymphocyte subpopulations migrate preferentially into different tissues, a process called homing (or trafficking). Discuss the roles of the three types of molecules that permit homing of lymphocytes.

4. Which three cytokines secreted by activated macrophages play a major role in mediating the localized and systemic effects associated with an acute inflammatory response?

5. An effective inflammatory response requires differentiation and proliferation of various nonlymphoid white blood cells. Explain how hematopoiesis in the bone marrow is induced by tissue injury or local infection.

6. For each pair of molecules listed below, indicate whether the molecules interact during the 1st, 2nd, 3rd, or 4th step in neutrophil extravasation at an inflammatory site. Use N to indicate any molecules that do not interact.

- _____ Chemokine and L-selectin
- _____ E-selectin and mucin-like CAM
- _____ IL-8 and E-selectin
- _____ Ig-superfamily CAM and integrin
- _____ ICAM and chemokine
- _____ Chemokine and G-protein-coupled receptor
- _____ ICAM and integrin

7. Discuss the main effects of IFN- γ and TNF- α during a chronic inflammatory response.

8. Five cytokines (IL-1, IL-6, TNF- α , LIF, and OSM) induce production of C-reactive protein and other acute-phase proteins by hepatocytes. Briefly explain how these different cytokines can exert the same effect on hepatocytes.

9. For each inflammation-related term (a–h), select the descriptions listed below (1–11) that apply. Each description may be used once, more than once, or not at all; more than one description may apply to some terms.

Terms

- _____ Tertiary extralymphoid tissue
- _____ P- and E-selectin
- _____ Prostaglandins
- _____ Nonsteroidal anti-inflammatory drugs
- _____ ICAM-1, -2, -3
- _____ MAdCAM
- _____ Bradykinin
- _____ Inflamed endothelium

Descriptions

- (1) Bind to sialylated carbohydrate moieties
- (2) Inhibit cyclooxygenase pathway
- (3) Induce expression of NF- κ B inhibitor
- (4) Has both Ig domains and mucin-like domains
- (5) Region of vascular endothelium found in postcapillary venules
- (6) Expressed by inflamed endothelium
- (7) Exhibits HEV-like vasculature in chronic inflammation
- (8) Belong to Ig-superfamily of CAMs
- (9) Exhibits increased expression of CAMs
- (10) Increase vascular permeability and induce fever
- (11) Induce fever

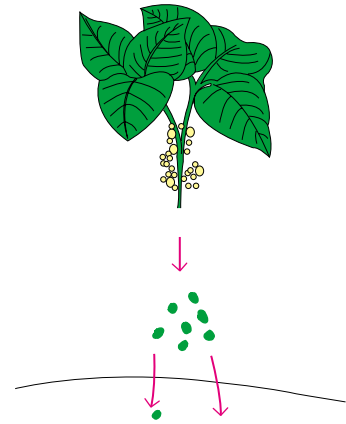
Hypersensitive Reactions

AN IMMUNE RESPONSE MOBILIZES A BATTERY OF effector molecules that act to remove antigen by various mechanisms described in previous chapters. Generally, these effector molecules induce a localized inflammatory response that eliminates antigen without extensively damaging the host's tissue. Under certain circumstances, however, this inflammatory response can have deleterious effects, resulting in significant tissue damage or even death. This inappropriate immune response is termed **hypersensitivity** or **allergy**. Although the word *hypersensitivity* implies an increased response, the response is not always heightened but may, instead, be an inappropriate immune response to an antigen. Hypersensitive reactions may develop in the course of either humoral or cell-mediated responses.

The ability of the immune system to respond inappropriately to antigenic challenge was recognized early in this century. Two French scientists, Paul Portier and Charles Richet, investigated the problem of bathers in the Mediterranean reacting violently to the stings of Portuguese Man of War jellyfish. Portier and Richet concluded that the localized reaction of the bathers was the result of toxins. To counteract this reaction, the scientists experimented with the use of isolated jellyfish toxins as vaccines. Their first attempts met with disastrous results. Portier and Richet injected dogs with the purified toxins, followed later by a booster of toxins. Instead of reacting to the booster by producing antibodies against the toxins, the dogs immediately reacted with vomiting, diarrhea, asphyxia, and, in some instances, death. Clearly this was an instance where the animals “overreacted” to the antigen. Portier and Richet coined the term *anaphylaxis*, loosely translated from Greek to mean the opposite of *prophylaxis*, to describe this overreaction. Richet was subsequently awarded the Nobel Prize in Physiology or Medicine in 1913 for his work on anaphylaxis.

We currently refer to anaphylactic reactions within the humoral branch initiated by antibody or antigen-antibody complexes as **immediate hypersensitivity**, because the symptoms are manifest within minutes or hours after a sensitized recipient encounters antigen. **Delayed-type hypersensitivity (DTH)** is so named in recognition of the delay of symptoms until days after exposure. This chapter examines the mechanisms and consequences of the four primary types of hypersensitive reactions.

chapter 16



A Second Exposure to Poison Oak May Result in Delayed-Type Hypersensitivity

- Gell and Coombs Classification
- IgE-Mediated (Type I) Hypersensitivity
- Antibody-Mediated Cytotoxic (Type II) Hypersensitivity
- Immune Complex-Mediated (Type III) Hypersensitivity
- Type IV or Delayed-Type Hypersensitivity (DTH)

Gell and Coombs Classification

Several forms of hypersensitive reaction can be distinguished, reflecting differences in the effector molecules generated in the course of the reaction. In immediate hypersensitive reactions, different antibody isotypes induce different immune effector molecules. IgE antibodies, for example, induce mast-cell degranulation with release of histamine and other biologically active molecules. IgG and IgM antibodies, on the other hand, induce hypersensitive reactions by activating complement. The effector molecules in the complement reactions are the membrane-attack complex and such complement split products as C3a, C4a, and C5a. In delayed-type hypersensitivity reactions, the effector molecules are various cytokines secreted by activated T_H or T_C cells.



VISUALIZING CONCEPTS

Type I	Type II	Type III	Type IV
IgE-Mediated Hypersensitivity	IgG-Mediated Cytotoxic Hypersensitivity	Immune Complex-Mediated Hypersensitivity	Cell-Mediated Hypersensitivity
Ag induces crosslinking of IgE bound to mast cells and basophils with release of vasoactive mediators	Ab directed against cell surface antigens mediates cell destruction via complement activation or ADCC	Ag-Ab complexes deposited in various tissues induce complement activation and an ensuing inflammatory response mediated by massive infiltration of neutrophils	Sensitized T _{H1} cells release cytokines that activate macrophages or T _C cells which mediate direct cellular damage
Typical manifestations include systemic anaphylaxis and localized anaphylaxis such as hay fever, asthma, hives, food allergies, and eczema	Typical manifestations include blood transfusion reactions, erythroblastosis fetalis, and autoimmune hemolytic anemia	Typical manifestations include localized Arthus reaction and generalized reactions such as serum sickness, necrotizing vasculitis, glomerulonephritis, rheumatoid arthritis, and systemic lupus erythematosus	Typical manifestations include contact dermatitis, tubercular lesions and graft rejection

FIGURE 16-1 The four types of hypersensitive responses.

As it became clear that several different immune mechanisms give rise to hypersensitive reactions, P. G. H. Gell and R. R. A. Coombs proposed a classification scheme in which hypersensitive reactions are divided into four types. Three types of hypersensitivity occur within the humoral branch and are mediated by antibody or antigen-antibody complexes: IgE-mediated (type I), antibody-mediated (type II), and immune complex-mediated (type III). A fourth type of hypersensitivity depends on reactions within the cell-mediated branch, and is termed delayed-type hypersensitivity, or DTH (type IV). Each type involves distinct mechanisms, cells, and mediator molecules (Figure 16-1). This classification scheme has served an important function in identifying the mechanistic differences among various hypersensitive reactions,

but it is important to point out that secondary effects blur the boundaries between the four categories.

IgE-Mediated (Type I) Hypersensitivity

A type I hypersensitive reaction is induced by certain types of antigens referred to as **allergens**, and has all the hallmarks of a normal humoral response. That is, an allergen induces a humoral antibody response by the same mechanisms as described in Chapter 11 for other soluble antigens, resulting in the generation of antibody-secreting plasma cells and memory cells. What distinguishes a type I hypersensitive response from a normal humoral response is that the plasma

cells secrete IgE. This class of antibody binds with high affinity to **Fc receptors** on the surface of tissue mast cells and blood basophils. Mast cells and basophils coated by IgE are said to be sensitized. A later exposure to the same allergen cross-links the membrane-bound IgE on sensitized mast cells and basophils, causing **degranulation** of these cells (Figure 16-2). The pharmacologically active mediators released from the granules act on the surrounding tissues. The principal effects—vasodilation and smooth-muscle contraction—may be either systemic or localized, depending on the extent of mediator release.

There Are Several Components of Type I Reactions

As depicted in Figure 16-2, several components are critical to development of type I hypersensitive reactions. This section will consider these components first and then describe the mechanism of degranulation.

ALLERGENS

The majority of humans mount significant IgE responses only as a defense against parasitic infections. After an individual has been exposed to a parasite, serum IgE levels in-

crease and remain high until the parasite is successfully cleared from the body. Some persons, however, may have an abnormality called **atopy**, a hereditary predisposition to the development of immediate hypersensitivity reactions against common environmental antigens. The IgE regulatory defects suffered by atopic individuals allow nonparasitic antigens to stimulate inappropriate IgE production, leading to tissue-damaging type I hypersensitivity. The term *allergen* refers specifically to nonparasitic antigens capable of stimulating type I hypersensitive responses in allergic individuals.

The abnormal IgE response of atopic individuals is at least partly genetic—it often runs in families. Atopic individuals have abnormally high levels of circulating IgE and also more than normal numbers of circulating eosinophils. These individuals are more susceptible to allergies such as hay fever, eczema, and asthma. The genetic propensity to atopic responses has been mapped to several candidate loci. One locus, on chromosome 5q, is linked to a region that encodes a variety of cytokines, including IL-3, IL-4, IL-5, IL-9, IL-13, and GM-CSF. A second locus, on chromosome 11q, is linked to a region that encodes the β chain of the high-affinity IgE receptor. It is known that inherited atopy is multigenic and that other loci probably also are involved. Indeed, as information from the Human Genome Project is analyzed, other candidate genes may be revealed.

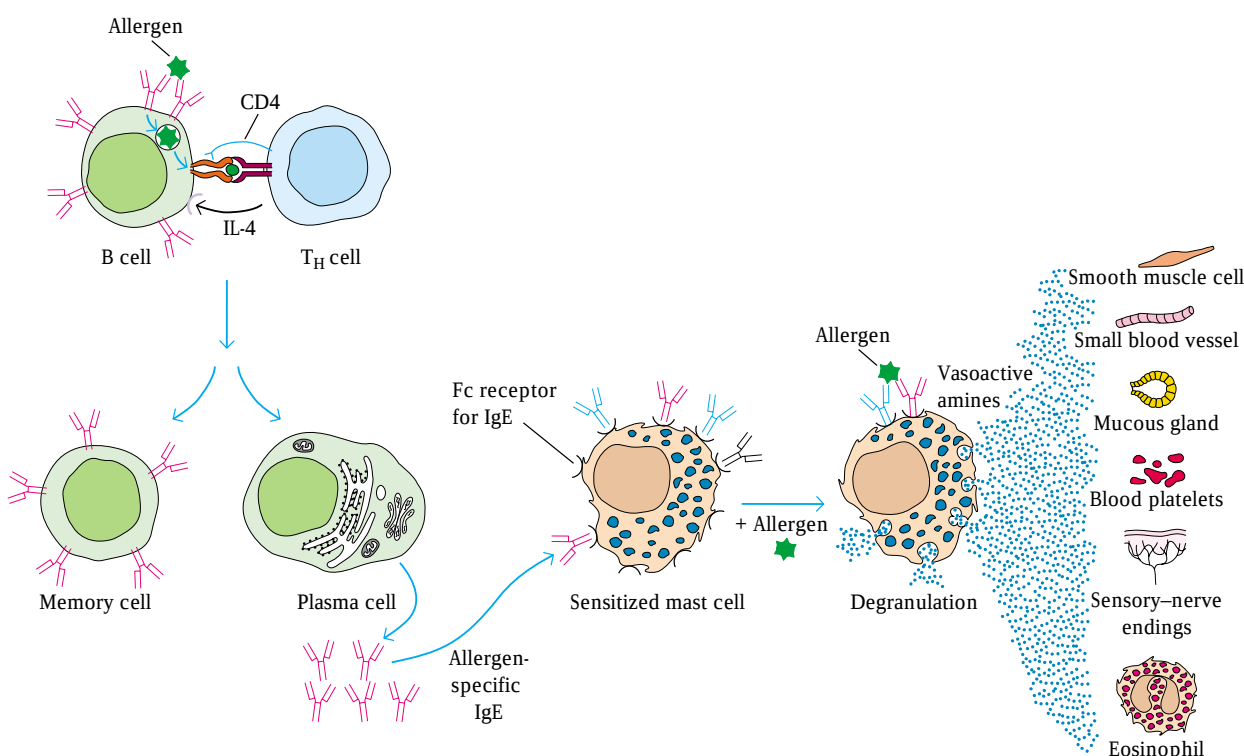


FIGURE 16-2 General mechanism underlying a type I hypersensitive reaction. Exposure to an allergen activates B cells to form IgE-secreting plasma cells. The secreted IgE molecules bind to IgE-specific Fc receptors on mast cells and blood basophils. (Many molecules of IgE with various specificities can bind to the IgE-Fc recep-

tor.) Second exposure to the allergen leads to crosslinking of the bound IgE, triggering the release of pharmacologically active mediators, vasoactive amines, from mast cells and basophils. The mediators cause smooth-muscle contraction, increased vascular permeability, and vasodilation.

TABLE 16-1 Common allergens associated with type I hypersensitivity

Proteins	Foods
Foreign serum	Nuts
Vaccines	Seafood
	Eggs
Plant pollens	Peas, beans
Rye grass	Milk
Ragweed	
Timothy grass	Insect products
Birch trees	Bee venom
	Wasp venom
Drugs	Ant venom
Penicillin	Cockroach calyx
Sulfonamides	Dust mites
Local anesthetics	
Salicylates	Mold spores
	Animal hair and dander

Most allergic IgE responses occur on mucous membrane surfaces in response to allergens that enter the body by either inhalation or ingestion. Of the common allergens listed in Table 16-1, few have been purified and characterized. Those that have include the allergens from rye grass pollen, ragweed pollen, codfish, birch pollen, timothy grass pollen, and bee venom. Each of these allergens has been shown to be a multi-antigenic system that contains a number of allergenic components. Ragweed pollen, a major allergen in the United States, is a case in point. It has been reported that a square mile of ragweed yields 16 tons of pollen in a single season. Indeed, all regions of the United States are plagued by ragweed pollen as well as pollen from trees indigenous to the region. The pollen particles are inhaled, and their tough outer wall is dissolved by enzymes in the mucous secretions, releasing the allergenic substances. Chemical fractionation of ragweed has revealed a variety of substances, most of which are not allergenic but are capable of eliciting an IgM or IgG response. Of the five fractions that are allergenic (i.e., able to induce an IgE response), two evoke allergic reactions in about 95% of ragweed-sensitive individuals and are called major allergens; these are designated the E and K fractions. The other three, called Ra3, Ra4, and Ra5, are minor allergens that induce an allergic response in only 20% to 30% of sensitive subjects.

Why are some pollens (e.g., ragweed) highly allergenic, whereas other equally abundant pollens (e.g., nettle) are rarely allergenic? No single physicochemical property seems to distinguish the highly allergenic E and K fractions of ragweed from the less allergenic Ra3, Ra4, and Ra5 fractions and from the nonallergenic fractions. Rather, allergens as a group appear to possess diverse properties. Some allergens, including foreign serum and egg albumin, are potent antigens; others, such as plant pollens, are weak antigens. Although most

allergens are small proteins or protein-bound substances having a molecular weight between 15,000 and 40,000, attempts to identify some common chemical property of these antigens have failed. It appears that allergenicity is a consequence of a complex series of interactions involving not only the allergen but also the dose, the sensitizing route, sometimes an adjuvant, and—most important, as noted above—the genetic constitution of the recipient.

REAGINIC ANTIBODY (IgE)

As described in Chapter 4, the existence of a human serum factor that reacts with allergens was first demonstrated by K. Prausnitz and H. Kustner in 1921. The local wheal and flare response that occurs when an allergen is injected into a sensitized individual is called the P-K reaction. Because the serum components responsible for the P-K reaction displayed specificity for allergen, they were assumed to be antibodies, but the nature of these P-K antibodies, or **reagins**, was not demonstrated for many years.

Experiments conducted by K. and T. Ishizaka in the mid-1960s showed that the biological activity of reaginic antibody in a P-K test could be neutralized by rabbit antiserum against whole atopic human sera but not by rabbit antiserum specific for the four human immunoglobulin classes known at that time (IgA, IgG, IgM, and IgD) (Table 16-2). In addition, when rabbits were immunized with sera from ragweed-sensitive individuals, the rabbit antiserum could inhibit (neutralize) a positive ragweed P-K test even after precipitation of the rabbit antibodies specific for the human IgG, IgA, IgM, and IgD isotypes. The Ishizakas called this new isotype IgE in reference to the E antigen of ragweed that they used to characterize it.

Serum IgE levels in normal individuals fall within the range of 0.1–0.4 $\mu\text{g/ml}$; even the most severely allergic individuals rarely have IgE levels greater than 1 $\mu\text{g/ml}$. These low levels made physicochemical studies of IgE difficult; it was not until the discovery of an IgE myeloma by S. G. O. Johansson and H. Bennich in 1967 that extensive chemical analysis of IgE could be undertaken. IgE was found to be composed of two heavy ϵ and two light chains with a combined molecular weight of 190,000. The higher molecular weight as compared with IgG (150,000) is due to the presence of an additional constant-region domain (see Figure 4-13). This additional domain (C_H4) contributes to an altered conformation of the Fc portion of the molecule that enables it to bind to glycoprotein receptors on the surface of basophils and mast cells. Although the half-life of IgE in the serum is only 2–3 days, once IgE has been bound to its receptor on mast cells and basophils, it is stable in that state for a number of weeks.

MAST CELLS AND BASOPHILS

The cells that bind IgE were identified by incubating human leukocytes and tissue cells with either ^{125}I -labeled IgE myeloma protein or ^{125}I -labeled anti-IgE. In both cases, autoradiography revealed that the labeled probe bound with high affinity to blood basophils and tissue mast cells. Basophils are

TABLE 16-2 Identification of IgE based on reactivity of atopic serum in P-K test

Serum	Treatment	Allergen added	P-K reaction at skin site
Atopic	None	—	—
Atopic	None	+	+
Nonatopic	None	+	—
Atopic	Rabbit antiserum to human atopic serum*	+	—
Atopic	Rabbit antiserum to human IgM, IgG, IgA, and IgD†	+	+

*Serum from an atopic individual was injected into rabbits to produce antiserum against human atopic serum. When this antiserum was reacted with human atopic serum, it neutralized the P-K reaction.

†Serum from an atopic individual was reacted with rabbit antiserum to the known classes of human antibody (IgM, IgA, IgG, and IgD) to remove these isotypes from the atopic serum. The treated atopic serum continued to give a positive P-K reaction, indicating that a new immunoglobulin isotype was responsible for this reactivity.

SOURCE: Based on K. Ishizaka and T. Ishizaka, 1967, *J. Immunol.* **99**:1187.

granulocytes that circulate in the blood of most vertebrates; in humans, they account for 0.5%–1.0% of the circulating white blood cells. Their granulated cytoplasm stains with basic dyes, hence the name basophil. Electron microscopy reveals a multilobed nucleus, few mitochondria, numerous glycogen granules, and electron-dense membrane-bound granules scattered throughout the cytoplasm that contain pharmacologically active mediators (see Figure 2-10c).

Mast-cell precursors are formed in the bone marrow during hematopoiesis and are carried to virtually all vascularized peripheral tissues, where they differentiate into mature cells. Mast cells are found throughout connective tissue, particularly near blood and lymphatic vessels. Some tissues, including the skin and mucous membrane surfaces of the respiratory and gastrointestinal tracts, contain high concentrations of mast cells; skin, for example, contains 10,000 mast cells per

mm³. Electron micrographs of mast cells reveal numerous membrane-bounded granules distributed throughout the cytoplasm, which, like those in basophils, contain pharmacologically active mediators (Figure 16-3). After activation, these mediators are released from the granules, resulting in the clinical manifestations of the type I hypersensitive reaction.

Mast cell populations in different anatomic sites differ significantly in the types and amounts of allergic mediators they contain and in their sensitivity to activating stimuli and cytokines. Mast cells also secrete a large variety of cytokines that affect a broad spectrum of physiologic, immunologic, and pathologic processes (see Table 12-1).

IgE-BINDING Fc RECEPTORS

The reaginic activity of IgE depends on its ability to bind to a receptor specific for the Fc region of the ϵ heavy chain. Two

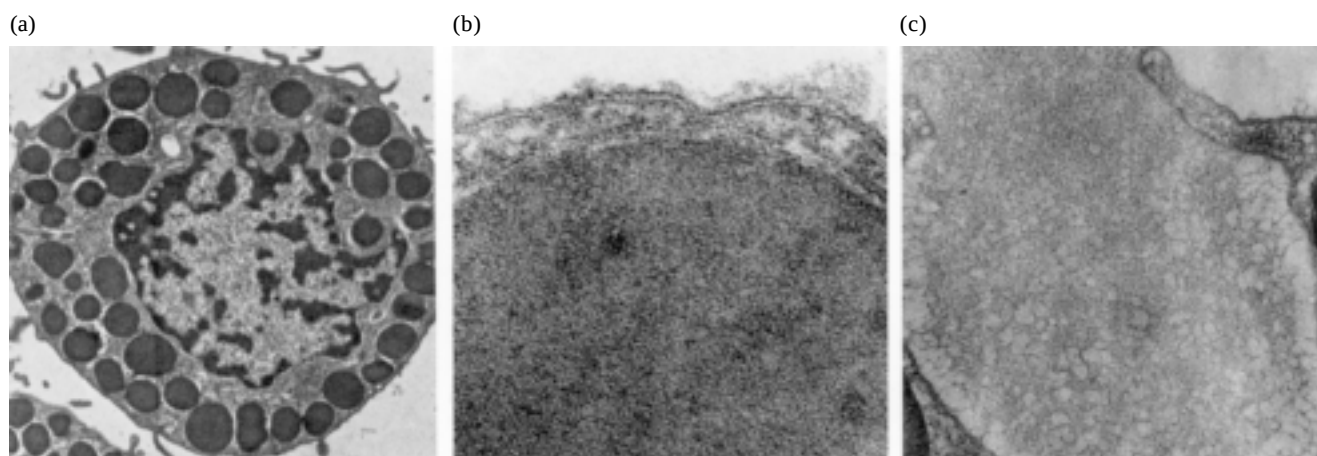


FIGURE 16-3 (a) Electron micrograph of a typical mast cell reveals numerous electron-dense membrane-bounded granules prior to degranulation. (b) Close-up of intact granule underlying the plasma

membrane of a mast cell. (c) Granule releasing its contents (towards top left) during degranulation. [From S. Burwen and B. Satir, 1977, *J. Cell Biol.* **73**:662.]

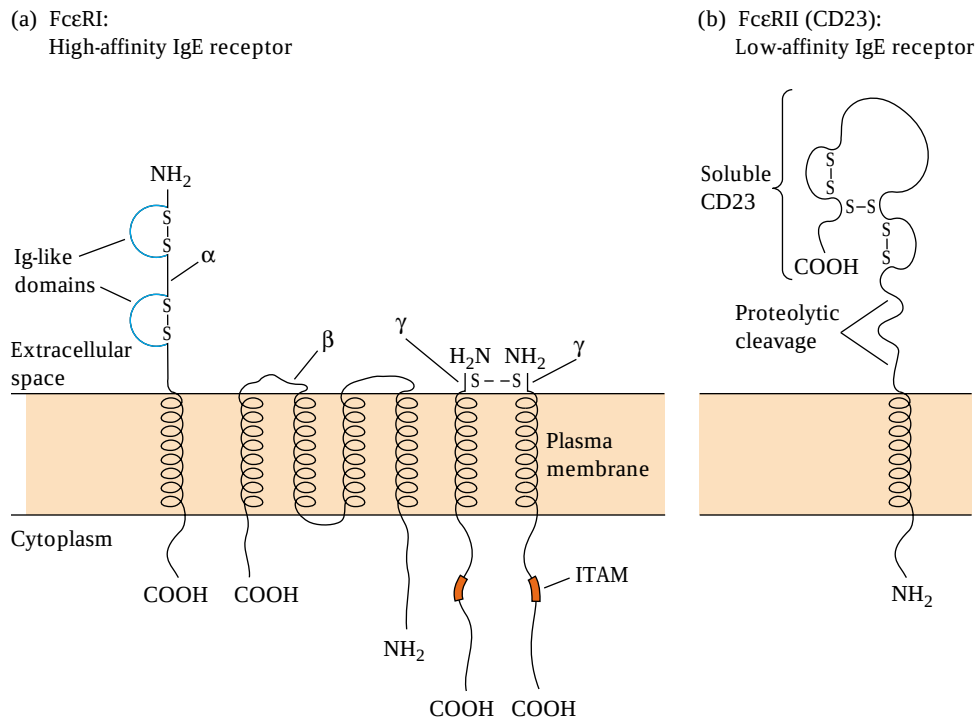


FIGURE 16-4 Schematic diagrams of the high-affinity FcεRI and low-affinity FcεRII receptors that bind the Fc region of IgE. (a) Each γ chain of the high-affinity receptor contains an ITAM, a motif also present in the Ig-α/Ig-β heterodimer of the B-cell receptor and in the

CD3 complex of the T-cell receptor. (b) The low-affinity receptor is unusual because it is oriented in the membrane with its NH₂-terminus directed toward the cell interior and its COOH-terminus directed toward the extracellular space.

classes of FcεR been identified, designated FcεRI and FcεRII, which are expressed by different cell types and differ by 1000-fold in their affinity for IgE.

HIGH-AFFINITY RECEPTOR (FcεRI) Mast cells and basophils express FcεRI, which binds IgE with a high affinity ($K_D = 1\text{--}2 \times 10^{-9}$ M). The high affinity of this receptor enables it to bind IgE despite the low serum concentration of IgE (1×10^{-7}). Between 40,000 and 90,000 FcεRI molecules have been shown to be present on a human basophil.

The FcεRI receptor contains four polypeptide chains: an α and a β chain and two identical disulfide-linked γ chains (Figure 16-4a). The external region of the α chain contains two domains of 90 amino acids that are homologous with the immunoglobulin-fold structure, placing the molecule in the immunoglobulin superfamily (see Figure 4-19). FcεRI interacts with the C_H3/C_H3 and C_H4/C_H4 domains of the IgE molecule via the two Ig-like domains of the α chain. The β chain spans the plasma membrane four times and is thought to link the α chain to the γ homodimer. The disulfide-linked γ chains extend a considerable distance into the cytoplasm. Each γ chain has a conserved sequence in its cytosolic domain known as an immunoreceptor tyrosine-based activation motif (ITAM). As described earlier, two other mem-

brane receptors that have this motif are CD3 and the associated ζ chains of the T-cell receptor complex (see Figure 10-10) and the Ig-α/Ig-β chains associated with membrane immunoglobulin on B cells (see Figure 11-7). The ITAM motif on these three receptors interacts with protein tyrosine kinases to transduce an activating signal to the cell. Allergen-mediated crosslinkage of the bound IgE results in aggregation of the FcεRI receptors and rapid tyrosine phosphorylation, which initiates the process of mast-cell degranulation. The role of FcεRI in anaphylaxis is confirmed by experiments conducted in mice that lack FcεRI. These mice have normal levels of mast cells but are resistant to localized and systemic anaphylaxis.

LOW-AFFINITY RECEPTOR (FcεRII) The other IgE receptor, designated FcεRII (or CD23), is specific for the C_H3/C_H3 domain of IgE and has a lower affinity for IgE ($K_D = 1 \times 10^{-6}$ M) than does FcεRI (Figure 16-4b). The FcεRII receptor appears to play a variety of roles in regulating the intensity of the IgE response. Allergen crosslinkage of IgE bound to FcεRII has been shown to activate B cells, alveolar macrophages, and eosinophils. When this receptor is blocked with monoclonal antibodies, IgE secretion by B cells is diminished. A soluble form of FcεRII (or sCD23), which is

generated by autoproteolysis of the membrane receptor, has been shown to enhance IgE production by B cells. Interestingly, atopic individuals have higher levels of CD23 on their lymphocytes and macrophages and higher levels of sCD23 in their serum than do nonatopic individuals.

IgE Crosslinkage Initiates Degranulation

The biochemical events that mediate degranulation of mast cells and blood basophils have many features in common. For simplicity, this section presents a general overview of mast-cell degranulation mechanisms without calling attention to the slight differences between mast cells and basophils. Although mast-cell degranulation generally is initiated by allergen crosslinkage of bound IgE, a number of other stimuli can also initiate the process, including the anaphylatoxins (C3a, C4a, and C5a) and various drugs. This section focuses on the biochemical events that follow allergen crosslinkage of bound IgE.

RECEPTOR CROSSLINKAGE

IgE-mediated degranulation begins when an allergen crosslinks IgE that is bound (fixed) to the FcεRI on the surface of a mast cell or basophil. In itself, the binding of IgE to FcεRI apparently has no effect on a target cell. It is only after allergen crosslinks the fixed IgE-receptor complex that degranulation proceeds. The importance of crosslinkage is indicated by the inability of monovalent allergens, which cannot crosslink the fixed IgE, to trigger degranulation.

Experiments have revealed that the essential step in degranulation is crosslinkage of two or more FcεRI molecules—with or without bound IgE. Although crosslinkage is normally effected by the interaction of fixed IgE with divalent or multivalent allergen, it also can be effected by a variety of experimental means that bypass the need for allergen and in some cases even for IgE (Figure 16-5).

Intracellular Events Also Regulate Mast-Cell Degranulation

The cytoplasmic domains of the β and γ chains of FcεRI are associated with protein tyrosine kinases (PTKs). Crosslinkage of the FcεRI receptors activates the associated PTKs, resulting in the phosphorylation of tyrosines within the ITAMs of the γ subunit as well as phosphorylation of residues on the β subunit and on phospholipase C. These phosphorylation events induce the production of a number of second messengers that mediate the process of degranulation (Figure 16-6).

Within 15 s after crosslinkage of FcεRI, methylation of various membrane phospholipids is observed, resulting in an increase in membrane fluidity and the formation of Ca²⁺ channels. An increase of Ca²⁺ reaches a peak within 2 min of FcεRI crosslinkage (Figure 16-7). This increase is due both to the uptake of extracellular Ca²⁺ and to a release of Ca²⁺ from

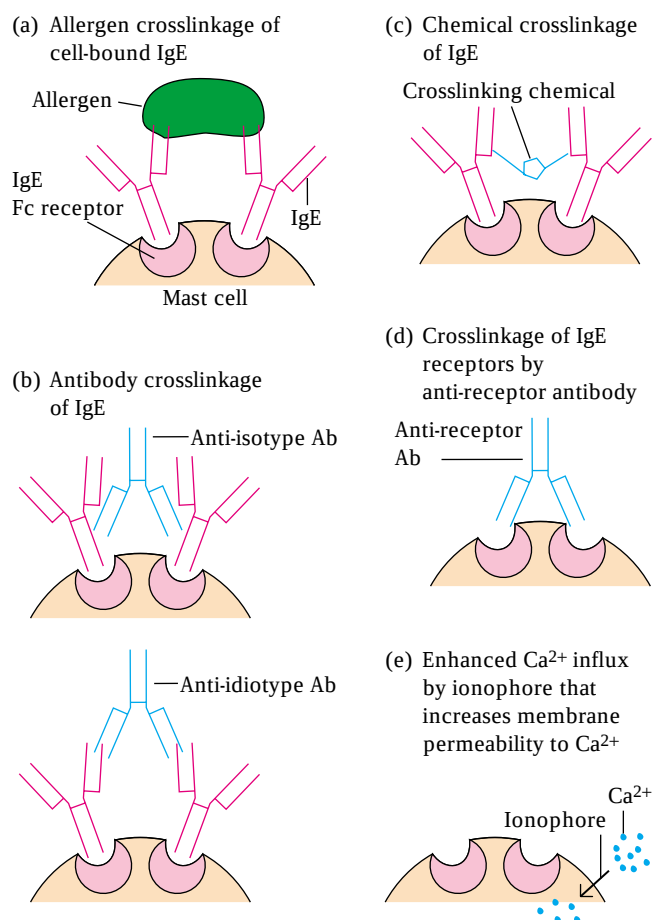


FIGURE 16-5 Schematic diagrams of mechanisms that can trigger degranulation of mast cells. Note that mechanisms (b) and (c) do not require allergen; mechanisms (d) and (e) require neither allergen nor IgE; and mechanism (e) does not even require receptor crosslinkage.

intracellular stores in the endoplasmic reticulum (see Figure 16-6). The Ca²⁺ increase eventually leads to the formation of arachidonic acid, which is converted into two classes of potent mediators: **prostaglandins** and **leukotrienes** (see Figure 16-6). The increase of Ca²⁺ also promotes the assembly of microtubules and the contraction of microfilaments, both of which are necessary for the movement of granules to the plasma membrane. The importance of the Ca²⁺ increase in mast-cell degranulation is highlighted by the use of drugs, such as disodium cromoglycate (cromolyn sodium), that block this influx as a treatment for allergies.

Concomitant with phospholipid methylation and Ca²⁺ increase, there is a transient increase in the activity of membrane-bound adenylate cyclase, with a rapid peak of its reaction product, cyclic adenosine monophosphate (cAMP), reached about 1 min after crosslinkage of FcεRI (see Figure 16-7). The effects of cAMP are exerted through the activation of cAMP-dependent

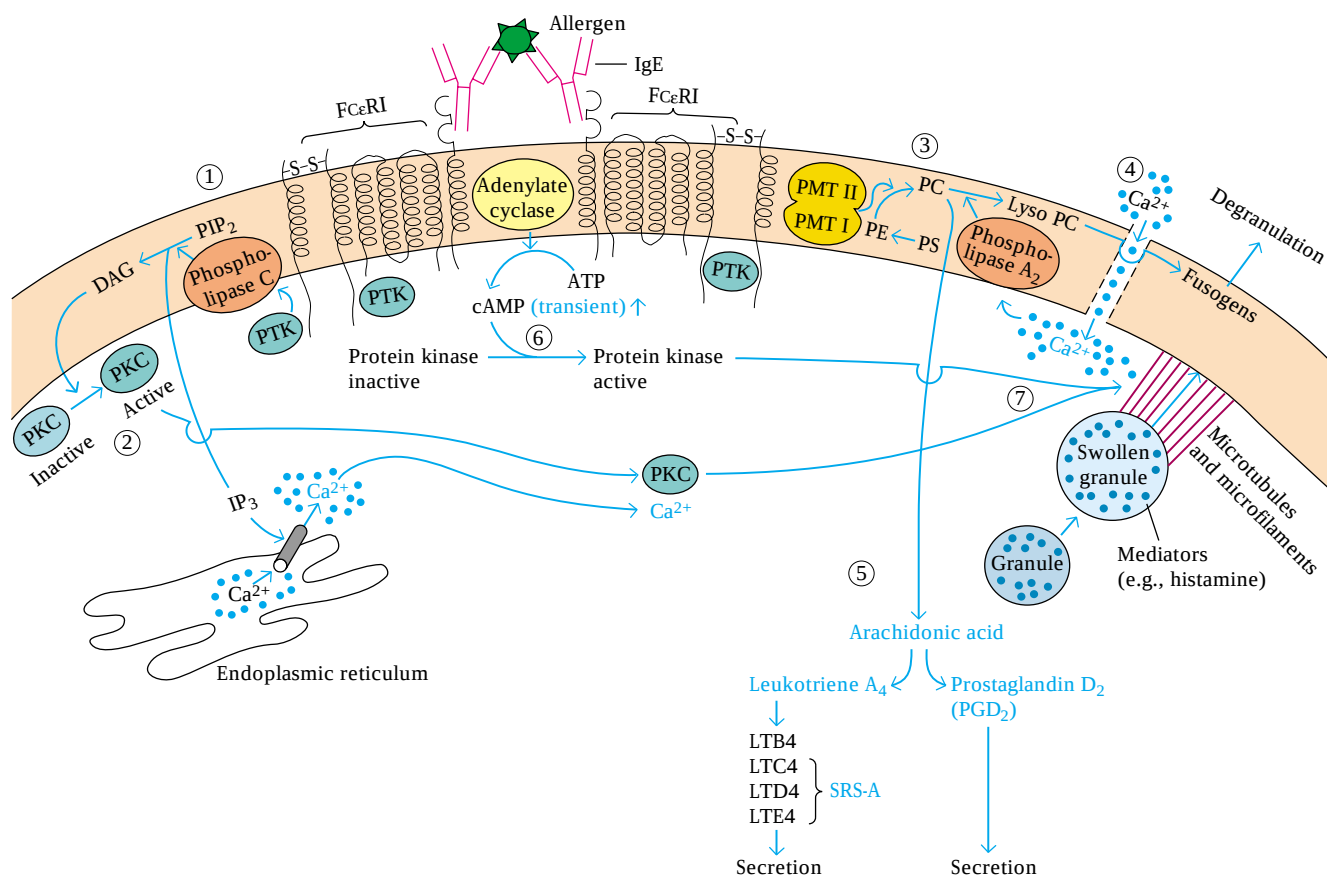


FIGURE 16-6 Diagrammatic overview of biochemical events in mast-cell activation and degranulation. Allergen crosslinkage of bound IgE results in FcεRI aggregation and activation of protein tyrosine kinase (PTK). (1) PTK then phosphorylates phospholipase C, which converts phosphatidylinositol-4,5 bisphosphate (PIP₂) into diacylglycerol (DAG) and inositol triphosphate (IP₃). (2) DAG activates protein kinase C (PKC), which with Ca²⁺ is necessary for microtubular assembly and the fusion of the granules with the plasma membrane. IP₃ is a potent mobilizer of intracellular Ca²⁺ stores. (3) Crosslinkage of FcεRI also activates an enzyme that converts phosphatidylserine (PS) into phosphatidylethanolamine (PE). Eventually, PE is methylated to form phosphatidylcholine (PC) by the phospholipid methyl transferase enzymes I and II (PMT I and II). (4) The accumulation of PC on the exterior sur-

face of the plasma membrane causes an increase in membrane fluidity and facilitates the formation of Ca²⁺ channels. The resulting influx of Ca²⁺ activates phospholipase A₂, which promotes the breakdown of PC into lysophosphatidylcholine (lyso PC) and arachidonic acid. (5) Arachidonic acid is converted into potent mediators: the leukotrienes and prostaglandin D₂. (6) FcεRI crosslinkage also activates the membrane adenylate cyclase, leading to a transient increase of cAMP within 15 s. A later drop in cAMP levels is mediated by protein kinase and is required for degranulation to proceed. (7) cAMP-dependent protein kinases are thought to phosphorylate the granule-membrane proteins, thereby changing the permeability of the granules to water and Ca²⁺. The consequent swelling of the granules facilitates fusion with the plasma membrane and release of the mediators.

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Several Pharmacologic Agents Mediate Type I Reactions

The clinical manifestations of type I hypersensitive reactions are related to the biological effects of the mediators released during mast-cell or basophil degranulation. These mediators are pharmacologically active agents that act on local tissues as well as on populations of secondary effector cells, including eosinophils, neutrophils, T lymphocytes, monocytes, and platelets. The mediators thus serve as an amplifying terminal effector mechanism, much as the complement system serves

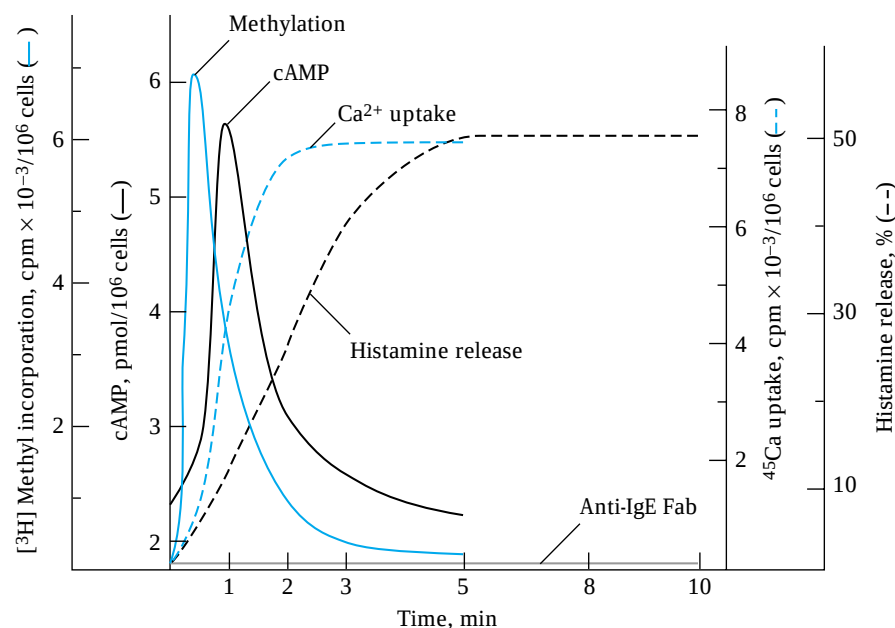


FIGURE 16-7 Kinetics of major biochemical events that follow crosslinkage of bound IgE on cultured human basophils with F(ab')₂ fragments of anti-IgE. Curves are shown for phospholipid methylation (solid blue), cAMP production (solid black), Ca²⁺ influx (dashed blue), and histamine release (dashed black). In control experiments with anti-IgE Fab fragments, no significant changes were observed. [Adapted from T. Ishizaka et al., 1985, *Int. Arch. Allergy Appl. Immunol.* **77**:137.]

as an amplifier and effector of an antigen-antibody interaction. When generated in response to parasitic infection, these mediators initiate beneficial defense processes, including vasodilation and increased vascular permeability, which brings an influx of plasma and inflammatory cells to attack the pathogen. On the other hand, mediator release induced by inappropriate antigens, such as allergens, results in unnecessary increases in vascular permeability and inflammation whose detrimental effects far outweigh any beneficial effect.

The mediators can be classified as either primary or secondary (Table 16-3). The primary mediators are produced before degranulation and are stored in the granules. The most significant primary mediators are histamine, proteases, eosinophil chemotactic factor, neutrophil chemotactic factor, and heparin. The secondary mediators either are synthesized after target-cell activation or are released by the breakdown of membrane phospholipids during the degranulation process. The secondary mediators include platelet-activating factor, leukotrienes, prostaglandins, bradykinins, and various cytokines. The differing manifestations of type I hypersensitivity in different species or different tissues partly reflect variations in the primary and secondary mediators present. The main biological effects of several of these mediators are described briefly in the next sections.

HISTAMINE

Histamine, which is formed by decarboxylation of the amino acid histidine, is a major component of mast-cell granules, accounting for about 10% of granule weight. Because it is stored—preformed—in the granules, its biological effects are observed within minutes of mast-cell activation. Once released from mast cells, histamine initially binds to specific

receptors on various target cells. Three types of histamine receptors—designated H₁, H₂, and H₃—have been identified; these receptors have different tissue distributions and mediate different effects when they bind histamine.

Most of the biologic effects of histamine in allergic reactions are mediated by the binding of histamine to H₁ receptors. This binding induces contraction of intestinal and bronchial smooth muscles, increased permeability of venules, and increased mucus secretion by goblet cells. Interaction of histamine with H₂ receptors increases vasopermeability and dilation and stimulates exocrine glands. Binding of histamine to H₂ receptors on mast cells and basophils suppresses degranulation; thus, histamine exerts negative feedback on the release of mediators.

LEUKOTRIENES AND PROSTAGLANDINS

As secondary mediators, the leukotrienes and prostaglandins are not formed until the mast cell undergoes degranulation and the enzymatic breakdown of phospholipids in the plasma membrane. An ensuing enzymatic cascade generates the prostaglandins and the leukotrienes (see Figure 16-6). It therefore takes a longer time for the biological effects of these mediators to become apparent. Their effects are more pronounced and longer lasting, however, than those of histamine. The leukotrienes mediate bronchoconstriction, increased vascular permeability, and mucus production. Prostaglandin D₂ causes bronchoconstriction.

The contraction of human bronchial and tracheal smooth muscles appears at first to be mediated by histamine, but, within 30–60 s, further contraction is mediated by the leukotrienes and prostaglandins. Being active at nanomole levels, the leukotrienes are as much as 1000 times more potent as

TABLE 16-3 Principal mediators involved in type I hypersensitivity

Mediator	Effects
PRIMARY	
Histamine, heparin	Increased vascular permeability; smooth-muscle contraction
Serotonin	Increased vascular permeability; smooth-muscle contraction
Eosinophil chemotactic factor (ECF-A)	Eosinophil chemotaxis
Neutrophil chemotactic factor (NCF-A)	Neutrophil chemotaxis
Proteases	Bronchial mucus secretion; degradation of blood-vessel basement membrane; generation of complement split products
SECONDARY	
Platelet-activating factor	Platelet aggregation and degranulation; contraction of pulmonary smooth muscles
Leukotrienes (slow reactive substance of anaphylaxis, SRS-A)	Increased vascular permeability; contraction of pulmonary smooth muscles
Prostaglandins	Vasodilation; contraction of pulmonary smooth muscles; platelet aggregation
Bradykinin	Increased vascular permeability; smooth-muscle contraction
Cytokines	
IL-1 and TNF- α	Systemic anaphylaxis; increased expression of CAMs on venular endothelial cells
IL-2, IL-3, IL-4, IL-5, IL-6, TGF- β , and GM-CSF	Various effects (see Table 12-1)

bronchoconstrictors than histamine is, and they are also more potent stimulators of vascular permeability and mucus secretion. In humans, the leukotrienes are thought to contribute to the prolonged bronchospasm and buildup of mucus seen in asthmatics.

CYTOKINES

Adding to the complexity of the type I reaction is the variety of cytokines released from mast cells and eosinophils. Some of these may contribute to the clinical manifestations of type I hypersensitivity. Human mast cells secrete IL-4, IL-5, IL-6, and TNF- α . These cytokines alter the local microenvironment, eventually leading to the recruitment of inflammatory cells such as neutrophils and eosinophils. IL-4 increases IgE production by B cells. IL-5 is especially important in the recruitment and activation of eosinophils. The high concentrations of TNF- α secreted by mast cells may contribute to shock in systemic anaphylaxis. (This effect may parallel the role of TNF- α in bacterial septic shock and toxic-shock syndrome described in Chapter 12.)

Type I Reactions Can Be Systemic or Localized

The clinical manifestations of type I reactions can range from life-threatening conditions, such as systemic anaphylaxis and asthma, to hay fever and eczema, which are merely annoying.

SYSTEMIC ANAPHYLAXIS

Systemic anaphylaxis is a shock-like and often fatal state whose onset occurs within minutes of a type I hypersensitive

reaction. This was the response observed by Portier and Richet in dogs after antigenic challenge. Systemic anaphylaxis can be induced in a variety of experimental animals and is seen occasionally in humans. Each species exhibits characteristic symptoms, which reflect differences in the distribution of mast cells and in the biologically active contents of their granules. The animal model of choice for studying systemic anaphylaxis has been the guinea pig. Anaphylaxis can be induced in guinea pigs with ease, and its symptoms closely parallel those observed in humans.

Active sensitization in guinea pigs is induced by a single injection of a foreign protein such as egg albumin. After an incubation period of about 2 weeks, the animal is usually challenged with an intravenous injection of the same protein. Within 1 min, the animal becomes restless, its respiration becomes labored, and its blood pressure drops. As the smooth muscles of the gastrointestinal tract and bladder contract, the guinea pig defecates and urinates. Finally bronchiole constriction results in death by asphyxiation within 2–4 min of the injection. These events all stem from the systemic vasodilation and smooth-muscle contraction brought on by mediators released in the course of the reaction. Post-mortem examination reveals that massive edema, shock, and bronchiole constriction are the major causes of death.

Systemic anaphylaxis in humans is characterized by a similar sequence of events. A wide range of antigens have been shown to trigger this reaction in susceptible humans, including the venom from bee, wasp, hornet, and ant stings; drugs, such as penicillin, insulin, and antitoxins; and seafood and nuts. If not treated quickly, these reactions can be fatal. Epinephrine is the drug of choice for systemic anaphylactic reactions. Epinephrine counteracts the effects of mediators such

as histamine and the leukotrienes by relaxing the smooth muscles and reducing vascular permeability. Epinephrine also improves cardiac output, which is necessary to prevent vascular collapse during an anaphylactic reaction. In addition, epinephrine increases cAMP levels in the mast cell, thereby blocking further degranulation.

LOCALIZED ANAPHYLAXIS (ATOPY)

In localized anaphylaxis, the reaction is limited to a specific target tissue or organ, often involving epithelial surfaces at the site of allergen entry. The tendency to manifest localized anaphylactic reactions is inherited and is called *atopy*. Atopic allergies, which afflict at least 20% of the population in developed countries, include a wide range of IgE-mediated disorders, including allergic rhinitis (hay fever), asthma, atopic dermatitis (eczema), and food allergies.

ALLERGIC RHINITIS The most common atopic disorder, affecting 10% of the U.S. population, is allergic rhinitis, commonly known as hay fever. This results from the reaction of airborne allergens with sensitized mast cells in the conjunctivae and nasal mucosa to induce the release of pharmacologically active mediators from mast cells; these mediators then cause localized vasodilation and increased capillary permeability. The symptoms include watery exudation of the conjunctivae, nasal mucosa, and upper respiratory tract, as well as sneezing and coughing.

ASTHMA Another common manifestation of localized anaphylaxis is asthma. In some cases, airborne or blood-borne allergens, such as pollens, dust, fumes, insect products, or viral antigens, trigger an asthmatic attack (allergic asthma); in other cases, an asthmatic attack can be induced by exercise or cold, apparently independently of allergen stimulation (intrinsic asthma). Like hay fever, asthma is triggered by degranulation of mast cells with release of mediators, but instead of occurring in the nasal mucosa, the reaction develops in the lower respiratory tract. The resulting contraction of the bronchial smooth muscles leads to bronchoconstriction. Airway edema, mucus secretion, and inflammation contribute to the bronchial constriction and to airway obstruction. Asthmatic patients may have abnormal levels of receptors for neuropeptides. For example, asthmatic patients have been reported to have increased expression of receptors for substance P, a peptide that contracts smooth muscles, and decreased expression of receptors for vasoactive intestinal peptide, which relaxes smooth muscles.

Most clinicians view asthma as primarily an inflammatory disease. The asthmatic response can be divided into early and late responses (Figure 16-8). The early response occurs within minutes of allergen exposure and primarily involves histamine, leukotrienes (LTC₄), and prostaglandin (PGD₂). The effects of these mediators lead to bronchoconstriction, vasodilation, and some buildup of mucus. The late response occurs hours later and involves additional mediators, including IL-4, IL-5, IL-16, TNF- α , eosinophil chemotactic factor (ECF),

and platelet-activating factor (PAF). The overall effects of these mediators is to increase endothelial cell adhesion as well as to recruit inflammatory cells, including eosinophils and neutrophils, into the bronchial tissue.

The neutrophils and eosinophils are capable of causing significant tissue injury by releasing toxic enzymes, oxygen radicals, and cytokines. These events lead to occlusion of the bronchial lumen with mucus, proteins, and cellular debris; sloughing of the epithelium; thickening of the basement membrane; fluid buildup (edema); and hypertrophy of the bronchial smooth muscles. A mucus plug often forms and adheres to the bronchial wall. The mucus plug contains clusters of detached epithelial-cell fragments, eosinophils, some neutrophils, and spirals of bronchial tissue known as Curschmann's spirals. Asthma is increasing in prevalence in the United States, particularly among children in inner-city environments (see Clinical Focus on page 376).

FOOD ALLERGIES Various foods also can induce localized anaphylaxis in allergic individuals. Allergen crosslinking of IgE on mast cells along the upper or lower gastrointestinal tract can induce localized smooth-muscle contraction and vasodilation and thus such symptoms as vomiting or diarrhea. Mast-cell degranulation along the gut can increase the permeability of mucous membranes, so that the allergen enters the bloodstream. Various symptoms can ensue, depending on where the allergen is deposited. For example, some individuals develop asthmatic attacks after ingesting certain foods. Others develop atopic urticaria, commonly known as hives, when a food allergen is carried to sensitized mast cells in the skin, causing swollen (edematous) red (erythematous) eruptions; this is the wheal and flare response, or P-K reaction, mentioned earlier.

ATOPIC DERMATITIS Atopic dermatitis (allergic eczema) is an inflammatory disease of skin that is frequently associated with a family history of atopy. The disease is observed most frequently in young children, often developing during infancy. Serum IgE levels are often elevated. The allergic individual develops skin eruptions that are erythematous and filled with pus. Unlike a delayed-type hypersensitive reaction, which involves T_H1 cells, the skin lesions in atopic dermatitis have T_H2 cells and an increased number of eosinophils.

Late-Phase Reactions Induce Localized Inflammatory Reactions

As a type I hypersensitive reaction begins to subside, mediators released during the course of the reaction often induce localized inflammation called the late-phase reaction. Distinct from the late response seen in asthma, the late-phase reaction begins to develop 4–6 h after the initial type I reaction and persists for 1–2 days. The reaction is characterized by infiltration of neutrophils, eosinophils, macrophages, lymphocytes, and basophils. The localized late-phase response also may be mediated partly by cytokines released from mast cells.

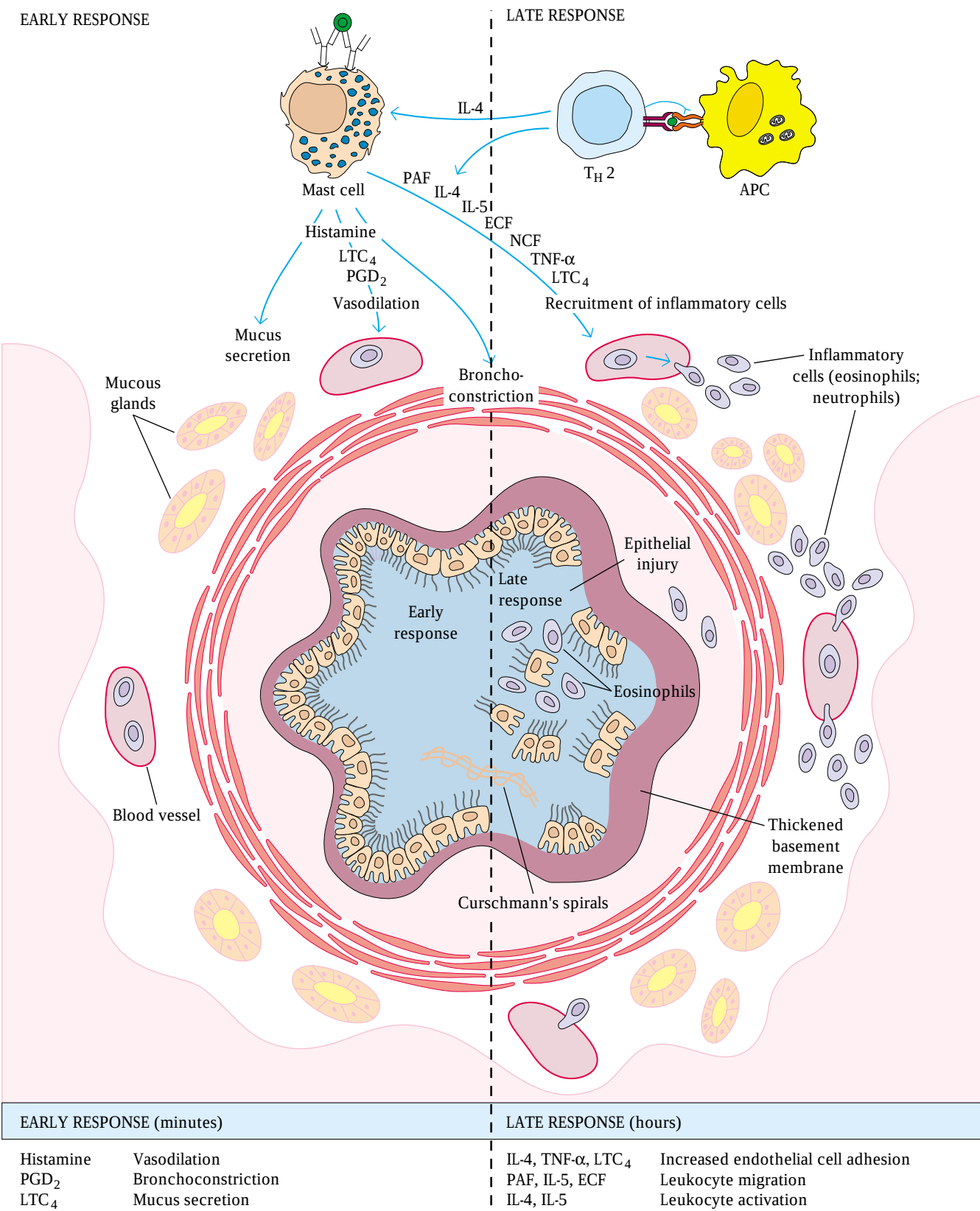


FIGURE 16-8 The early and late inflammatory responses in asthma. The immune cells involved in the early and late responses are repre-

sented at the top. The effects of various mediators on an airway, represented in cross section, are illustrated in the center.

Both TNF- α and IL-1 increase the expression of cell-adhesion molecules on venular endothelial cells, thus facilitating the buildup of neutrophils, eosinophils, and monocytes that characterizes the late-phase response.

Eosinophils play a principal role in the late-phase reaction, accounting for some 30% of the cells that accumulate. Eosinophil chemotactic factor, released by mast cells during the initial reaction, attracts large numbers of eosinophils to the affected site. Various cytokines released at the site, including IL-3, IL-5, and GM-CSF, contribute to the growth and differentiation of the eosinophils. Eosinophils express Fc receptors for IgG and IgE isotypes and bind directly to antibody-coated allergen. Much as in mast-cell degranulation, binding of antibody-coated antigen activates eosinophils, leading to their degranulation and release of inflammatory mediators, including leukotrienes, major basic protein, platelet-activation factor, eosinophil cationic protein (ECP), and eosinophil-derived neurotoxin. The release of these eosinophil-derived mediators may play a protective role in parasitic infections. However, in response to allergens, these mediators contribute to extensive tissue damage in the late-phase reaction. The influx of eosinophils in the late-phase response has been shown to contribute to the chronic inflammation of the bronchial mucosa that characterizes persistent asthma.

Neutrophils are another major participant in late-phase reactions, accounting for another 30% of the inflammatory cells. Neutrophils are attracted to the area of a type I reaction by neutrophil chemotactic factor, released from degranulating mast cells. In addition, a variety of cytokines released at the site, including IL-8, have been shown to activate neutrophils, resulting in release of their granule contents, including lytic enzymes, platelet-activating factor, and leukotrienes.

Type I Responses Are Regulated by Many Factors

As noted earlier, the antigen dose, mode of antigen presentation, and genetic constitution of an animal influence the level of the IgE response induced by an antigen (i.e., its allergenicity). Breeding experiments with mice have shown that this genetic variation is not linked to the MHC. A genetic component also has been shown to influence susceptibility to type I hypersensitive reactions in humans. If both parents are allergic, there is a 50% chance that a child will also be allergic; when only one parent is allergic, there is a 30% chance that a child will manifest some kind of type I reaction.

The effect of antigen dosage on the IgE response is illustrated by immunization of BDF1 mice. Repeated low doses of an appropriate antigen induce a persistent IgE response in these mice, but higher antigen doses result in transient IgE production and a shift toward IgG. The mode of antigen presentation also influences the development of the IgE response. For example, immunization of Lewis-strain rats with keyhole limpet hemocyanin (KLH) plus aluminum hydrox-

ide gel or *Bordetella pertussis* as an adjuvant induces a strong IgE response, whereas injection of KLH with complete Freund's adjuvant produces a largely IgG response. Infection of mice with the nematode *Nippostrongylus brasiliensis* (Nb), like certain adjuvants, preferentially induces an IgE response. For example, Nb-infected mice develop higher levels of IgE specific for an unrelated antigen than do uninfected control mice.

The relative levels of the T_H1 and T_H2 subsets also are key to the regulation of type I hypersensitive responses. T_H1 cells reduce the response, whereas T_H2 cells enhance it. Cytokines secreted by T_H2 cells—namely, IL-3, IL-4, IL-5, and IL-10—stimulate the type I response in several ways. IL-4 enhances class switching to IgE and regulates the clonal expansion of IgE-committed B cells; IL-3, IL-4, and IL-10 enhance mast-cell production; and IL-3 and IL-5 enhance eosinophil maturation, activation, and accumulation. In contrast, T_H1 cells produce IFN- γ which inhibits the type I response.

The pivotal role of IL-4 in regulation of the type I response was demonstrated in experiments by W. E. Paul and coworkers. When these researchers activated normal, unprimed B cells in vitro with the bacterial endotoxin lipopolysaccharide (LPS), only 2% of the cells expressed membrane IgG1 and only 0.05% expressed membrane IgE. However, when unprimed B cells were incubated with LPS plus IL-4, the percentage of cells expressing IgG1 increased to 40%–50% and the percentage expressing IgE increased to 15%–25%. In an attempt to determine whether IL-4 plays a role in regulating IgE production in vivo, Paul primed Nb-infected mice with the harmless antigen TNP-KLH in the presence and absence of monoclonal antibody to IL-4. The antibody to IL-4 reduced the production of IgE specific for TNP-KLH in these Nb-infected mice to 1% of the level in control animals.

Further support for the role of IL-4 in the IgE response comes from the experiments of K. Rajewsky and coworkers with IL-4 knockout mice. These IL-4-deficient mice were unable to mount an IgE response to helminthic antigens. Furthermore, increased levels of $CD4^+$ T_H2 cells and increased levels of IL-4 have been detected in atopic individuals. When allergen-specific $CD4^+$ T cells from atopic individuals are cloned and added to an autologous B-cell culture, the B cells synthesize IgE, whereas allergen-specific $CD4^+$ T cells from nonatopic individuals do not support IgE production.

In contrast to IL-4, IFN- γ decreases IgE production, suggesting that the balance of IL-4 and IFN- γ may determine the amount of IgE produced (Figure 16-9). Since IFN- γ is secreted by the T_H1 subset and IL-4 by the T_H2 subset, the relative activity of these subsets may influence an individual's response to allergens. According to this proposal, atopic and nonatopic individuals would exhibit qualitatively different type I responses to an allergen: the response in atopic individuals would involve the T_H2 subset and result in production of IgE; the response in nonatopic individuals would involve the T_H1 subset and result in production of IgM or IgG. To test this hypothesis, allergen-specific T cells were cloned from atopic and nonatopic individuals. The cloned T cells from the

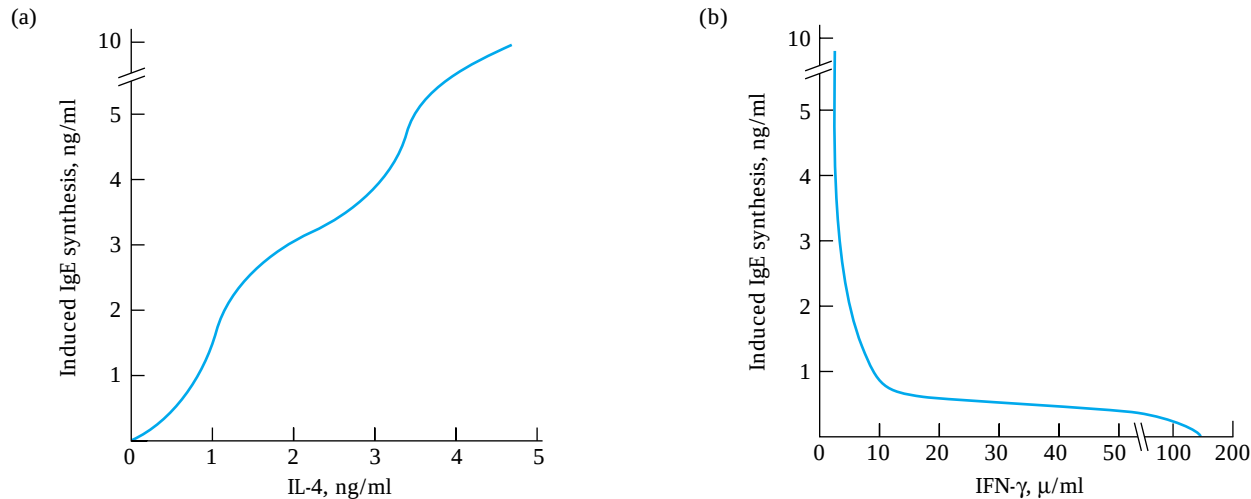


FIGURE 16-9 Effect of IL-4 and IFN- γ on in vitro production of IgE. These plots show the amount of IgE produced by plasma cells cul-

tured in the presence of various concentrations of IL-4 (a) or IFN- γ (b). [Adapted from G. Del Prete, 1988, *J. Immunol.* **140**:4193.]

atopic individuals were predominantly of the T_H2 phenotype (secreting IL-4), whereas the cloned T cells from nonatopic individuals were predominantly of the T_H1 phenotype (secreting IFN- γ). Needless to say, there is keen interest in down-regulating IL-4 as a possible treatment for allergic individuals.

Several Methods Are Used to Detect Type I Hypersensitivity Reactions

Type I hypersensitivity is commonly identified and assessed by skin testing. Small amounts of potential allergens are introduced at specific skin sites either by intradermal injection or by superficial scratching. A number of tests can be applied to sites on the forearm or back of an individual at one time. If a person is allergic to the allergen, local mast cells degranulate and the release of histamine and other mediators produces a wheal and flare within 30 min (Figure 16-10). The advantage of skin testing is that it is relatively inexpensive and allows screening of a large number of allergens at one time. The disadvantage of skin testing is that it sometimes sensitizes the allergic individual to new allergens and in some rare cases may induce systemic anaphylactic shock. A few individuals also manifest a late-phase reaction, which comes 4–6 h after testing and sometimes lasts for up to 24 h. As noted already, eosinophils accumulate during a late-phase reaction, and release of eosinophil-granule contents contributes to the tissue damage in a late-phase reaction site.

Another method of assessing type I hypersensitivity is to determine the serum level of total IgE antibody by the radioimmunosorbent test (RIST). This highly sensitive technique, based on the radioimmunoassay, can detect nanomolar levels of total IgE. The patient's serum is reacted with agarose beads or paper disks coated with rabbit anti-IgE. After the beads or disks are washed, ^{125}I -labeled rabbit anti-IgE is added. The radioactivity of the beads or disks, mea-

sured with a gamma counter, is proportional to the level of IgE in the patient's serum (Figure 16-11a).

The similar radioallergosorbent test (RAST) detects the serum level of IgE specific for a given allergen. The allergen is coupled to beads or disks, the patient's serum is added, and



FIGURE 16-10 Skin testing by intradermal injection of allergens into the forearm. In this individual, a wheal and flare response developed within a few minutes at the site where grass was injected, indicating that the individual is allergic to grass. [From L. M. Lichtenstein, 1993, *Sci. Am.* **269**(2):117. Used with permission.]

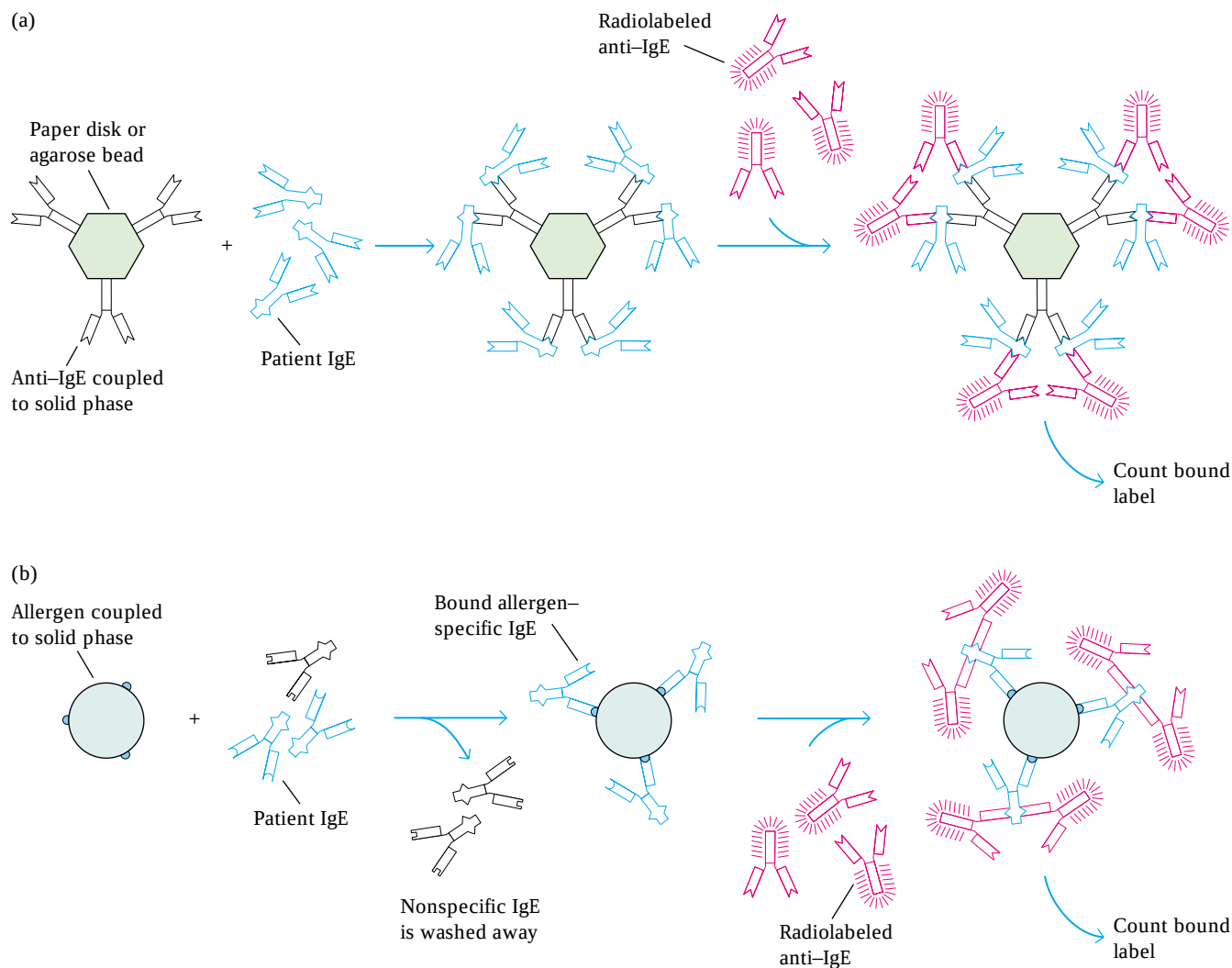


FIGURE 16-11 Procedures for assessing type I hypersensitivity. (a) Radioimmunosorbent test (RIST) can quantify nanogram amounts

of total serum IgE. (b) Radioallergen sorbent test (RAST) can quantify nanogram amounts of serum IgE specific for a particular allergen.

unbound antibody is washed away. The amount of specific IgE bound to the solid-phase allergen is then measured by adding ^{125}I -labeled rabbit anti-IgE, washing the beads, and counting the bound radioactivity (Figure 16-11b).

Type I Hypersensitivities Can Be Controlled Medically

The obvious first step in controlling type I hypersensitivities is to avoid contact with known allergens. Often the removal of house pets, dust-control measures, or avoidance of offending foods can eliminate a type I response. Elimination of inhalant allergens (such as pollens) is a physical impossibility, however, and other means of intervention must be pursued.

Immunotherapy with repeated injections of increasing doses of allergens (hyposensitization) has been known for some time to reduce the severity of type I reactions, or even

eliminate them completely, in a significant number of individuals suffering from allergic rhinitis. Such repeated introduction of allergen by subcutaneous injections appears to cause a shift toward IgG production or to induce T-cell-mediated suppression (possibly by a shift to the T_H1 subset and $\text{IFN-}\gamma$ production) that turns off the IgE response (Figure 16-12). In this situation, the IgG antibody is referred to as *blocking antibody* because it competes for the allergen, binds to it, and forms a complex that can be removed by phagocytosis; as a result, the allergen is not available to crosslink the fixed IgE on the mast-cell membranes, and allergic symptoms decrease.

Another form of immunotherapy is the use of humanized monoclonal anti-IgE. These antibodies bind to IgE, but only if IgE is not already bound to $\text{Fc}\epsilon\text{RI}$; the latter would lead to histamine release. In fact, the monoclonal antibodies are specifically selected to bind membrane IgE on IgE-expressing B cells.



CLINICAL FOCUS

The Genetics of Asthma

Asthma affects almost 5% of the population of the United States. For reasons that are still unclear, the incidence of asthma recently has increased dramatically in developed countries. Even more alarming is that the severity of the disease also appears to be increasing. The increase in asthma mortality is highest among children, and in the United States the mortality is highest among African-American children of the inner city. In 1999, 7.7 million children had asthma and more than 2000 of them died of the disease. These statistics are increasing each year. In addition to its human costs, asthma imposes high financial costs on society. During 2000, the cost for the treatment of asthma in the United States was more than \$12 billion.

Asthma is commonly defined as an inflammatory disease of the airway, and it is characterized by bronchial hyperrespon-

siveness. Atopic individuals, those with a predisposition to the type I hypersensitive response, are most susceptible to the development of bronchial hyperresponsiveness and asthma, but only 10%–30% of atopic individuals actually develop asthma. The evidence that asthma has a genetic component originally was derived from family studies, which estimated that the relative contribution of genetic factors to atopy and asthma is 40%–60%. While genetic factors are important, further studies have indicated that environmental factors also play a large role. Additionally, asthma is a complex genetic disease, controlled by several genes, so that susceptibility to it is likely to involve the interaction of multiple genetic and environmental factors.

How do we determine which genes contribute to a complex multigenic disease such as this? One approach is the candidate-gene approach, in which a hypothesis suggests that a particular gene or set

of genes may have some relation to the disease. After such a gene has been identified, families with apparent predisposition to the disease are examined for polymorphic alleles of the gene in question. Comparing family members who do or do not have the disease allows correlation between a particular allele and the presence of the disease. The problem with this approach is its bias toward identification of genes already suspected to play a role in the disease, which precludes identification of new genes. A good example of the use of the candidate-gene approach is the identification of a region on chromosome 5, region 5q31–33, that appears to be linked to the development of asthma. Using a candidate-gene approach, this region was investigated because it includes a cluster of cytokine genes, among them the genes that encode IL-3, -4, -5, -9, and -13, as well as the gene that encodes granulocyte-macrophage colony-stimulating factor. *IL-4* is thought to be a good candidate gene, since it induces the Ig class-switch to IgE. Several groups of investigators have examined this region in different populations and concluded that there is a polymorphism associ-

These antibodies are humanized by the genetic engineering of the genes encoding the H and L chains; mouse framework regions are replaced with human framework sequences and

the end result is a mouse/human chimeric monoclonal that is not likely to be recognized as foreign by the human immune system. When injected into people suffering from allergy, these antibodies can bind free IgE as well as down-regulate IgE production in B cells. This results in lower serum IgE concentration which, in turn, reduces the sensitivity of basophils. This form of immunotherapy is useful in treating many forms of allergies, especially crippling food allergies.

Another approach for treating allergies stems from the finding that soluble antigens tend to induce a state of anergy by activating T cells in the absence of the necessary co-stimulatory signal (see Figure 10-15). Presumably, a soluble

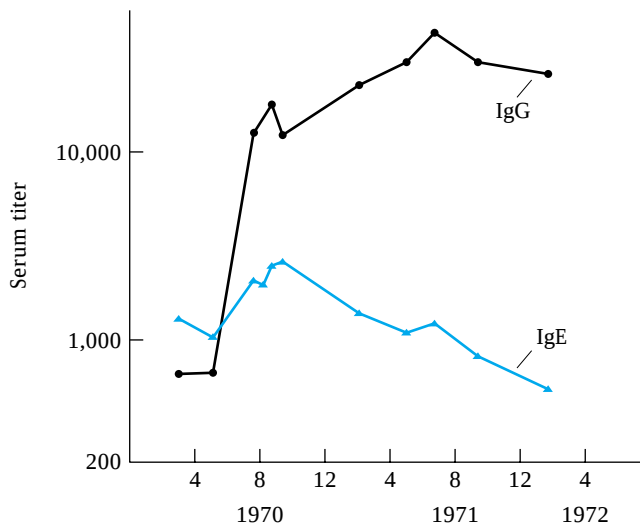


FIGURE 16-12 Hyposensitization treatment of type I allergy. Injection of ragweed antigen periodically for 2 years into a ragweed-sensitive individual induced a gradual decrease in IgE levels and a dramatic increase in IgG. Both antibodies were measured by a radioimmunoassay. [Adapted from K. Ishizaka and T. Ishizaka, 1973, in *Asthma Physiology, Immunopharmacology and Treatment*, K. F. Austen and L. M. Lichtenstein (eds.), Academic Press.]

ated with predisposition to asthma that maps to the promotor region of *IL-4*. Additionally, two alleles of *IL-9* associated with atopy have been identified.

Another approach to identifying genes associated with a particular disease is a random genomic search. In this method, the entire genome is scanned for polymorphisms associated with the disease in question. Using the random genomic approach, a British study (Lympay et al., 1992) identified a linkage between a polymorphism on chromosome 11—more specifically, region 11q13—associated with atopy in British families. This region maps to the vicinity of the β subunit of the high-affinity IgE receptor (Fc ϵ RI β). This association is exciting, since we know how important IgE is in mediating type I reactions. However, some caution in interpreting these results is necessary. This study looked at associations between polymorphisms and atopy, but most individuals who are atopic do not develop asthma. Therefore this association, while important in identifying factors in developing atopy, may not be relevant to the development of asthma.

More recently, a large genome-wide screen for loci linked to asthma susceptibility was conducted in ethnically diverse populations that included Caucasians, Hispanics, and African-Americans. This study, published by a large collaborative group from medical centers throughout the United States identified many candidate loci associated with asthma. One locus on chromosome 5 coincided with the already identified region at 5q31–33. Interestingly, however, this locus was associated with asthma in Caucasians but not in Hispanics or African-Americans. Similarly, some loci appeared to have a high correlation with asthma in Hispanics only, and other loci were identified as unique to African-Americans. Another interesting conclusion was that the association between chromosome 11q and atopy did not appear to be correlated with asthma. This could indicate that asthma and atopy have different molecular bases. More important, it suggests that genetic linkage to atopy should not be confused with genetic linkage to asthma. Overall, this study identified several genes linked to asthma and found that the number and relative

importance of these genes may differ among ethnic groups. This suggests that genetic differences as well as differences in environment may be the underlying basis of the differences observed in the prevalence as well as the severity of the disease among ethnic groups in the United States.

It is well documented that a higher than average percentage of African-American inner-city children have serious complications with asthma. This has raised the question whether there is a genetic predisposition for asthma in African-Americans. Recently, however, a report from Rosenstreich and colleagues has indicated an important environmental linkage to asthma in the inner city. This group assessed the role of allergies to the cockroach in the development of asthma; they found that a combination of cockroach allergy and exposure to high levels of cockroach allergen can help explain the high frequency of asthma-related health problems in inner-city children. These data also point to defects in the public-health systems in large cities. Clearly, a concerted effort by public agencies to eradicate insect infestations would benefit the health of those who live in inner-city communities.

antigen is internalized by endocytosis, processed, and presented with class II MHC molecules, but fails to induce expression of the requisite co-stimulatory ligand (B7) on antigen-presenting cells.

Knowledge of the mechanism of mast-cell degranulation and the mediators involved in type I reactions opened the way to drug therapy for allergies. Antihistamines have been the most useful drugs for symptoms of allergic rhinitis. These drugs act by binding to the histamine receptors on target cells and blocking the binding of histamine. The H_1 receptors are blocked by the classical antihistamines, and the H_2 receptors by a newer class of antihistamines.

Several drugs block release of allergic mediators by interfering with various biochemical steps in mast-cell activation and degranulation (Table 16-4). Disodium cromoglycate (cromolyn sodium) prevents Ca^{2+} influx into mast cells. Theophylline, which is commonly administered to asthmatics orally or through inhalers, blocks phosphodiesterase, which catalyzes the breakdown of cAMP to 5'-AMP. The resulting prolonged increase in cAMP levels blocks degranulation. A number of drugs stimulate the β -adrenergic system by stimulating β -adrenergic receptors. As mentioned earlier,

TABLE 16-4

Mechanism of action of some drugs used to treat type I hypersensitivity

Drug	Action
Antihistamines	Block H_1 and H_2 receptors on target cells
Cromolyn sodium	Blocks Ca^{2+} influx into mast cells
Theophylline	Prolongs high cAMP levels in mast cells by inhibiting phosphodiesterase, which cleaves cAMP to 5'-AMP*
Epinephrine (adrenalin)	Stimulates cAMP production by binding to β -adrenergic receptors on mast cells*
Cortisone	Reduces histamine levels by blocking conversion of histidine to histamine and stimulates mast-cell production of cAMP*

*Although cAMP rises transiently during mast-cell activation, degranulation is prevented if cAMP levels remain high.

epinephrine (also known as adrenaline) is commonly administered during anaphylactic shock. It acts by binding to β -adrenergic receptors on bronchial smooth muscles and mast cells, elevating the cAMP levels within these cells. The increased levels of cAMP promote relaxation of the bronchial muscles and decreased mast-cell degranulation. A number of epinephrine analogs have been developed that bind to selected β -adrenergic receptors and induce cAMP increases with fewer side effects than are seen with epinephrine. Cortisone and various other anti-inflammatory drugs also have been used to reduce type I reactions.

Antibody-Mediated Cytotoxic (Type II) Hypersensitivity

Type II hypersensitive reactions involve antibody-mediated destruction of cells. Antibody can activate the complement system, creating pores in the membrane of a foreign cell (see Figure 13-5), or it can mediate cell destruction by antibody-dependent cell-mediated cytotoxicity (ADCC). In this process, cytotoxic cells with Fc receptors bind to the Fc region of antibodies on target cells and promote killing of the cells (see Figure 14-12). Antibody bound to a foreign cell also can serve as an opsonin, enabling phagocytic cells with Fc or C3b receptors to bind and phagocytose the antibody-coated cell (see Figure 13-12).

This section examines three examples of type II hypersensitive reactions. Certain autoimmune diseases involve autoantibody-mediated cellular destruction by type II mechanisms. These diseases are described in Chapter 20.

Transfusion Reactions Are Type II Reactions

A large number of proteins and glycoproteins on the membrane of red blood cells are encoded by different genes, each of which has a number of alternative alleles. An individual possessing one allelic form of a blood-group antigen can recognize other allelic forms on transfused blood as foreign and mount an antibody response. In some cases, the antibodies have already been induced by natural exposure to similar antigenic determinants on a variety of microorganisms present in the normal flora of the gut. This is the case with the ABO blood-group antigens (Figure 16-13a).

Antibodies to the A, B, and O antigens, called isohemagglutinins, are usually of the IgM class. An individual with blood type A, for example, recognizes B-like epitopes on intestinal microorganisms and produces isohemagglutinins to the B-like epitopes. This same individual does not respond to A-like epitopes on the same intestinal microorganisms because these A-like epitopes are too similar to self and a state of self-tolerance to these epitopes should exist (Figure 16-13b). If a type A individual is transfused with blood containing type B cells, a **transfusion reaction** occurs in which the anti-B isohemagglutinins bind to the B blood cells and mediate their

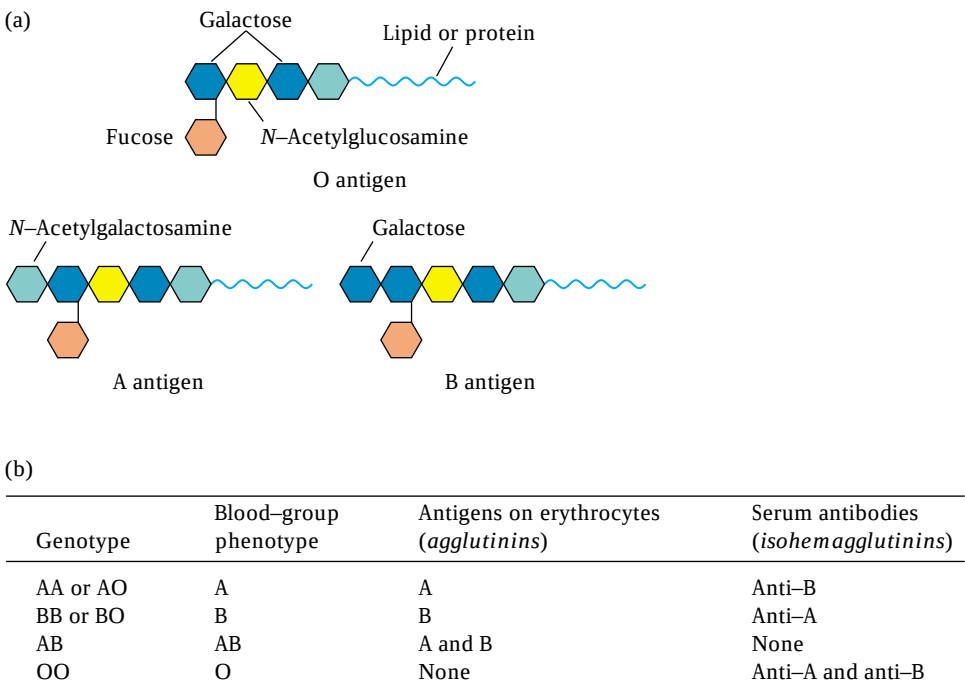


FIGURE 16-13 ABO blood group. (a) Structure of terminal sugars, which constitute the distinguishing epitopes, in the A, B, and O

blood antigens. (b) ABO genotypes and corresponding phenotypes, agglutinins, and isohemagglutinins.

destruction by means of complement-mediated lysis. Antibodies to other blood-group antigens may result from repeated blood transfusions because minor allelic differences in these antigens can stimulate antibody production. These antibodies are usually of the IgG class.

The clinical manifestations of transfusion reactions result from massive intravascular hemolysis of the transfused red blood cells by antibody plus complement. These manifestations may be either immediate or delayed. Reactions that begin immediately are most commonly associated with ABO blood-group incompatibilities, which lead to complement-mediated lysis triggered by the IgM isohemagglutinins. Within hours, free hemoglobin can be detected in the plasma; it is filtered through the kidneys, resulting in hemoglobinuria. Some of the hemoglobin gets converted to bilirubin, which at high levels is toxic. Typical symptoms include fever, chills, nausea, clotting within blood vessels, pain in the lower back, and hemoglobin in the urine. Treatment involves prompt termination of the transfusion and maintenance of urine flow with a diuretic, because the accumulation of hemoglobin in the kidney can cause acute tubular necrosis.

Delayed hemolytic transfusion reactions generally occur in individuals who have received repeated transfusions of ABO-compatible blood that is incompatible for other blood-group antigens. The reactions develop between 2 and 6 days after transfusion, reflecting the secondary nature of these reactions. The transfused blood induces clonal selection and production of IgG against a variety of blood-group membrane antigens, most commonly Rh, Kidd, Kell, and Duffy. The predominant isotype involved in these reactions is IgG, which is less effective than IgM in activating complement. For this reason, complement-mediated lysis of the transfused red blood cells is incomplete, and many of the transfused cells are destroyed at extravascular sites by agglutination, opsonization, and subsequent phagocytosis by macrophages. Symptoms include fever, low hemoglobin, increased bilirubin, mild jaundice, and anemia. Free hemoglobin is usually not detected in the plasma or urine in these reactions because RBC destruction occurs in extravascular sites.

Hemolytic Disease of the Newborn Is Caused by Type II Reactions

Hemolytic disease of the newborn develops when maternal IgG antibodies specific for fetal blood-group antigens cross the placenta and destroy fetal red blood cells. The consequences of such transfer can be minor, serious, or lethal. Severe hemolytic disease of the newborn, called **erythroblastosis fetalis**, most commonly develops when an Rh⁺ fetus expresses an **Rh antigen** on its blood cells that the Rh⁻ mother does not express.

During pregnancy, fetal red blood cells are separated from the mother's circulation by a layer of cells in the placenta called the trophoblast. During her first pregnancy with an Rh⁺ fetus, an Rh⁻ woman is usually not exposed to enough fetal red blood cells to activate her Rh-specific B cells. At the

time of delivery, however, separation of the placenta from the uterine wall allows larger amounts of fetal umbilical-cord blood to enter the mother's circulation. These fetal red blood cells activate Rh-specific B cells, resulting in production of Rh-specific plasma cells and memory B cells in the mother. The secreted IgM antibody clears the Rh⁺ fetal red cells from the mother's circulation, but the memory cells remain, a threat to any subsequent pregnancy with an Rh⁺ fetus. Activation of these memory cells in a subsequent pregnancy results in the formation of IgG anti-Rh antibodies, which cross the placenta and damage the fetal red blood cells (Figure 16-14). Mild to severe anemia can develop in the fetus, sometimes with fatal consequences. In addition, conversion of hemoglobin to bilirubin can present an additional threat to the newborn because the lipid-soluble bilirubin may accumulate in the brain and cause brain damage.

Hemolytic disease of the newborn caused by Rh incompatibility in a subsequent pregnancy can be almost entirely prevented by administering antibodies against the Rh antigen to the mother within 24–48 h after the first delivery. These antibodies, called **Rhogam**, bind to any fetal red blood cells that enter the mother's circulation at the time of delivery and facilitate their clearance before B-cell activation and ensuing memory-cell production can take place. In a subsequent pregnancy with an Rh⁺ fetus, a mother who has been treated with Rhogam is unlikely to produce IgG anti-Rh antibodies; thus, the fetus is protected from the damage that would occur when these antibodies crossed the placenta.

The development of hemolytic disease of the newborn caused by Rh incompatibility can be detected by testing maternal serum at intervals during pregnancy for antibodies to the Rh antigen. A rise in the titer of these antibodies as pregnancy progresses indicates that the mother has been exposed to Rh antigens and is producing increasing amounts of antibody. The presence of maternal IgG on the surface of fetal red blood cells can be detected by a Coombs test. Isolated fetal red blood cells are incubated with the Coombs reagent, goat antibody to human IgG antibody. If maternal IgG is bound to the fetal red blood cells, the cells agglutinate with the Coombs reagent.

If hemolytic disease caused by Rh incompatibility is detected during pregnancy, the treatment depends on the severity of the reaction. For a severe reaction, the fetus can be given an intrauterine blood-exchange transfusion to replace fetal Rh⁺ red blood cells with Rh⁻ cells. These transfusions are given every 10–21 days until delivery. In less severe cases, a blood-exchange transfusion is not given until after birth, primarily to remove bilirubin; the infant is also exposed to low levels of UV light to break down the bilirubin and prevent cerebral damage. The mother can also be treated during the pregnancy by **plasmapheresis**. In this procedure, a cell-separation machine is used to separate the mother's blood into two fractions, cells and plasma. The plasma containing the anti-Rh antibody is discarded, and the cells are reinfused into the mother in an albumin or fresh-plasma solution.

The majority of cases (65%) of hemolytic disease of the newborn have minor consequences and are caused by ABO

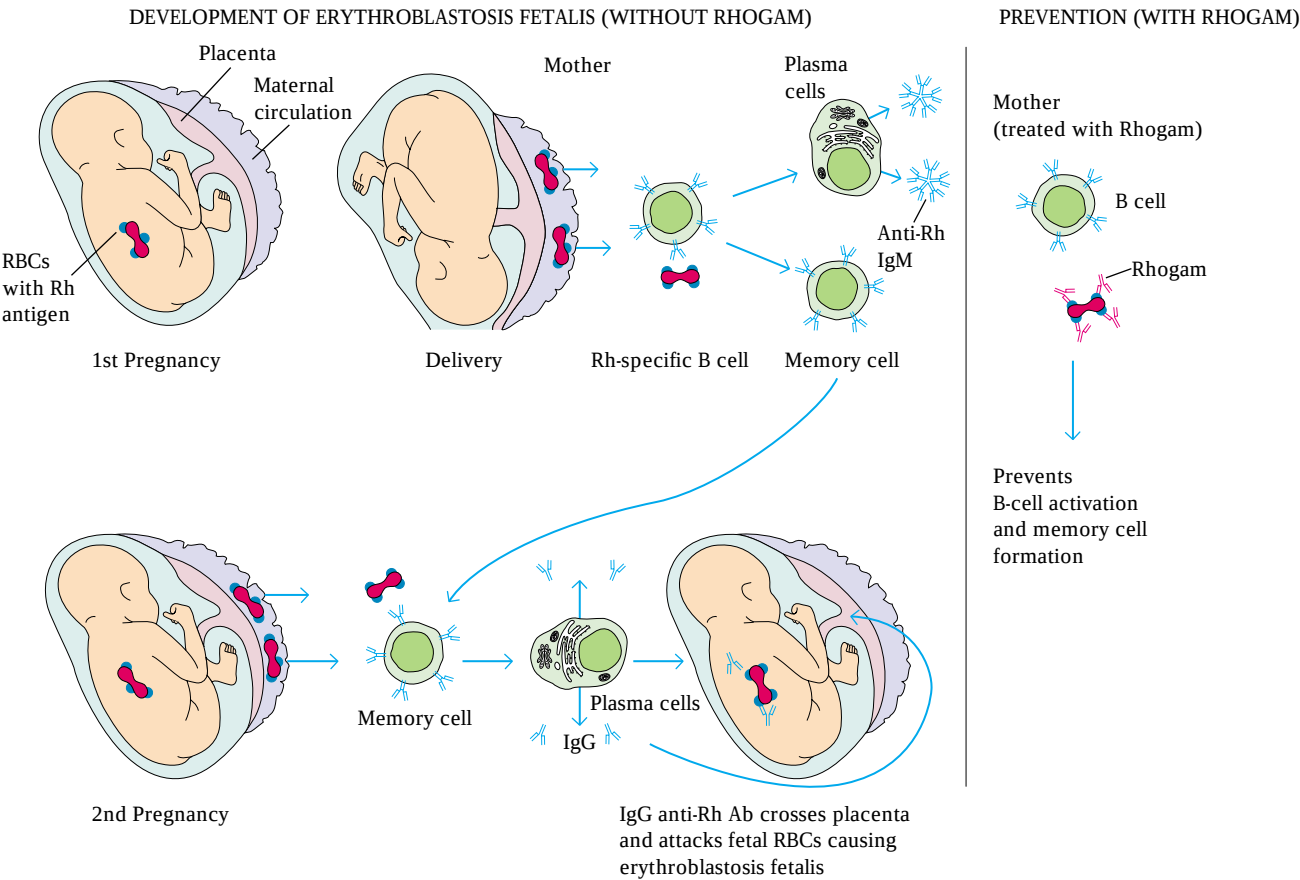


FIGURE 16-14 Development of erythroblastosis fetalis (hemolytic disease of the newborn) caused when an Rh⁻ mother carries an Rh⁺

fetus (left), and effect of treatment with anti-Rh antibody, or Rhogam (right).

blood-group incompatibility between the mother and fetus. Type A or B fetuses carried by type O mothers most commonly develop these reactions. A type O mother is most likely to develop IgG antibody to the A or B blood-group antigens either through natural exposure or through exposure to fetal blood-group A or B antigens in successive pregnancies. Usually the fetal anemia resulting from this incompatibility is mild; the major clinical manifestation is a slight elevation of bilirubin, with jaundice. Depending on the severity of the anemia and jaundice, a blood-exchange transfusion may be required in these infants. In general the reaction is mild, however, and exposure of the infant to low levels of UV light is enough to break down the bilirubin and avoid cerebral damage.

Drug-Induced Hemolytic Anemia Is a Type II Response

Certain antibiotics (e.g., penicillin, cephalosporin, and streptomycin) can adsorb nonspecifically to proteins on RBC membranes, forming a complex similar to a hapten-carrier complex. In some patients, such drug-protein complexes induce formation of antibodies, which then bind to the

adsorbed drug on red blood cells, inducing complement-mediated lysis and thus progressive anemia. When the drug is withdrawn, the hemolytic anemia disappears. Penicillin is notable in that it can induce all four types of hypersensitivity with various clinical manifestations (Table 16-5).

TABLE 16-5 Penicillin-induced hypersensitive reactions		
Type of reaction	Antibody or lymphocytes induced	Clinical manifestations
I	IgE	Urticaria, systemic anaphylaxis
II	IgM, IgG	Hemolytic anemia
III	IgG	Serum sickness, glomerulonephritis
IV	T _{DTH} cells	Contact dermatitis

Immune Complex–Mediated (Type III) Hypersensitivity

The reaction of antibody with antigen generates immune complexes. Generally this complexing of antigen with antibody facilitates the clearance of antigen by phagocytic cells. In some cases, however, large amounts of immune complexes can lead to tissue-damaging type III hypersensitive reactions. The magnitude of the reaction depends on the quantity of immune complexes as well as their distribution within the body. When the complexes are deposited in tissue very near the site of antigen entry, a localized reaction develops. When the complexes are formed in the blood, a reaction can develop wherever the complexes are deposited. In particular, complex deposition is frequently observed on blood-vessel walls, in the synovial membrane of joints, on the glomerular basement membrane of the kidney, and on the choroid plexus of the brain. The deposition of these complexes initiates a reaction that results in the recruitment of neutrophils to the site. The tissue there is injured as a consequence of granular release from the neutrophil.

Type III hypersensitive reactions develop when immune complexes activate the complement system's array of immune effector molecules (see Figure 13-2). As explained in Chapter 13, the C3a, C4a, and C5a complement split products are anaphylatoxins that cause localized mast-cell degranulation and consequent increase in local vascular permeability. C3a, C5a, and C5b67 are also chemotactic factors for neutrophils, which can accumulate in large numbers at the site of immune-complex deposition. Larger immune complexes are deposited on the basement membrane of blood-vessel walls or kidney glomeruli, whereas smaller complexes may pass through the basement membrane and be deposited in the subepithelium. The type of lesion that results depends on the site of deposition of the complexes.

Much of the tissue damage in type III reactions stems from release of lytic enzymes by neutrophils as they attempt to phagocytose immune complexes. The C3b complement component acts as an opsonin, coating immune complexes. A neutrophil binds to a C3b-coated immune complex by means of the type I complement receptor, which is specific for C3b. Because the complex is deposited on the basement-membrane surface, phagocytosis is impeded, so that lytic enzymes are released during the unsuccessful attempts of the neutrophil to ingest the adhering immune complex. Further activation of the membrane-attack mechanism of the complement system can also contribute to the destruction of tissue. In addition, the activation of complement can induce aggregation of platelets, and the resulting release of clotting factors can lead to formation of microthrombi.

Type III Reactions Can Be Localized

Injection of an antigen intradermally or subcutaneously into an animal that has high levels of circulating antibody specific for that antigen leads to formation of localized immune

complexes, which mediate an acute Arthus reaction within 4–8 h (Figure 16-15). Microscopic examination of the tissue reveals neutrophils adhering to the vascular endothelium and then migrating into the tissues at the site of immune-complex deposition. As the reaction develops, localized tissue and vascular damage results in an accumulation of fluid (edema) and red blood cells (erythema) at the site. The severity of the reaction can vary from mild swelling and redness to tissue necrosis.

After an insect bite, a sensitive individual may have a rapid, localized type I reaction at the site. Often, some 4–8 h later, a typical Arthus reaction also develops at the site, with pronounced erythema and edema. Intrapulmonary Arthus-type

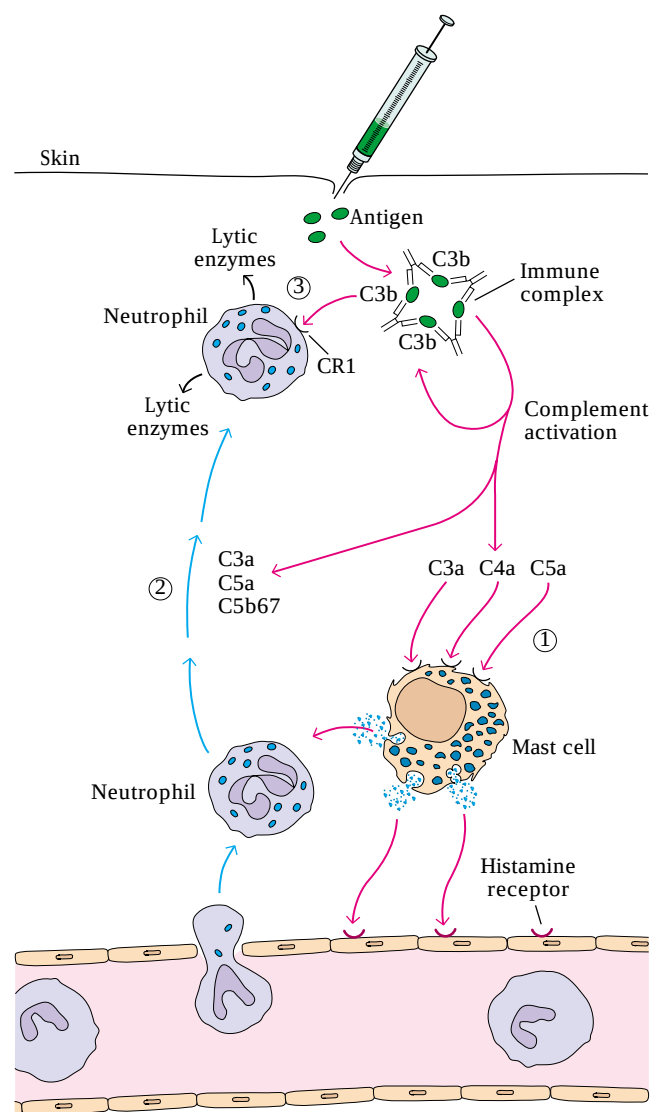


FIGURE 16-15 Development of a localized Arthus reaction (type III hypersensitive reaction). Complement activation initiated by immune complexes (classical pathway) produces complement intermediates that (1) mediate mast-cell degranulation, (2) chemotactically attract neutrophils, and (3) stimulate release of lytic enzymes from neutrophils trying to phagocytose C₃b-coated immune complexes.

reactions induced by bacterial spores, fungi, or dried fecal proteins can also cause pneumonitis or alveolitis. These reactions are known by a variety of common names reflecting the source of the antigen. For example, “farmer’s lung” develops after inhalation of thermophilic actinomycetes from moldy hay, and “pigeon fancier’s disease” results from inhalation of a serum protein in dust derived from dried pigeon feces.

Type III Reactions Can Also Be Generalized

When large amounts of antigen enter the bloodstream and bind to antibody, circulating immune complexes can form. If antigen is in excess, small complexes form; because these are not easily cleared by the phagocytic cells, they can cause tissue-damaging type III reactions at various sites. Historically, generalized type III reactions were often observed after the administration of antitoxins containing foreign serum, such as horse antitetanus or antidiphtheria serum. In such cases, the recipient of a foreign antiserum develops antibodies specific for the foreign serum proteins; these antibodies then form circulating immune complexes with the foreign serum antigens. Typically, within days or weeks after exposure to foreign serum antigens, an individual begins to manifest a combination of symptoms that are called **serum sickness** (Figure 16-16). These symptoms include fever, weakness, generalized vasculitis (rashes) with edema and erythema, lymphadenopathy, arthritis, and sometimes glomerulonephritis. The precise manifestations of serum sickness depend on the quantity of immune complexes formed as well as the overall size of the complexes, which determine the site of their deposition. As mentioned above, the sites of deposition vary but, in general, complexes accumulate in tissues where filtration of plasma occurs. This explains the high incidence of glomerulonephritis (complex deposition in the kidney) and vasculitis (deposition in the arteries) and arthritis (deposition in the synovial joints) caused by serum sickness.

Formation of circulating immune complexes contributes to the pathogenesis of a number of conditions other than serum sickness. These include the following:

- **Autoimmune Diseases**
 - Systemic lupus erythematosus
 - Rheumatoid arthritis
 - Goodpasture’s syndrome
- **Drug Reactions**
 - Allergies to penicillin and sulfonamides
- **Infectious Diseases**
 - Poststreptococcal glomerulonephritis
 - Meningitis
 - Hepatitis
 - Mononucleosis
 - Malaria
 - Trypanosomiasis

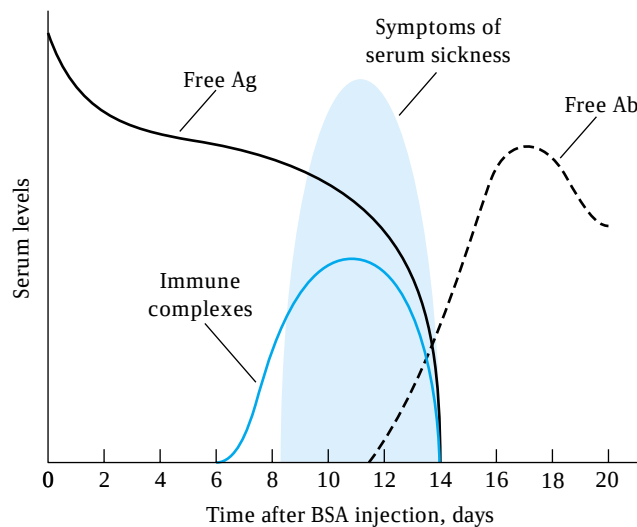


FIGURE 16-16 Correlation between formation of immune complexes and development of symptoms of serum sickness. A large dose of antigen (BSA) was injected into a rabbit at day 0. As antibody formed, it complexed with the antigen and was deposited in the kidneys, joints, and capillaries. The symptoms of serum sickness (light blue area) corresponded to the peak in immune-complex formation. As the immune complexes were cleared, free circulating antibody (dashed black curve) was detected and the symptoms of serum sickness subsided. [Based on F. G. Germuth, Jr., 1953, *J. Exp. Med.* **97**:257.]

Complexes of antibody with various bacterial, viral, and parasitic antigens have been shown to induce a variety of type III hypersensitive reactions, including skin rashes, arthritic symptoms, and glomerulonephritis. Poststreptococcal glomerulonephritis, for example, develops when circulating complexes of antibody and streptococcal antigens are deposited in the kidney and damage the glomeruli. A number of autoimmune diseases stem from circulating complexes of antibody with self-proteins, with glycoproteins, or even with DNA. In systemic lupus erythematosus, complexes of DNA and anti-DNA antibodies accumulate in synovial membranes, causing arthritic symptoms, or accumulate on the basement membrane of the kidney, causing progressive kidney damage.

Type IV or Delayed-Type Hypersensitivity (DTH)

When some subpopulations of activated T_H cells encounter certain types of antigens, they secrete cytokines that induce a localized inflammatory reaction called delayed-type hypersensitivity (DTH). The reaction is characterized by large influxes of nonspecific inflammatory cells, in particular, macrophages. This type of reaction was first described in 1890 by

Robert Koch, who observed that individuals infected with *Mycobacterium tuberculosis* developed a localized inflammatory response when injected intradermally with a filtrate derived from a mycobacterial culture. He called this localized skin reaction a “tuberculin reaction.” Later, as it became apparent that a variety of other antigens could induce this response (Table 16-6), its name was changed to delayed-type or type IV hypersensitivity in reference to the delayed onset of the reaction and to the tissue damage (hypersensitivity) that is often associated with it. The term *hypersensitivity* is somewhat misleading, for it suggests that a DTH response is always detrimental. Although in some cases a DTH response does cause extensive tissue damage and is in itself pathologic, in many cases tissue damage is limited, and the response plays an important role in defense against intracellular pathogens and contact antigens. The hallmarks of a type IV reaction are the delay in time required for the reaction to develop and the recruitment of macrophages as opposed to neutrophils, as found in a type III reaction. Macrophages are the major component of the infiltrate that surrounds the site of inflammation.

There Are Several Phases of the DTH Response

The development of the DTH response begins with an initial sensitization phase of 1–2 weeks after primary contact with an antigen. During this period, T_H cells are activated and clonally expanded by antigen presented together with the requisite class II MHC molecule on an appropriate antigen-presenting cell (Figure 16-17a). A variety of antigen-presenting cells have been shown to be involved in the activation of a DTH response, including Langerhans cells and macrophages. Langerhans cells are dendritic cells found in the epidermis. These cells are thought to pick up antigen that enters through the skin and transport it to regional lymph nodes, where

T cells are activated by the antigen. In some species, including humans, the vascular endothelial cells express class II MHC molecules and also function as antigen-presenting cells in the development of the DTH response. Generally, the T cells activated during the sensitization phase are $CD4^+$, primarily of the T_H1 subtype, but in a few cases $CD8^+$ cells have also been shown to induce a DTH response. The activated T cells previously were called T_{DTH} cells to denote their function in the DTH response, although in reality they are simply a subset of activated T_H1 cells (or, in some cases, T_C cells).

A subsequent exposure to the antigen induces the effector phase of the DTH response (see Figure 16-17b). In the effector phase, T_H1 cells secrete a variety of cytokines that recruit and activate macrophages and other nonspecific inflammatory cells. A DTH response normally does not become apparent until an average of 24 h after the second contact with the antigen; the response generally peaks 48–72 h after second contact. The delayed onset of this response reflects the time required for the cytokines to induce localized influxes of macrophages and their activation. Once a DTH response begins, a complex interplay of nonspecific cells and mediators is set in motion that can result in tremendous amplification. By the time the DTH response is fully developed, only about 5% of the participating cells are antigen-specific T_H1 cells; the remainder are macrophages and other nonspecific cells.

Macrophages are the principal effector cells of the DTH response. Cytokines elaborated by T_H1 cells induce blood monocytes to adhere to vascular endothelial cells and migrate from the blood into the surrounding tissues. During this process the monocytes differentiate into activated macrophages. As Chapter 2 described, activated macrophages exhibit increased levels of phagocytosis and an increased ability to kill microorganisms through various cytotoxic mediators. In addition, activated macrophages express increased levels of class II MHC molecules and cell-adhesion molecules and therefore function more effectively as antigen-presenting cells.

The influx and activation of macrophages in the DTH response is important in host defense against parasites and bacteria that live within cells, where circulating antibodies cannot reach them. The heightened phagocytic activity and the buildup of lytic enzymes from macrophages in the area of infection lead to nonspecific destruction of cells, and thus of the intracellular pathogen. Generally, the pathogen is cleared rapidly with little tissue damage. However, in some cases, especially if the antigen is not easily cleared, a prolonged DTH response can itself become destructive to the host as the intense inflammatory response develops into a visible granulomatous reaction. A granuloma develops when continuous activation of macrophages induces the macrophages to adhere closely to one another, assuming an epithelioid shape and sometimes fusing to form multinucleated giant cells (Figure 16-18). These giant cells displace the normal tissue cells, forming palpable nodules, and release high concentrations of lytic enzymes, which destroy surrounding tissue. In these cases, the response can damage blood vessels and lead

TABLE 16-6

Intracellular pathogens and contact antigens that induce delayed-type (type IV) hypersensitivity

Intracellular bacteria	Intracellular viruses
<i>Mycobacterium tuberculosis</i>	Herpes simplex virus
<i>Mycobacterium leprae</i>	Variola (smallpox)
<i>Listeria monocytogenes</i>	Measles virus
<i>Brucella abortus</i>	
Intracellular fungi	Contact antigens
<i>Pneumocystis carinii</i>	Picrylchloride
<i>Candida albicans</i>	Hair dyes
<i>Histoplasma capsulatum</i>	Nickel salts
<i>Cryptococcus neoformans</i>	Poison ivy
Intracellular parasites	Poison oak
<i>Leishmania</i> sp.	



VISUALIZING CONCEPTS

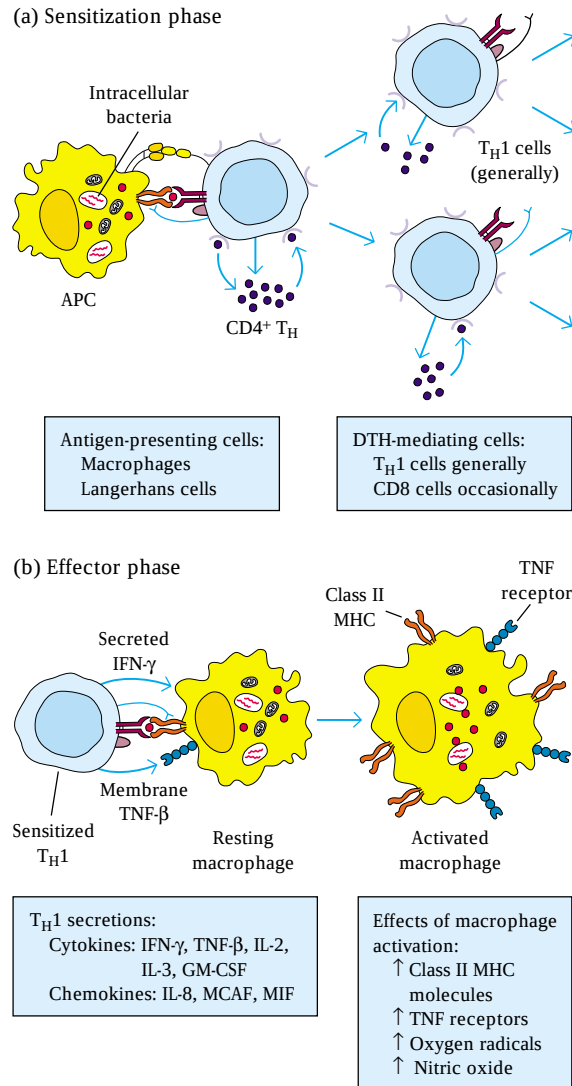


FIGURE 16-17 Overview of the DTH response. (a) In the sensitization phase after initial contact with antigen (e.g., peptides derived from intracellular bacteria), T_H cells proliferate and differentiate into T_H1 cells. Cytokines secreted by these T cells are indicated by the dark blue balls. (b) In the effector phase after subsequent exposure of sen-

sitized T_H1 cells to antigen, the T_H1 cells secrete a variety of cytokines and chemokines. These factors attract and activate macrophages and other nonspecific inflammatory cells. Activated macrophages are more effective in presenting antigen, thus perpetuating the DTH response, and function as the primary effector cells in this reaction.

to extensive tissue necrosis. The response to *Mycobacterium tuberculosis* illustrates the double-edged nature of the DTH response. Immunity to this intracellular bacterium involves a DTH response in which activated macrophages wall off the organism in the lung and contain it within a granuloma-type lesion called a tubercle. Often, however, the concentrated release of lytic enzymes from the activated macrophages within tubercles damages lung tissue. Some examples of truly hypersensitive conditions, in which tissue damage far outweighs any beneficial effects, are described in Chapter 17.

Numerous Cytokines Participate in the DTH Reaction

Among the cytokines produced by T_H1 cells are a number that attract and activate macrophages to the site of infection. IL-3 and GM-CSF induce localized hematopoiesis of the granulocyte-monocyte lineage. IFN- γ and TNF- β (together with macrophage-derived TNF- α and IL-1) act on nearby endothelial cells, inducing a number of changes that facilitate extravasation of monocytes and other nonspecific inflam-

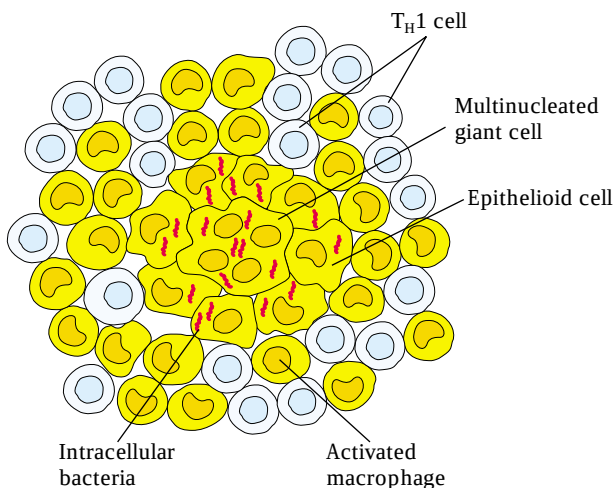


FIGURE 16-18 A prolonged DTH response can lead to formation of a granuloma, a nodule-like mass. Lytic enzymes released from activated macrophages in a granuloma can cause extensive tissue damage.

matory cells. Circulating neutrophils and monocytes adhere to the adhesion molecules displayed on the vascular endothelial cells and extravasate into the tissue spaces. Neutrophils appear early in the reaction, peaking by about 6 h and then declining in numbers. The monocyte infiltration occurs between 24 and 48 h after antigen exposure.

As the monocytes enter the tissues to become macrophages, they are chemotactically drawn to the site of the DTH response by chemokines such as monocyte chemoattractant and activating factor (MCAF). Another chemokine called migration-inhibition factor (MIF) inhibits macrophages from migrating beyond the site of a DTH reaction. As macrophages accumulate at the site of a DTH reaction, they are activated by cytokines, particularly IFN- γ and membrane-bound TNF- β produced by T_H1 cells. As noted earlier, macrophages become more effective as antigen-presenting cells upon activation. Thus, the activated macrophages can efficiently mediate activation of more T cells, which in turn secrete more cytokines that recruit and activate even more macrophages. This self-perpetuating response, however, is a double-edged sword, with a fine line existing between a beneficial, protective response and a detrimental response characterized by extensive tissue damage.

A report of experiments with knockout mice that could not produce IFN- γ demonstrated the importance of this cytokine in the DTH response. When these knockout mice were infected with an attenuated strain of *Mycobacterium bovis* known as BCG (Bacille Calmette Guérin), nearly all the animals died within 60 days, whereas wild-type mice survived (Figure 16-19). Macrophages from the IFN- γ knockout mice were shown to have reduced levels of class II MHC molecules and of bactericidal metabolites such as nitric oxide and superoxide anion.

The DTH Reaction Is Detected with a Skin Test

The presence of a DTH reaction can be measured experimentally by injecting antigen intradermally into an animal and observing whether a characteristic skin lesion develops at the injection site. A positive skin-test reaction indicates that the individual has a population of sensitized T_H1 cells specific for the test antigen. For example, to determine whether an individual has been exposed to *M. tuberculosis*, PPD, a protein derived from the cell wall of this mycobacterium, is injected intradermally. Development of a red, slightly swollen, firm lesion at the site between 48 and 72 h later indicates previous exposure. The skin lesion results from intense infiltration of cells to the site of injection during a DTH reaction; 80%–90% of these cells are macrophages. Note, however, that a positive test does not allow one to conclude whether the exposure was to a pathogenic form of *M. tuberculosis* or to a vaccine form received through immunization, which is performed in some parts of the world.

Contact Dermatitis Is a Type of DTH Response

Many contact-dermatitis reactions, including the responses to formaldehyde, trinitrophenol, nickel, turpentine, and active agents in various cosmetics and hair dyes, poison oak, and poison ivy, are mediated by T_H1 cells. Most of these substances are small molecules that can complex with skin proteins.

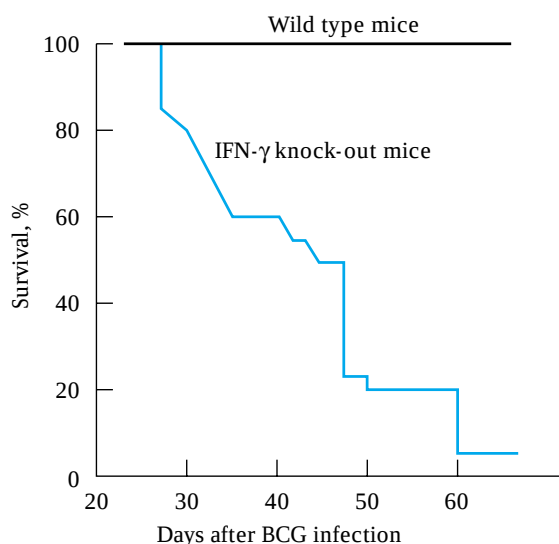


FIGURE 16-19 Experimental demonstration of the role of IFN- γ in host defense against intracellular pathogens. Knockout mice were produced by introducing a targeted mutation in the gene encoding IFN- γ . The mice were then infected with 10^7 colony-forming units of attenuated *Mycobacterium bovis* (BCG) and their survival monitored. [Adapted from D. K. Dalton et al., 1993, Science **259**:1739.]

This complex is internalized by antigen-presenting cells in the skin (e.g., Langerhans cells), then processed and presented together with class II MHC molecules, causing activation of sensitized T_H1 cells. In the reaction to poison oak, for example, a pentadecacatechol compound from the leaves of the plant forms a complex with skin proteins. When T_H cells react with this compound appropriately displayed by local antigen-presenting cells, they differentiate into sensitized T_H1 cells. A subsequent exposure to pentadecacatechol will elicit activation of T_H1 cells and induce cytokine production (Figure 16-20). Approximately 48–72 h after the second exposure, the secreted cytokines cause macrophages to accumulate at the site. Activation of these macrophages and release of lytic enzymes result in the redness and pustules that characterize a reaction to poison oak.

SUMMARY

- Hypersensitive reactions are inflammatory reactions within the humoral or cell-mediated branches of the immune system that lead to extensive tissue damage or even death. The four types of hypersensitive reaction generate characteristic effector molecules and clinical manifestations.
- A type I hypersensitive reaction is mediated by IgE antibodies, whose Fc region binds to receptors on mast cells or blood basophils. Crosslinkage of the fixed IgE by allergen leads to mast cell or basophil degranulation with release of pharmacologically active mediators. The principal effects of these mediators are smooth-muscle contraction and vasodilation. Clinical manifestations of type I reactions include potentially life-threatening systemic anaphylaxis and localized responses such as hay fever and asthma.
- A type II hypersensitive reaction occurs when antibody reacts with antigenic determinants present on the surface of cells, leading to cell damage or death through complement-mediated lysis or antibody-dependent cell-mediated cytotoxicity (ADCC). Transfusion reactions and hemolytic disease of the newborn are type II reactions.
- A type III hypersensitive reaction is mediated by the formation of immune complexes and the ensuing activation of complement. Complement split products serve as immune effector molecules that elicit localized vasodilation and chemotactically attract neutrophils. Deposition of immune complexes near the site of antigen entry can induce an Arthus reaction, in which lytic enzymes released by the accumulated neutrophils and the complement membrane-attack complex cause localized tissue damage.
- A type IV hypersensitive reaction involves the cell-mediated branch of the immune system. Antigen activation of sensitized T_H1 cells induces release of various cytokines that cause macrophages to accumulate and become activated. The net effect of the activation of macrophages is to release lytic enzymes that cause localized tissue damage.

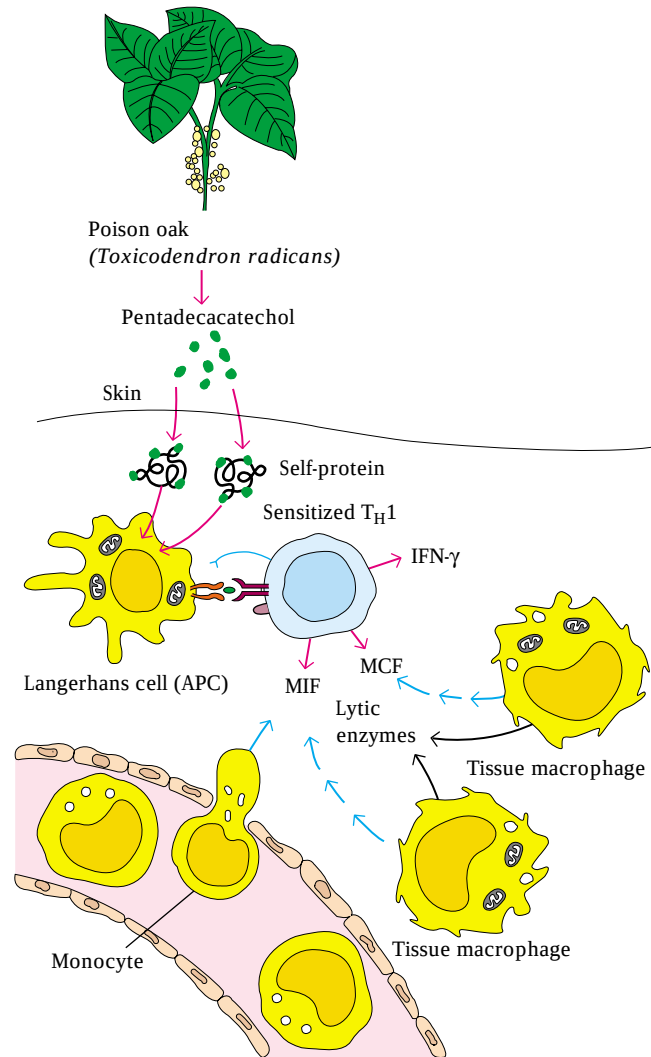


FIGURE 16-20 Development of delayed-type hypersensitivity reaction after a second exposure to poison oak. Cytokines such as IFN- γ , macrophage-chemotactic factor (MCF), and migration-inhibition factor (MIF) released from sensitized T_H1 cells mediate this reaction. Tissue damage results from lytic enzymes released from activated macrophages.

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USEFUL WEB SITES

<http://www.niaid.nih.gov/>

National Institute of Allergy and Infectious Diseases homepage. NIAID is the NIH Institute that sponsors research in infectious diseases. Their Web site provides a number of links to other relevant sites.

<http://allergy.mcg.edu/home.html>

A site maintained by the American College of Allergy, Asthma & Immunology. An excellent source of patient information about many allergies. This site contains many valuable links.

<http://www.glaxowellcome.co.uk/health/actiontb/>

Action TB—A review of tuberculosis for a general audience that appears on the Glaxo-Wellcome company's Web site.

<http://www.aaaai.org/>

The American Association of Allergy, Asthma and Immunology Web site. A good site for exploring the many aspects of asthma.

Study Questions

CLINICAL FOCUS QUESTION Discuss why IL-4 and FcεRIβ are excellent candidate genes involved in the genetic susceptibility to asthma.

1. Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.
 - a. Mice infected with *Nippostrongylus brasiliensis* exhibit decreased production of IgE.
 - b. IL-4 decreases IgE production by B cells.
 - c. The initial step in the process of mast-cell degranulation is crosslinking of Fc receptors.
 - d. Antihistamines are effective for the treatment of type III hypersensitivity.
 - e. Most pollen allergens contain a single allergenic component.
 - f. Babies can acquire IgE-mediated allergies by passive transfer of maternal antibody.
 - g. Transfusion reactions are a manifestation of type II hypersensitivity.
2. In an immunology laboratory exercise, you are studying the response of mice injected intradermally with complete antibodies to the IgE Fc receptor (FcεR1) or with Fab fragments of such antibodies.
 - a. Predict the response expected with each type of antibody.
 - b. Would the responses observed depend on whether the mice were allergic? Explain.



3. Serum sickness can result when an individual is given a large dose of antiserum such as a mouse antitoxin to snake venom. How could you take advantage of recent technological advances to produce an antitoxin that would not produce serum sickness in patients who receive it?

4. What immunologic mechanisms most likely account for a person's developing each of the following reactions after an insect bite?

- Within 1–2 min after being bitten, swelling and redness appear at the site and then disappear by 1 h.
- 6–8 h later, swelling and redness again appear and persist for 24 h.
- 72 h later, the tissue becomes inflamed, and tissue necrosis follows.

5. Indicate which type(s) of hypersensitive reaction (I–IV) apply to the following characteristics. Each characteristic can apply to one, or more than one, type.

- Is an important defense against intracellular pathogens.
- Can be induced by penicillin.
- Involves histamine as an important mediator.
- Can be induced by poison oak in sensitive individuals.
- Can lead to symptoms of asthma.
- Occurs as result of mismatched blood transfusion.
- Systemic form of reaction is treated with epinephrine.
- Can be induced by pollens and certain foods in sensitive individuals.
- May involve cell destruction by antibody-dependent cell-mediated cytotoxicity.
- One form of clinical manifestation is prevented by Rhogam.
- Localized form characterized by wheal and flare reaction.

6. In the table below, indicate whether each immunologic event listed does (+) or does not (–) occur in each type of hypersensitive response.

Immunologic event	Hypersensitivity			
	Type I	Type II	Type III	Type IV
IgE-mediated degranulation of mast cells				
Lysis of antibody-coated blood cells by complement				
Tissue destruction in response to poison oak				
C3a- and C5a-mediated mast-cell degranulation				
Chemotaxis of neutrophils				
Chemotaxis of eosinophils				
Activation of macrophages by IFN- γ				
Deposition of antigen-antibody complexes on basement membranes of capillaries				
Sudden death due to vascular collapse (shock) shortly after injection or ingestion of antigen				

Immune Response to Infectious Diseases

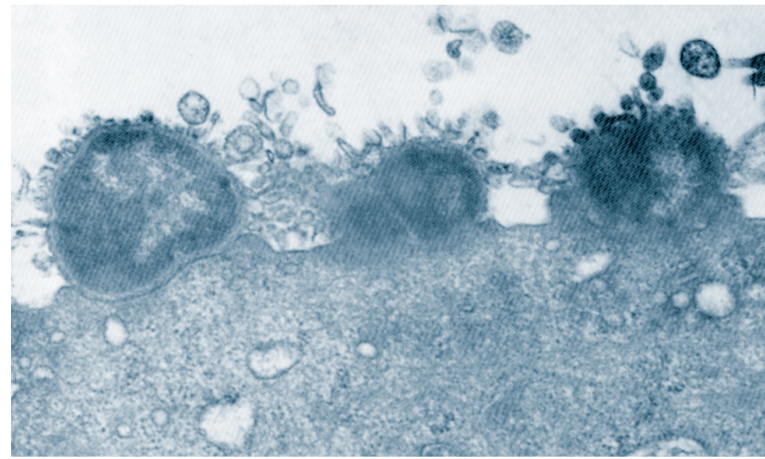
IF A PATHOGEN IS TO ESTABLISH AN INFECTION IN A susceptible host, a series of coordinated events must circumvent both innate and adaptive immunity. One of the first and most important features of host innate immunity is the barrier provided by the epithelial surfaces of the skin and the lining of the gut. The difficulty of penetrating these epithelial barriers ensures that most pathogens never gain productive entry into the host. In addition to providing a physical barrier to infection, the epithelia also produce chemicals that are useful in preventing infection. The secretion of gastric enzymes by specialized epithelial cells lowers the pH of the stomach and upper gastrointestinal tract, and other specialized cells in the gut produce antibacterial peptides.

A major feature of innate immunity is the presence of the normal gut flora, which can competitively inhibit the binding of pathogens to gut epithelial cells. Innate responses can also block the establishment of infection. For example, the cell walls of some gram-positive bacteria contain a peptidoglycan that activates the alternative complement pathway, resulting in the generation of C3b, which opsonizes bacteria and enhances phagocytosis (see Chapter 13). Some bacteria produce endotoxins such as LPS, which stimulate the production of cytokines such as TNF- α , IL-1, and IL-6 by macrophages or endothelial cells. These cytokines can activate macrophages. Phagocytosis of bacteria by macrophages and other phagocytic cells is another highly effective line of innate defense. However, some types of bacteria that commonly grow intracellularly have developed mechanisms that allow them to resist degradation within the phagocyte.

Viruses are well known for the stimulation of innate responses. In particular, many viruses induce the production of interferons, which can inhibit viral replication by inducing an antiviral response. Viruses are also controlled by NK cells. As described in Chapter 14, NK cells frequently form the first line of defense against viral infections.

Generally, pathogens use a variety of strategies to escape destruction by the adaptive immune system. Many pathogens reduce their own antigenicity either by growing within host cells, where they are sequestered from immune attack, or by shedding their membrane antigens. Other pathogens camouflage themselves by mimicking the surfaces of host cells, either by expressing molecules with amino acid sequences similar to those of host cell-membrane molecules or by acquiring a covering of host membrane molecules. Some pathogens are able to suppress the immune response selec-

chapter 17



Neisseria gonorrhoeae Attaching to Urethral Epithelial Cells

- Viral Infections
- Bacterial Infections
- Protozoan Diseases
- Diseases Caused by Parasitic Worms (Helminths)
- Emerging Infectious Diseases

tively or to regulate it so that a branch of the immune system is activated that is ineffective against the pathogen. Continuous variation in surface antigens is another strategy that enables a pathogen to elude the immune system. This antigenic variation may be due to the gradual accumulation of mutations, or it may involve an abrupt change in surface antigens.

Both innate and adaptive immune responses to pathogens provide critical defense, but infectious diseases, which have plagued human populations throughout history, still cause the death of millions each year. Although widespread use of vaccines and drug therapy has drastically reduced mortality from infectious diseases in developed countries, such diseases continue to be the leading cause of death in the Third World. It is estimated that over 1 billion people are infected worldwide, resulting in more than 11 million deaths every year (Figure 17-1). Despite these alarming numbers, estimated expenditures for research on infectious diseases prevalent in the Third World are less than 5% of total health-research expenditures worldwide. Not only is this a tragedy for these countries, but some of these diseases are beginning to emerge or re-emerge in developed countries. For

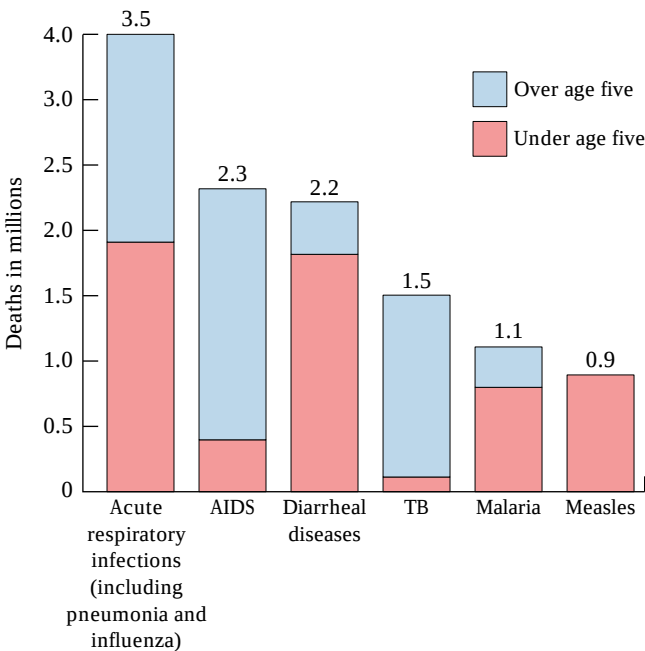


FIGURE 17-1 Leading infectious disease killers. Data collected and compiled by the World Health Organization in 2000 for deaths in 1998. HIV-infected individuals who died of TB are included among AIDS deaths.

example, some United States troops returned from the Persian Gulf with leishmaniasis; cholera cases have recently increased worldwide, with more than 100,000 cases reported in KwaZulu-Natal, South Africa, during the summer of 2001;

and a new drug-resistant strain of *Mycobacterium tuberculosis* is spreading at an alarming rate in the United States.

In this chapter, the concepts described in earlier chapters, antigenicity (Chapter 3) and immune effector mechanisms (Chapters 12–16), as well as vaccine development (which will be considered in Chapter 18) are applied to selected infectious diseases caused by viruses, bacteria, protozoa, and helminths—the four main types of pathogens.

Viral Infections

A number of specific immune effector mechanisms, together with nonspecific defense mechanisms, are called into play to eliminate an infecting virus (Table 17-1). At the same time, the virus acts to subvert one or more of these mechanisms to prolong its own survival. The outcome of the infection depends on how effectively the host’s defensive mechanisms resist the offensive tactics of the virus.

The innate immune response to viral infection is primarily through the induction of type I interferons (IFN- α and IFN- β) and the activation of NK cells. Double stranded RNA (dsRNA) produced during the viral life cycle can induce the expression of IFN- α and IFN- β by the infected cell. Macrophages, monocytes, and fibroblasts also are capable of synthesizing these cytokines, but the mechanisms that induce the production of type I interferons in these cells are not completely understood. IFN- α and IFN- β can induce an antiviral response or resistance to viral replication by binding to the IFN α/β receptor. Once bound, IFN- α and IFN- β activate the JAK-STAT pathway, which in turn induces the transcription of several genes. One of these genes encodes an

TABLE 17-1 Mechanisms of humoral and cell-mediated immune responses to viruses

Response type	Effector molecule or cell	Activity
Humoral	Antibody (especially, secretory IgA)	Blocks binding of virus to host cells, thus preventing infection or reinfection
	IgG, IgM, and IgA antibody	Blocks fusion of viral envelope with host-cells plasma membrane
	IgG and IgM antibody	Enhances phagocytosis of viral particles (opsonization)
	IgM antibody	Agglutinates viral particles
	Complement activated by IgG or IgM antibody	Mediates opsonization by C3b and lysis of enveloped viral particles by membrane-attack complex
Cell-mediated	IFN- γ secreted by T _H or T _C cells	Has direct antiviral activity
	Cytotoxic T lymphocytes (CTLs)	Kill virus-infected self-cells
	NK cells and macrophages	Kill virus-infected cells by antibody-dependent cell-mediated cytotoxicity (ADCC)

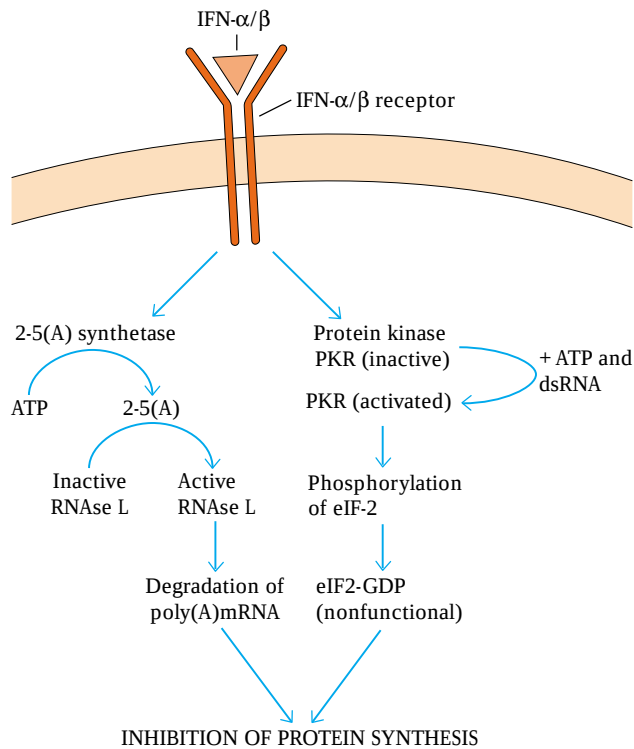


FIGURE 17-2 Induction of antiviral activity by IFN- α and - β . These interferons bind to the IFN receptor, which in turn induces the synthesis of both 2-5(A) synthetase and protein kinase (PKR). The action of 2-5(A) synthetase results in the activation of RNase L, which can degrade mRNA. PKR inactivates the translation initiation factor eIF-2 by phosphorylating it. Both pathways thus result in the inhibition of protein synthesis and thereby effectively block viral replication.

enzyme known as 2'-5'-oligo-adenylate synthetase [2-5(A) synthetase], which activates a ribonuclease (RNase L) that degrades viral RNA. Other genes activated by IFN- α/β binding to its receptor also contribute to the inhibition of viral replication. For example, IFN- α/β binding induces a specific protein kinase called dsRNA-dependent protein kinase (PKR), which inactivates protein synthesis, thus blocking viral replication in infected cells (Figure 17-2).

The binding of IFN- α and IFN- β to NK cells induces lytic activity, making them very effective in killing virally infected cells. The activity of NK cells is also greatly enhanced by IL-12, a cytokine that is produced very early in a response to viral infection.

Many Viruses are Neutralized by Antibodies

Antibodies specific for viral surface antigens are often crucial in containing the spread of a virus during acute infection and in protecting against reinfection. Antibodies are particularly effective in protecting against infection if they are localized at the site of viral entry into the body. Most viruses express sur-

face receptor molecules that enable them to initiate infection by binding to specific host-cell membrane molecules. For example, influenza virus binds to sialic acid residues in cell-membrane glycoproteins and glycolipids; rhinovirus binds to intercellular adhesion molecules (ICAMs); and Epstein-Barr virus binds to type 2 complement receptors on B cells. If antibody to the viral receptor is produced, it can block infection altogether by preventing the binding of viral particles to host cells. Secretory IgA in mucous secretions plays an important role in host defense against viruses by blocking viral attachment to mucosal epithelial cells. The advantage of the attenuated oral polio vaccine, considered in Chapter 18, is that it induces production of secretory IgA, which effectively blocks attachment of poliovirus along the gastrointestinal tract.

Viral neutralization by antibody sometimes involves mechanisms that operate after viral attachment to host cells. In some cases, antibodies may block viral penetration by binding to epitopes that are necessary to mediate fusion of the viral envelope with the plasma membrane. If the induced antibody is of a complement-activating isotype, lysis of enveloped virions can ensue. Antibody or complement can also agglutinate viral particles and function as an opsonizing agent to facilitate Fc- or C3b-receptor-mediated phagocytosis of the viral particles.

Cell-Mediated Immunity is Important for Viral Control and Clearance

Although antibodies have an important role in containing the spread of a virus in the acute phases of infection, they are not usually able to eliminate the virus once infection has occurred—particularly if the virus is capable of entering a latent state in which its DNA is integrated into host chromosomal DNA. Once an infection is established, cell-mediated immune mechanisms are most important in host defense. In general, CD8⁺ T_C cells and CD4⁺ T_H1 cells are the main components of cell-mediated antiviral defense, although in some cases CD4⁺ T_C cells have also been implicated. Activated T_H1 cells produce a number of cytokines, including IL-2, IFN- γ , and TNF, that defend against viruses either directly or indirectly. IFN- γ acts directly by inducing an antiviral state in cells. IL-2 acts indirectly by assisting in the recruitment of CTL precursors into an effector population. Both IL-2 and IFN- γ activate NK cells, which play an important role in host defense during the first days of many viral infections until a specific CTL response develops.

In most viral infections, specific CTL activity arises within 3–4 days after infection, peaks by 7–10 days, and then declines. Within 7–10 days of primary infection, most virions have been eliminated, paralleling the development of CTLs. CTLs specific for the virus eliminate virus-infected self-cells and thus eliminate potential sources of new virus. The role of CTLs in defense against viruses is demonstrated by the ability of virus-specific CTLs to confer protection for the specific virus on nonimmune recipients by adoptive transfer. The viral specificity of the CTL as well can be demonstrated with

adoptive transfer: adoptive transfer of a CTL clone specific for influenza virus strain X protects mice against influenza virus X but not against influenza virus strain Y.

Viruses Can Evade Host Defense Mechanisms

Despite their restricted genome size, a number of viruses have been found to encode proteins that interfere at various levels with specific or nonspecific host defenses. Presumably, the advantage of such proteins is that they enable viruses to replicate more effectively amidst host antiviral defenses. As described above, the induction of IFN- α and IFN- β is a major innate defense against viral infection, but some viruses have developed strategies to evade the action of IFN- α/β . These include hepatitis C virus, which has been shown to overcome the antiviral effect of the interferons by blocking or inhibiting the action of PKR (see Figure 17-2).

Another mechanism for evading host responses, utilized in particular by herpes simplex viruses (HSV) is inhibition of antigen presentation by infected host cells. HSV-1 and HSV-2 both express an immediate-early protein (a protein synthesized shortly after viral replication) called ICP47, which very effectively inhibits the human transporter molecule needed for antigen processing (TAP; see Figure 8-8). Inhibition of TAP blocks antigen delivery to class I MHC receptors on HSV-infected cells, thus preventing presentation of viral antigen to CD8⁺ T cells. This results in the trapping of empty class I MHC molecules in the endoplasmic reticulum and effectively shuts down a CD8⁺ T-cell response to HSV-infected cells.

The targeting of MHC molecules is not unique to HSV. Other viruses have been shown to down-regulate class I MHC expression shortly after infection. Two of the best-characterized examples, the adenoviruses and cytomegalovirus (CMV), use distinct molecular mechanisms to reduce the surface expression of class I MHC molecules, again inhibiting antigen presentation to CD8⁺ T cells. Some viruses—CMV, measles virus, and HIV—have been shown to reduce levels of class II MHC molecules on the cell surface, thus blocking the function of antigen-specific antiviral helper T cells.

Antibody-mediated destruction of viruses requires complement activation, resulting either in direct lysis of the viral particle or opsonization and elimination of the virus by phagocytic cells. A number of viruses have strategies for evading complement-mediated destruction. Vaccinia virus, for example, secretes a protein that binds to the C4b complement component, inhibiting the classical complement pathway; and herpes simplex viruses have a glycoprotein component that binds to the C3b complement component, inhibiting both the classical and alternative pathways.

A number of viruses escape immune attack by constantly changing their antigens. In the influenza virus, continual antigenic variation results in the frequent emergence of new infectious strains. The absence of protective immunity to

these newly emerging strains leads to repeated epidemics of influenza. Antigenic variation among rhinoviruses, the causative agent of the common cold, is responsible for our inability to produce an effective vaccine for colds. Nowhere is antigenic variation greater than in the human immunodeficiency virus (HIV), the causative agent of AIDS. Estimates suggest that HIV accumulates mutations at a rate 65 times faster than does influenza virus. Because of the importance of AIDS, a section of Chapter 19 addresses this disease.

A large number of viruses evade the immune response by causing generalized immunosuppression. Among these are the paramyxoviruses that cause mumps, the measles virus, Epstein-Barr virus (EBV), cytomegalovirus, and HIV. In some cases, immunosuppression is caused by direct viral infection of lymphocytes or macrophages. The virus can then either directly destroy the immune cells by cytolytic mechanisms or alter their function. In other cases, immunosuppression is the result of a cytokine imbalance. For example, EBV produces a protein, called BCRF1, that is homologous to IL-10; like IL-10, BCRF1 suppresses cytokine production by the T_H1 subset, resulting in decreased levels of IL-2, TNF, and IFN- γ .

Influenza Has Been Responsible for Some of the Worst Pandemics in History

The influenza virus infects the upper respiratory tract and major central airways in humans, horses, birds, pigs, and even seals. In 1918–19, an influenza pandemic (worldwide epidemic) killed more than 20 million people, a toll surpassing the number of casualties in World War I. Some areas, such as Alaska and the Pacific Islands, lost more than half of their population during that pandemic.

PROPERTIES OF THE INFLUENZA VIRUS

Influenza viral particles, or virions, are roughly spherical or ovoid in shape, with an average diameter of 90–100 nm. The virions are surrounded by an outer envelope—a lipid bilayer acquired from the plasma membrane of the infected host cell during the process of budding. Inserted into the envelope are two glycoproteins, **hemagglutinin (HA)** and **neuraminidase (NA)**, which form radiating projections that are visible in electron micrographs (Figure 17-3). The hemagglutinin projections, in the form of trimers, are responsible for the attachment of the virus to host cells. There are approximately 1000 hemagglutinin projections per influenza virion. The hemagglutinin trimer binds to sialic acid groups on host-cell glycoproteins and glycolipids by way of a conserved amino acid sequence that forms a small groove in the hemagglutinin molecule. Neuraminidase, as its name indicates, cleaves *N*-acetylneuraminic (sialic) acid from nascent viral glycoproteins and host-cell membrane glycoproteins, an activity that presumably facilitates viral budding from the infected host cell. Within the envelope, an inner layer of matrix protein surrounds the nucleocapsid, which consists of eight dif-

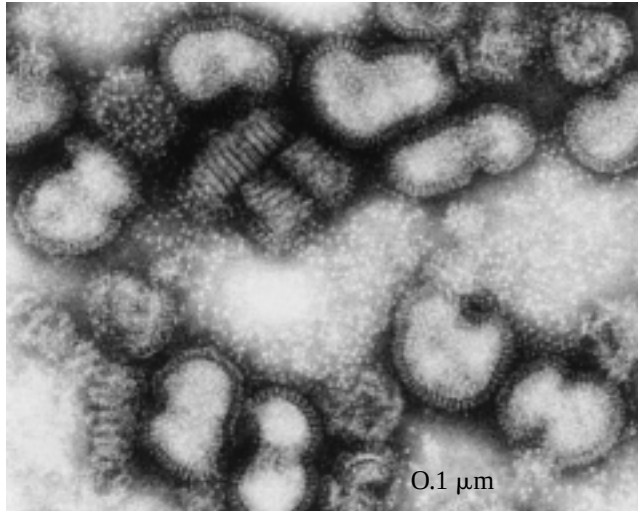


FIGURE 17-3 Electron micrograph of influenza virus reveals roughly spherical viral particles enclosed in a lipid bilayer with protruding hemagglutinin and neuraminidase glycoprotein spikes. [Courtesy of G. Murti, Department of Virology, St. Jude Children's Research Hospital, Memphis, Tenn.]

ferent strands of single-stranded RNA (ssRNA) associated with protein and RNA polymerase (Figure 17-4). Each RNA strand encodes one or more different influenza proteins.

Three basic types of influenza (A, B, and C), can be distinguished by differences in their nucleoprotein and matrix proteins. Type A, which is the most common, is responsible for the major human pandemics. Antigenic variation in hemagglutinin and neuraminidase distinguishes subtypes of type A influenza virus. According to the nomenclature of the World Health Organization, each virus strain is defined by its animal host of origin (specified, if other than human), geographical origin, strain number, year of isolation, and antigenic description of HA and NA (Table 17-2). For example, A/Sw/Iowa/15/30 (H1N1) designates strain-A isolate 15 that arose in swine in Iowa in 1930 and has antigenic subtypes 1 of HA and NA. Notice that the H and N spikes are antigenically distinct in these two strains. There are 13 different hemagglutinins and 9 neuraminidases among the type A influenza viruses.

The distinguishing feature of influenza virus is its variability. The virus can change its surface antigens so completely that the immune response to infection with the virus that caused a previous epidemic gives little or no protection against the virus causing a subsequent epidemic. The antigenic variation results primarily from changes in the hemagglutinin and neuraminidase spikes protruding from the viral envelope (Figure 17-5). Two different mechanisms generate antigenic variation in HA and NA: antigenic drift and antigenic shift. **Antigenic drift** involves a series of spontaneous point mutations that occur gradually, resulting in minor changes in HA and NA. **Antigenic shift** results in the sudden

emergence of a new subtype of influenza whose HA and possibly also NA are considerably different from that of the virus present in a preceding epidemic.

The first time a human influenza virus was isolated was in 1934; this virus was given the subtype designation H0N1 (where H is hemagglutinin and N is neuraminidase). The H0N1 subtype persisted until 1947, when a major antigenic shift generated a new subtype, H1N1, which supplanted the previous subtype and became prevalent worldwide until 1957, when H2N2 emerged. The H2N2 subtype prevailed for the next decade and was replaced in 1968 by H3N2. Antigenic shift in 1977 saw the re-emergence of H1N1. The most recent antigenic shift, in 1989, brought the re-emergence of H3N2, which remained dominant throughout the next several years. However, an H1N1 strain re-emerged in Texas in 1995, and current influenza vaccines contain both H3N2 and H1N1 strains. With each antigenic shift, hemagglutinin and neuraminidase undergo major sequence changes, resulting in major antigenic variations for which the immune system lacks memory. Thus, each antigenic shift finds the population immunologically unprepared, resulting in major outbreaks of influenza, which sometimes reach pandemic proportions.

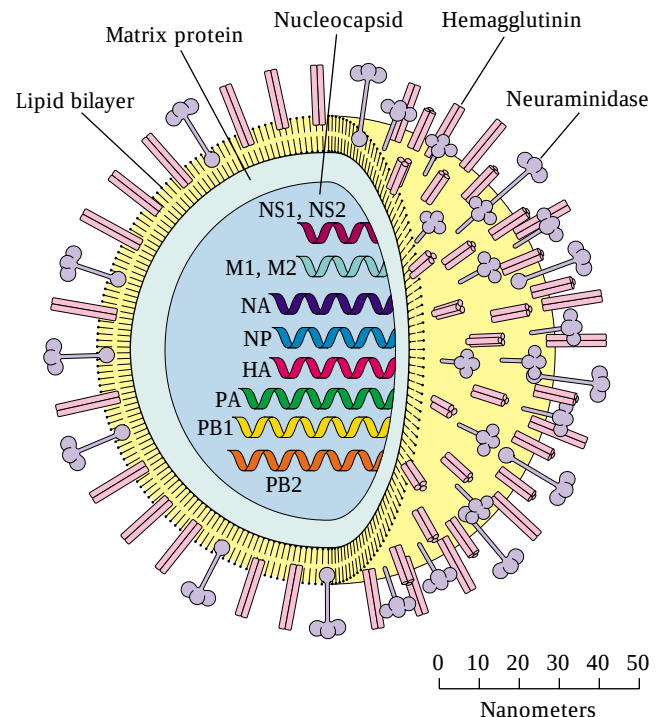


FIGURE 17-4 Schematic representation of influenza structure. The envelope is covered with neuraminidase and hemagglutinin spikes. Inside is an inner layer of matrix protein surrounding the nucleocapsid, which consists of eight ssRNA molecules associated with nucleoprotein. The eight RNA strands encode ten proteins: PB1, PB2, PA, HA (hemagglutinin), NP (nucleoprotein), NA (neuraminidase), M1, M2, NS1, and NS2.

TABLE 17-2

Some influenza A strains and their hemagglutinin (H) and neuraminidase (N) subtype

Species	Virus strain designation	Antigenic subtype
Human	A/Puerto Rico/8/34	H0N1
	A/Fort Monmouth/1/47	H1N1
	A/Singapore/1/57	H2N2
	A/Hong Kong/1/68	H3N2
	A/USSR/80/77	H1N1
	A/Brazil/11/78	H1N1
	A/Bangkok/1/79	H3N2
	A/Taiwan/1/86	H1N1
	A/Shanghai/16/89	H3N2
	A/Johannesburg/33/95	H3N2
	A/Wuhan/359/95	H3N2
	A/Texas/36/95	H1N1
	A/Hong Kong/156/97	H5N1
Swine	A/Sw/Iowa/15/30	H1N1
	A/Sw/Taiwan/70	H3N2
Horse (equine)	A/Eq/Prague/1/56	H7N7
	A/Eq/Miami/1/63	H3N8
Birds	A/Fowl/Dutch/27	H7N7
	A/Tern/South America/61	H5N3
	A/Turkey/Ontario/68	H8N4
	A/Chicken/Hong Kong/258/97	H5N1

Between pandemic-causing antigenic shifts, the influenza virus undergoes antigenic drift, generating minor antigenic variations, which account for strain differences within a subtype. The immune response contributes to the emergence of these different influenza strains. As individuals infected with a given influenza strain mount an effective immune response, the strain is eliminated. However, the accumulation of point mutations sufficiently alters the antigenicity of some variants so that they are able to escape immune elimination (Figure 17-6a). These variants become a new strain of influenza, causing another local epidemic cycle. The role of antibody in such immunologic selection can be demonstrated in the laboratory by mixing an influenza strain with a monoclonal antibody specific for that strain and then culturing the virus in cells. The antibody neutralizes all unaltered viral particles and only those viral particles with mutations resulting in altered antigenicity escape neutralization and are able to continue the infection. Within a short time in culture, a new influenza strain can be shown to emerge.

Antigenic shift is thought to occur through genetic reassortment between influenza virions from humans and from various animals, including horses, pigs, and ducks (Figure 17-6b). The fact that influenza contains eight separate strands of ssRNA makes possible the reassortment of the RNA strands of human and animal virions within a single cell infected with both viruses. Evidence for *in vivo* genetic reassortment between influenza A viruses from humans and domestic pigs was obtained in 1971. After infecting a pig simultaneously with human Hong Kong influenza (H3N2) and with swine influenza (H1N1), investigators were able to recover virions expressing H3N1. In some cases, an apparent antigenic shift may represent the re-emergence of a previous strain that has remained hidden for several decades. In May of 1977, a strain of influenza, A/USSR/77 (H1N1), appeared that proved to be identical to a strain that had caused an epidemic 27 years earlier. The virus could have been preserved over the years in a frozen state or in an animal reservoir. When such a re-emergence occurs, the HA and NA antigens expressed are not really new; however, they will be seen by the immune system of anyone not previously exposed to that strain (people under the age of twenty-seven in the 1977 epidemic, for example) as if they were new because no memory cells specific for these antigenic subtypes will exist in the susceptible population. Thus, from an immunologic point of view, the re-emergence of an old influenza A strain

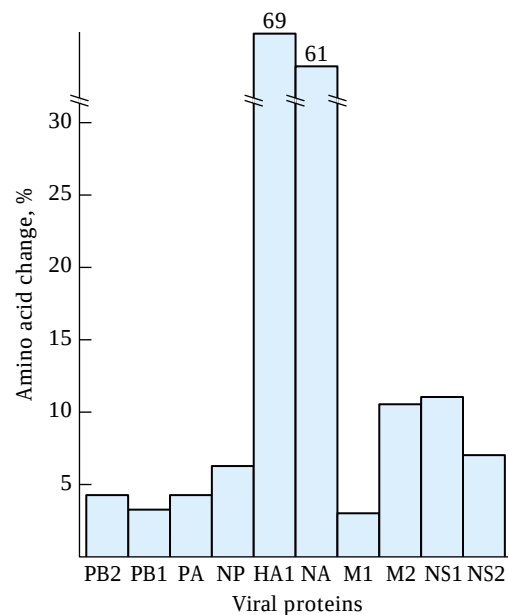


FIGURE 17-5 Amino acid sequence variation in 10 influenza viral proteins from two H3N2 strains and one H1N1 strain. The surface glycoproteins hemagglutinin (HA1) and neuraminidase (NA) show significant sequence variation; in contrast, the sequences of internal viral proteins, such as matrix proteins (M1 and M2) and nucleoprotein (NP), are largely conserved. [Adapted from G. G. Brownlee, 1986, in *Options for the Control of Influenza*, Alan R. Liss.]

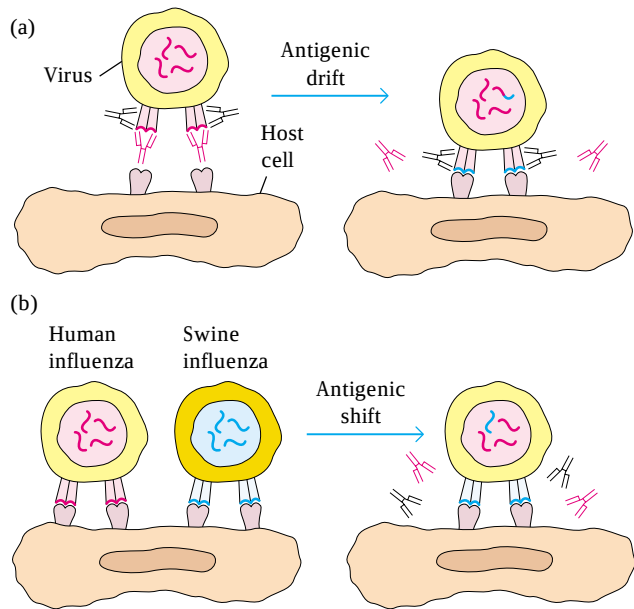


FIGURE 17-6 Two mechanisms generate variations in influenza surface antigens. (a) In antigenic drift, the accumulation of point mutations eventually yields a variant protein that is no longer recognized by antibody to the original antigen. (b) Antigenic shift may occur by re-assortment of an entire ssRNA between human and animal virions infecting the same cell. Only four of the eight RNA strands are depicted.

can have the same effect as an antigenic shift that generates a new subtype.

HOST RESPONSE TO INFLUENZA INFECTION

Humoral antibody specific for the HA molecule is produced during an influenza infection. This antibody confers protection against influenza, but its specificity is strain-specific and is readily bypassed by antigenic drift. Antigenic drift in the HA molecule results in amino acid substitutions in several antigenic domains at the molecule's distal end (Figure 17-7). Two of these domains are on either side of the conserved sialic-acid-binding cleft, which is necessary for binding of virions to target cells. Serum antibodies specific for these two regions are important in blocking initial viral infectivity. These antibody titers peak within a few days of infection and then decrease over the next 6 months; the titers then plateau and remain fairly stable for the next several years. This antibody does not appear to be required for recovery from influenza, as patients with agammaglobulinemia recover from the disease. Instead, the serum antibody appears to play a significant role in resistance to reinfection by the same strain. When serum-antibody levels are high for a particular HA molecule, both mice and humans are resistant to infection by virions expressing that HA molecule. If mice are infected with influenza virus and antibody production is experimentally suppressed, the mice recover from the infection but can be reinfected with the same viral strain. In addition to

humoral responses, CTLs can play a role in immune responses to influenza.

Bacterial Infections

Immunity to bacterial infections is achieved by means of antibody unless the bacterium is capable of intracellular growth, in which case delayed-type hypersensitivity has an important role. Bacteria enter the body either through a number of natural routes (e.g., the respiratory tract, the gastrointestinal tract, and the genitourinary tract) or through normally inaccessible routes opened up by breaks in mucous membranes or skin. Depending on the number of organisms

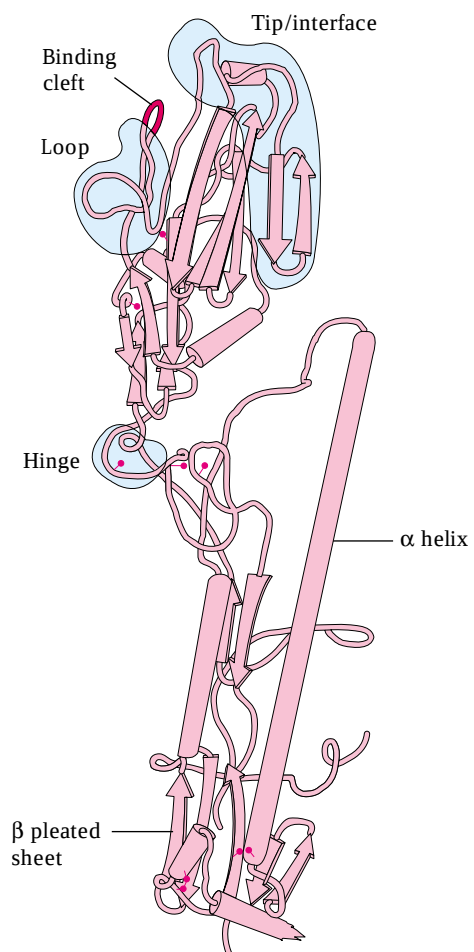


FIGURE 17-7 Structure of hemagglutinin molecule. Sialic acid on host cells interacts with the binding cleft, which is bounded by regions—designated the loop and tip/interface—where antigenic drift is prevalent (blue areas). Antibodies to these regions are important in blocking viral infections. Continual changes in amino acid residues in these regions allow the influenza virus to evade the antibody response. Small red dots represent residues that exhibit a high degree of variation among virus strains. [Adapted from D. C. Wiley et al., 1981, *Nature* **289**:373.]

entering and their virulence, different levels of host defense are enlisted. If the inoculum size and the virulence are both low, then localized tissue phagocytes may be able to eliminate the bacteria with an innate, nonspecific defense. Larger inoculums or organisms with greater virulence tend to induce an adaptive, specific immune response.

Immune Responses to Extracellular and Intracellular Bacteria Can Differ

Infection by extracellular bacteria induces production of humoral antibodies, which are ordinarily secreted by plasma cells in regional lymph nodes and the submucosa of the respiratory and gastrointestinal tracts. The humoral immune response is the main protective response against extracellular bacteria. The antibodies act in several ways to protect the host from the invading organisms, including removal of the bacteria and inactivation of bacterial toxins (Figure 17-8). Extracellular bacteria can be pathogenic because they induce a localized inflammatory response or because they produce toxins. The toxins, endotoxin or exotoxin, can be cytotoxic but also may cause pathogenesis in other ways. An excellent example of this is the toxin produced by diphtheria, which exerts a toxic effect on the cell by blocking protein synthesis. Endotoxins, such as lipopolysaccharides (LPS), are generally components of bacterial cell walls, while exotoxins, such as diphtheria toxin, are secreted by the bacteria.

Antibody that binds to accessible antigens on the surface of a bacterium can, together with the C3b component of complement, act as an opsonin that increases phagocytosis and thus clearance of the bacterium (see Figure 17-8). In the case of some bacteria—notably, the gram-negative organisms—complement activation can lead directly to lysis of the organism. Antibody-mediated activation of the complement system can also induce localized production of immune effector molecules that help to develop an amplified and more effective inflammatory response. For example, the complement split products C3a, C4a, and C5a act as anaphylatoxins, inducing local mast-cell degranulation and thus vasodilation and the extravasation of lymphocytes and neutrophils from the blood into tissue space (see Figure 17-8). Other complement split products serve as chemotactic factors for neutrophils and macrophages, thereby contributing to the buildup of phagocytic cells at the site of infection. Antibody to a bacteria toxin may bind to the toxin and neutralize it; the antibody-toxin complexes are then cleared by phagocytic cells in the same manner as any other antigen-antibody complex.

While innate immunity is not very effective against intracellular bacterial pathogens, intracellular bacteria can activate NK cells, which, in turn, provide an early defense against these bacteria. Intracellular bacterial infections tend to induce a cell-mediated immune response, specifically, delayed-type hypersensitivity. In this response, cytokines secreted by CD4⁺ T cells are important—notably IFN- γ , which activates

macrophages to kill ingested pathogens more effectively (see Figure 14-15).

Bacteria Can Effectively Evade Host Defense Mechanisms

There are four primary steps in bacterial infection:

- Attachment to host cells
- Proliferation
- Invasion of host tissue
- Toxin-induced damage to host cells

Host-defense mechanisms act at each of these steps, and many bacteria have evolved ways to circumvent some of these host defenses (Table 17-3).

Some bacteria have surface structures or molecules that enhance their ability to attach to host cells. A number of gram-negative bacteria, for instance, have pili (long hairlike projections), which enable them to attach to the membrane of the intestinal or genitourinary tract (Figure 17-9). Other bacteria, such as *Bordetella pertussis*, secrete adhesion molecules that attach to both the bacterium and the ciliated epithelial cells of the upper respiratory tract.

Secretory IgA antibodies specific for such bacterial structures can block bacterial attachment to mucosal epithelial cells and are the main host defense against bacterial attachment. However, some bacteria (e.g., *Neisseria gonorrhoeae*, *Haemophilus influenzae*, and *Neisseria meningitidis*) evade the IgA response by secreting proteases that cleave secretory IgA at the hinge region; the resulting Fab and Fc fragments have a shortened half-life in mucous secretions and are not able to agglutinate microorganisms.

Some bacteria evade the IgA response of the host by changing these surface antigens. In *N. gonorrhoeae*, for example, pilin, the protein component of the pili, has a highly variable structure. Variation in the pilin amino acid sequence is generated by gene rearrangements of its coding sequence. The pilin locus consists of one or two expressed genes and 10–20 silent genes. Each gene is arranged into six regions called *minicassettes*. Pilin variation is generated by a process of gene conversion, in which one or more minicassettes from the silent genes replace a minicassette of the expression gene. This process generates enormous antigenic variation, which may contribute to the pathogenicity of *N. gonorrhoeae* by increasing the likelihood that expressed pili will bind firmly to epithelial cells. In addition, the continual changes in the pilin sequence allow the organism to evade neutralization by IgA.

Some bacteria possess surface structures that serve to inhibit phagocytosis. A classic example is *Streptococcus pneumoniae*, whose polysaccharide capsule prevents phagocytosis very effectively. There are 84 serotypes of *S. pneumoniae* that differ from one another by distinct capsular polysaccharides.





VISUALIZING CONCEPTS

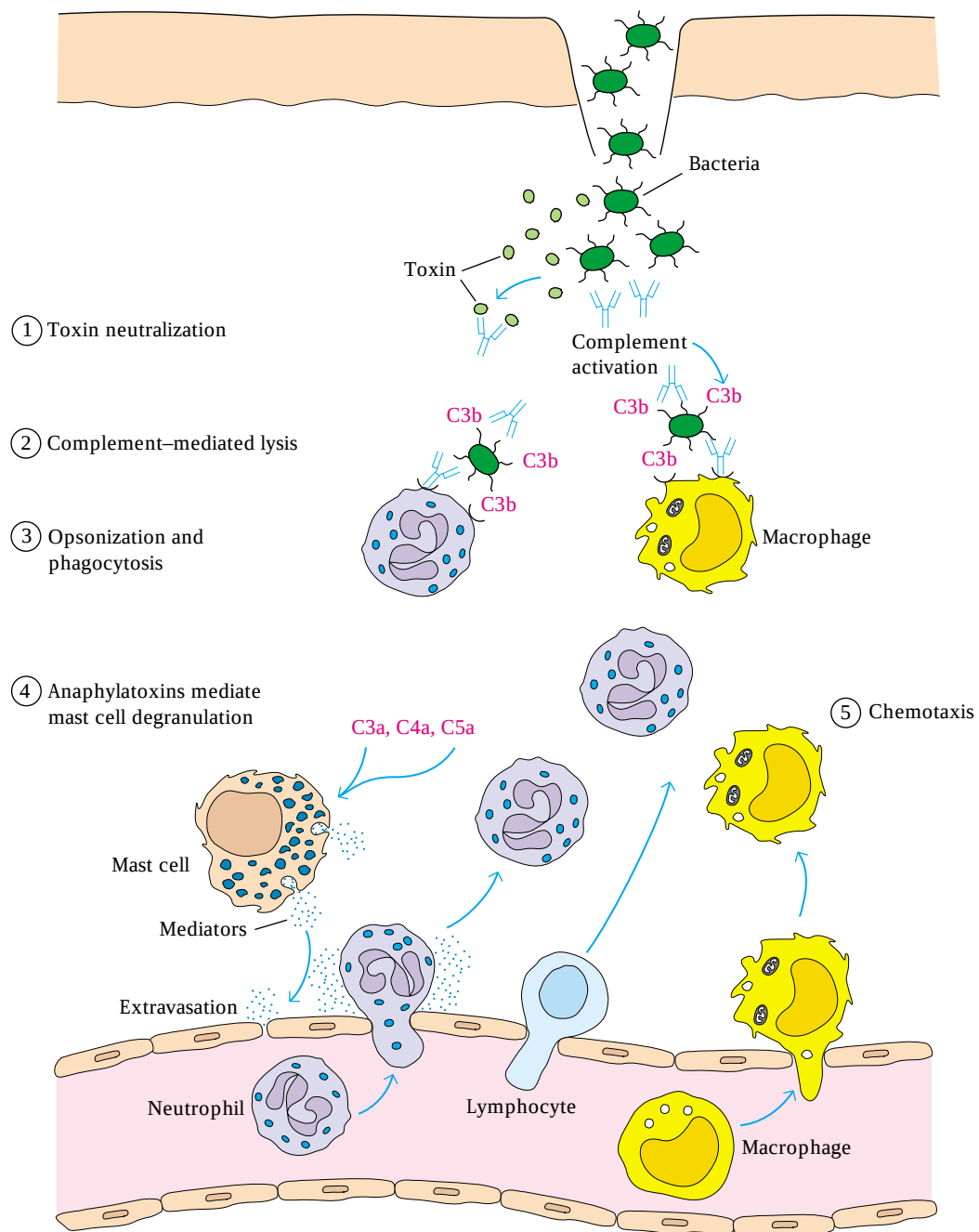


FIGURE 17-8 Antibody-mediated mechanisms for combating infection by extracellular bacteria. (1) Antibody neutralizes bacterial toxins. (2) Complement activation on bacterial surfaces leads to complement-mediated lysis of bacteria. (3) Antibody and the complement split product C₃b bind to bacteria, serving as opsonins to

increase phagocytosis. (4) C3a, C4a, and C5a, generated by antibody-initiated complement activation, induce local mast cell degranulation, releasing substances that mediate vasodilation and extravasation of lymphocytes and neutrophils. (5) Other complement split products are chemotactic for neutrophils and macrophages.

TABLE 17-3 Host immune responses to bacterial infection and bacterial evasion mechanisms

Infection process	Host defense	Bacterial evasion mechanisms
Attachment to host cells	Blockage of attachment by secretory IgA antibodies	Secretion of proteases that cleave secretory IgA dimers (<i>Neisseria meningitidis</i> , <i>N. gonorrhoeae</i> , <i>Haemophilus influenzae</i>) Antigenic variation in attachment structures (pili of <i>N. gonorrhoeae</i>)
Proliferation	Phagocytosis (Ab- and C3b-mediated opsonization) Complement-mediated lysis and localized inflammatory response	Production of surface structures (polysaccharide capsule, M protein, fibrin coat) that inhibit phagocytic cells Mechanisms for surviving within phagocytic cells Induction of apoptosis in macrophages (<i>Shigella flexneri</i>) Generalized resistance of gram-positive bacteria to complement-mediated lysis Insertion of membrane-attack complex prevented by long side chain in cell-wall LPS (some gram-negative bacteria)
Invasion of host tissues	Ab-mediated agglutination	Secretion of elastase that inactivates C3a and C5a (<i>Pseudomonas</i>)
Toxin-induced damage to host cells	Neutralization of toxin by antibody	Secretion of hyaluronidase, which enhances bacterial invasiveness

During infection, the host produces antibody against the infecting serotype. This antibody protects against reinfection with the same serotype but will not protect against infection by a different serotype. In this way, *S. pneumoniae* can cause disease many times in the same individual. On other bacteria, such as *Streptococcus pyogenes*, a surface protein projection called the M protein inhibits phagocytosis. Some pathogenic staphylococci are able to assemble a protective coat from host proteins. These bacteria secrete a coagulase enzyme that precipitates a fibrin coat around them, shielding them from phagocytic cells.

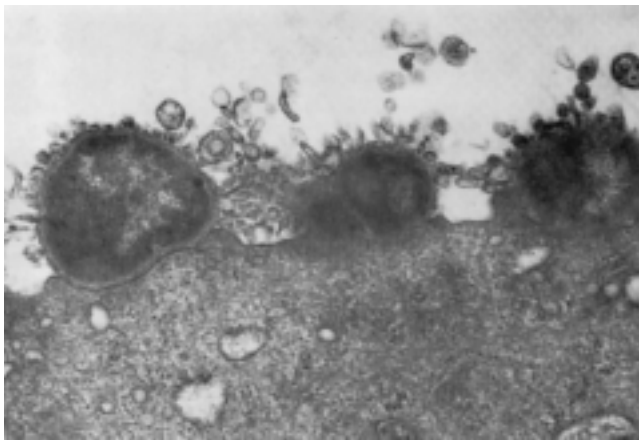


FIGURE 17-9 Electron micrograph of *Neisseria gonorrhoeae* attaching to urethral epithelial cells. Pili (P) extend from the gonococcal surface and mediate the attachment. [From M. E. Ward and P. J. Watt, 1972, J. Inf. Dis. **126**:601.]

Mechanisms for interfering with the complement system help other bacteria survive. In some gram-negative bacteria, for example, long side chains on the lipid A moiety of the cell-wall core polysaccharide help to resist complement-mediated lysis. *Pseudomonas* secretes an enzyme, elastase, that inactivates both the C3a and C5a anaphylatoxins, thereby diminishing the localized inflammatory reaction.

A number of bacteria escape host defense mechanisms by their ability to survive within phagocytic cells. Some, such as *Listeria monocytogenes*, do this by escaping from the phagolysosome to the cytoplasm, which is a more favorable environment for their growth. Other bacteria, such as *Mycobacterium avium*, block lysosomal fusion with the phagolysosome; and some mycobacteria are resistant to the oxidative attack that takes place within the phagolysosome.

Immune Responses Can Contribute to Bacterial Pathogenesis

In some cases, disease is caused not by the bacterial pathogen itself but by the immune response to the pathogen. As described in Chapter 12, pathogen-stimulated overproduction of cytokines leads to the symptoms of bacterial septic shock, food poisoning, and toxic-shock syndrome. For instance, cell-wall endotoxins of some gram-negative bacteria activate macrophages, resulting in release of high levels of IL-1 and TNF- α , which can cause septic shock. In staphylococcal food poisoning and toxic-shock syndrome, exotoxins produced by the pathogens function as superantigens, which can activate all T cells that express T-cell receptors with a particular V_{β} domain (see Table 10-4). The resulting overproduction of cytokines by activated T_H cells causes many of the symptoms of these diseases.

The ability of some bacteria to survive intracellularly within infected cells can result in chronic antigenic activation of CD4⁺ T cells, leading to tissue destruction by a cell-mediated response with the characteristics of a delayed-type hypersensitivity reaction (see Chapter 14). Cytokines secreted by these activated CD4⁺ T cells can lead to extensive accumulation and activation of macrophages, resulting in formation of a **granuloma**. The localized concentrations of lysosomal enzymes in these granulomas can cause extensive tissue necrosis. Much of the tissue damage seen with *M. tuberculosis* is due to a cell-mediated immune response.

Diphtheria (*Corynebacterium diphtheriae*) May Be Controlled by Immunization with Inactivated Toxoid

Diphtheria is the classic example of a bacterial disease caused by a secreted exotoxin to which immunity can be induced by immunization with an inactivated **toxoid**. The causative agent, a gram-positive, rodlike organism called *Corynebacterium diphtheriae*, was first described by Klebs in 1883 and was shown a year later by Loeffler to cause diphtheria in guinea pigs and rabbits. Autopsies on the infected animals revealed that, while bacterial growth was limited to the site of inoculation, there was widespread damage to a variety of organs, including the heart, liver, and kidneys. This finding led Loeffler to speculate that the neurologic and cardiologic manifestations of the disease were caused by a toxic substance elaborated by the organism.

Loeffler's hypothesis was validated in 1888, when Roux and Yersin produced the disease in animals by injecting them with a sterile filtrate from a culture of *C. diphtheriae*. Two years later, von Behring showed that an antiserum to the toxin was able to prevent death in infected animals. He prepared a toxoid by treating the toxin with iodine trichloride and demonstrated that it could induce protective antibodies in animals. However, the toxoid was still quite toxic and therefore unsuitable for use in humans. In 1923, Ramon found that exposing the toxin to heat and formalin rendered it nontoxic but did not destroy its antigenicity. Clinical trials showed that formaldehyde-treated toxoid conferred a high level of protection against diphtheria.

As immunizations with the toxoid increased, the number of cases of diphtheria decreased rapidly. In the 1920s, there were approximately 200 cases of diphtheria per 100,000 population in the United States. In 1989, the Centers for Disease Control reported only three cases of diphtheria in the entire United States. Recently in the former Soviet Union, there has been an alarming epidemic of diphtheria due to a reduction in vaccinations.

Natural infection with *C. diphtheriae* occurs only in humans. The disease is spread from one individual to another by airborne respiratory droplets. The organism colonizes the nasopharyngeal tract, remaining in the superficial layers of the respiratory mucosa. Growth of the organism itself causes

little tissue damage, and only a mild inflammatory reaction develops. The virulence of the organism is due completely to its potent exotoxin. The toxin causes destruction of the underlying tissue, resulting in the formation of a tough fibrinous membrane ("pseudomembrane") composed of fibrin, white blood cells, and dead respiratory epithelial cells. The membrane itself can cause suffocation. The exotoxin also is responsible for widespread systemic manifestations. Pronounced myocardial damage (often leading to congestive heart failure) and neurologic damage (ranging from mild weakness to complete paralysis) are common.

The exotoxin that causes diphtheria symptoms is encoded by the *tox* gene carried by phage β . Within some strains of *C. diphtheriae*, phage β can exist in a state of **lysogeny**, in which the β -prophage DNA persists within the bacterial cell. Only strains that carry lysogenic phage β are able to produce the exotoxin. The diphtheria exotoxin contains two disulfide-linked chains, a binding chain and toxin chain. The binding chain interacts with ganglioside receptors on susceptible cells, facilitating internalization of the exotoxin. Toxicity results from the inhibitory effect of the toxin chain on protein synthesis. The diphtheria exotoxin is extremely potent; a single molecule has been shown to kill a cell. Removal of the binding chain prevents the exotoxin from entering the cell, thus rendering the exotoxin nontoxic. As described in Chapter 4, an immunotoxin can be prepared by replacing the binding chain with a monoclonal antibody specific for a tumor-cell surface antigen; in this way the toxin chain can be targeted to tumor cells (see Figure 4-23).

Today, diphtheria toxoid is prepared by treating diphtheria toxin with formaldehyde. The reaction with formaldehyde cross-links the toxin, resulting in an irreversible loss in its toxicity while enhancing its antigenicity. The toxoid is administered together with tetanus toxoid and inactivated *Bordetella pertussis* in a combined vaccine that is given to children beginning at 6–8 weeks of age. Immunization with the toxoid induces the production of antibodies (antitoxin), which can bind to the toxin and neutralize its activity. Because antitoxin levels decline slowly over time, booster doses are recommended at 10-year intervals to maintain antitoxin levels within the protective range. Interestingly, antibodies specific for epitopes on the binding chain of the diphtheria exotoxin are critical for toxin neutralization because these antibodies block internalization of the active toxin chain.

Tuberculosis (*Mycobacterium tuberculosis*) Is Primarily Controlled by CD4⁺ T Cells

Tuberculosis is the leading cause of death in the world from a single infectious agent, killing about 3 million individuals every year and accounting for 18.5% of all deaths in adults between the ages of 15 and 59. About 1.79 billion people, roughly one-third of the world's population, are infected with the causative agent *M. tuberculosis* and are at risk of developing the disease. Long thought to have been eliminated as a

public health problem in the United States, tuberculosis re-emerged in the early 1990s, particularly in the inner cities and in areas where HIV-infection levels are high (see the last section of this chapter). In 2000, approximately 17,000 individuals were diagnosed with tuberculosis in the United States.

Although several *Mycobacterium* species can cause tuberculosis, *M. tuberculosis* is the principal causative agent. This organism is spread easily, and pulmonary infection usually results from inhalation of small droplets of respiratory secretions containing a few bacilli. The inhaled bacilli are ingested by alveolar macrophages and are able to survive and multiply intracellularly by inhibiting formation of phagolysosomes. When the infected macrophages lyse, as they eventually do, large numbers of bacilli are released. A cell-mediated response involving $CD4^+$ T cells, which is required for immunity to tuberculosis, may be responsible for much of the tissue damage in the disease. $CD4^+$ T-cell activity is the basis for the tuberculin skin test to the purified protein derivative (PPD) from *M. tuberculosis* (see Chapter 14).

Upon infection with *M. tuberculosis*, the most common clinical pattern, termed pulmonary tuberculosis, appears in about 90% of those infected. In this pattern, $CD4^+$ T cells are activated within 2–6 weeks after infection, inducing the infiltration of large numbers of activated macrophages. These cells wall off the organism inside a granulomatous lesion called a tubercle (Figure 17-10). A tubercle consists of a few small lymphocytes and a compact collection of activated macrophages, which sometimes differentiate into epithelioid cells or multinucleated giant cells. The massive activation of macrophages that occurs within tubercles often results in the concentrated release of lytic enzymes. These enzymes destroy nearby healthy cells, resulting in circular regions of necrotic tissue, which eventually form a lesion with a caseous (cheese-like) consistency (see Figure 17-10). As these caseous lesions heal, they become calcified and are readily visible on x-rays, where they are called Ghon complexes.

Because the activated macrophages suppress proliferation of the phagocytosed bacilli, infection is contained. Cytokines produced by $CD4^+$ T cells (T_H1 subset) play an important role in the response by activating macrophages, so that they are able to kill the bacilli or inhibit their growth. The role of $IFN-\gamma$ in the immune response to mycobacteria has been demonstrated with knockout mice lacking $IFN-\gamma$. These mice died when they were infected with an attenuated strain of mycobacteria (BCG), whereas $IFN-\gamma^+$ normal mice survive.

Recent studies have revealed high levels of IL-12 in the pleural effusions of tuberculosis patients. The high levels of IL-12, produced by activated macrophages, are not surprising, given the decisive role of IL-12 in stimulating T_H1 -mediated responses (see Figure 12-12). In mouse models of tuberculosis, IL-12 has been shown to increase resistance to the disease. Not only does IL-12 stimulate development of T_H1 cells, but it also may contribute to resistance by inducing the production of chemokines that attract macrophages to the site of infection. When IL-12 is neutralized by antibody to IL-12, granuloma formation in tuberculous mice is blocked.

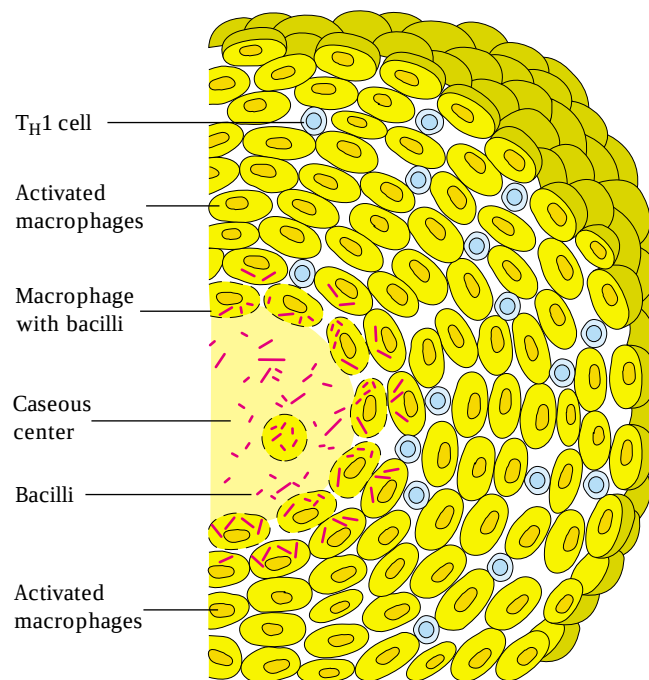


FIGURE 17-10 A tubercle formed in pulmonary tuberculosis. [Modified from A. M. Dannenberg, 1993, *Hosp. Prac.* (Jan. 15):51.]

The $CD4^+$ T-cell-mediated immune response mounted by the majority of people exposed to *M. tuberculosis* thus controls the infection and later protects against reinfection. However, about 10% of individuals infected with *M. tuberculosis* follow a different clinical pattern: the disease progresses to chronic pulmonary tuberculosis or extrapulmonary tuberculosis. This progression may occur years after the primary infection. In this clinical pattern, accumulation of large concentrations of mycobacterial antigens within tubercles leads to extensive and continual chronic $CD4^+$ T-cell activation and ensuing macrophage activation. The resulting high concentrations of lytic enzymes cause the necrotic caseous lesions to liquefy, creating a rich medium that allows the tubercle bacilli to proliferate extracellularly. Eventually the lesions rupture, and the bacilli disseminate in the lung and/or are spread through the blood and lymphatic vessels to the pleural cavity, bone, urogenital system, meninges, peritoneum, or skin.

Tuberculosis is treated with several drugs used in combination, including isoniazid, rifampin, streptomycin, pyrazinamide, and ethambutol. The combination therapy of isoniazid and rifampin has been particularly effective. The intracellular growth of *M. tuberculosis* makes it difficult for drugs to reach the bacilli. For this reason, drug therapy must be continued for at least 9 months to eradicate the bacteria. Some patients with tuberculosis do not exhibit any clinical symptoms, and some patients with symptoms begin to feel better within 2–4 weeks

after treatment begins. To avoid the side effects associated with the usual antibiotic therapy, many patients, once they feel better, stop taking the medications long before the recommended treatment period is completed. Because briefer treatment may not eradicate organisms that are somewhat resistant to the antibiotics, a multidrug-resistant strain can emerge. Noncompliance with required treatment regimes, one of the most troubling aspects of the large number of current tuberculosis cases, clearly compromises efforts to contain the spread of the disease.

Presently, the only vaccine for *M. tuberculosis* is an attenuated strain of *M. bovis* called BCG (Bacillus Calmette-Guerin). The vaccine appears to provide fairly effective protection against extrapulmonary tuberculosis but has been inconsistent against pulmonary tuberculosis. In different studies, BCG has provided protection in anywhere from 0% to 80% of vaccinated individuals; in some cases, BCG vaccination has even increased the risk of infection. Moreover, after BCG vaccination, the tuberculin skin test cannot be used as an effective monitor of exposure to *M. tuberculosis*. Because of the variable effectiveness of the BCG vaccine and the inability to monitor for exposure with the skin test after vaccination, this vaccine is not used in the United States. However, the alarming increase in multidrug-resistant strains has stimulated renewed efforts to develop a more effective tuberculosis vaccine.

Protozoan Diseases

Protozoans are unicellular eukaryotic organisms. They are responsible for several serious diseases in humans, including amoebiasis, Chagas' disease, African sleeping sickness, malaria, leishmaniasis, and toxoplasmosis. The type of immune response that develops to protozoan infection and the effectiveness of the response depend in part on the location of the parasite within the host. Many protozoans have life-cycle stages in which they are free within the bloodstream, and it is during these stages that humoral antibody is most effective. Many of these same pathogens are also capable of intracellular growth; during these stages, cell-mediated immune reactions are effective in host defense. In the development of vaccines for protozoan diseases, the branch of the immune system that is most likely to confer protection must be carefully considered.

Malaria (*Plasmodium* Species) Infects 600 Million People Worldwide

Malaria is one of the most devastating diseases in the world today, infecting nearly 10% of the world population and causing 1–2 million deaths every year. Malaria is caused by various species of the genus *Plasmodium*, of which *P. falciparum* is the most virulent and prevalent. The alarming development of multiple-drug resistance in *Plasmodium* and the increased resistance of its vector, the *Anopheles* mosquito, to DDT underscore the importance of developing new strategies to hinder the spread of malaria.

PLASMODIUM LIFE CYCLE AND PATHOGENESIS OF MALARIA

Plasmodium progresses through a remarkable series of developmental and maturational stages in its extremely complex life cycle. Female *Anopheles* mosquitoes, which feed on blood meals, serve as the vector for *Plasmodium*, and part of the parasite's life cycle takes place within the mosquito. (Because male *Anopheles* mosquitoes feed on plant juices, they do not transmit *Plasmodium*.)

Human infection begins when sporozoites, one of the *Plasmodium* stages, are introduced into an individual's bloodstream as an infected mosquito takes a blood meal (Figure 17-11). Within 30 min, the sporozoites disappear from the

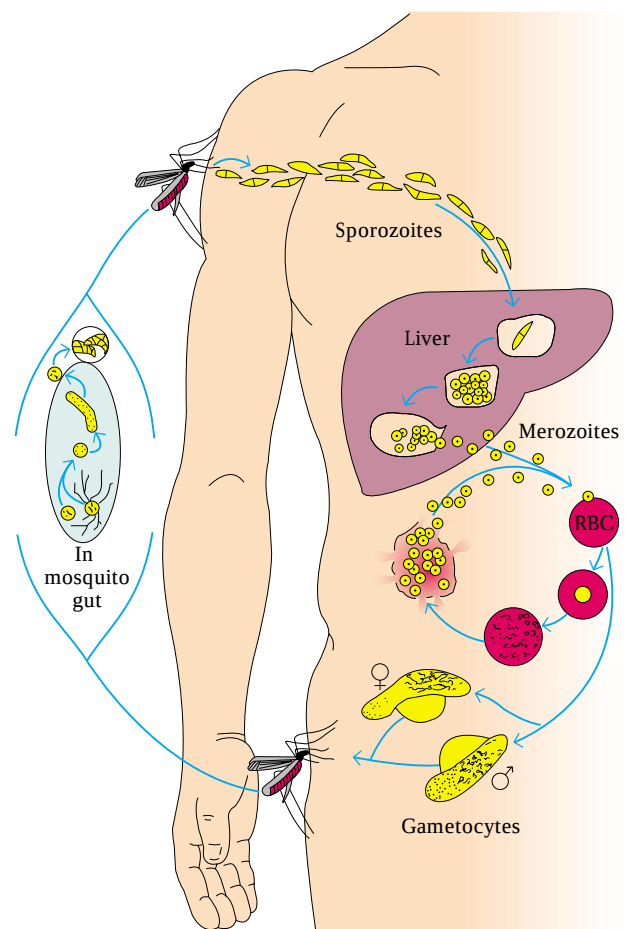


FIGURE 17-11 The life cycle of *Plasmodium*. Sporozoites enter the bloodstream when an infected mosquito takes a blood meal. The sporozoites migrate to the liver, where they multiply, transforming liver hepatocytes into giant multinucleate schizonts, which release thousands of merozoites into the bloodstream. The merozoites infect red blood cells, which eventually rupture, releasing more merozoites. Eventually some of the merozoites differentiate into male and female gametocytes, which are ingested by a mosquito and differentiate into gametes. The gametes fuse to form a zygote that differentiates to the sporozoite stage within the salivary gland of the mosquito.

blood as they migrate to the liver, where they infect hepatocytes. Sporozoites are long, slender cells that are covered by a 45-kDa protein called circumsporozoite (CS) antigen, which appears to mediate their adhesion to hepatocytes. The binding site on the CS antigen is a conserved region in the carboxyl-terminal end (called region II) that has a high degree of sequence homology with known cell-adhesion molecules.

Within the liver, the sporozoites multiply extensively and undergo a complex series of transformations that culminate in the formation and release of merozoites in about a week. It has been estimated that a liver hepatocyte infected with a single sporozoite can release 5,000–10,000 merozoites. The released merozoites infect red blood cells, initiating the symptoms and pathology of malaria. Within a red blood cell, merozoites replicate and undergo successive differentiations; eventually the cell ruptures and releases new merozoites, which go on to infect more red blood cells. Eventually some of the merozoites differentiate into male and female gametocytes, which may be ingested by a female *Anopheles* mosquito during a blood meal. Within the mosquito's gut, the male and female gametocytes differentiate into gametes that fuse to form a zygote, which multiplies and differentiates into sporozoites within the salivary gland. The infected mosquito is now set to initiate the cycle once again.

The symptoms of malaria are recurrent chills, fever, and sweating. The symptoms peak roughly every 48 h, when successive generations of merozoites are released from infected red blood cells. An infected individual eventually becomes weak and anemic and shows splenomegaly. The large numbers of merozoites formed can block capillaries, causing intense headaches, renal failure, heart failure, or cerebral damage—often with fatal consequences. There is speculation that some of the symptoms of malaria may be caused not by *Plasmodium* itself but instead by excessive production of cytokines. This hypothesis stemmed from the observation that cancer patients treated in clinical trials with recombinant tumor necrosis factor (TNF) developed symptoms that mimicked malaria. The relation between TNF and malaria symptoms was studied by infecting mice with a mouse-specific strain of *Plasmodium*, which causes rapid death by cerebral malaria. Injection of these mice with antibodies to TNF was shown to prevent the rapid death.

HOST RESPONSE TO PLASMODIUM INFECTION

In regions where malaria is endemic, the immune response to *Plasmodium* infection is poor. Children less than 14 years old mount the lowest immune response and consequently are most likely to develop malaria. In some regions, the childhood mortality rate for malaria reaches 50%, and worldwide the disease kills about a million children a year. The low immune response to *Plasmodium* among children can be demonstrated by measuring serum antibody levels to the sporozoite stage. Only 22% of the children living in endemic areas have detectable antibodies to the sporozoite stage,

whereas 84% of the adults have such antibodies. Even in adults, the degree of immunity is far from complete, however, and most people living in endemic regions have lifelong low-level *Plasmodium* infections.

A number of factors may contribute to the low levels of immune responsiveness to *Plasmodium*. The maturational changes from sporozoite to merozoite to gametocyte allow the organism to keep changing its surface molecules, resulting in continual changes in the antigens seen by the immune system. The intracellular phases of the life cycle in liver cells and erythrocytes also reduce the degree of immune activation generated by the pathogen and allow the organism to multiply while it is shielded from attack. Furthermore, the most accessible stage, the sporozoite, circulates in the blood for only about 30 min before it infects liver hepatocytes; it is unlikely that much immune activation can occur in such a short period of time. And even when an antibody response does develop to sporozoites, *Plasmodium* has evolved a way of overcoming that response by sloughing off the surface CS-antigen coat, thus rendering the antibodies ineffective.

DESIGN OF MALARIA VACCINES

An effective vaccine for malaria should maximize the most effective immune defense mechanisms. Unfortunately, little is known of the roles that humoral and cell-mediated responses play in the development of protective immunity to this disease. Current approaches to design of malaria vaccines focus largely on the sporozoite stage. One experimental vaccine, for example, consists of *Plasmodium* sporozoites attenuated by x-irradiation. In one study, nine volunteers were repeatedly immunized by the bite of *P. falciparum*-infected, irradiated mosquitoes. Later challenge by the bites of mosquitoes infected with virulent *P. falciparum* revealed that six of the nine recipients were completely protected. These results are encouraging, but translating these findings into mass immunization remains problematic. Sporozoites do not grow well in cultured cells, so an enormous insectory would be required to breed mosquitoes in which to prepare enough irradiated sporozoites to vaccinate just one small village.

Current vaccine strategies are aimed at producing synthetic subunit vaccines consisting of epitopes that can be recognized by T cells and B cells. While no effective vaccine has been developed, this is an active area of investigation.

African Sleeping Sickness (*Trypanosoma* Species)

Two species of African trypanosomes, which are flagellated protozoans, can cause sleeping sickness, a chronic, debilitating disease transmitted to humans and cattle by the bite of the tsetse fly. In the bloodstream, a trypanosome differentiates into a long, slender form that continues to divide every 4–6 h. The disease progresses through several stages, beginning with an early (systemic) stage in which trypanosomes multiply in the blood and progressing to a neurologic stage in which the



VISUALIZING CONCEPTS

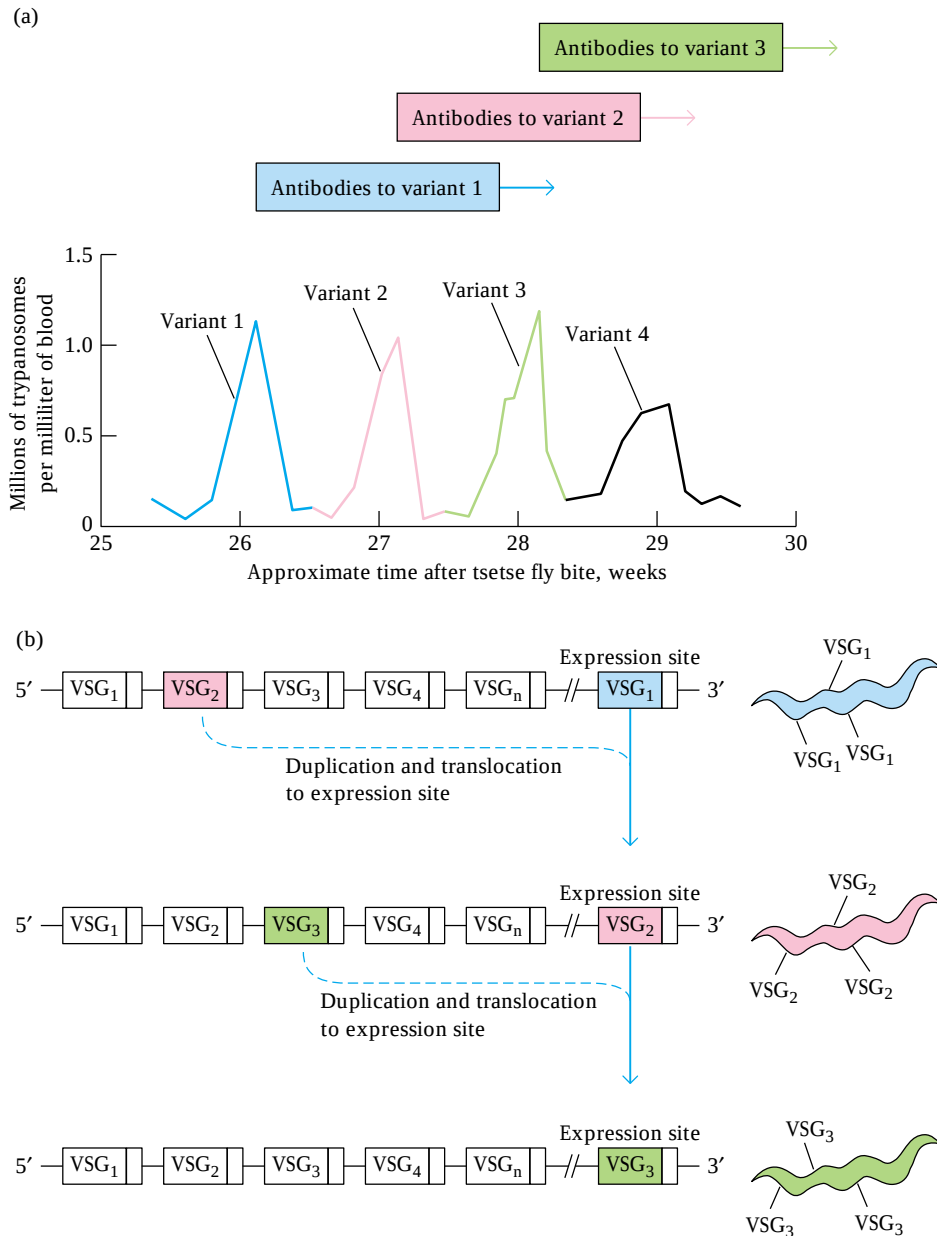


FIGURE 17-12 (a) Successive waves of parasitemia after infection with *Trypanosoma* result from antigenic shifts in the parasite's variant surface glycoprotein (VSG). Each variant that arises is unaffected by the humoral antibodies induced by the previous variant. (b) Anti-

genic shifts in trypanosomes occur by the duplication of gene segments encoding variant VSG molecules and their translocation to an expression site located close to the telomere. [Part (a) adapted from J. Donelson, 1988, *The Biology of Parasitism*, Alan R. Liss.]

parasite infects the central nervous system, causing meningoencephalitis and eventually the loss of consciousness.

As parasite numbers increase after infection, an effective humoral antibody response develops to the glycoprotein

coat, called variant surface glycoprotein (VSG), that covers the trypanosomal surface (Figure 17-12). These antibodies eliminate most of the parasites from the bloodstream, both by complement-mediated lysis and by opsonization and

subsequent phagocytosis. However, about 1% of the organisms, which bear an antigenically different VSG, escape the initial antibody response. These surviving organisms now begin to proliferate in the bloodstream, and a new wave of parasitemia is observed. The successive waves of parasitemia reflect a unique mechanism of antigenic shift by which the trypanosomes can evade the immune response to their glycoprotein antigens. This process is so effective that each new variant that arises in the course of a single infection is able to escape the humoral antibodies generated in response to the preceding variant, so that waves of parasitemia recur (Figure 17-12a).

Several unusual genetic processes generate the extensive variation in trypanosomal VSG that enables the organism to escape immunologic clearance. An individual trypanosome carries a large repertoire of VSG genes, each encoding a different VSG primary sequence. *Trypanosoma brucei*, for example, contains more than 1000 VSG genes in its genome, clustered at multiple chromosomal sites. A trypanosome expresses only a single VSG gene at a time. Activation of a VSG gene results in duplication of the gene and its transposition to a transcriptionally active expression site (ES) at the telomeric end of specific chromosomes (Figure 17-12b). Activation of a new VSG gene displaces the previous gene from the telomeric expression site. A number of chromosomes in the trypanosome have transcriptionally active expression sites at the telomeric ends, so that a number of VSG genes can potentially be expressed, but unknown control mechanisms limit expression to a single VSG expression site at a time.

There appears to be some order in the VSG variation during infection. Each new variant arises not by clonal outgrowth from a single variant cell but instead from the growth of multiple cells that have activated the same VSG gene in the current wave of parasite growth. It is not known how this process is regulated among individual trypanosomes. The continual shifts in epitopes displayed by the VSG make the development of a vaccine for African sleeping sickness extremely difficult.

Leishmaniasis Is a Useful Model for Demonstrating Differences in Host Responses

The protozoan parasite *Leishmania major* provides a powerful and illustrative example of how host responses can differ between individuals. These differences can lead to either clearance of the parasite or fatality from the infection. *Leishmania* is a protozoan that lives in the phagosomes of macrophages. Resistance to the infection correlates well with the production of IFN- γ and the development of a T_H1 response. Elegant studies in mice have demonstrated that strains that are resistant to *Leishmania* develop a T_H1 response and produce IFN- γ upon infection. Such strains of mice become highly susceptible to *Leishmania*-induced fatal-

ity if they lose either IFN- γ or the IFN- γ receptor, further underscoring the importance of IFN- γ in containing the infection. A few strains of mice, such as BALB/c, are highly susceptible to *Leishmania*, and these animals frequently succumb to infection. These mice mount a T_H2 -type response to *Leishmania* infection; they produce high levels of IL-4 and essentially no IFN- γ . Thus, one difference between an effective and an ineffective defense against the parasite is the development of a T_H1 response or a T_H2 response. Recent studies demonstrate that one difference between the resistant strains of mice and BALB/c is that a small restricted subset of BALB/c $CD4^+$ T cells are capable of recognizing a particular epitope on *L. major*, and this subset produces high levels of IL-4 early in the response to the parasite. This skews the response toward a predominantly T_H2 type. Understanding how these different T-helper responses affect the outcome of infection could provide a more rational approach to the design of effective treatments and successful vaccines for other pathogens.

Diseases Caused by Parasitic Worms (Helminths)

Unlike protozoans, which are unicellular and often grow within human cells, helminths are large, multicellular organisms that reside in humans but do not ordinarily multiply there and are not intracellular pathogens. Although helminths are more accessible to the immune system than protozoans, most infected individuals carry few of these parasites; for this reason, the immune system is not strongly engaged and the level of immunity generated to helminths is often very poor.

Parasitic worms are responsible for a wide variety of diseases in both humans and animals. More than a billion people are infected with *Ascaris*, a parasitic roundworm that infects the small intestine, and more than 300 million people are infected with *Schistosoma*, a trematode worm that causes a chronic debilitating infection. Several helminths are important pathogens of domestic animals and invade humans who ingest contaminated food. These helminths include *Taenia*, a tapeworm of cattle and pigs, and *Trichinella*, the roundworm of pigs that causes trichinosis.

Several *Schistosoma* species are responsible for the chronic, debilitating, and sometimes fatal disease **schistosomiasis** (formerly known as *bilharzia*). Three species, *S. mansoni*, *S. japonicum*, and *S. haematobium*, are the major pathogens in humans, infecting individuals in Africa, the Middle East, South America, the Caribbean, China, Southeast Asia, and the Philippines. A rise in the incidence of schistosomiasis in recent years has paralleled the increasing worldwide use of irrigation, which has expanded the habitat of the freshwater snail that serves as the intermediate host for schistosomes.

Infection occurs through contact with free-swimming infectious larvae, called cercariae, which are released from an infected snail at the rate of 300–3000 per day. When cercariae contact human skin, they secrete digestive enzymes that help them to bore into the skin, where they shed their tails and are transformed into schistosomules. The schistosomules enter the capillaries and migrate to the lungs, then to the liver, and finally to the primary site of infection, which varies with the species. *S. mansoni* and *S. japonicum* infect the intestinal mesenteric veins; *S. haematobium* infects the veins of the urinary bladder. Once established in their final tissue site, schistosomules mature into male and female adult worms. The worms mate and the females produce at least 300 spiny eggs a day. Unlike protozoan parasites, schistosomes and other helminths do not multiply within their hosts. The eggs produced by the female worm do not mature into adult worms in humans; instead, some of them pass into the feces or urine and are excreted to infect more snails. The number of worms in an infected individual increases only through repeated exposure to the free-swimming cercariae, and so most infected individuals carry rather low numbers of worms.

Most of the symptoms of schistosomiasis are initiated by the eggs. As many as half of the eggs produced remain in the host, where they invade the intestinal wall, liver, or bladder and cause hemorrhage. A chronic state can then develop in which the adult worms persist and the unexcreted eggs induce cell-mediated delayed-type hypersensitive reactions, resulting in large granulomas that are gradually walled off by fibrous tissue. Although the eggs are contained by the formation of the granuloma, often the granuloma itself obstructs the venous blood flow to the liver or bladder.

Although an immune response does develop to the schistosomes, in most individuals it is not sufficient to eliminate the adult worms, even though the intravascular sites of schistosome infestation should make the worm an easy target for immune attack. Instead, the worms survive for up to 20 years. The schistosomules would appear to be the forms most susceptible to attack, but because they are motile, they can evade the localized cellular buildup of immune and inflammatory cells. Adult schistosome worms also have several unique mechanisms that protect them from immune defenses. The adult worm has been shown to decrease the expression of antigens on its outer membrane and also to enclose itself in a glycolipid and glycoprotein coat derived from the host, masking the presence of its own antigens. Among the antigens observed on the adult worm are the host's own ABO blood-group antigens and histocompatibility antigens! The immune response is, of course, diminished by this covering made of the host's self-antigens, which must contribute to the lifelong persistence of these organisms.

The relative importance of the humoral and cell-mediated responses in protective immunity to schistosomiasis is controversial. The humoral response to infection with *S. mansoni* is characterized by high titers of antischistosome IgE antibodies, localized increases in mast cells and their sub-

sequent degranulation, and increased numbers of eosinophils (Figure 17-13, *top*). These manifestations suggest that cytokines produced by a T_H2 -like subset are important for the immune response: IL-4, which induces B cells to class-switch to IgE production; IL-5, which induces bone-marrow precursors to differentiate into eosinophils; and IL-3, which (along with IL-4) stimulates growth of mast cells. Degranulation of mast cells releases mediators that increase the infiltration of such inflammatory cells as macrophages and eosinophils. The eosinophils express Fc receptors for IgE and IgG and bind to the antibody-coated parasite. Once bound to the parasite, an eosinophil can participate in antibody-dependent cell-mediated cytotoxicity (ADCC), releasing mediators from its granules that damage the parasite (see Figure 14-12). One eosinophil mediator, called basic protein, is particularly toxic to helminths.

Immunization studies with mice, however, suggest that this humoral IgE response may not provide protective immunity. When mice are immunized with *S. mansoni* vaccine, the protective immune response that develops is not an IgE response, but rather a T_H1 response characterized by IFN- γ production and macrophage accumulation (Figure 17-13, *bottom*). Furthermore, inbred strains of mice with deficiencies in mast cells or IgE develop protective immunity from vaccination, whereas inbred strains with deficiencies in cell-mediated $CD4^+$ T-cell responses fail to develop protective immunity in response to the vaccine. These studies suggest that the $CD4^+$ T-cell response may be the most important in immunity to schistosomiasis. It has been suggested that the ability to induce an ineffective T_H2 -like response may have evolved in schistosomes as a clever defense mechanism to ensure that T_H2 cells produced sufficient levels of IL-10 to inhibit protective immunity mediated by the T_H1 -like subset in the $CD4^+$ T response.

Antigens present on the membrane of cercariae and young schistosomules are promising vaccine components because these stages appear to be most susceptible to immune attack. Injecting mice and rats with monoclonal antibodies to cercariae and young schistosomules passively transferred resistance to infection with live cercariae. When these protective antibodies were used in affinity columns to purify schistosome membrane antigens from crude membrane extracts, it was found that mice immunized and boosted with these purified antigens exhibited increased resistance to a later challenge with live cercariae. Schistosome cDNA libraries were then established and screened with the protective monoclonal antibodies to identify those cDNAs encoding surface antigens. Experiments using cloned cercariae or schistosomule antigens are presently under way to assess their ability to induce protective immunity in animal models. However, in developing an effective vaccine for schistosomiasis, a fine line separates a beneficial immune response, which at best limits the parasite load, from a detrimental response, which in itself may become pathologic.



VISUALIZING CONCEPTS

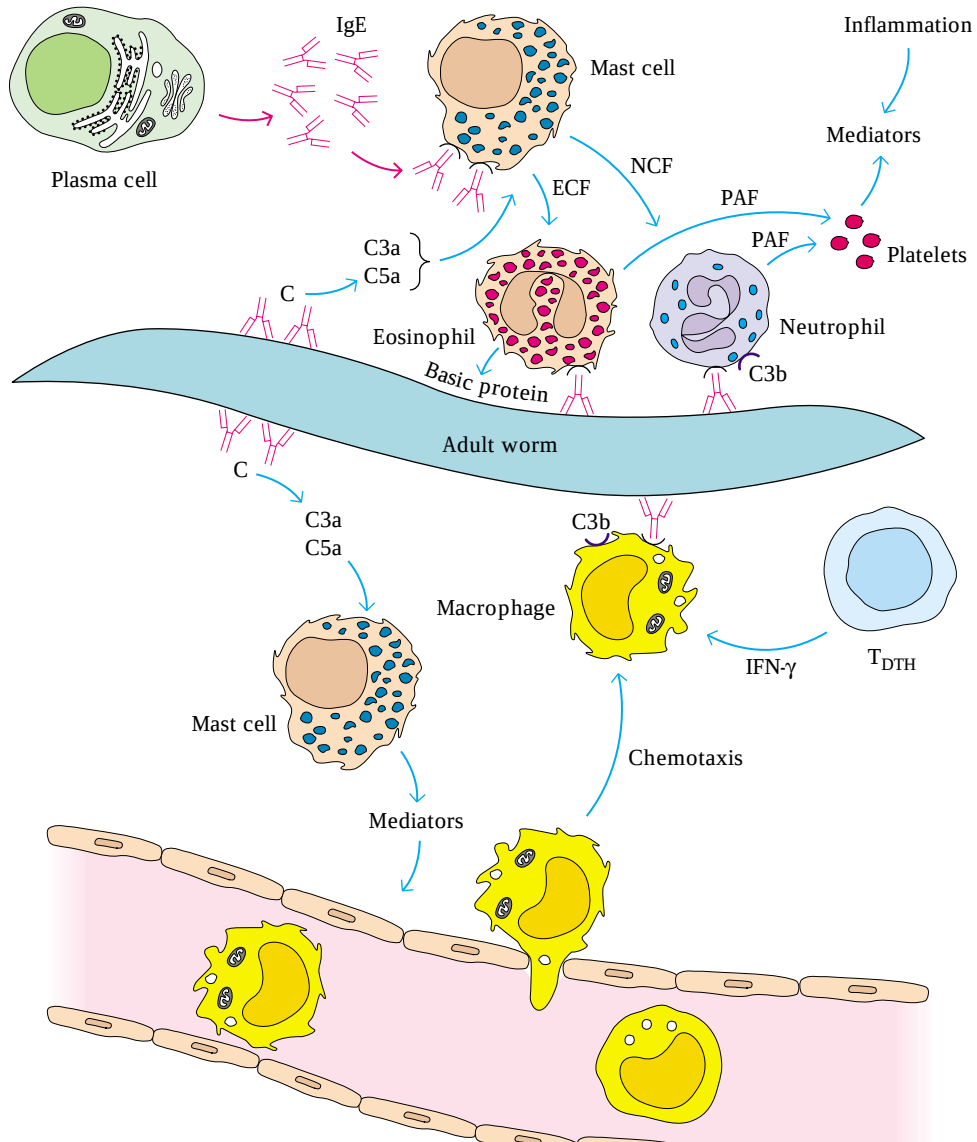


FIGURE 17-13 Overview of the immune response generated against *Schistosoma mansoni*. The response includes an IgE humoral component (top) and a cell-mediated component involv-

ing CD4⁺ T cells (bottom). C = complement; ECF = eosinophil chemotactic factor; NCF = neutrophil chemotactic factor; PAF = platelet-activating factor.

Emerging Infectious Diseases

A cursory glance at the current offerings in your local bookstore or video rental store brings into focus the preoccupation of the public and the press with new infectious agents. Several times a year, it seems, we hear about a new virus or bacterium that arises in a particular location and causes severe illness or death in a population. Newly described

pathogens are referred to as emerging pathogens. Some of the emerging pathogens that have been described since the early 1970s appear in Table 17-4. These new pathogens are thought to have emerged within the recent past. HIV is an example of a newly emerged pathogen.

In other instances, diseases that were no longer causing widespread infection suddenly began to infect an ever-larger number of individuals. These are referred to as “re-emerging”

TABLE 17-4 Emerging pathogens recognized since 1973

Year	Pathogen	Disease
1973	Rotavirus	Major cause of infantile diarrhea globally
1974	Hepatitis C	Non-A, non-B hepatitis commonly transmitted via transfusions
1976	<i>Cryptosporidium parvum</i>	Acute chronic diarrhea
1977	Ebola virus	Ebola haemorrhagic fever
	<i>Legionella pneumophila</i>	Legionnaires' disease
	Hantavirus	Haemorrhagic fever with renal syndrome
	<i>Campylobacter jejuni</i>	Enteric diseases distributed globally
1980	Human T-lymphotrophic virus I (HTLV-1)	T-cell lymphoma
1981	Toxin-producing strains of <i>Staphylococcus aureus</i>	Toxic shock syndrome
1982	<i>Escherichia coli</i> 0157:H7	Haemorrhagic colitis
	HTLV-II	Hairy cell leukemia
	<i>Borrelia burgdorferi</i>	Lyme disease
1983	HIV	AIDS
	<i>Helicobacter pylori</i>	Peptic ulcers
1988	Hepatitis E	Enteric non-A, non-B hepatitis
1990	Guanarito virus	Venezuelan haemorrhagic fever
1991	<i>Encephalitozoon hellem</i>	Conjunctivitis, disseminated disease
1992	<i>Vibrio cholerae</i> 0139	New strain of epidemic cholera
	<i>Bartonella henselae</i>	Cat scratch disease
1994	Sabia virus	Brazilian haemorrhagic fever
1995	Human herpes virus-8	Associated with Kaposi sarcoma in AIDS patients
1996	TSE causing agent	New variant of Creutzfeldt-Jakob disease (mad cow disease)
1997	Influenza A subtype H5N1	Avian influenza
1999	Influenza A subtype H9N2	New strain of human influenza
	Nipah virus	Encephalitis
	West Nile virus	Encephalitis

SOURCE: Adapted from M. F. Good et. al., 1988, *Annual Review of Immunology*, Vol. 6.

infectious diseases. The re-emergence of these diseases should not be surprising if we consider that bacteria can adapt to living in almost any environment. If they can adapt to living at the high temperatures of the thermal vents deep within the oceans, it is not difficult to accept that they can evolve to evade antimicrobial drugs. (An additional risk from intentionally disseminated diseases is discussed in the Clinical Focus.)

Tuberculosis is a well-known re-emerging disease. Fifteen years ago, public health officials were convinced that tuberculosis would soon disappear as a major health consideration in the United States. Then, because of a number of events, including the AIDS epidemic, thousands of infected individuals developed TB strains resistant to the conventional battery of antibiotics. These individuals then passed on the newly emerged, antibiotic-resistant strains of *M. tuberculosis* to others. While the rate of infection with *M. tuberculosis* in the United States increased sharply during the early part of the 1990s, by 1995 the incidence had begun to decline again. However, the worldwide incidence of the disease is still in-

creasing, and the World Health Organization predicts that, between 1998 and 2020, one billion more people will become infected and over 70 million will die from this disease if preventive measures are not adopted.

Another re-emerging disease is diphtheria. This disease was almost non-existent throughout Europe in recent years because of vaccination; in 1994, however, scattered cases were reported in some of the republics of the former Soviet Union. By 1995, there were over 50,000 cases reported in the same region, and thousands died from diphtheria infection. The social upheaval and instability that came with the breakup of the Soviet Union was almost certainly a major factor in the re-emergence of this disease, because of the resultant lapses in public health measures—perhaps most important was the loss of immunization programs. Since 1995, immunization programs have been re-established and the trend has reversed, with only 13,687 cases of diphtheria reported in Russian republics in 1996, 6932 in 1998, and 1573 in 2000.

Other diseases have appeared seemingly from nowhere and, as far as we know, are new pathogens. These include



CLINICAL FOCUS

The Threat of Infection from Potential Agents of Bioterrorism

The use of human pathogens as weapons has a long history. Lord Jeffery Amherst used smallpox against native American populations before the Revolutionary War, and there are reports of attempts to spread plague and anthrax in both the distant and recent past. A few years ago, members of a dissident cult in Oregon introduced salmonella into the salad bars of several restaurants in an attempt cause sickness and death. The more recent discovery of anthrax spores mailed to congressmen and news offices accelerates our interest in possible agents of bioterrorism.

Pathogens and toxins with potential for use as weapons are called “select agents” and include bacteria, bacterial toxins, and certain viruses (see table). The threat from such agents depends on both the severity of the disease it causes and the ease with which it can be disseminated. For example, Ebola virus causes a fulminating hemorrhagic disease, but

spread of the virus requires direct contact with infected fluids. More worrisome are pathogens that can be spread by aerosol contact, such as anthrax, and toxins that can be added to food or water supplies, such as botulinum toxin.

It is ironic that one of the most feared bioterrorism agents is smallpox, the target of the first vaccine. Smallpox is caused by the virus *Variola major*; 30% or more of those infected with this virus die within a month of exposure. Survivors may be horribly scarred. Smallpox can spread rapidly, even before symptoms are visible. As described in Chapter 1, the vaccine for smallpox is a virus (*Vaccinia*) related to variola, which in most cases causes a localized pustule that resolves within 3 weeks. Smallpox disappeared as a consequence of widespread vaccination—the last reported case of natural infection was in 1977. As the disease was eradicated, vaccination was discontinued. In the United States, vaccination ceased in 1972. Production of the vaccine ceased and the remaining doses were put into storage.

Reasons for discontinuing smallpox vaccination include side effects that affect approximately 40 individuals per million vaccinees. These can be life threatening and take the form of encephalitis or disseminated skin infection. In addition, recently vaccinated individuals can spread the virus to others, especially those with compromised immunity. The occasional negative reactions to vaccinia can be treated by the administration of immunoglobulin isolated from sera of persons previously vaccinated, but this so-called

Vaccinia IG, or VIG, is no longer produced and little remains available. Facing the threat of smallpox as a bioterrorism agent means that vaccination must be reconsidered. It is unlikely that the vaccine produced today will be the same one used earlier. Vaccine was produced by infection of the scarified skin of calves and virus was collected by scraping the infected area. Most likely a new vaccine candidate will be produced under controlled conditions in a tissue-cultured cell line that is certified free of any contaminating viruses. Furthermore, the actual virus used may be a more highly attenuated form of vaccinia. Stocks of VIG must be replenished before a mass vaccination effort is begun.

Most of the viruses on the select agent list are not easy to disseminate. Agents of bioterrorism prepared in a form that allows easy dispersal are referred to as *weaponized*. While nightmare scenarios include customized viral agents engineered in the laboratory, the more likely weaponized pathogens are bacteria. An accidental release of anthrax (*Bacillus anthracis*) in Sverdlovsk in the former Soviet Union infected 79 persons, of whom 68 died, pointing to the deadly potential of this organism. In late 2001, mail containing anthrax (see the accompanying figure) infected a number of persons in multiple postal centers as the letters progressed to their destinations, giving a glimpse of how widely and rapidly a bioweapon might be spread through modern infrastructure.

Bacillus anthracis is a common veterinary pathogen, and like smallpox was the subject of early vaccine efforts, in this case by Louis Pasteur. Human infection was found mainly in those working with hair or hides from animals, especially goats. Infection occurs via three different routes:

- Inhalation causes severe flu-like illness with high mortality unless diagnosed and treated immediately

Category A agents of bioterrorism

Anthrax (*Bacillus anthracis*)
 Botulism (*Clostridium botulinum* toxin)
 Plague (*Yersinia pestis*)
 Smallpox (*Variola major*)
 Tularemia (*Francisella tularensis*)
 Viral hemorrhagic fevers
 (filoviruses [e.g., Ebola, Marburg]
 and arenaviruses [e.g., Lassa,
 Machupo])

such pathogens as the widely publicized Ebola virus and *Legionella pneumophila*, the bacterial causative agent for Legionnaires' disease. Ebola was first recognized after an outbreak in Africa, in 1976. By 1977, the virus that causes this

disease had been isolated and classified as a filovirus, a type of RNA virus that includes Marburg virus, a close relative of Ebola. Ebola causes a particularly severe hemorrhagic fever that kills more than 50% of those infected. Because of the

with antibiotics such as penicillin, doxycycline or ciprofloxacin.

- Cutaneous exposure results in skin lesions with characteristic black deep eschar. Cutaneous anthrax has a 20% mortality if untreated, but usually responds to antibiotics.
- Gastrointestinal exposure results in ulcers in the ileum or cecum, bloody diarrhea, and sepsis, and is nearly always fatal because of difficulty in diagnosis.

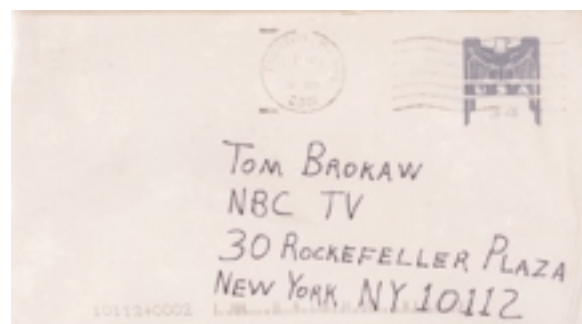
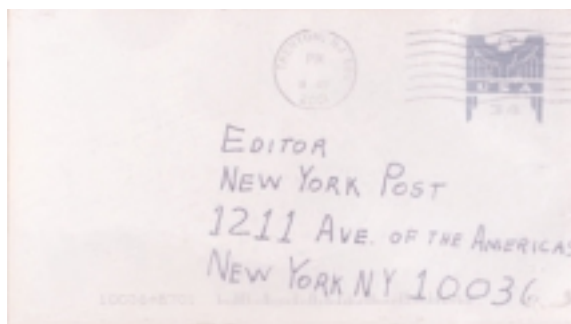
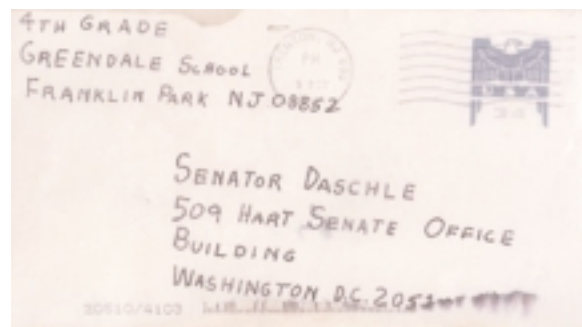
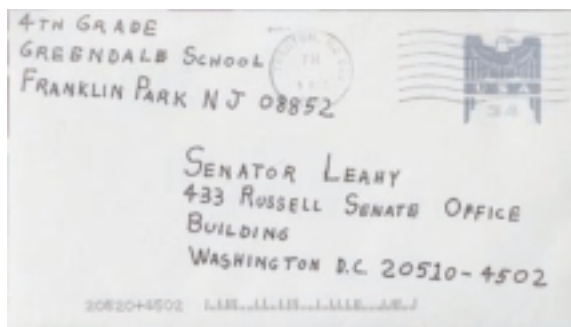
B. anthracis is particularly deadly because the bacillus forms spores that are quite stable to heat, dryness, sunlight, and other factors that normally limit pathogen viability. It is relatively simple to induce spore formation, and it is spores that are

used as bioweapons. Primate studies suggest that inhalation of 2500 to 55,000 spores will cause fatal disease, although the number is controversial. Victims may have flu-like symptoms; a chest x-ray will reveal a characteristic widening of the mediastinum, and blood smears will show gram-positive bacilli. Since prompt diagnosis and treatment is required for survival it is essential that medical personnel recognize the disease.

A vaccine has been developed for anthrax, but its use has been limited to the military. The present preparation is a filtrate from cultures of a non-spore-forming strain of *B. anthracis*. Newly proposed vaccines take advantage of the information gained from basic studies of the mechanism used by the organism to infect target cells, as well as our understanding of the

structure and function of anthrax-derived proteins. The major protein involved in infection is the so-called protective antigen, or PA, which pairs with either edema factor (EF) or lethal factor (LF) to cause productive infection. Antibodies that target the binding site on PA for either LF or EF are being developed as the next generation of vaccines against anthrax.

The threat from select agents of bioterrorism, like that from emerging diseases, is being addressed by careful attention to unusual infection events, and by increased study of agents that lend themselves to weaponization. Research to determine the efficacy of various treatments and the windows of immunity that result from administration of antitoxins have risen to top priority in the U.S. following the events of September 11, 2001.



Letters to congressmen and news agencies that contained anthrax spores.

Courtesy of the Federal Bureau of Investigation.

severity of disease and the rapid progression to death after the initial appearance of symptoms, this virus has received a great deal of attention. However, while the risk of death is very high if you are infected with Ebola, it is fairly easy to

control the spread of the virus. Through isolation of infected individuals, hospital workers and medical personnel can be protected. In such ways, the spread of Ebola virus has been contained during the two most recent outbreaks.

Another emerging disease recently described is Legionnaires' disease, a virulent pneumonia first reported in 221 individuals who had attended an American Legion convention in Philadelphia in 1976. Of the 221 afflicted, 34 died from the infection. The organism causing the disease was not known, but further investigation led to the identification of a bacterium that was named *Legionella pneumophila*. This bacterium proliferates in cool, damp areas and can be found in the condensing units of large commercial air-conditioning systems. The air-conditioning system can produce an aerosol that contains the bacteria, thus spreading the infection throughout the area served by the unit. This was determined to be the source of the bacteria at the 1976 convention in Philadelphia. Because the hazard of such aerosols is now recognized, improved design of air-conditioning and plumbing systems has greatly reduced the incidence of the disease.

In 1999, a new virus emerged in the Western Hemisphere. West Nile virus was first isolated in Uganda in 1937, but until recently it was not found outside Africa and western Asia. In 1999, West Nile virus was found in the New York City metropolitan area and by summer 2002, incidence of West Nile virus was reported in all but a few states in the Northwest, indicating a rapid spread of this virus in a short period of time. West Nile virus belongs to a group of viruses known as flaviviruses, a group of viruses spread by insects, usually mosquitoes. The most common reservoir of the virus is birds. Crows are particularly sensitive to infection by this virus. Mosquitoes bite an infected bird and, most commonly, the virus-infected mosquito passes the virus to another bird. However, on occasion, the mosquito bites a human, infecting that individual with the virus. Since West Nile is not contagious between humans, it cannot be spread among human populations. In all but a small proportion of humans, West Nile infection does not cause disease. Only in individuals with compromised immune function is the virus a health hazard. Because this virus can cross the blood–brain barrier in compromised individuals, it can cause life threatening encephalitis or meningitis and this is the usual cause of death. Between 1999 and 2001, West Nile caused 18 deaths and sickened 131 others. By September 6, 2002, 954 cases of West Nile had been reported to CDC and 43 people had died in the year 2002. These statistics indicate that West Nile is spreading and is a virus to monitor carefully. Current public health control mechanisms include education of the public regarding mosquito control.

Why are these new diseases emerging and others re-emerging? One reason suggested by public-health officials is the crowding of the world's poorest populations into very small places within huge cities. Another factor is the great increase in international travel; it is now easy to traverse the globe in a very short time, making it possible for an individual to become infected on one continent and then spread the disease to another continent tens of thousands of miles distant. Other features of modern life that may contribute include mass distribution of food, which exposes large populations to potentially contaminated food, and unhygienic food preparation. The World Health Organization and the U. S. Center for

Disease Control both actively monitor new infections and work together closely to detect and identify new infectious agents and to provide up-to-date information for travelers to parts of the world where such agents may pose a risk.

SUMMARY

- Innate immune responses form the initial defense against pathogens. These include physical barriers, such as skin, as well as the nonspecific production of complement components and certain cytokines in response to infection by various pathogens.
- The immune response to viral infections involves both humoral and cell-mediated components. Antibody to a viral receptor can block viral infections of host cells. However, a number of viruses, including influenza, are able to mutate their receptor molecules and thus evade the humoral antibody response (see Figure 17-6). Once a viral infection has been established, cell-mediated immunity appears to be more important than humoral.
- The immune response to extracellular bacterial infections is generally mediated by antibody. Antibody can induce localized production of immune effector molecules of the complement system, thus facilitating development of an inflammatory response. Antibody can also activate complement-mediated lysis of the bacterium, neutralize toxins, and serve as an opsonin to increase phagocytosis. Some bacteria secrete protease enzymes that cleave IgA dimers, thus reducing the effectiveness of IgA in the mucous secretions. Other bacteria escape phagocytosis by producing surface capsules or proteins that inhibit adherence to phagocytes, by secreting toxins that kill phagocytes, or by their ability to survive within phagocytes. Host defense against intracellular bacteria depends largely on CD4⁺ T-cell-mediated responses.
- Both humoral and cell-mediated immune responses have been implicated in immunity to protozoan infections. In general, humoral antibody is effective against blood-borne stages of the protozoan life-cycle, but once protozoans have infected host cells, cell-mediated immunity is necessary. Protozoans escape the immune response through several mechanisms. Some—notably, *Trypanosoma brucei*—are covered by a glycoprotein coat that is constantly changed by a genetic-switch mechanism (see Figure 17-12). Others (including *Plasmodium*, the causative agent of malaria) slough off their glycoprotein coat after antibody has bound to it.
- Helminths are large parasites that normally do not multiply within cells. Because few of these organisms are carried by an affected individual, immune-system exposure to helminths is limited; consequently, only a low level of immunity is induced. Although helminths generally are attacked by antibody-mediated defenses, these may be ineffective. A cell-mediated response by CD4⁺ T cells plays a critical role in the response to *Schistosoma*.

- Emerging and re-emerging pathogens include some that are newly described and others that had been thought to be controlled by public-health practices. Factors leading to the emergence of such pathogens include increased travel and intense crowding of some populations.

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USEFUL WEB SITES

<http://www.cdc.gov/ncidod/>

National Center for Infectious Diseases home page—a superb site for monitoring emerging diseases. This is a subdivision of the Centers for Disease Control (CDC), and links to CDC are found at this site.

<http://www.niaid.nih.gov/>

National Institute of Allergy and Infectious Diseases home page—NIAID is the NIH institute that sponsors research in infectious diseases, and its Web site provides a number of links to other relevant sites.

<http://www.who.int/>

World Health Organization home page—the international organization that monitors infectious diseases worldwide.

<http://www.hopkins-biodefense.org/>

The Johns Hopkins University Center for Civilian Biodefense Strategies; in particular, the link entitled “Dark Winter: A bioterrorism exercise” is excellent.

Study Questions

CLINICAL FOCUS QUESTION VIG is used to treat individuals who display complications following administration of the smallpox vaccine. Where is VIG obtained and why is it frequently an effective treatment?

1. The effect of the MHC on the immune response to peptides of the influenza virus nucleoprotein was studied in H-2^b mice that had been previously immunized with live influenza virions. The CTL activity of primed lymphocytes was determined by in vitro CML assays using H-2^k fibroblasts as target cells. The target cells had been transfected with different H-2^b class I MHC genes and were infected either with live influenza or incubated with nucleoprotein synthetic peptides. The results of these assays are shown in the table below.
 - a. Why was there no killing of the target cells in system A even though the target cells were infected with live influenza?
 - b. Why was a CTL response generated to the nucleoprotein in system C, even though it is an internal viral protein?



- c. Why was there a good CTL response in system C to peptide 365–380, whereas there was no response in system D to peptide 50–63?
- d. If you were going to develop a synthetic peptide vaccine for influenza in humans, how would these results obtained in mice influence your design of a vaccine?

Target cell (H-2 ^K fibroblast)	Test antigen	CTL activity of influenza- primed H-2 ^b lymphocytes (% lysis)
(A) Untransfected	Live influenza	0
(B) Transfected with class I D ^b	Live influenza	60
(C) Transfected with class I D ^b	Nucleoprotein peptide 365–380	50
(D) Transfected with class I D ^b	Nucleoprotein peptide 50–63	2
(E) Transfected with class I K ^b	Nucleoprotein peptide 365–380	0.5
(F) Transfected with class I K ^b	Nucleoprotein peptide 50–63	1

2. Describe the nonspecific defenses that operate when a disease-producing microorganism first enters the body.
3. Describe the various specific defense mechanisms that the immune system employs to combat various pathogens.
4. What is the role of the humoral response in immunity to influenza?
5. Describe the unique mechanisms each of the following pathogens has for escaping the immune response: (a) African trypanosomes, (b) *Plasmodium* species, and (c) influenza virus.
6. M. F. Good and coworkers analyzed the effect of MHC haplotype on the antibody response to a malarial circumsporozoite (CS) peptide antigen in several recombinant congenic mouse strains. Their results are shown in the table below.
- a. Based on the results of this study, which MHC molecule(s) serve(s) as restriction element(s) for this peptide antigen?

- b. Since antigen recognition by B cells is not MHC restricted, why is the humoral antibody response influenced by the MHC haplotype?

Strain	H-2 alleles					Antibody response to CS peptide
	K	IA	IE	S	D	
B10.BR	k	k	k	k	k	<1
B10.A (4R)	k	k	b	b	b	<1
B10.HTT	s	s	k	k	d	<1
B10.A (5R)	b	b	k	d	d	67
B10	b	b	b	b	b	73
B10.MBR	b	k	k	k	q	<1

SOURCE: Adapted from M. F. Good et al., 1988, *Annu. Rev. Immunol.* 6:633.

7. Fill in the blanks in the following statements.
- a. The current vaccine for tuberculosis consists of an attenuated strain of *M. bovis* called _____.
- b. Variation in influenza surface proteins is generated by _____ and _____.
- c. Variation in pilin, which is expressed by many gram-negative bacteria, is generated by the process of _____.
- d. The mycobacteria causing tuberculosis are walled off in granulomatous lesions called _____, which contain a small number of _____ and many _____.
- e. The diphtheria vaccine is a formaldehyde-treated preparation of the exotoxin, called a _____.
- f. A major contribution to nonspecific host defense against viruses is provided by _____ and _____.
- g. The primary host defense against viral and bacterial attachment to epithelial surfaces is _____.
- h. Two cytokines of particular importance in the response to infection with *M. tuberculosis* are _____, which stimulates development of T_H1 cells, and _____, which promotes activation of macrophages.
8. Discuss the factors that contribute to the emergence of new pathogens or the re-emergence of pathogens previously thought to be controlled in human populations.

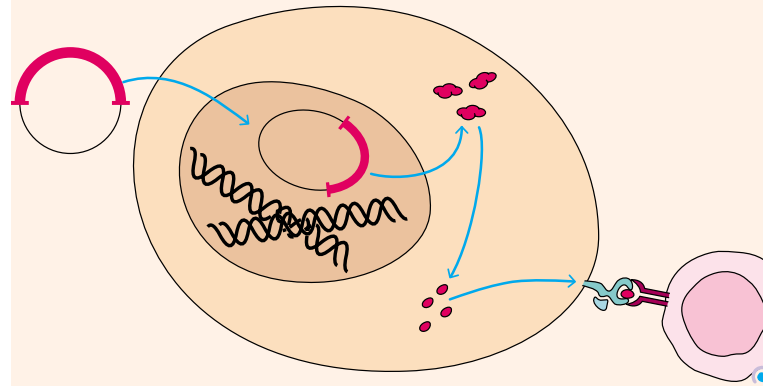
chapter 18

Vaccines

THE DISCIPLINE OF IMMUNOLOGY HAS ITS ROOTS IN the early vaccination trials of Edward Jenner and Louis Pasteur. Since those pioneering efforts, vaccines have been developed for many diseases that were once major afflictions of mankind. The incidence of diseases such as diphtheria, measles, mumps, pertussis (whooping cough), rubella (German measles), poliomyelitis, and tetanus has declined dramatically as vaccination has become more common. Clearly, vaccination is a cost-effective weapon for disease prevention. Perhaps in no other case have the benefits of vaccination been as dramatically evident as in the eradication of smallpox, one of mankind's long-standing and most terrible scourges. Since October 1977, not a single naturally acquired smallpox case has been reported anywhere in the world. Equally encouraging is the predicted eradication of polio. The last recorded case of naturally acquired polio in the Western Hemisphere occurred in Peru in 1991, and the World Health Organization (WHO) predicts that paralytic polio will be eradicated throughout the world within the next few years. A new addition to the weapons against childhood disease is a vaccine against bacterial pneumonia, a major cause of infant death.

A crying need remains for vaccines against other diseases. Every year, millions throughout the world die from malaria, tuberculosis, and AIDS, diseases for which there are no effective vaccines. It is estimated by the World Health Organization that 16,000 individuals a day, or 5.8 million a year, become infected with HIV-1, the virus that causes AIDS. An effective vaccine could have an immense impact on the control of this tragic spread of death and disaster. In addition to the challenges presented by diseases for which no vaccines exist, there remains the need to improve the safety and efficacy of present vaccines and to find ways to lower their cost and deliver them efficiently to all who need them, especially in developing countries of the world. The WHO estimates that millions of infant deaths in the world are due to diseases that could be prevented by existing vaccines (see Clinical Focus).

The road to successful development of a vaccine that can be approved for human use, manufactured at reasonable cost, and efficiently delivered to at-risk populations is costly, long, and tedious. Procedures for manufacture of materials that can be tested in humans and the ways they are tested in clinical trials are regulated closely. Even those candidate vaccines that survive initial scrutiny and are approved for use in human trials are not guaranteed to find their way into



Vaccination with DNA

- Active and Passive Immunization
- Designing Vaccines for Active Immunization
- Whole-Organism Vaccines
- Purified Macromolecules as Vaccines
- Recombinant-Vector Vaccines
- DNA Vaccines
- Multivalent Subunit Vaccines

common usage. Experience has shown that not every vaccine candidate that was successful in laboratory and animal studies prevents disease in humans. Some potential vaccines cause unacceptable side effects, and some may even worsen the disease they were meant to prevent. Live virus vaccines pose a special threat to those with primary or acquired immunodeficiency (see Chapter 19). Stringent testing is an absolute necessity, because vaccines will be given to large numbers of well persons. Adverse side effects, even those that occur at very low frequency, must be balanced against the potential benefit of protection by the vaccine.

Vaccine development begins with basic research. Recent advances in immunology and molecular biology have led to effective new vaccines and to promising strategies for finding new vaccine candidates. Knowledge of the differences in epitopes recognized by T cells and B cells has enabled immunologists to begin to design vaccine candidates to maximize activation of both arms of the immune system. As differences in antigen-processing pathways became evident, scientists began to design vaccines and to use adjuvants that maximize antigen presentation with class I or class II MHC molecules.